

A Three-Dimensional Hydrodynamic and Transport Model for Lake Environments

By

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Dedicated to my wife Carmen

ABSTRACT

The goal of this work was to construct a comprehensive, efficient and accurate three-dimensional (3-D) computational tool that could be used to improve our understanding of the physical processes of water circulation, transport and mixing in lakes. The availability of a relatively large number of three-dimensional flow solvers suggested at the outset that there was a more pressing need for formulating a methodology by which such codes could be tested and evaluated, rather than the development of a new code. That methodology, once developed, was applied to compare a finite difference and a finite element flow solver for their suitability for applications in lakes. The 3-D semi-implicit leapfrog-trapezoidal finite difference solver was selected as the more suitable on the basis of the results of the test methodology. The model was extensively modified to incorporate features needed for the accurate simulation of lake hydrodynamic phenomena. The extended model uses spatially and temporally variable wind fields and calculates mixing coefficients with a high-order turbulence closure model. A transport model for temperature was implemented, which uses flux limiters and directional splitting methods to discretize the 3-D advection operator and calculates heat fluxes at the free surface based on meteorological information and prognostic variables. A stand-alone Lagrangian particle-tracking model was also developed to visualize circulation patterns and simulate the transport of particulate matter.

The model was applied to two lakes of very contrasting internal dynamics. The first, Clear Lake, is a multi-basin, shallow, polymictic and eutrophic lake in northern

California. The second, Lake Tahoe, is a deep monomictic and oligotrophic lake straddling the border between California and Nevada. The internal dynamics of Clear Lake is characterized by a cycle of setup and relaxation of horizontal density gradients controlled by strong, periodic wind events. The model was used in Clear Lake to describe, together with field observations, its large-scale circulation, with particular emphasis in the Oaks arm. The dominant features in the internal dynamics of Lake Tahoe are the basin-scale internal waves. In Lake Tahoe, the model was used to study the internal waves in winter 1999-2000. In both cases, the model produced an excellent match to the measured data. As a tool, the 3-D model provided a means of interpreting the data sets for both lakes, for hypothesis testing and for the confirmation of previously unidentified transport phenomena.

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CHAPTER 1

INTRODUCTION

1.1. Background

Water, with its contained substances and physical properties, provides an environment for a great number of plants and animals and is a vital resource for human use and consumption. Most of the water on the planet exists in liquid form in the oceans. Of the remaining portion, 1.7% is in the form of ice at the poles, 1.7% in groundwater sources and only 0.1% in surface and atmospheric systems. The word “lake” refers to large inland bodies of fresh or salty water occupying depressions, called basins, of the Earth’s surface. Even though they only contain 0.014% of the total water on the planet, lakes supply probably 80% of the world’s drinking water, constitute major tourist attractions and support important fisheries (Imberger 1998).

Lakes, like mountains and rivers, have life histories that exhibit varying stages from youth through maturity to old age. As they age, they undergo a natural process of enrichment called *eutrophication* that occurs over geologic time spans. A ‘young’ lake is generally *oligotrophic*, as it contains few nutrients and sustains only a small biomass. An ‘old’ lake is eutrophic and will suffer from such symptoms as high concentrations of dissolved materials, excessive plant production, reduction in the clarity of the water or the depletion of oxygen depletion in the deeper waters. Man’s concern is not primarily with this natural process but with the anthropogenic acceleration of eutrophication. This

so-called *cultural eutrophication* occurs when the activity of humans causes excessive amounts of nutrients¹ to be added to the lake system and is a problem of excessive biomass (Henderson-Sellers 1984). Man's activity may also result in elevated levels of toxic substances, such as heavy metals, in lakes and the consequent deterioration of the ecosystem. Through the slow process of *bioaccumulation* and *biomagnification*, minute and apparently harmless concentrations of such substances can engender toxic levels in organisms that are high in the food chain.

An evaluation of the environmental effects of the anthropogenic activities on lakes requires a deep understanding of the physical processes of water circulation and mixing. Water motions in large natural lakes² primarily result from the combined forcing of wind stress, rotation of the Earth, and density gradients. In fresh lakes³ the density field is mainly determined by the temperature distribution, which represents a balance among many processes, including air-water exchange, transport by currents and vertical turbulent mixing. In reservoirs and lakes with low residence times, the inflow and outflow dynamics may additionally exert a considerable influence on the physical behavior of the lake. The vertical thermal structure of a lake, which shall be referred to as stratification, is probably the most important physical/chemical/biological factor that determines the quality of its waters. Its importance is that it induces mechanical stability,

¹ Nutrients are defined as anything that is necessary for the promotion of growth, the repair of tissue, or is an energy source for an organism.

² Lakes may be classified according to the origin of their basins into natural, and manmade lakes or reservoirs. *Natural lakes* are the focus of this thesis, and they originate from glacial activities, as craters of extinct volcanoes, or through the various activities of rivers, to name just a few. The term *reservoir*, on the other hand, refers to manmade lakes created by the construction of dams across rivers or streams. This classification entails another fundamental difference in that the residence time, defined as the volume of the lake divided by the inflow rate, is usually shorter for reservoirs (Henderson-Sellers 1984).

³ Lakes can be alternatively classified according to the chemical composition of their waters into fresh and saline lakes. In saline lakes, both water and salt concentration determine the density.

by which it damps the turbulent mixing, restricting the vertical exchange of oxygen, nutrients, etc that impact the biomass productivity of the lake ecosystem.

The understanding of the processes of water circulation and mixing has developed considerably during the last few years, partly because of the increased power of observation by engineers and scientists using new and sophisticated instruments. Field measurement, however, cannot describe completely those processes, given their high degree of spatial and temporal variability. The modeling approach, conjunctively used with field observations, provides a powerful tool for studying the spatial and temporal structure of lake hydrodynamic processes. In the past, and even nowadays, many studies on lake circulation have used two-dimensional models in which the three-dimensional (hereafter referred to as 3-D) governing equations are averaged either in the vertical direction (depth-averaged) or the lateral direction (laterally-averaged). They are simple to program and not very computationally demanding to use, though their applicability is limited by fairly restrictive assumptions (Smith 1997). Vertically averaged models, for example, require the vertical density profile be well mixed. Most lakes, however, undergo periods of time when they are stratified⁴. Even if the lake is isothermal, wind-driven currents can show significant variations in the vertical direction that cannot be described with depth-averaged models. Laterally averaged models, on the other hand, require homogeneity of the hydrodynamic variables across the lake, which entails the neglect of wind forcing variability and the influence of the Earth's rotation. Advances in the design

⁴ Most lakes undergo a regular pattern of seasonal stratification. Lakes are classified as dimictic, monomictic, polymictic, ... depending on the number of isothermal periods in the annual stratification cycle. A lake will be amictic if it rarely stratifies. Meromictic lakes remain unmixed, and this state is usually due to the presence of a layer of water that contains a high percentage of dissolved salts (e.g. Mono Lake in Northern California).

of numerical algorithms and the availability of more powerful computers in recent years has made possible the development and extensive use of three-dimensional models for lake hydrodynamic studies.

1.2. Review of three-dimensional models

The earliest 3-D circulation models were developed in the 1960s for application to ocean and lakes. They were based on the assumption referred to as the *rigid lid* approximation in which the free surface location is constant and the surface gravity waves (including ocean tides and lake seiches) are filtered out of the solutions. For purposes of studying large-scale oceanic circulation on a coarse numerical grid, rigid-lid models are generally adequate and are still being developed for modern applications. The correct simulation of the transient response of a lake, however, requires a model that solves directly for the variable free surface (*free surface model*). Sheng et al. (Sheng *et al.* 1978) compared a rigid-lid model with a free surface model and showed that the rigid-lid model gave poor results during periods of active seiching of Lake Erie under spatially and temporally varying winds. The rigid-lid model developed by Huang and Sloss (Huang and Sloss 1981) for Lake Ontario gave reasonably good results for the mean monthly circulation. Due to the rigid lid approximation, however, the model showed a much faster lake response to wind events and exaggerated inertial motions. The predicted instantaneous temperature distributions in the near-surface layers were generally 1 to 3 °C higher than observations, indicating weaker upwelling events (Sheng 1986).

Early free surface models solved the primitive equations⁵ with an *explicit time-marching scheme*⁶ (Freeman *et al.* 1972, Leendertse *et al.* 1973, Leendertse and Liu 1975). Although the numerical algorithm is straightforward, the numerical time step is severely restricted by the time for the surface gravity wave to propagate between two adjacent horizontal grid points, a limitation known as the Courant-Friedrich-Lewy (CFL or Courant) stability condition for the gravity waves. Leendertse and Liu (Leendertse and Liu 1975), for example, used a time step of 15 seconds in their study of San Francisco Bay. Their model was described in a series of Rand Corporation reports (Leendertse *et al.* 1973, Leendertse and Liu 1975), and it was noteworthy because it was very complete in the equation formulation, including non-linear friction and advection, the effects of Earth's rotation, horizontal shear stresses, arbitrary bottom topography, coupled transport of salt with the hydrodynamics, density-gradient forcing and a physically realistic vertical turbulent parameterization (Leendertse and Liu 1975). The model of Freeman and others (Freeman *et al.* 1972), developed for application to the Great Lakes, is similar to that of the Rand Corporation, but it used a “normalized” or “stretched” vertical coordinate, referred to in the literature as the σ -coordinate. The σ -transformation maps the physical domain with a variable depth into one with a uniform depth ranging from $\sigma = 0$ at the surface to $\sigma = -1$ at the bottom, and it is nowadays widely used in 3-D models of flows over variable bathymetry (e.g. Blumberg and Mellor 1987, Uittenbogaard *et al.* 1992). It

⁵ The primitive equations consist of the momentum equations, the continuity equation and active scalar transport equation supplemented with an equation of state linking density with the active scalar concentration, as described in chapter 2.

⁶ An explicit numerical scheme is one in which the unknown hydrodynamic variables at any spatial point at time level $n+1$ are calculated entirely from known values at neighboring points from one or more previous time levels.

can lead, however, to severe numerical errors in regions of rapidly changing depth unless a high level of grid refinement is used (Smith 1997).

In attempting to reduce the computational cost of 3-D, explicit, finite difference models while retaining free-surface effects, the so-called *mode-splitting technique* was introduced in the 1970s. Mode splitting consists of separating the 3-D governing equations into a set of vertically averaged equations (that describe the external or barotropic mode) and a set of equations governing the vertical structure of the flow (internal or baroclinic mode). The propagation of surface gravity waves is governed by the external-mode equations. The time discretization for the gravity-wave terms in the external-mode can be either explicit or implicit⁷. The internal mode equations are on the other hand treated explicitly, except for the vertical diffusion terms that are usually treated implicitly to avoid a time-step limitation in shallow water. Simons (Simons 1974) was probably the first to use the mode-splitting technique when he modeled Lake Ontario. This method has since been used in many other applications (e.g. Blumberg and Mellor 1987), and it is used in the majority of 3-D shallow water models that are now operational (Smith 1997). The mode-splitting approach, though widely accepted in 3-D modeling, has several important drawbacks (Smith 1997). If an explicit time discretization is used for both modes with time-splitting, care must be taken to ensure that the external mode (2-D) velocities are the exact depth average of the internal mode (3-D) velocities to avoid instabilities. If an implicit time stepping scheme is used for the external mode, the time-splitting errors can be eliminated if the same time step is used in

⁷ Implicit numerical schemes are those in which the unknown hydrodynamic variables at a given spatial point at time level $n+1$ depend on the unknown variables at neighboring points at the same time level $n+1$. The unknown variables are then obtained by the simultaneous solution of a system of algebraic equations.

the external and internal modes. Another problem is associated with the consistent representation of the bottom frictional stress between the internal and external modes (see Smith 1997).

An alternative to mode splitting, adopted for example in the latest versions of the Rand Corporation 3-D model (Leendertse 1989) or the 3-D TRISULA model from Delft Hydraulics of the Netherlands (Uittenbogaard *et al.* 1992), is the *alternating-direction-implicit* (ADI) method. These models are 3-D extensions of the well-known ADI methods that have been successfully applied in 2-D (see page 149 in Vreugdenhil 1994). These models are formulated with second-order accuracy and treat the vertical diffusion term implicitly. Given that they are unconditionally stable, one may use large time steps. Other alternative approaches to the mode splitting is the time-splitting method of de Goede (de Goede 1991) and the semi-implicit scheme of Casulli and Cheng (Casulli and Cheng 1992), later incorporated in other 3-D models (Smith 1997, Hodges *et al.* 2000).

The *time-splitting* scheme of de Goede (de Goede 1991) consists of two stages, the first of which requires the solution of a large number of independent tri-diagonal systems of equations involving the implicit treatment of vertical diffusion. In the second stage, the terms describing the propagation of surface gravity waves (the water surface pressure gradient in the momentum equations and the velocity divergence in the continuity equation) are treated implicitly, resulting in a five-diagonal matrix system for the water surface elevation. Once the water surface elevation is computed the velocities are calculated explicitly.

The *semi-implicit scheme* (Casulli and Cheng 1992, Smith 1997, Hodges *et al.* 2000) is similar to the scheme of de Goede in that only some of the terms in the governing equations (gravity wave and vertical diffusion terms) are treated implicitly, while the others (Coriolis, horizontal friction, advection and baroclinic pressure term) are treated explicitly. The semi-implicit models, however, do not use any kind of splitting procedure. TRIM3D (Casulli and Cheng 1992, Casulli and Cattani 1994, Gross *et al.* 1999) and ELCOM (Hodges *et al.* 2000) are two-level schemes⁸. The advection terms in the momentum equations are discretized by using an unconditionally stable, semi-Lagrangian or Euler-Lagrangian method (ELM) with linear interpolation in TRIM and quadratic interpolation in ELCOM. The schemes solve the governing equations in a non-conservative form that is consistent with the use of an ELM. The semi-implicit model SI3D (Smith 1997) uses a three-level leapfrog-trapezoidal scheme to solve the conservative form of the governing equations, and it has second order accuracy in both time and space. The leapfrog scheme is used in the first iteration and is followed by as many trapezoidal steps as desired. The objective of the trapezoidal iterations is, in part, to remove the well-known computational mode associated with the leapfrog discretization (Durran 1999), and to increase the stability of the code (Smith 1997).

All the semi-implicit schemes mentioned above (TRIM3D, ELCOM and SI3D) use uniform finite-difference grids to discretize the domains of interest. The main problem with a uniform grid lies in the representation of the boundaries, in that the grid normally

⁸ A two-level scheme is one where the solution at time level $n+1$ only requires knowledge of the prognostic variables at time n . In a multi-level scheme the calculation of the solution at time $n+1$ will require information from level n , $n-1$, ...

can not conform to the shoreline. Instead the shore is approximated by a series of line elements and associated corners that can lead to spurious circulations unless a high level of grid refinement is used. This type of problem has been overcome by using *boundary fitted coordinates*⁹ (e.g. Song and Haidvogel 1994) or *unstructured grids* (Casulli and Walters 2000).

Though research has been weighted toward the finite-difference method¹⁰, there has been some use of the *finite element*, the *pseudo-spectral* and the *spectral techniques* applied to the solution of the 3-D shallow water equations. One of the earliest 3-D spectral models, by Heaps (Heaps 1972), appeared in the 1970s. In this and subsequent spectral (see Davies *et al.* 1997 for a review) and semi-spectral models (Haidvogel *et al.* 1991) the prognostic variables (generically represented by Φ) are given as linear combinations of a finite number of terms,

$$\Phi(x, y, \sigma, t) = \sum_{j=1}^n A_j(x, y, t) \cdot N_j(\sigma) \quad (1.1)$$

The $N_j(\sigma)$ in Eq. 1.1 are orthogonal functions (Legendre or Chebyshev polynomials) representing the vertical structure (in σ -coordinates), and the set of coefficients A_j vary with horizontal position and time. The spectral method constrains the residual error¹¹ to be orthogonal to each of the orthogonal expansion functions, and the problem is

⁹ The grid in boundary fitted coordinates is transformed to fit the coastline in a smooth regular manner. The use of boundary fitted coordinates improves the representation of irregular boundaries, but the governing equations involve additional terms, and hence additional computation due to the mapping of the grid.

¹⁰ All the models mentioned above are finite difference models.

¹¹ The residual error refers to the error that results from substituting the assumed form of the prognostic variables (equation 1.1) into the governing equation.

effectively reduced to one of finding the 2-D horizontal distribution of the expansion coefficients. In the semi-spectral techniques the residual error is required to be zero at every point on some fixed mesh (Durrant 1999). The classical approach used in the horizontal has been a uniform finite difference grid (Davies *et al.* 1997), though some of the models use boundary-fitted coordinates (e.g. Haidvogel *et al.* 1991). The spectral method can also be used in the horizontal dimension, but this approach is attractive only if the region of interest has regular boundaries or can be transformed into one with regular boundaries (Davies *et al.* 1997).

The advantage of the finite element method is its considerable flexibility in allowing unstructured grid networks¹² that can be employed to represent the shoreline curvature better or to increase the grid-point density where needed or desired. The finite element method, however, naturally generates implicit equations for the unknown variables at each new time level¹³, which are computationally expensive to solve. The model developed by King (King 1998, King 1993, King 1985) uses a fully implicit approach to discretize the non-linear 3-D set of primitive equations without any kind of splitting. In order to reduce the computational burden, the domain can be discretized using 1-D, 2-D and/or 3-D elements, so that the 3-D equations are only solved where they are needed. Lynch and Werner (Lynch and Werner 1991) adopted a mode-splitting technique in their 3-D non-linear finite element model. The semi-implicit method is adopted for the solution of the external mode equations that are arranged in wave-equation form. This

¹² The unstructured grid can also be used in conjunction with finite difference methods, as is the case of the model by Casulli and Walters (Casulli and Walters 2000).

¹³ Nevertheless, early 3-D finite element models (e.g. Kawahara *et al.* 1982) used explicit methods. The model of Kawahara and others adopted a mixed spatial discretization method, using finite elements in the horizontal dimensions and a multi-level finite-difference approach in the vertical.

arrangement of the equations is a well-known method in 2-D modeling for its ability to suppress short wavelength spurious oscillations appearing in other finite element models based on the primitive equations (Gray and Lynch 1979). The review paper by Davies and others (Davies *et al.* 1997) includes additional references on finite element models.

1.3. Objective and organization of this work

The objective of this thesis is to test alternative three-dimensional modeling approaches to the solution of the shallow water equations and extend the most suitable one to construct a 3-D numerical model for lake environments that is *comprehensive* (containing the appropriate physics) and *generalized* (requiring minimal tuning, and adaptable to various applications) in terms of its dynamic features, while *accurate* (containing little numerical damping and dispersion) and *economical* (computationally efficient) in terms of its numerical features.

While many hydrodynamic models do exist, three-dimensional hydrodynamic models have generally been used with overly simplified bathymetry, boundary forcing, and lake conditions. A lack of efficient computational engines has also limited the duration of the simulation period to days or weeks. The present attempt to extend three-dimensional models to more realistic conditions (i.e., complex bathymetry, complex wind field, and stratified water column) is an essential step in developing a fully integrated 3-D transport and fate model. Once developed, such a model could be used to examine the fate of a range of water quality parameters by inserting appropriate speciation and transformation sub-models. Such sub-models could be readily updated and improved as one's understanding changed.

The governing equations and boundary conditions for the modeling of lake environments are introduced in chapter 2. The complexity of the system of coupled differential equations governing geophysical flows is such that no analytical solutions are known, and numerical procedures are needed to solve them. A series of benchmark problems was defined as a means of quantifying the suitability of any given numerical algorithm for the solution of the governing equations and the simulation of lake hydrodynamic processes. In chapter 3 those problems are introduced, and two different modeling alternatives are applied to their solution. The two models compared are considered to be plausible candidates to provide the skeleton for the development of a 3-D model for lakes, and they represent two distinctly different examples in the spectrum of 3-D numerical hydrodynamic codes. The first, referred to as RMA10, is a three-dimensional finite element model (King 1998) and is particularly suitable to represent domains with complex geometry. The second solves the 3-D governing equations using a semi-implicit finite difference discretization on an orthogonal structured grid, and it is referred to as SI3D.

SI3D was selected, based on the results of the tests, as the skeleton of the 3-D model for lake environments. It was extended to use temperature instead of salinity as the active scalar responsible for stratification, to use variable wind fields, to calculate heat fluxes based on the prognostic variables, and to calculate the momentum and scalar turbulent transfer coefficients with a high-order turbulence closure (Mellor-Yamada Level 2.5). The transport algorithm was modified to eliminate spurious oscillations in the

temperature field that could interfere in the calculation of turbulent transfer coefficients. Additional modules have been developed to simulate the transport of tracer plumes within the code, and a separate post-processing module for particle tracking has also been developed. The hydrodynamic model SI3D and its extensions are described and tested in chapters 4 through 6. The extended version of SI3D will be referred to in this thesis as SI3D-L.

In successive chapters two applications of SI3D-L in two very different lake environments are presented. The first, Clear Lake, is a multi-basin, shallow, polymictic and highly eutrophic lake that suffers from high mercury concentrations. The initial motivation of this work comes from the involvement of the author in modeling the transport paths of mercury from its main source, the Sulphur Bank Mercury Mine at the end of one the basins of Clear Lake, to the rest of the water body. The mathematical model product of this thesis was thought to be a primary tool in assessing the “environmental fate” of mercury in Clear Lake, allowing for extensive hypothesis testing while minimizing the need for costly experiments. In Clear Lake the wind field is shaped by the surrounding topography and shows a complex and very characteristic spatial structure. The quasi-periodic and spatially variable wind field interacts with the density stratification and drives a very characteristic and dynamic circulation with horizontal gyres at a range of spatial scales. The internal dynamics are characterized by the creation of horizontal temperature gradients that relax during calm periods. Internal waves do not exist as the stratification conditions rapidly change and even disappear in a fraction of the time the waves would take to complete one cycle.

The second application, described in chapter 8, consists of the simulation of internal dynamics in Lake Tahoe, a deep monomictic¹⁴ and oligotrophic lake. Lake Tahoe in the Sierra Nevada Mountains of California and Nevada is unique for its depth and water clarity, although long-term monitoring has identified a trend of decreasing transparency in its water. The stratification in this lake undergoes slow seasonal changes, and the internal dynamics is characterized by the excitation of internal waves, which are periodic motions causing oscillations of the isopycnals¹⁵. Several investigators have noted evidence of internal waves in Lake Tahoe, but the internal wave climate is largely undocumented (Thompson 2000). This thesis analyzes the internal wave environment early in the winter of 1999, and it then describes the temporal and spatial structure of the dominant internal waves with the aid of the 3-D hydrodynamic model.

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¹⁴ Lake Tahoe could be alternatively described as meromictic, as the winter mixing depth is quite variable from year to year.

¹⁵ The term isopycnal refers to lines or surfaces of equal density. In fresh water lakes those surfaces are identified by having a uniform temperature (isotherms).

CHAPTER 2

GOVERNING EQUATIONS AND BOUNDARY CONDITIONS

2.1. Introduction

The objective of this chapter is to introduce the equations that form the basis of geophysical fluid dynamics. They describe the velocity and pressure fields, and the salinity and temperature fields, in large water bodies such as estuaries or lakes. As presented here¹⁶, these are the equations that, with appropriate boundary conditions, are numerically solved in most of the 3-D hydrodynamic models mentioned in chapter 1. A very detailed analysis and derivation of the governing equations for geophysical fluids and its simplifying approximations can be found in Smith 1997, and hence this detail will not be pursued here.

2.2. Dynamic and thermodynamic equations

The governing equations that form the basis of 3-D circulation models are the continuity (or mass conservation) equation for incompressible fluids, the Reynolds-averaged form of the Navier-Stokes equations for momentum, and the transport equation for active scalar fields (temperature, salinity and sediment concentration). These are supplemented with an equation of state, such as those presented in table 2.1 that are used in the 3-D hydrodynamic model RMA10 (King 1993), that relate active scalar concentrations to fluid density.

¹⁶ This form is referred to as the primitive form of the equations.

a. Salinity

$$\rho(S) = 1.940 \cdot \frac{4.906 \cdot S - 11.7}{6511.7 - 1.906 \cdot S} \cdot 516.$$

b. Temperature

$$\rho(T) = 516 \cdot (1.939933 + T(5.88599 \cdot 10^{-5} - 1.108539 \cdot 10^{-5} \cdot T))$$

c. Sediment concentration

$$\rho(c) = 516 \cdot (1.940 + c \cdot 10^{-3})$$

Table 2.1. Equations of state used in the 3-D finite element hydrodynamic model RMA10. S is salinity in parts per thousand, T is the temperature in °C, c is sediment concentration in mg/l, and the density ρ is in kg/m³

Two approximations are further introduced for geophysical flows (Kundu 1990) that drastically simplify the governing equations¹⁷. First, it is assumed that the weight of the fluid identically balances the pressure (the *hydrostatic assumption* which allows one to replace the pressure in the momentum equations by the water surface elevation as the unknown); and second, density differences are neglected unless the differences are multiplied by gravity (*Boussinesq approximation*). If one further incorporates the well-known *Boussinesq eddy viscosity concept* to model the turbulent transport terms, the following set of governing equations arise (Blumberg and Mellor 1987, Smith 1997):

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \tag{2.1}$$

¹⁷ The simplifying approximations can be justified based on order of magnitude analyses as presented in Cushman-Roisin 1994 for general geophysical flows or in Lynch 1986 for lakes.

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} - f v = -\frac{1}{\rho_0} \frac{\partial p}{\partial x} + \frac{\partial}{\partial x} \left(K_H \frac{\partial u}{\partial x} \right) + \frac{\partial}{\partial y} \left(K_H \frac{\partial u}{\partial y} \right) + \frac{\partial}{\partial z} \left(K_V \frac{\partial u}{\partial z} \right) \quad (2.2)$$

$$\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} + f u = -\frac{1}{\rho_0} \frac{\partial p}{\partial y} + \frac{\partial}{\partial x} \left(K_H \frac{\partial v}{\partial x} \right) + \frac{\partial}{\partial y} \left(K_H \frac{\partial v}{\partial y} \right) + \frac{\partial}{\partial z} \left(K_V \frac{\partial v}{\partial z} \right) \quad (2.3)$$

$$0 = -\frac{\rho}{\rho_0} g - \frac{1}{\rho_0} \frac{\partial p}{\partial z} \quad (2.4)$$

$$\frac{\partial s}{\partial t} + u \frac{\partial s}{\partial x} + v \frac{\partial s}{\partial y} + w \frac{\partial s}{\partial z} = \frac{\partial}{\partial x} \left(D_H \frac{\partial s}{\partial x} \right) + \frac{\partial}{\partial y} \left(D_H \frac{\partial s}{\partial y} \right) + \frac{\partial}{\partial z} \left(D_V \frac{\partial s}{\partial z} \right) + \Delta_s \quad (2.5)$$

$$\rho = \rho(s) \quad (2.6)$$

Here u , v , and w are the velocities in the x , y , and z directions, f is the Coriolis parameter, g is the acceleration of gravity, ρ is water density variations with respect to its mean reference value ρ_0 , s stands for active scalar (temperature, salt or suspended solids), and p denotes pressure. The coefficients K_H and K_V represent the horizontal and vertical turbulent momentum transfer coefficients (or kinematic eddy viscosity), and D_H and D_V are the horizontal and vertical turbulent transfer coefficients (eddy diffusivity) for the active scalar s . The last term in Eq. 2.6 is a source-sink term. In the particular case of the temperature transport equation the source term Δ_s is represented by the divergence of the downward solar irradiance¹⁸ I and has the form,

$$\Delta_s = \frac{1}{\rho c_p} \frac{\partial I}{\partial z} \quad (2.7)$$

¹⁸ The solar irradiance refers to the short-wave component of the solar radiation that reaches the water free surface. See section 4 for further details.

where c_p is the heat capacity of water. The pressure gradient term in the momentum equations (Eqs. 2.2 and 2.3) can be reformulated by using the hydrostatic Eq. 2.4 as

$$\frac{1}{\rho_0} \frac{\partial p}{\partial x} = \frac{1}{\rho_0} \frac{\partial p_a}{\partial x} + g \frac{\partial \zeta}{\partial x} + g \frac{1}{\rho_0} \int_z^\zeta \frac{\partial \rho}{\partial x} dz' \quad (2.8)$$

and

$$\frac{1}{\rho_0} \frac{\partial p}{\partial y} = \frac{1}{\rho_0} \frac{\partial p_a}{\partial y} + g \frac{\partial \zeta}{\partial y} + g \frac{1}{\rho_0} \int_z^\zeta \frac{\partial \rho}{\partial y} dz' \quad (2.9)$$

Here p_a is the atmospheric pressure, and ζ is the water surface elevation above an undisturbed level that can be chosen as $z = 0$. The integral terms in (2.8) and (2.9) are referred to as the baroclinic pressure terms and couple the hydrodynamic solution to the solution of the active scalar field. The water surface slope terms, on the other hand, are referred to as the barotropic pressure terms and are constant with depth.

An equation for ζ is obtained by integrating the continuity equation (Eq. 2.1) over the water column and combining it with the kinematic boundary condition at the free surface, outlined in section 2.3. That equation is

$$\frac{\partial \zeta}{\partial t} + \frac{\partial}{\partial x} \left[\int_{-H}^\zeta u dz \right] + \frac{\partial}{\partial y} \left[\int_{-H}^\zeta v dz \right] = 0 \quad (2.10)$$

where $z = H(x,y)$ is the depth of the bottom boundary measured from the undisturbed free surface $z = 0$.

2.3. Boundary conditions

Hydrodynamic boundary conditions at the free surface require that kinematic and dynamic conditions be satisfied (Smith 1997). The *kinematic boundary condition* at the free surface is

$$\frac{\partial \zeta}{\partial t} + u \frac{\partial \zeta}{\partial x} + v \frac{\partial \zeta}{\partial y} = w \Big|_{z=\zeta} \quad (2.11)$$

This condition requires that a fluid parcel at the free surface remains on the free surface, and thus there is no advective flux through the free surface. The dynamic boundary conditions specify the wind stresses (τ_{0x}, τ_{0y}) at the free surface

$$\rho_0 K_v \left(\frac{\partial u}{\partial z}, \frac{\partial v}{\partial z} \right) = (\tau_{0x}, \tau_{0y}) \quad (2.12)$$

No advective fluxes of temperature or salt are allowed through the free surface. In the particular case of temperature, the boundary condition is given by

$$\rho_0 D_v \left(\frac{\partial s}{\partial z} \right) = H_s \quad (2.13)$$

where H_s represent the net effect of the non-penetrative components of the heat flux¹⁹ (evaporation, conduction and long-wave radiation).

¹⁹ The non-penetrative components of the heat flux are the long-wave radiation, evaporation and heat conduction. A complete description of the thermal exchange at the air-water interface is given in section 4.1.

At the lower boundary the formal hydrodynamic condition is zero velocity, or a *no-slip condition*. However, this condition is not used in numerical models since the vertical resolution is not adequate to resolve the strong velocity gradients in the viscous sub-layers of the bottom boundary layer (Tennekes and Lumley 1972). Because the flow in the immediate vicinity of the bottom boundary is important only insofar as it determines the overall stress exerted on the boundary, it is preferable to base the bottom boundary condition on the specification of the stress (Smith 1997), such that

$$\rho_0 K_v \left(\frac{\partial u}{\partial z}, \frac{\partial v}{\partial z} \right) = (\tau_{bx}, \tau_{by}) \quad (2.14)$$

where the bottom frictional stress (τ_{bx}, τ_{by}) is normally determined from the bottom velocity (u_b, v_b) by a quadratic law of the form

$$(\tau_{bx}, \tau_{by}) = C_d \sqrt{u_b^2 + v_b^2} (u_b, v_b) \quad (2.15)$$

The kinematic boundary condition at the bottom obeys the expression

$$w|_{z=-H} = -u \frac{\partial H}{\partial x} - v \frac{\partial H}{\partial y} \quad (2.16)$$

where H is the depth of the water column relative to mean water level. The kinematic condition entails, in this case, no flow normal to the bottom boundary. Similarly, the boundary condition at the bottom for the scalar transport equation is zero scalar flux.

Along the shoreline, in the horizontal direction, either a no-slip or quadratic-stress-law boundary condition can be applied. However, given that the horizontal grid size in a typical application is on the order of 100 m, the velocity field close to the shoreline boundaries can be significantly distorted. Instead a *perfect slip* condition is preferred, in which the water is allowed to move parallel to the shoreline without any resistance. No flux of any scalar is allowed across lateral boundaries.

CHAPTER 3

BENCHMARK PROBLEMS FOR MODEL EVALUATION AND COMPARISON

3.1. Introduction

This chapter introduces a series of benchmark test problems (Table 3.1) with analytical solutions that will allow for the quantification of the suitability of a model to represent relevant processes in homogenous and stratified lakes. Those test problems will be used as the basis for a detailed comparison of two codes, RMA10 (King 1998) and SI3D (Smith 1997). These models represent two examples well apart in the spectrum of 3-D numerical hydrodynamic algorithms. The first uses a classical finite element approach and is particularly suited to represent domains with a complex geometry. SI3D, on the other hand, solves the 3-D governing equations using a highly efficient, semi-implicit, finite difference discretization on an orthogonal structured grid.

Numerical experimentation offers a straightforward method to evaluate the performance of 3-D algorithms in the solution of lake hydrodynamics problems. The experiments not only constitute an alternative, and sometimes unique, means of evaluating classical measures of accuracy, but they further allow one to analyze the conservation properties of the schemes (mass and energy conservation), the effects of complex geometries and forcing, and the computational expense involved in the solution of problems. Section 3.2 is a brief description of the algorithms used in the two codes for the solution of the

equations governing the flow in isothermal waters, which have already been introduced in chapter 2. A theoretical analysis of the algorithms applied to 1-D problems is presented and the results of this analysis are compared with those obtained by means of numerical experimentation in higher dimensions in sections 3.3 and 3.4. The test cases in this first part consist of the simulation of oscillatory motions in homogenous basins (barotropic seiches) of rectangular shape that are excited by conveniently chosen initial conditions. The effect of curved lateral and vertical boundaries is analyzed in sections 3.6 and 3.7 through the simulation of free oscillatory motions in homogenous basins of circular shape in plan view and rectangular basins with a parabolic bottom. Section 3.8 focuses on the representation of wind-driven motions in lakes, and simulations are presented of 2-D oscillatory motions and a 3-D steady-state circulation forced by wind in simple rectangular and isothermal basins. Finally the computational expense incurred in the solution of the test problems in homogeneous basins is examined as one of the key elements in the comparison of two numerical models. The benchmark problems in a stratified basin are introduced (section 3.11) and the performance of the models is again analyzed (sections 3.12 through 3.15), after the algorithms used by SI3D and RMA10 for the solution of the advection diffusion equation are described (section 3.10). The equations of state incorporated in SI3D are the same as those used in RMA10 and shown in Table 2.1. In the presentation of results considerable use has been made of the abbreviations that have been gathered in appendix A (List of abbreviations).

TEST CASE & Reference	ANALYTICAL SOLUTION
Free barotropic oscillations in rectangular basin with flat bottom (Neuman and Pierson 1966, p. 291)	$h = h_0 \cos\left(\frac{\pi}{L_0} x\right) \cos\left(\frac{\pi}{L_0} ct\right)$ $u = \frac{h_0 c}{H} \sin\left(\frac{\pi}{L_0} x\right) \sin\left(\frac{\pi}{L_0} ct\right)$
Free barotropic oscillations in circular basin with flat bottom (Lamb 1945, p. 285)	$h = A_s J_s(kr) \cos(s\theta) \cos(\sqrt{gH}kt), \text{ k defined by } \left. \frac{\partial J_s(kr)}{\partial r} \right _{r=R_0} = 0;$ $u = \frac{ig}{k\sqrt{gH}} \frac{\partial h}{\partial x}; \quad v = \frac{ig}{k\sqrt{gH}} \frac{\partial h}{\partial y}$
Free barotropic oscillations in rectangular basin with truncated parabolic bottom (Chrystal 1904)	$h = -h_0 \cdot \frac{d}{dw} \text{COSF}(c_p, \eta) \cdot \sin(\sigma t) \cdot c_p \text{ and } \sigma \text{ defined by}$ $\text{COSF}(c_p, \chi) \text{SINF}(c_p, \chi) = 0 \text{ and } c_p = (\sigma a)^2 / (g H_{\max})$ $\left\{ \int_0^t u \, dt \right\} \cdot H_{\max} (1 - \eta^2) = A \cdot \text{COSF}(c_p, \eta) \cdot \sin(\sigma t)$
Wind forced oscillations in a rectangular basin (Csanady 1982, p. 36)	$h = \frac{u_*^2}{c^2} x_c - \frac{4L_0 u_*^2}{\pi^2 c^2} \left\{ \sum_{n=0}^{\infty} (-1)^n \cdot \cos \frac{(2n+1)\pi c t}{L_0} \cdot \sin \frac{(2n+1)\pi c x_c}{L_0} \right\}$ $U = \frac{4L_0 u_*^2}{\pi^2 c} \left\{ \sum_{n=0}^{\infty} (-1)^n \cdot \sin \frac{(2n+1)\pi c t}{L_0} \cdot \cos \frac{(2n+1)\pi c x_c}{L_0} \right\}$
Steady-state vertical velocity profile forced by wind (Heaps 1984)	$u(z) = \frac{u_*^2}{4K_z H} z(3z - 2H)$

Table 3.1. Test cases for homogeneous flows. All are solutions to the linearized shallow water equations. All but the last case assume no friction in the system. The Coriolis forces are likewise assumed negligible.

3.2 Numerical algorithms for homogeneous flow

3.2.1 SI3D

The model SI3D (Smith and Larock 1993, Smith 1997, Smith and Larock 1997) uses a finite difference discretization of the layered averaged form of the governing equations (i.e. it is a multilevel model based on a z-coordinate system). The layer-integrated volume transport replaces velocity as the dependent variable so that the depth-integrated continuity equation is effectively linear. The solution is calculated at a finite number of points, called nodes, that in SI3D are arranged in a regularly spaced grid in the physical domain. Those nodes within the boundaries of a water body are called wet points to distinguish them from the dry points or nodes located outside the boundaries of the computational domain. The control volume surrounding each node in a regularly spaced 3-D grid is a rectangular prism and is referred to as a computational cell. The distance between nodal points is equal in the horizontal directions ($\Delta x = \Delta y = \Delta s$) and, in general, different from the nodal spacing in the vertical direction (Δz). The velocities are staggered such that $u_{i+1/2,j,k}$, $v_{i,j+1/2,k}$ and $w_{i,j,k+1/2}$ are displaced $(\Delta s/2, 0, 0)$, $(0, \Delta s/2, 0)$ and $(0, 0, \Delta z/2)$ away from the grid point where scalars (water surface elevation, salinity or temperature) are defined.

The scheme used in the code is a semi-implicit leapfrog-trapezoidal algorithm, and it produces second order accuracy in both time and space. The leapfrog scheme is used in the first iteration and is followed by as many trapezoidal steps as desired. The objective of the trapezoidal iterations is, in part, to remove the well-known computational mode associated with the leapfrog discretization (Durrant 1999). In addition, an increase in the

number of trapezoidal iterations increases the stability of the code (Smith 1997). The semi-implicit approach is based on treating the gravity wave and vertical diffusion terms implicitly to avoid time-step limitations due to gravity-wave CFL conditions, and to guarantee stability of the method. All other terms, including advection, are treated explicitly. One of the most appealing features of the code is its ability to deal with 2-D or 3-D situations using the same algorithm, depending on the number of layers specified.

3.2.2. RMA10

Over the past twenty years a series of finite element models has been developed for simulation of flow and quality in water bodies: the RMA set of models. These models have been extended from a two-dimensional depth-averaged model, RMA2, to the simulation of flow and sediment transport in stratified three-dimensional systems, RMA10 (King 1985; King 1993; King 1998; and see Heath *et al.* 1999 for a partial bibliography of technical reports prepared by the U.S. Army Corps of Engineers that describes applications of adapted versions of the RMA models to surface water problems). The latter versions have been designed to analyze systems using the 3-D shallow water and advection-diffusion equations.

RMA10 uses a standard Galerkin weighted residual formulation for the finite element method (FEM). In the FEM the computational domain (Ω) is represented as a collection of simple domains called elements. Over each element an approximate solution to the problem is constructed in the form

$$\Phi(x, y, z, t) = \sum_{j=1}^n \Phi_j(t) \cdot N_j(x, y, z), \quad (3.1)$$

where $N_j(x,y,z)$ are basis functions (also known as shape functions in the engineering literature) and $\Phi_j(t)$ are the approximate values of the variable Φ at n points called nodes. In 2-D either triangular or quadrilateral elements can be used (with 6 and 8 nodes respectively), while in 3-D four types of elements are allowed in RMA10: bricks, prisms, pyramids and tetrahedra, having 20, 15, 13 and 10 nodes. The nodes are located at the vertices of the polygonal elements (corner nodes) and at the center of each edge (mid-side nodes). The elements are defined using an isoparametric approximation (i.e. the basis or interpolation functions are used to map an irregular element in the physical domain into a “master” element of fixed and regular shape). The interpolation scheme within the elements has a mixed character. In the hydrodynamic equations, the velocity is interpolated using quadratic basis functions (serendipity elements in quadrilateral elements) while water surface elevation is interpolated linearly. The active scalar concentration (temperature, salinity or sediment) is interpolated with quadratic basis functions (serendipity elements in quadrilateral elements) within the elements. Gaussian quadrature is used in RMA10 for the evaluation of the integrals in the FEM.

The discretization of the unsteady terms follows a weighted scheme of the form

$$\left(\frac{\partial u}{\partial t} \right) = (1 - \alpha) \left(\frac{\partial u}{\partial t} \right)_0 + \alpha \frac{u - u_0}{\Delta t} \quad (3.2)$$

where the subscript 0 stands for the variables evaluated at the beginning of the time step, while the absence of a subscript denotes the variable evaluated at the end of the time step of size Δt . The value of the parameter α indicates how implicit the scheme is. For a value

$\alpha = 1.0$ the scheme is fully implicit, while for $\alpha = 2.0$ the equivalent of the Crank-Nicholson scheme is obtained. RMA10 uses $\alpha = 1.8$ (King 1993). Once discretized, the full non-linear equations are solved using the Newton-Raphson method.

The geometry of the domain where the equations must be solved changes with time as the water surface elevation varies. To develop an Eulerian form for the solution, it is desirable to transform the varying geometry into a domain with fixed geometry. King (King 1985) pointed out that at locations where a sharp break in bottom profile occurs the classical sigma transformation was not unique and momentum in the component directions might not be correctly preserved. As an alternative, a modified transformation is used in RMA10 that preserves the original bottom profile but transforms the water surface to a constant elevation.

3.3 Theoretical analysis of numerical schemes for homogeneous flows

The accuracy and stability properties of a numerical scheme can be studied through the eigenvalues of its amplification matrix. The amplification matrix relates the magnitudes of the unknown variables at time $t + \Delta t$ with those at time t , and its eigenvalues obey the expression (Fletcher 1991, Cathers and O'Connor 1985)

$$\lambda_{num} = \frac{u(t + \Delta t)}{u(t)} = \exp[I\omega_{num}\Delta t] = \exp[-\nu_{num}\Delta t]\exp[I\Omega_{num}\Delta t] \quad (3.3)$$

$\omega_{num}(= \Omega_{num} + I \nu_{num})$ is the representation of the complex frequency $\omega (= \Omega + I \nu)$ of a given wave component, and $I = (-1)^{1/2}$. According to the theory of numerical analysis (Fletcher 1991, Pinder and Gray 1977) a necessary condition for stability of a scheme is

that λ must satisfy $|\lambda| \leq 1 + O(\Delta t)$ for all wave numbers. We call $|\lambda|$ the dissipation-amplification factor (ϵ_A). A method with $|\lambda| < 1$ is said to be dissipative, and if $|\lambda| = 1$ the method is known as neutrally stable. The ratio $\Omega_{\text{num}}/\Omega$ is the velocity or celerity ratio (Q) and characterizes the phase error introduced by a numerical algorithm.

Numerical scheme	Celerity ratio
SI3D (SI-Leapfrog) (Eliason and Bourgeois 1997)	$Q = \frac{\tan^{-1} [2\gamma / (1 - \gamma^2)]}{\beta \text{ CN}}$
SI3D (SI-Trapezoidal) (Casulli and Cattani 1994)	$Q = \frac{\tan^{-1} [\gamma / (1 - \gamma^2 / 4)]}{\beta \text{ CN}}$
RMA10 ($\alpha = 2.0$) (Pinder and Gray 1977)	$Q = \frac{N}{2\pi} \cos^{-1} \left(\frac{1 - A_{HO}}{[(1 + A_{HO})(1 + A_{HO})]^{1/2}} \right) + 1$

Table 3.2. Theoretical celerity ratio for the numerical schemes used in SI3D and RMA10. CN is the gravity wave Courant number ($= (gh)^{1/2} \Delta t / \Delta x$), L is the wavelength, $\gamma = 2 \text{ CN} \sin(\beta/2)$, $\beta = \text{non-dimensional wave number } (2\pi \Delta x / L)$, $N = 2\pi / (\Delta t \sigma gh)$, and

$$A_{HO} = \frac{1}{4} \left\{ \frac{8 - 2 \cos(2\sigma \Delta x)}{(3 - \cos(2\sigma \Delta x))(2 + \cos(2\sigma \Delta x))} \right\} CN^2 \sin^2(\sigma \Delta x).$$

Table 3.2 lists the celerity ratio Q for the algorithms used in SI3D (semi-implicit leapfrog and semi-implicit trapezoidal) and RMA10 (centered time-stepping scheme, $\alpha = 2.0$), applied to the solution of the 1-D linearized shallow water equations. In the absence of frictional forces all of these schemes are neutrally stable, and the computed and physical waves should have the same amplitude. The agreement on the celerity of the computed and physical waves Q will depend on the relative magnitude of the wavelength and the grid size ($L/\Delta x$ or non-dimensional wavelength), which is closely related the non-dimensional wavenumber β . Figure 3.1 illustrates the dependence of Q on $L/\Delta x$ for two different values of the gravity wave Courant number ($CN = 0.5$ and $CN = 5$).

Figure 3.1 shows that the SI-trapezoidal scheme has better propagation characteristics than the SI-leapfrog. That difference is more noticeable as CN increases, and it explains the improvements in phase propagation found by Smith (Smith 1997) when using a trapezoidal step after the leapfrog. Further iterations will not improve the phase characteristics of the scheme, though it can modify its stability properties. The propagation characteristics of RMA10 are slightly better than those of SI3D at higher values of $L/\Delta x$. The differences, however, vanish with higher resolution. These differences vary from 10^{-2} to 10^{-4} over the range of $L/\Delta x$ from 10 to 100 with $CN \sim 5$. For $CN < 1$ and low values of the non-dimensional wavelength, the scheme used in RMA10 introduces a phase lead, and the celerity ratio asymptotically approaches one from above.

The plots illustrate that RMA10 does not propagate $2\Delta x$ waves in the domain of the solution. Those waves will show up as inter-node oscillations in the model results. SI3D does propagate these waves but very poorly. Owing to the mixed interpolation scheme used in RMA10, the inter-node oscillation mode in surface elevation is not allowed (Walters and Carey 1983, Walters 1983). The velocity solution in RMA10, on the other hand, will be seriously affected by those inter-nodal oscillations. Alternative methods for the control of the spurious oscillations in the finite element solution of the shallow water equations include (1) modification of the governing equations (Gray and Lynch 1979), (2) use of dissipative time-stepping schemes, (3) introduction of horizontal diffusion terms in the momentum equations, and, of course, (4) grid refinement. RMA10 makes use of a non-centered time-stepping scheme ($\alpha = 1.8$) and horizontal diffusion terms as a means of controlling spurious

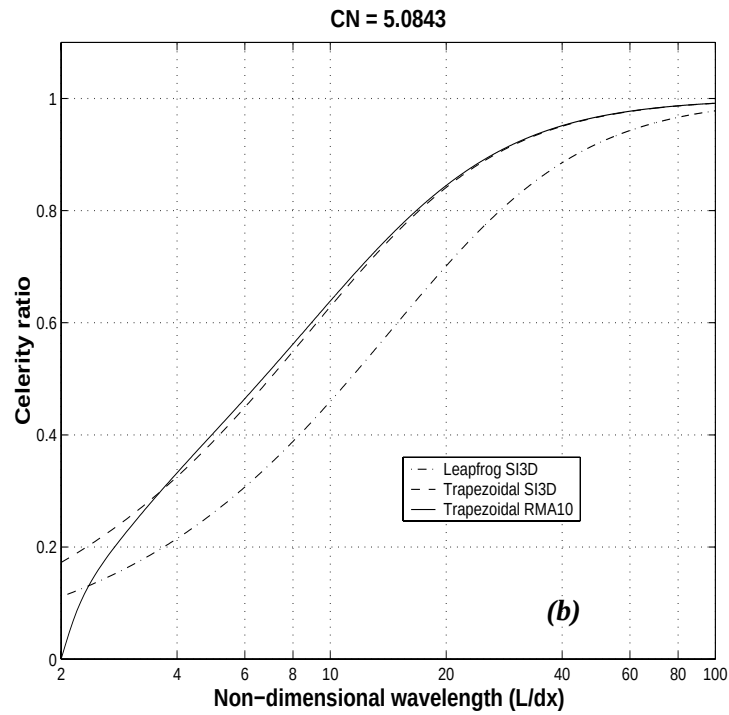
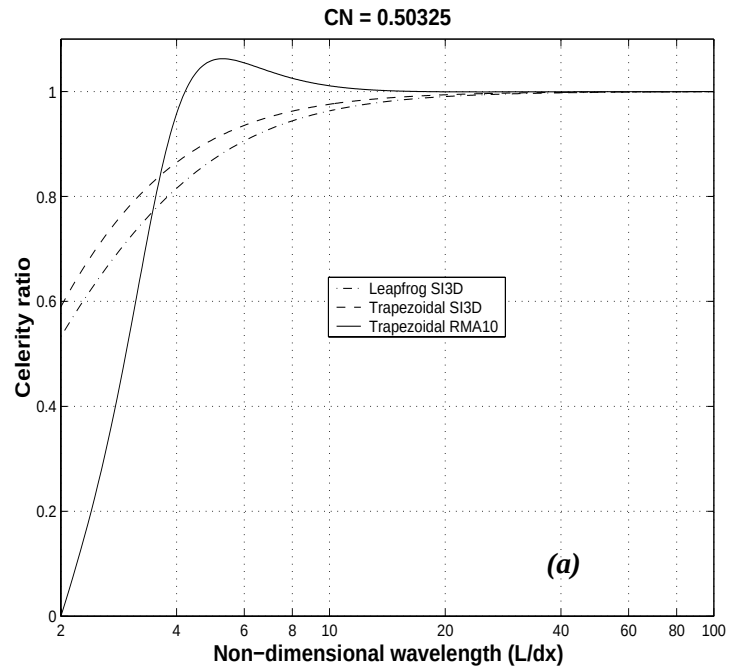


Figure 3.1. Theoretical phase portraits for the numerical algorithms used in SI3D and RMA10. (a) $CN = 0.5$; (b) $CN = 5$.

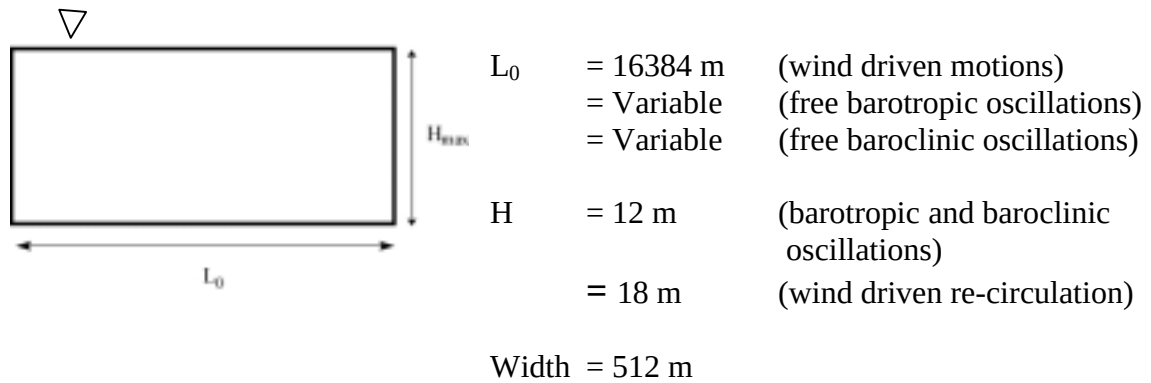
oscillations. With these methods, however, the damping effects are not localized to small wavelengths and long waves will also be affected.

3.4. Methodology: experiment setup and evaluation of errors

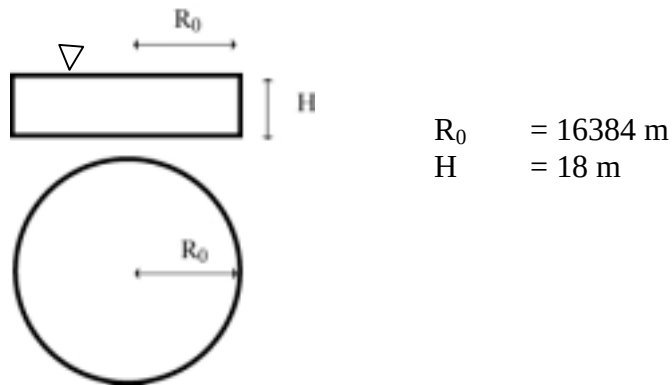
Both codes were set up to solve the linearized form of the governing equations. RMA10 was further modified to use a non-dissipative centered time-stepping scheme. All other sources of dissipation were eliminated. From here on, all results for RMA10 refer to the linear version using $\alpha = 2.0$ and no friction terms in the solution, unless otherwise noted. All further references to SI3D presume the use of the leapfrog-trapezoidal scheme. The modified versions of the codes were then applied to the solution of the problems in Table 3.1. Table 3.3 contains a description of the test basins where the numerical experiments have been performed. The fluid in all cases was isothermal.

To obtain a quantitative estimate of the phase and amplitude errors introduced by the numerical schemes, from time series of water surface elevation or velocity at a given computational node, complex trace analysis (Taner *et al.* 1979) was used. Complex trace analysis (CTA) allows a definition of an instantaneous amplitude, phase and frequency in ways that are logical extensions of the definitions of these terms for simple harmonic oscillations. By comparing the instantaneous phase and amplitude of the numerical signal against the phase and amplitude of the time series obtained with the analytical solution, it is possible to develop estimates of Q and a dissipation factor ($\varepsilon_A = \exp[-\nu_{num} \cdot \Delta t]$). The value of Q obtained from the analysis of the simulated and analytical time series will be called Q_e to distinguish from the theoretical ratio Q_t given in Table 3.2.

Rectangular basin with flat bottom



Circular basin with flat bottom



Rectangular basin with truncated parabolic bottom

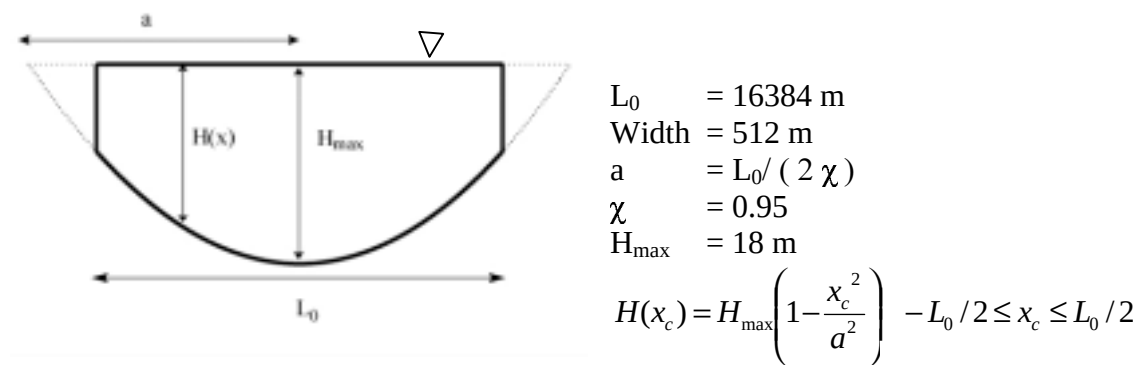


Table 3.3. Geometry of test basins used in numerical experiments

The ability of the technique to quantify and separate rates of amplitude reduction and phase distortion was examined with a series of simple tests. A reference signal (y_1) was generated by using a sinusoidal function with amplitude A and period T : $y_1(t) = A \cdot \cos(2 \cdot \pi \cdot t / T)$. A second signal y_2 was then constructed by introducing a known phase error ε_p and dissipation factor ε_A in the reference signal and adding noise ϵ , $y_2(t) = A \cdot (\varepsilon_A)^{t/\Delta t} \cdot \cos(2 \cdot \pi \cdot t / T + \varepsilon_p \cdot t) + \epsilon$, where Δt stands for the discretization step used in the generation of the signal. The exponent on ε_A is the index of the time series. The signal to noise ratio was set at 100:1 in the experiments.

The CTA was applied to the reference (y_1) and perturbed signals (y_2) using a range of dissipation factors and phase errors to evaluate the ability of the method to discriminate between two different values of Q and ε_A . The experiments show that (1) the ability to discriminate between two different values of Q deteriorates as the dissipation error ($1 - \varepsilon_A$) increases, (2) with dissipation errors of $O(10^{-4})$ one can expect to discriminate between values of Q differing by $O(10^{-4})$, and (3) with signal to noise ratios of 100:1 one is able to discriminate between dissipation factors differing by $O(10^{-4})$.

3.5. Barotropic oscillations in rectangular flat bottom basin

The simplest geometry for testing a numerical algorithm is that of a rectangular basin with a flat bottom. Numerical experiments performed in such a simple geometry provide a straightforward method to evaluate the phase, dissipation and conservation properties of

3-D algorithms. In contrast, in Table 3.2 the analysis is applied to the algorithms in 1-D. In a series of tests, the codes were given the problem of simulating the evolution of the water surface elevation in such a basin where the initial conditions are such that first mode barotropic seiches are excited. The length of the basin was modified while keeping the horizontal spacing between nodes constant as a means of attaining different values of non-dimensional wavelength $L/\Delta x$. In SI3D the size of the cells was fixed at $\Delta x = 256$ m. In RMA10 the quadrilateral (2-D) or brick (3-D) elements are 512×512 m in the horizontal direction, and hence the horizontal separation between consecutive nodes is 256 m. Changes in CN were obtained by changing Δt . Four different time steps were used in the tests: 15, 30, 45 and 90 seconds. The corresponding Courant numbers are 0.64, 1.27, 1.90 and 3.81. The codes were initially set to run in 2-D and later in 3-D.

3.5.1. Phase and dissipation errors

The values of Q resulting from CTA applied to the numerical solution of the problem in 2-D are shown in Table 3.4. The time series of water surface elevation at nodes located at $x = L_0/4$ was used in the analysis. In all cases the dissipation factor was within 0.0001 of unity; hence they have not been included in the tables. The results show a deterioration of the phase error as the CN increases and the level of discretization (measured by the ratio $L/\Delta x$) decreases. For $CN < 1$ RMA10 introduces a phase lead, while for $CN > 1$ the numerical solution lags in phase. These features also appear in the theoretical analysis of Table 3.2 and Fig. 3.1. The values of the celerity ratio Q_t from Table 2, however, underestimate Q_e shown in Table 3.4. The differences decrease slightly over the range of non-dimensional wavelength $L/\Delta x$ studied but asymptotes to a value of $O(10^{-4})$.

	L/dx	CN =0.64	CN=1.27	CN=1.91	CN=3.81
SI3D	8	0.95622	0.90726	0.84211	0.64794
	16	0.98884	0.97423	0.95151	0.85449
	32	0.99741	0.99358	0.98731	0.95573
	64	0.99958	0.99861	0.99700	0.98847
	96	0.99998	0.99955	0.99883	0.99499
	128	1.00012	0.99988	0.99948	0.99730
	256	1.00026	1.00020	1.00009	0.99956
RMA10	8	1.02678	0.96792	0.89003	0.67083
	16	1.00112	0.98621	0.96252	0.86253
	32	0.99979	0.99599	0.98963	0.95784
	64	1.00024	0.99928	0.99766	0.98912
	96	1.00038	0.99995	0.99923	0.99538
	128	1.00043	1.00019	0.99978	0.99761
	256	1.00049	1.00043	1.00033	0.99977

Table 3.4. 2-D celerity ratio (Q_e) in SI3D and RMA10.

Neither the dissipation factor nor the phase errors introduced in the solution by SI3D or RMA10, working with multiple nodes in the vertical direction, differ by more than $O(10^{-4})$ from those obtained in the analysis in 2-D. These results confirm the validity of the analysis in 2-D and even 1-D as a means to characterize the behavior in 3-D, within the accuracy of CTA to differentiate between phase and amplitude errors.

RMA10, however, uses a dissipative non-centered time-stepping scheme with $\alpha = 1.8$ instead of the trapezoidal time-stepping scheme. Table 3.5 shows the values of ε_A obtained from the simulated time series obtained with RMA10 and $\alpha = 1.8$. For comparative purposes, the magnitude of ε_A with a horizontal eddy viscosity (needed for stability) of $O(10^4)$ is not even detectable by CTA method. The dissipation introduced, hence, by the non-centered time-stepping scheme will depend on the discretization level but is in general higher than that introduced through horizontal diffusion. Given that the

ability of CTA to differentiate between celerity ratios is limited and deteriorates with the level of dissipation, the experiments did not show significant differences in the phase error when the non-centered scheme is used.

L/dx	CN =0.64	CN=1.27	CN=1.91	CN=3.81
64	0.99978	0.99914	0.99807	0.99249
96	0.99990	0.99962	0.99914	0.99660
128	0.99994	0.99978	0.99951	0.99807
256	1.00000	0.99994	0.99987	0.99951

Table 3.5. Dissipation factor in RMA10 ($\alpha = 1.8$).

In general, the numerical experiments show that the phase propagation properties of SI3D are better than those of RMA10 at high values of $L/\Delta x$ and low gravity Courant number CN. For low $L/\Delta x$ and high gravity Courant numbers (the worst scenario) the phase error introduced by RMA10 is smaller than in SI3D.

3.5.2. Mass and energy conservation

The codes were used to yield estimates of potential energy, PE , kinetic energy, KE and total energy, E . The datum for the calculation of PE was the bottom of the rectangular basin. The integral of h from its equilibrium position over the whole surface was also computed; this sum should equal zero at all times if the schemes conserve mass. The variation of mass with respect to its theoretical value (volume of water in the domain) was normalized on the initial volume of water in the basin. The variation of total energy $\Delta E (= E - E_0)$ in the system at a given time was similarly normalized on the initial energy content in the system E_0 . These fractions should be zero and remain constant in a closed

basin where no sources or sinks of energy are present, unless numerical diffusion or instabilities are introduced in the solution.

Figure 3.2 shows the time series of fractional change of mass and energy for a range of CN and $L/\Delta x$. Figures 3.2.a and 3.2.b ($L/\Delta x = 16$) are plotted with a different scale from Figs. 3.2.c and 3.2.d ($L/\Delta x = 64$), so that the differences between SI3D and RMA10 (shown later) are made clear. The total energy change in the system during the 24-hour simulations with SI3D is shown in Table 3.6. The absolute magnitudes of the changes are $O(10^{-4})$ and generally grew with grid resolution. For higher values of CN the energy levels decrease, while for values of $CN < 1$ the level of energy actually builds up linearly with time, and after 24 hours the total available energy had increased by a factor of 1.0008 with respect to its original level. The results indicate that SI3D has a good ability to conserve mass, with errors that in all cases are $O(10^{-8})$.

In RMA10 ($\alpha = 2$) the available potential energy evolves very closely to its theoretical value but KE drifts with time. The total energy in the system increases with time in all cases. At the end of the simulation period (after 24 hours), the total energy in the system has increased, depending on the values of CN and $L/\Delta x$, between 3 and 0.001 percent of its initial value. In Table 3.6, the variation of energy in simulations with RMA10 is displayed as functions of CN and $L/\Delta x$ for $\alpha = 2$. The total energy grows with time

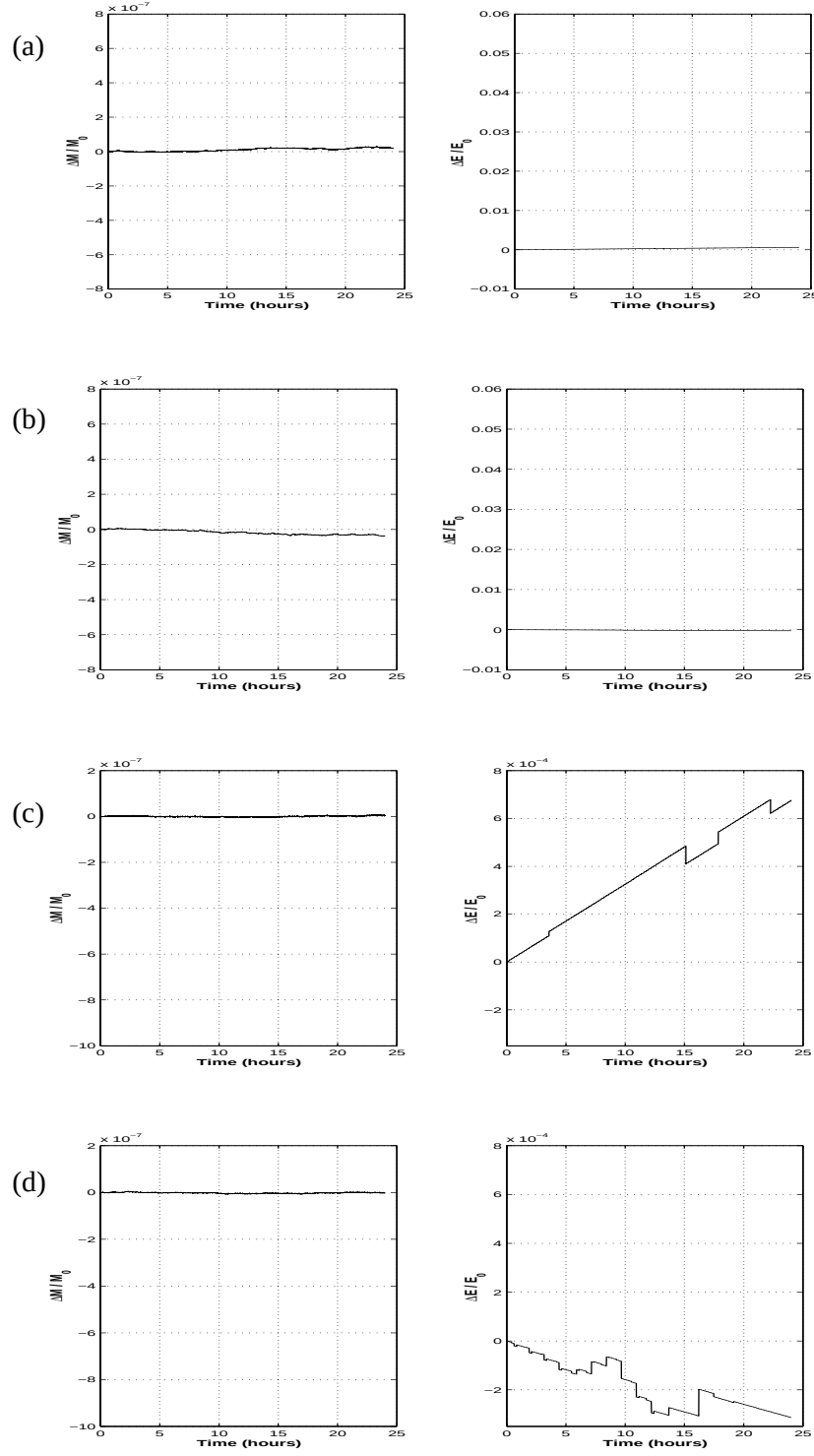


Figure 3.2. Fractional change of mass ($\Delta M/M_0$) and energy ($\Delta E/E_0$) in SI3D. (a) $CN = 0.64, L/\Delta x = 16$; (b) $CN = 1.90, L/\Delta x = 16$; (c) $CN = 0.64, L/\Delta x = 16$; (d) $CN = 1.90, L/\Delta x = 64$.

at a rate faster than exponential (Fig. 3.3). The rate of increase seemed specially noticeable for low values of $L/\Delta x$ and could be explained in terms of energy being drained to the zero mode or $2\Delta x$ waves that remain in the domain. RMA10 does not have a suitable mechanism to dissipate that energy, other than the use of bottom friction, horizontal eddy viscosity or a dissipative time-stepping scheme. The resolution of the grid turns out to be the best means of removing the energy excess in the system. In Fig. 3.3 it can be seen that the code preserves mass very well, with a variation that is only $O(10^{-7})$ from its theoretical value.

	SI3D		RMA10 ($\alpha = 2.0$)	
$L/\Delta x$	CN=1.91	CN=0.64	CN=1.91	CN=0.64
8	-1.49e-4	1.17e-3	0.0015	0.0018
16	-2.79e-4	6.28e-4	0.0506	0.0574
32	-3.95e-4	7.55e-4	0.0029	0.0029
64	-3.13e-4	6.76e-4	1.9e-4	1.3e-4
96	-4.62e-4	7.23e-4	4.8e-5	1.4e-4
128	-5.47e-4	7.29e-4	----	1.0e-4

Table 3.6. Magnitude of relative change of total energy in 24 hours.

Due to the dissipative character of a non-centered time-stepping scheme, the choice of $\alpha = 1.8$ in RMA10 imposed on the total energy a marked decreasing trend (not shown). For low $L/\Delta x$ the dissipation introduced with the time-stepping scheme is extraordinary, and in very few periods the energy is totally removed. The rate of dissipation also increased with CN. Mass conservation is not affected at all by the choice of α , and the variations are $O(10^{-7})$ of its theoretical value.

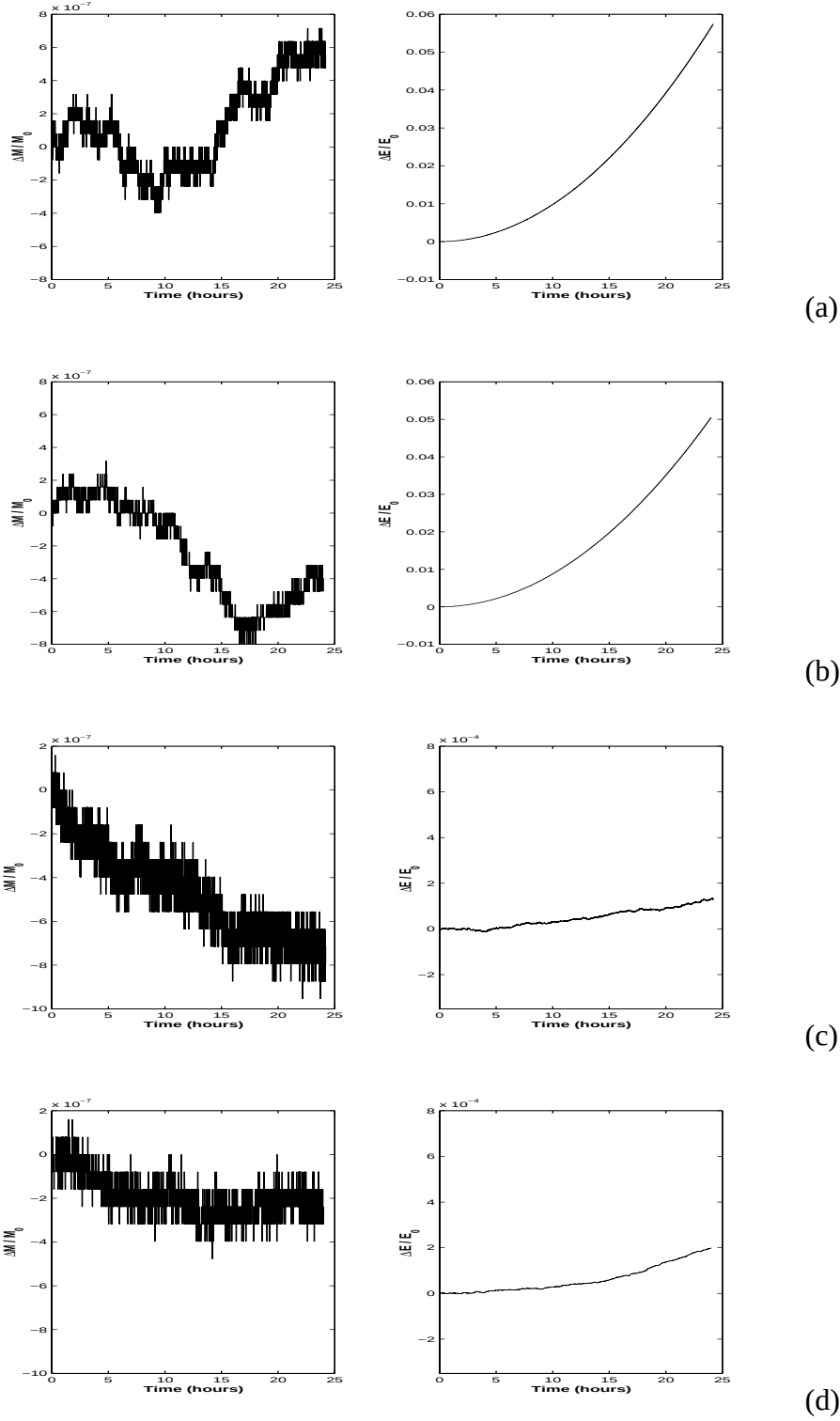


Figure 3.3. Fractional change of mass ($\Delta M/M_0$) and energy ($\Delta E/E_0$) in RMA10 ($\alpha = 2.0$).
 (a) $CN = 0.64$, $L/\Delta x = 16$; (b) $CN = 1.90$, $L/\Delta x = 16$; (c) $CN = 0.64$, $L/\Delta x = 64$; (d) $CN = 1.90$, $L/\Delta x = 64$

In summary, the tests show that (1) RMA10 ($\alpha = 2$) exhibits a gain of energy in the numerical solution at all levels of discretization, and the energy grows at a rate that is faster than exponential; (2) the numerical solution calculated by SI3D, however, only grows for values of Courant number that are below 1, and it does so linearly; (3) the introduction of dissipation through a non-centered time scheme in RMA10 is able to prevent the growing tendency in total energy, but it unduly damps all the energy in the solution; and (4) the variations of total mass in SI3D are $O(10^{-8})$ while in RMA10 are $O(10^{-7})$.

3.6. Barotropic free oscillations in uniform depth circular basin

To study the effects of representing curved boundaries with structured grids (used in SI3D) or unstructured grids (used in RMA10), a series of tests dealing with the simulation of barotropic free oscillations in a circular basin is conducted. The initial conditions given to the codes were such that either modes of barotropic oscillations with one nodal circle (NC-simulation) or one nodal diameter (ND-simulation) were excited in the domain. The nodal diameter was aligned with the y-axis. Two different configurations were tested in SI3D (Figs. 3.4a and 3.4b); the first used 512 meter grid cells to discretize the circular domain, and the second used 256 meter grid cells. In 2-D the total numbers of computational nodes were 3080 and 12596, respectively. The configuration in RMA10 consisted of a total of 673 computational nodes arranged in $2048 \text{ n} \times 2048 \text{ m}$ elements, giving an approximate resolution of 1024 meters. The elements were modified at the

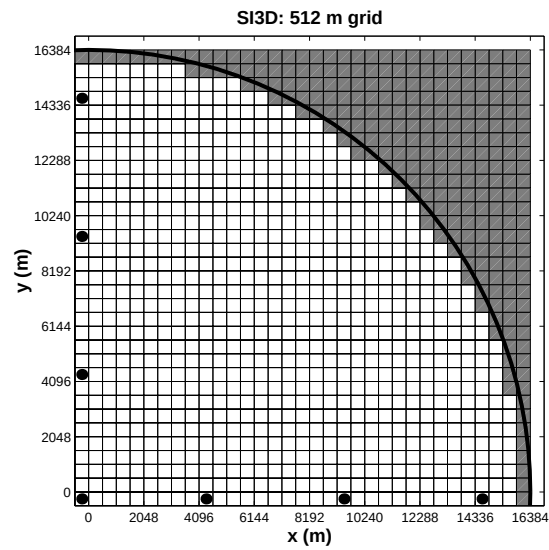


Figure 3.4.a.. Low resolution grid used in SI3D ($\Delta x=512$ m). Only the first quadrant is shown. Dry cells are indicated in gray. The dots represent test nodes.

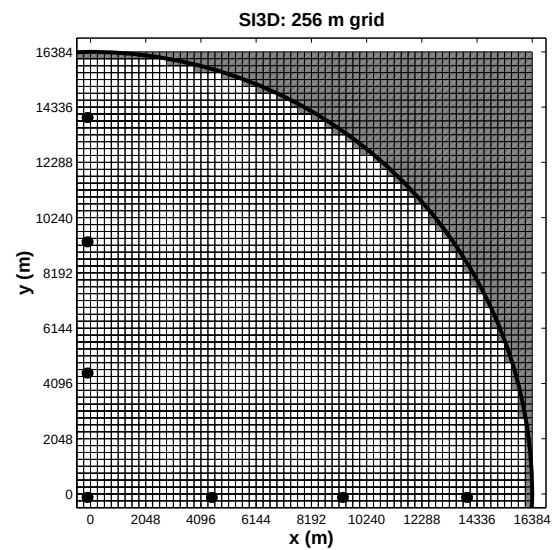


Figure 3.4.b. High resolution grid used in SI3D ($\Delta x=256$ m). Only the first quadrant is shown.

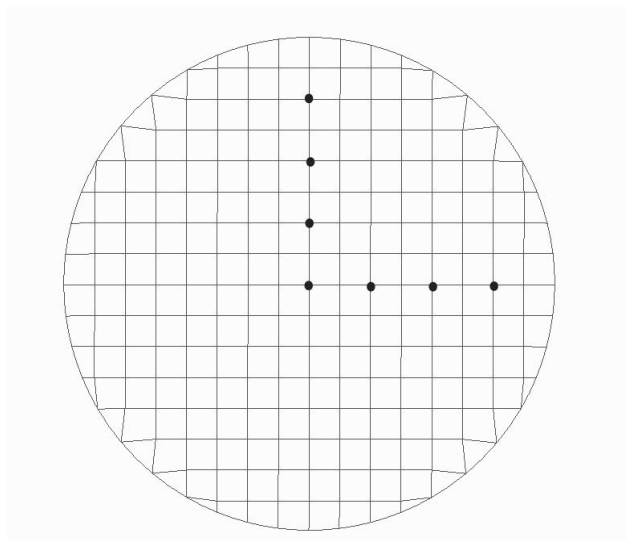


Figure 3.4.c. Computational grid used in RMA10. The dots are test nodes in the grid (all corner nodes).

Error Measures	Definition
Phase Lag($\Delta\Omega$)	$\Omega_{\text{num}} = \Omega + \Delta\Omega$
Index of agreement	$d = 1 - \frac{\sum_{n=1}^{N_{\text{max}}} (A_n - N_n)^2}{\sum_{n=1}^{N_{\text{max}}} \left(\left A_n' \right - \left N_n' \right \right)^2}$
ℓ_1 norm	$l_1 = \frac{\sum_{n=1}^{N_{\text{max}}} N_n - A_n }{\sum_{n=1}^{N_{\text{max}}} A_n }$
ℓ_2 norm	$l_2 = \frac{\left[\sum_{n=1}^{N_{\text{max}}} (N_n - A_n)^2 \right]^{1/2}}{\left[\sum_{n=1}^{N_{\text{max}}} (A_n)^2 \right]^{1/2}}$
ℓ_∞ norm	$l_\infty = \frac{\max_{\forall n} N_n - A_n }{\max_{\forall n} A_n }$
Root Mean Square Error	$RMSE = \frac{\sum_{n=1}^{N_{\text{max}}} (N_n - A_n)^2}{N_{\text{max}}}$

Table 3.7. Definition of error measures used to evaluate the performance of RMA10 and SI3D. A_n = analytical solution, N_n = numerical result. Primes denote deviations from the mean analytical solution. N_{max} = number of data points.

boundaries to represent the circular boundary as accurately as possible (Fig. 3.4c). The time step in all cases was 90 seconds.

To compare the results obtained with the two codes, the set of error measures defined in Table 3.7 was applied. Seven computational nodes (called test nodes) were selected where the time series of surface displacement h and velocity u from the code would be compared with the analytical solutions. The nodes were located at the center and along

two radials of the circular domain (aligned with the x and y axes). In the case of the NC-simulations, given the symmetry of the problem, the two radials contain the same information, and only four test nodes were used in the analysis. The location of the test nodes is shown in Fig. 3.4. Next the numerical results were compared with the analytical solution in the entire domain at each time step to obtain time series for global measures of the accuracy. The same error measures defined in Table 3.7 were used. In this case N_{max} denotes the total number of computational points within the circular domain. The average value of the global error measures for the whole time of simulation is used as a means of comparing the codes.

3.6.1. Analysis of time series at selected computational nodes

3.6.1.1. Mode with one nodal diameter

In Table 3.8 are gathered the error measures evaluated at the test nodes for the ND-simulations. In SI3D the time series of h , as one moves out from the center along a line parallel to the nodal diameter, shows an increase in the phase lead and the error measures. No deterioration in phase or any of the other error measures is observed along the sequence of nodes perpendicular to the nodal diameter. The numerical and analytical solutions for u remain equally separated in all the test points visited in the analysis. Those effects, however, are not observed in the results from RMA10.

Table 3.8 in general shows a superior accuracy for RMA10 relative to SI3D (the error measures are about one order of magnitude smaller for RMA10). These results are due to the fact that a better representation of the domain implies a better representation of the

wave period. The frequency of the barotropic oscillations is primarily determined by the geometry of the basin, and the finite element code does very well in representing that geometry.

Water surface elevation								
		1	2	3	4	5	6	7
SI3D	A	1.0000	1.0000	1.0000	1.0000	1.0000	1.0001	1.0001
	Q	1.0077	1.0077	1.0077	1.0077	1.0087	1.0098	1.0107
	d	0.9188	0.9185	0.9185	0.9185	0.8944	0.8662	0.8365
	l_1	0.4843	0.4849	0.4842	0.4832	0.5499	0.6197	0.6872
	l_2	0.5559	0.5569	0.5563	0.5553	0.6325	0.7116	0.7874
	l_∞	0.9367	0.9363	0.9368	0.9376	1.0614	1.2254	1.3583
RMA10	A	-----	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	Q	-----	0.9991	0.9991	0.9991	0.9993	0.9993	0.9993
	d	-----	0.9988	0.9988	0.9988	0.9930	0.9966	0.9984
	l_1	-----	0.0596	0.0595	0.0595	0.1362	0.0942	0.0658
	l_2	-----	0.0687	0.0687	0.0687	0.1682	0.1166	0.0815
	l_∞	-----	0.1176	0.1180	0.1175	0.5147	0.3228	0.2391
EW Velocity component								
SI3D	A	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	Q	1.0077	1.0077	1.0077	1.0077	1.0077	1.0077	1.0077
	d	0.9163	0.9163	0.9163	0.9151	0.9163	0.9163	0.9163
	l_1	0.4880	0.4877	0.4863	0.4812	0.4878	0.4871	0.4864
	l_2	0.5614	0.5610	0.5588	0.5471	0.5612	0.5602	0.5592
	l_∞	0.9458	0.9434	0.9391	0.9196	0.9457	0.9431	0.9397
RMA10	A	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	Q	0.9991	0.9991	0.9991	0.9991	0.9991	0.9991	0.9991
	d	0.9987	0.9987	0.9987	0.9987	0.9987	0.9987	0.9987
	l_1	0.0627	0.0626	0.0626	0.0631	0.0627	0.0628	0.0632
	l_2	0.0703	0.0702	0.0704	0.0720	0.0703	0.0702	0.0707
	l_∞	0.1251	0.1216	0.1264	0.1376	0.1246	0.1189	0.1251

Table 3.8. Error measures at test nodes for ND-simulations.

3.6.1.2. Mode with one nodal circle

The error measures for the NC-simulations are shown in Table 3.9. The results indicate that SI3D ($\Delta x = 256$ m) introduced smaller errors than RMA10 in the NC-simulations;

there is no indication of deterioration of the numerical solution as one moves from the center. All error measures and for all nodes (no one to one comparison to avoid differences in the location of the nodes) indicates that SI3D performed better than RMA10 in this test case. Neither the representation of the domain nor the time step have been improved in either of the codes, but in this case one has a wave with shorter wavelength, and hence the discretization used in RMA10 suddenly becomes much worse than that used in SI3D ($\Delta x = 256$ m).

<i>Water surface elevation</i>					
		1	2	3	4
SI3D	A	1.0000	1.0000	1.0000	1.0000
	Q	1.0043	1.0043	1.0043	1.0043
	d	0.8891	0.8891	0.8819	0.8888
	l_1	0.5599	0.5583	0.5570	0.5687
	l_2	0.6431	0.6410	0.6273	0.6518
	l_∞	1.0836	1.0930	1.0942	1.1103
RMA10	A	1.0000	1.0000	1.0000	1.0000
	Q	0.9946	0.9946	0.9946	0.9946
	d	0.8292	0.8292	0.8292	0.8292
	l_1	0.6901	0.6899	0.6898	0.6902
	l_2	0.7902	0.7902	0.7900	0.7906
	l_∞	1.3139	1.3133	1.3084	1.3160
<i>EW Velocity component</i>					
SI3D	A	-----	1.0000	1.0000	1.0000
	Q	-----	1.0043	1.0043	1.0043
	d	-----	0.8904	0.8904	0.8891
	l_1	-----	0.5598	0.5559	0.5466
	l_2	-----	0.6426	0.6381	0.6213
	l_∞	-----	1.0786	1.0737	1.0588
RMA10	A	-----	1.0000	1.0000	1.0000
	Q	-----	0.9946	0.9946	0.9946
	d	-----	0.8246	0.8265	0.8292
	l_1	-----	0.6780	0.6768	0.6902
	l_2	-----	0.7746	0.7713	0.7906
	l_∞	-----	1.3716	1.2822	1.3160

Table 3.9. Error measures at test nodes for NC-simulations.

3.6.2. Space-integrated error measures

For the NC-simulations SI3D ($\Delta x = 256$ m) performed slightly better than RMA10 due in part to the higher spatial resolution that translates into a smaller phase error. The values of the global error measures for h are $(\ell_1, \ell_2, \ell_\infty, \text{RMSE}) = (3.9906, 3.9894, 4.0283, 17.5863)$ for RMA10, and $(\ell_1, \ell_2, \ell_\infty, \text{RMSE}) = (3.4032, 3.3785, 3.3248, 14.5125)$ for SI3D ($\Delta x = 256$ m). For the velocity solution, the error from SI3D was about 80% of the error from RMA10.

The ND oscillations, on the other hand, are represented considerably better with RMA10. The error measures for h are $(\ell_1, \ell_2, \ell_\infty, \text{RMSE}) = (0.2237, 0.2237, 0.2276, 1.3346)$ for RMA10, while the same error measures for SI3D ($\Delta x = 256$ m) are from 6 to 8 times bigger $(\ell_1, \ell_2, \ell_\infty, \text{RMSE}) = (1.7603, 1.7605, 1.7564, 10.6044)$. The index of agreement for h reveals as well the inferior performance of SI3D ($d = 0.9634$ for RMA10 and $d = 0.7485$ for SI3D). The error in the computation of the velocity components shows similar differences. These differences are mainly due to the differences in phase error caused by a deficient representation of the curved boundaries with a structured grid.

The error measures in RMA10 deteriorate when, instead of using the centered time-stepping scheme, one uses $\alpha = 1.8$. The mean values of the error measures for h are $(\ell_1, \ell_2, \ell_\infty, \text{RMSE}) = (0.4233, 0.4232, 0.4332, 8.1608)$. For u the index of agreement goes from $d = 0.7785$ for $\alpha = 2.0$ to $d = 0.3684$ for $\alpha = 1.8$. In any case, the performance of RMA10 is better than SI3D.

3.6.3. Phase-corrected results

To avoid the influence of the phase-lag error and isolate other effects of the boundary representation error, the frequency used in the generation of the analytical solution was that evaluated from the numerical results at the center of the computational domain. The mean values of this new set of global errors are shown in Table 3.10.

		SI3D		RMA10	
		<i>ND-simul.</i>	<i>NC-simul</i>	<i>NC-simul.</i>	<i>ND-simul.</i>
Water surface elevation	d	0.9993	0.9791	0.9986	0.9999
	ℓ_1	0.0128	0.1431	0.0243	0.0049
	ℓ_2	0.0134	0.1276	0.0269	0.0057
	ℓ_∞	0.0429	0.0947	0.0320	0.0109
	RMSE	0.1042	1.0083	0.3782	0.0742
EW velocity component	d	0.9983	0.9971	0.9845	0.9837
	ℓ_1	0.0155	0.0436	0.1826	0.0452
	ℓ_2	0.0429	0.0459	0.1960	0.0650
	ℓ_∞	0.4475	0.0709	0.2795	0.2218
	RMSE	0.6425	0.3590	1.1789	0.4751
NS velocity component	d	0.9916	0.9971	0.9846	0.9853
	ℓ_1	0.0890	0.0436	0.1830	0.1255
	ℓ_2	0.1591	0.0459	0.1956	0.1532
	ℓ_∞	0.6488	0.0709	0.2763	0.3421
	RMSE	0.4518	0.3590	1.1858	0.2374

Table 3.10. Mean value of time series of space- integrated error measures for ND-simulations and NC-simulations.

The error measures in Table 3.10 indicate that for the NC-simulations a much more accurate h is calculated with RMA10 than with SI3D, according to the ℓ measures of error, d and $RMSE$. On the other hand the velocity components in SI3D are estimated much more accurately than in RMA10, even when using the lower resolution grid. These

results, together with those from section 3.6.2, indicate that, besides a lower phase error, there exist other reasons for the better performance of SI3D. The lack of resolution in the grid used in RMA10 excites inter-nodal oscillations that degrade the quality of the numerical results. It was observed in the simulations (not shown) that the incorporation of friction degrades the quality of the h solution while it improves the agreement of the velocity components with their theoretical counterparts.

For the ND-simulations, the error measures (Table 3.10) indicate again that h calculated with RMA10 agrees better with the analytical solutions. The results for (u, v) are contradictory. Some error measures (ℓ_1 , ℓ_2 and d) point towards SI3D with the highest resolution grid as the best, while, according to other measures, the agreement with the analytical solution is better in RMA10. Hence the velocity solutions in RMA10 and SI3D have similar degrees of accuracy when the analytical results are corrected for phase. These results must be interpreted together with the fact that without phase correction in the analytical solution the errors in RMA10 are 8 times smaller. The better performance of RMA10 in the ND-simulations is mainly attributable to the better representation of the curved domain and hence a better representation of the phase of the oscillations.

For all three hydrodynamic variables, plots were generated showing the magnitude of the difference at each computational node in the domain between the numerical solution and the phase-corrected analytical solution. The error in these plots is represented by the intensity of a gray scale. The plots define the regions where the error is more pronounced. Figures 3.5 and 3.6 show those error plots obtained 16.85 hours after the beginning of the

simulation, and they have been generated using SI3D with the higher resolution ($\Delta x = 256$ m) and for RMA10 with $\alpha = 2.0$ and no friction added, respectively. Only the results for the test case with a nodal circle are shown. The structure of the error in the velocity field in SI3D was somewhat expected, with the largest magnitudes of disagreement concentrated near the boundaries. However, here the velocities are smaller. On the other hand, the error structure in RMA10 shows the maximum disagreement to be closer to the nodal circle, where one would expect maximum oscillations in the velocity. The error structure of the water surface elevation shows a very well organized pattern in the case of SI3D, with concentric circular regions of higher and lower error magnitudes, while in the case of RMA10 this pattern is not so clear and the error structure is patchy. The maximum disagreements are located near the boundaries where the quadrilateral elements suffer considerable deformation to represent the circular domain and where triangular elements were used.

3.7. Barotropic oscillations in basins with a truncated parabolic bottom

Real systems do not have a flat bottom, as has been assumed in the previous test cases. To evaluate the effects introduced into the solution by a numerical algorithm when dealing with systems of spatially-varying depth, the next simulations consider barotropic oscillations in a rectangular basin in plan that has a concave truncated parabolic bottom (see Table 3.3). In particular, the questions that are addressed here are (1) do the models introduce additional phase error beyond that for flat-bottom rectangular basins, and (2) does a multiple-layer simulation retain the uniformity of the horizontal velocities along vertical profiles? The time step was set to 45 seconds, and 24 hours were simulated.

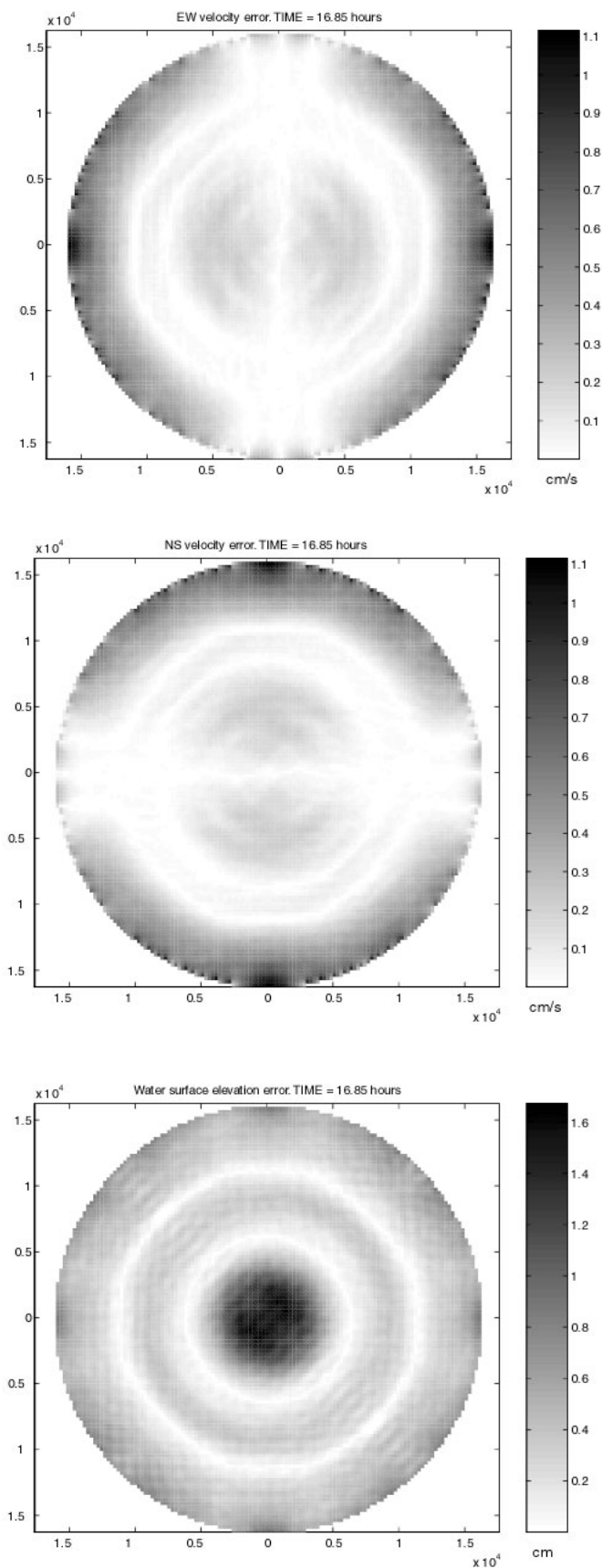


Figure 3.5. Magnitude of the difference at each computational node between the numerical solution calculated with SI3D and the phase-corrected analytical solution.

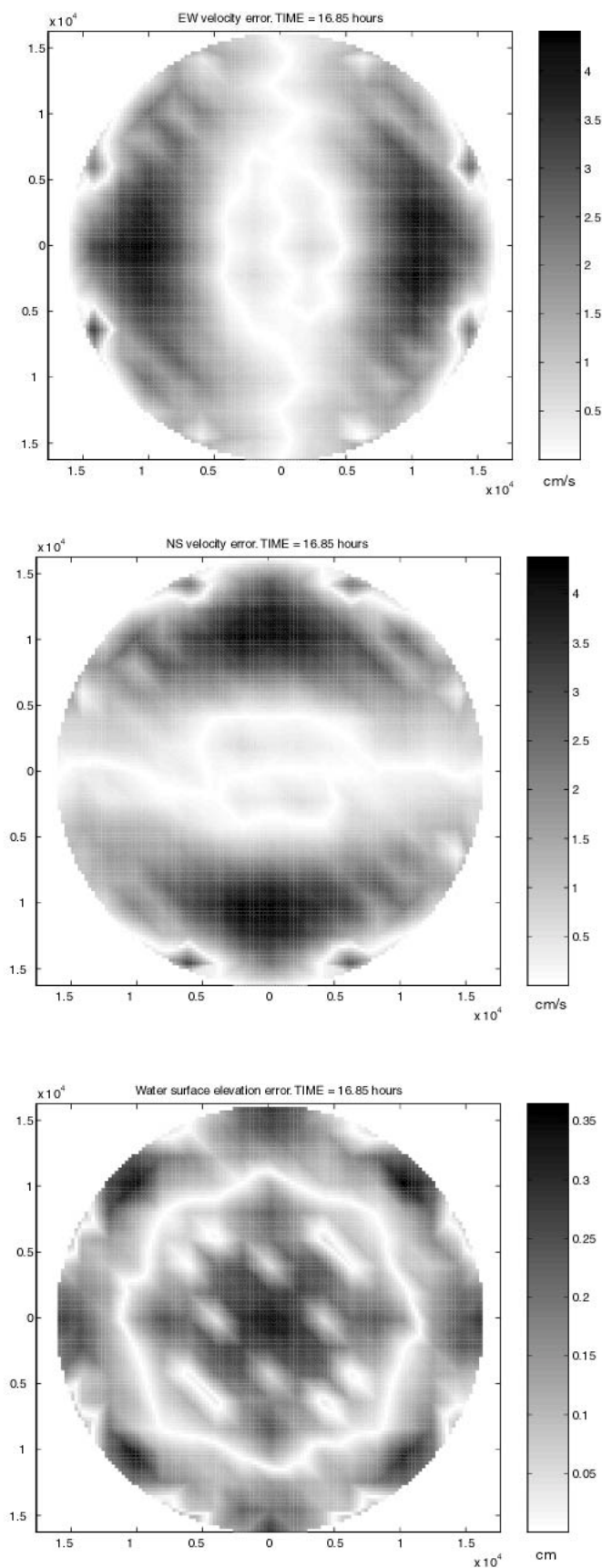


Figure 3.6. Magnitude of the difference at each computational node in the domain between the numerical solution calculated with RMA10 and the phase-corrected analytical solution. The error is calculated at each computational node (see Fig. 3.4.c) and interpolated to a structured grid with 256 m resolution.

The basin was discretized in RMA10 using a total of 32 quadrilateral elements with dimensions 512 x 512 meters. This choice gives a distance between nodes of 256 m and an effective control over $2\Delta x$ waves. A total of 64 grid cells (256 x 256 m) in the horizontal plane were used in SI3D to discretize the test domain. In order to solve the problem in 3-D, two elements were used in RMA10 and 6 computational points were used in SI3D to discretize the vertical direction. The first layer of elements in RMA10 had a uniform thickness of 9 meters, and the second varied in order to capture the bathymetry.

3.7.1. Phase distortion introduced by varying bathymetry

The time series of h at a node located at $x = 12288$ m was used in the evaluation of Q and ε_A with CTA, and in the case of the 3-D run this node was located at the surface. The celerity ratio in RMA10 ($\alpha = 2.0$) in 2-D was 0.9992. The value of Q_e encountered in the simulations with RMA10 ($\alpha = 1.8$) in 2-D was 0.9992 and is not significantly different from that found for $\alpha = 2.0$. The dissipation factor ε_A was in this case 0.9994. RMA10 ($\alpha = 1.8$) in 3-D introduced a phase lead ($Q_e = 1.0025$). The dissipation factor was greater than that in the 2-D simulations (0.9993 vs. 0.9994) and is a consequence of the introduction of horizontal eddy viscosity. In fact no stable solutions were found unless values of K_H of $O(10^4)$ were used in the 3-D simulations. The observed celerity ratios for the 2-D and 3-D runs with SI3D are 0.9992 and 0.9986 respectively, and ε_A in both cases is unity to within 10^{-4} . These results suggest that a varying bathymetry will distort and slightly modify the behavior of the 3-D models with respect to their behavior in 2-D.

3.7.2. Uniformity of velocity field along a vertical profile

A measure of the uniformity of a vertical velocity profile was defined as

$$\vartheta = \left(\sum_{k=1}^{N_v} |u_k - \langle u \rangle| \right) / N_v \quad (3.4)$$

Here N_v is the number of nodes along the vertical profile, u_k is the velocity magnitude at node k along a profile at a given time step, and $\langle u \rangle$ is the depth-averaged velocity at that same time step. The error measure for all time steps was evaluated at a node at $x = 12288$ m. RMA10 did become unstable unless K_H of $O(10^{+3})$ was added to the scheme. The average value of the error time series is 2 mm/s in the case of RMA10. It is identically equal to zero, within 0.1 of mm/s, for SI3D. The convergence criterion for the Newton-Raphson iteration procedure in RMA10 was set at 1 mm/s, and this might in part explain the value of the vertical uniformity error measure.

3.8. Wind-driven hydrodynamics in homogeneous basin

In the test cases considered to this point, the system has been excited through the imposition of particular initial conditions. However, in natural lakes water motions are driven by the boundary conditions. In the absence of inflows and outflows, wind is the main driving force. The first test problem in this section is that of oscillatory motions forced by wind in a rectangular basin and sufficiently narrow as to suppress any significant cross-basin transport. The second test case looks at the steady state velocity

profile in a rectangular homogeneous basin, subject to constant uniform wind stress at the surface and a no-slip boundary condition at the bottom with constant K_z .

3.8.1. Oscillations forced by wind in rectangular uniform depth basins

The first term in the analytical solution for h (Table 3.1) corresponds to the wind setup, and the other terms describe a series of standing oscillations known as seiches. The different harmonics included in the analytical solution will be propagated with different celerities by the numerical models, and small oscillations will appear in the numerical results. For the generation of the analytical solutions 1000 terms in the infinite series are considered in the computations. This simplification implies a maximum error of $O(10^{-7})$. Δt was set to 1.5 minutes, and a total of 24 hours was simulated. The wind is applied uniformly over the basin beginning at the first time step, and it has a magnitude of 10 m/s.

		SI3D	<i>RMA10</i>
Water surface elevation	ℓ_1	0.1194	0.1088
	ℓ_2	0.1174	0.1044
	ℓ_∞	0.1845	0.1975
	RMSE	0.1777	0.1535
Along wind velocity component	ℓ_1	0.1739	0.1450
	ℓ_2	0.1897	0.1681
	ℓ_∞	0.3171	0.4345
	RMSE	0.1589	0.1409

Table 3.11. Error measures for numerical solutions obtained with SI3D and RMA10 in the simulation of wind-forced oscillations in a rectangular basin.

The test basin is discretized with meshes that give a distance between computational points of 256 meters. The difference between the numerical and the analytical solutions

was evaluated by means of ℓ -norms and RMSE and is shown in Table 3.11 for RMA10 and SI3D. It shows that RMA10 performs better than SI3D according to all error measures except for the ℓ_∞ norm. Those differences are mainly due to the magnitude of disagreement around the peaks of the analytical solution.

3.8.2. Wind-driven steady state vertical velocity profile with constant K_z

The test is discretized with grids that offer a resolution in the horizontal of 256 meters. In the vertical, 6 elements (13 computational points) are used in RMA10 and 12 grid cells are used in SI3D. This gives a vertical resolution of 1.5 meters in RMA10 and 1 m in SI3D. A 10 m/s wind was imposed uniformly over the surface. The value of the dynamic transfer coefficient was set to 180 Pascal second. RMA10 was solved in steady state mode. SI3D was solved in dynamic mode until steady state conditions were reached.

The disagreement between the numerical and analytical solutions is quantified using the ℓ -norm error measures and RMSE (Table 3.7). In this case the number of data points is the number of nodes in the vertical. For RMA10 the error measures are $(\ell_1, \ell_2, \ell_\infty, \text{RMSE}) = (0.0114, 0.0115, 0.0085, 0.0038)$, while for SI3D these are $(\ell_1, \ell_2, \ell_\infty, \text{RMSE}) = (0.0120, 0.0125, 0.0092, 0.0038)$. While the ℓ -norm error measures point towards RMA10 as more accurate, the RMSE is equal for both codes. The ℓ -norm error measures depend, however, on the actual location of the points, and hence they are not directly comparable since the computational points are located at different locations within the water column for the finite difference and finite element codes. The same experiment was repeated, using the same forcing, but now coarsening the grids in the vertical direction. Now only 4 elements

were used in RMA10 (9 nodes separated 2.5 m), and 9 grid cells ($\Delta z = 2.0\text{m}$) were used in SI3D. The error measures for this second configuration are $(\ell_1, \ell_2, \ell_\infty, \text{RMSE}) = (0.0301, 0.0266, 0.0174, 0.0093)$ for RMA10 and $(\ell_1, \ell_2, \ell_\infty, \text{RMSE}) = (0.0229, 0.0206, 0.0157, 0.0060)$ for SI3D. They indicate a deterioration of the accuracy (as one would expect) and a higher sensitivity of RMA10 to a reduction in the vertical resolution. In this case SI3D seemed to outperform RMA10.

3.9. Computational cost in homogeneous basins

The accuracy of the models is not the only element to be considered when comparing the performance of codes. The cost of an accurate solution needs to be taken into account and this cost can be measured in terms of the number of nodes and the time to obtain an “acceptable” solution. To compare their computational cost, the codes were set up to run similar problems in similar environments. All output routines were turned off, and both codes were compiled using a MIPSpro f90 compiler version 7.3. No optimization options were used in the compilation. The tests were run on an SGI Octane with a 175 MHz IP30 processor and 386 Mbytes of memory. All the advective, baroclinic and horizontal momentum diffusion terms were turned on in SI3D, and the full version of RMA10 was used in this series of tests. No scalar transport equation was solved with either code.

The first test consisted of the simulation of barotropic free oscillations in a rectangular basin during 24 hours, using $\Delta t = 45$ s. Four different basins of lengths 8192, 12288, 16384 and 32768 meters were discretized with meshes with a spatial resolution of 256 m in the horizontal direction. For RMA10, in the vertical dimension, a single element was used in an initial set of runs. In SI3D, to obtain a similar number of computational points

in the vertical, 3 grid cells each with a thickness of 4m were used. The total number of computational points in the tests is given in Table 3.12. To increase the resolution in the vertical direction, 6 elements were used in RMA10 and 1-meter thick grid cells in SI3D, so that a total number of 3041 and 3072 points, respectively, were involved in the computations.

SI3D			RMA10		
<i>Number of nodes</i>	<i>Total time (seconds)</i>	<i>Time per node (sec)</i>	<i>Number of nodes</i>	<i>Total time (seconds)</i>	<i>Time per node (sec)</i>
<i>Rectangular basin</i>					
192	75.239	0.392	200	1744.32	8.72
288	110.557	0.384	296	2476.17	8.36
384	145.637	0.379	392	4540.88	11.58
768	391.830	0.510	776	7703.40	9.93
3072	903.010	0.294	3041	156955	51.62
<i>Circular basin</i>					
12596	4079.38	0.32	673	5227.89	7.77
75576	13951.42	0.1800	3379	$4.476 \cdot 10^5$	132.46

Table 3.12. Computational costs involved in the solution of test problems with SI3D and RMA10.

The time required for each run is given in Table 3.12. Since the number of nodes/cells used in the solution was slightly different, it was desired to compare the computational cost associated with the simulations between the two codes, the cost was expressed per computational point. The results show a 30:1 ratio in speed between the codes, with the higher speeds corresponding to SI3D finite difference code with the lower vertical resolution. That ratio increases further to 175:1 when dealing with 6 elements in the vertical in RMA10 and a similar resolution in SI3D.

In a second series of tests the codes were timed during the solution of the problem of barotropic oscillations in a circular basin, both in 2-D and 3-D. The problem was simulated during 24 hours, using $\Delta t = 90$ s. The horizontal grid was the same as that used in section 3.7. In the case of SI3D only the higher resolution mesh was used ($\Delta x = 256$ m). When working in 3-D, the grids produced a 3 meter vertical spacing. In RMA10 this means using three elements in the vertical and a total of 3379 nodal points; while in order to get a 3 m resolution for SI3D six computational cells are needed in the vertical direction, for a total of 75576 computational points. The computational time is the same for the ND-simulations and NC-simulations. Hence, no distinction is made. The results of the timings are given in Table 3.12.

Table 3.12 shows a considerable increase in the computational time when going from 2-D to 3-D in RMA10. It is almost 20 times more expensive even though it only uses 5 times the number of nodes involved in the 2-D computations. That increase in computational time is not observed, however, in SI3D where the time per node does not increase significantly when multiplying by 6 the number of nodes and increasing the number of layers in the vertical. The finite difference code is still, using 20 times the number of nodes in 2-D, 1.3 times faster than RMA10. In 3-D the difference in speeds of the finite difference code relative to the speed of RMA10 is even more pronounced.

3.10. Algorithms for transport equation. The baroclinic term

3.10.1. RMA10

RMA10 uses a spatial quadratic interpolation scheme for the active scalars in the solution of the advection-diffusion equation (Eq. 2.5). The discretization of the unsteady terms

follows a weighted finite-difference scheme of the form shown in Eq. 3.2 for the velocities. Gresho and Sani (Gresho and Sani 1998) provide a very detailed analysis of the solution of the one-dimensional advection-diffusion equation with constant velocity using a standard Galerkin weighted residual formulation and quadratic interpolating functions. The analysis proceeds by defining the analog finite difference set and applying the von Neuman method. The eigenvalues λ_n for the solution scheme in RMA10, applied to the pure advection equation, are given by

$$i\lambda_n = \frac{\omega_n \Delta x}{u} = \frac{-2 \sin 2\theta_n + \sqrt{(1 - \cos 2\theta_n)(19 - \cos 2\theta_n)}}{3 - \cos 2\theta_n} \quad (3.5)$$

where ω_n is the frequency corresponding to the n th mode, u is the advection velocity, and θ_n is a dimensionless wavenumber ($=2\pi n\Delta x/L$, L being the wave length and Δx the inter-nodal spacing). This expression is zero for waves with $L = 2\Delta x$ and shows that the finite element method with quadratic shape functions suffers from spurious $2\Delta x$ waves when applied to the solution of the advection (hyperbolic) equation. They are not propagated in the numerical domain and are not (unless a dissipative time stepping scheme is adopted) dissipated. Gresho and Sani (Gresho and Sani 1998) show the results of numerical experiments where different waveforms are propagated at a constant velocity, both in one- and two-dimensional domains, using different degrees of resolution. They demonstrate that the finite element scheme is excellent for the advection equation when "sufficiently smooth" data are involved in the problem. However, the presence of

discontinuities or rough data or boundary conditions will excite the spurious modes that the theoretical analysis shows.

The presence of a diffusion term in the advection-diffusion equation will create a system of mixed character, hyperbolic/parabolic, and will remove the high wave-number components of the solution. The excitation/dissipation of those "rough" elements will depend, though, on the Peclet number, defined as the ratio between advection and diffusion processes. For low Peclet number the system is parabolic and the finite element method is very appropriate. For high Peclet number problems, however, the advection diffusion equation has a hyperbolic character, and the solution with the finite element method is prone to developing wiggles.

Gresho and Sani (Gresho and Sani 1998) define two major causes of wiggles in the solution of the advection diffusion equation: (1) poorly resolved or rough initial conditions and (2) hard boundary conditions. They ascribe hard boundary conditions to Dirichlet or Robin BC, and easy boundary conditions to Neuman or periodic BC. Special care should be taken with the grid resolution if sharp variations in concentrations are to be simulated or used as initial conditions.

Even though the scalars are modeled with quadratic interpolation functions, the code uses linear interpolating functions to represent the density field. According to King (King 1999), with this feature the code seeks to be consistent with the linear interpolation function in the water surface elevation. The momentum and transport equations are

solved separately until, through iterations, convergence is achieved. During the first iterations for the solution at time $n+1$ of the velocity field, the code uses the density information at time n projected forward in time to step $n+1$. Once the scalar transport has been solved, the information from time $n+1$ from the previous iteration is used in the evaluation of the baroclinic forcing term in the momentum equation.

3.10.2. SI3D

The solution of the advection-diffusion equation follows the solution of the momentum and continuity equations and uses the leapfrog-trapezoidal scheme. This combination of schemes effectively suppresses the computational mode associated with the leapfrog scheme that would lead to oscillations of the solution. In the initial leapfrog step the advection and horizontal diffusion terms are treated explicitly (at time n and $n-1$ respectively), while the vertical diffusion terms are averaged between time $n+1$ and $n-1$ (for stability purposes). In the trapezoidal iterations the velocity information at the new time step and the estimate of the scalar at time $n+1$ from the previous iteration are incorporated in the calculation of the advection term; the horizontal diffusion is treated at time n , and the vertical diffusion term is discretized implicitly. The scheme hence requires the solution of one tri-diagonal system of equations in each water column using the Thomas algorithm (Fletcher 1991). The explicit treatment of the advection term leads to stability constraints in the time step, referred to as the Courant-Friedrichs-Lewy (CFL) condition

$$CN = u \frac{\Delta t}{\Delta x} < 1 \quad (3.6)$$

Very simple tests in 1-D were conducted where the advection algorithm used in SI3D was applied to the solution of the problem of linear convection of a truncated sine wave or a step in concentration (Fletcher 1991, page 284). The numerical scheme, as expected for a space-centered difference algorithm, introduces significant oscillations in the solution, especially at low Courant number and in regions where the gradients are not well resolved. Dissipation, measured as the reduction in the peak concentration, is augmented slightly with additional trapezoidal steps and, partly, explains the increased stability reported in Smith (Smith 1997) with the incorporation of trapezoidal iterations. The analysis of 1-D and 2-D test problems with the advection scheme further indicates that the scheme is conservative, though it produces oscillations in the solution if the spatial resolution is not able to resolve adequately the gradients in the scalar field (see chapter 5 for a detailed presentation of the algorithms and the results of these 1-D and 2-D simple test problems).

The baroclinic term in the momentum equations is treated in each leapfrog step at time n , while in the subsequent trapezoidal iterations this term uses information at time n and $n+1$ from previous iterations. In other semi-implicit codes (Hodges *et al.* 2000 and Gross *et al.* 1999) there exists a stability constraint to the time step due to the fact that the baroclinic term in the governing equation is treated explicitly. The stability constraint is described by the internal wave gravity Courant number IWCN, given by

$$IWCN = \sqrt{g'D} \frac{\Delta t}{\Delta x} < 1 \quad (3.7)$$

where g' is the reduced gravity due to stratification ($g\Delta\rho/\rho_0$), $\Delta\rho$ represents the density difference between the top and the bottom of the water column, ρ_0 is the density at the surface, D is effective depth and $\sqrt{g'D}$ is an approximation of the wave speed of an internal wave. However, given that during the trapezoidal steps information from time $n+1$ is incorporated into the solution, this stability constraint is not strict and can be violated.

3.11. Simulating internal waves

An accurate representation of the baroclinic forcing in the momentum equations and the transport of active scalars is critical for the correct simulation of the internal dynamics of stratified basins. In most lakes the internal dynamics are characterized by the occurrence of oscillatory motions with some well-defined frequency, referred to as internal waves, that cause isopycnals to tilt and oscillate around an equilibrium position (Imberger 1998). The next problem aims at revealing the ability of a 3-D model to simulate basin-scale internal waves, and it is thus a valid test for the correct representation of the transport of scalars and the baroclinic forcing. These basin-scale waves in natural systems can be identified as Kelvin or Poincaré waves or, as in the case of narrow basins with a negligible influence of the Earth's rotation, simple gravitational seiching.

The test problem consists of the simulation of a baroclinic or internal seiche in an enclosed rectangular and narrow flat-bottom basin with a background linear density profile. Analytical solutions to this problem exist if (1) the motion is effectively 2-D in a vertical plane, (2) the fluid is assumed inviscid, and (3) non-linear terms are neglected.

The components of the velocity solution driven by variations in the density field, referred to as baroclinic components, obey the expressions (Eliason and Bourgeois 1997)

$$u' = u_0 \sin\left(\frac{l \pi x}{L}\right) \cos\left(\frac{m \pi z}{H}\right) \sin(\omega t) \quad (3.8)$$

$$w' = -\frac{l H}{m L} u_0 \cos\left(\frac{l \pi x}{L}\right) \sin\left(\frac{m \pi z}{H}\right) \sin(\omega t) \quad (3.9)$$

and the evolution of the density at any given location within the basin is described by

$$\rho' = \frac{\rho_0 N u_0}{g} \cos\left(\frac{l \pi x}{L}\right) \sin\left(\frac{m \pi z}{H}\right) \cos(\omega t) \quad (3.10)$$

The primes in these equations indicate variables associated with the baroclinic component of the fluid motion. The baroclinic velocity components are denoted by u' and w' , ρ' represents the variations of density relative to the mean density profile, and ρ_0 is a reference density (the mean density at the surface). In Eq. 3.10 N is the constant background buoyancy frequency ($N^2 = (g/\rho_0) \cdot \partial\rho/\partial z$), L and H are the length and depth of the basin, m and l represent the horizontal and vertical mode numbers respectively, and ω is the frequency of the oscillations, related to the stratification and the geometry of the basin via

$$\omega = \frac{l H N}{m L} \quad (3.11)$$

The comparison of the analytical solution to that obtained with a numerical model provides a very valuable means of analyzing the characteristics of the algorithms in the model. A series of numerical experiments was performed, aiming at examining and comparing (1) the introduction of phase lag and dissipation, (2) the conservation of mass and energy, (3) the presence of diapycnal mixing, (4) the existence of any stability constraints on the time step selection, (5) the computational cost, and (6) the need for additional parameters (e.g. viscous terms for stabilization of the solution) in the representation of internal seiches by the numerical algorithms used in RMA10 and SI3D. The basin adopted in the first experiments is 16,384 m long, 512 m wide and 12 m deep. The computational grids used in SI3D and RMA10 for the discretization of the 3-D domain had a resolution of 256 m in the horizontal and 1 m in the vertical direction. The test consists of the simulation of a baroclinic seiche during a total of 144 hours, using different time steps ranging from 360 to 1080 seconds. In later experiments the size of the basin was modified to achieve different levels of spatial resolution, expressed by the ratio of the wavelength to the longitudinal spacing between nodes.

In natural systems isopycnals initially tilt in response to wind forcing at the free surface. In the test problem, however, the oscillations of the isopycnals are excited by conveniently defining initial conditions. The initial conditions are obtained from the analytical solutions by setting the time in Eqs. 3.8 through 3.10 to zero. The surface temperature was set to 25 °C, and the bottom temperature was set to 15 °C in the undisturbed state, with the density varying linearly with depth from 998.15 kg/m³ at the surface to 1000.15 kg/m³ at the bottom. The corresponding buoyancy frequency for the

linearly varying background density distribution in this basin is 0.0404 s^{-1} . On this density distribution was added a sinusoidal perturbation of the density field that could excite baroclinic oscillations. The initial temperature field is shown in Fig. 3.7.

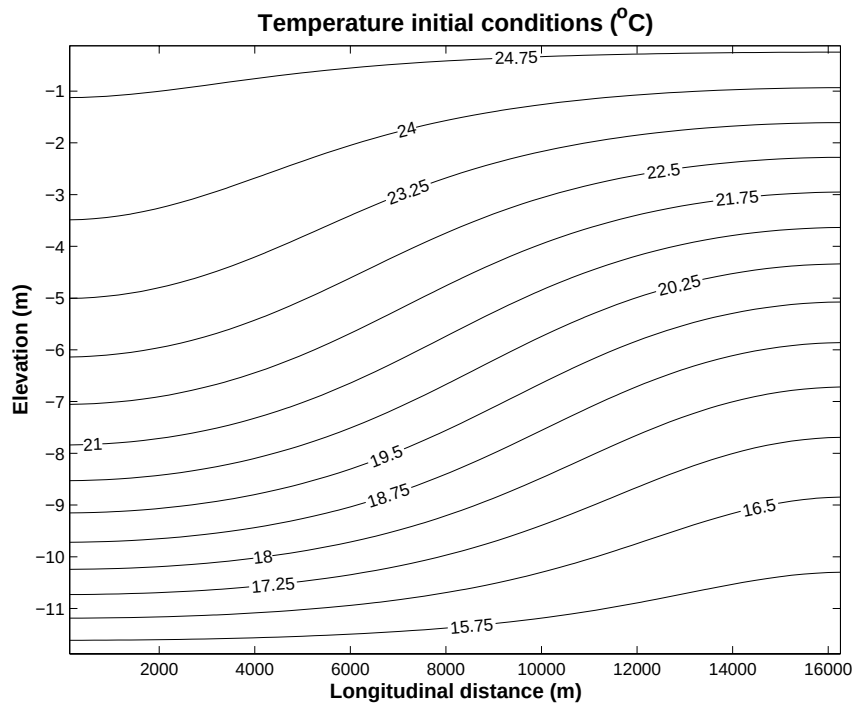


Figure 3.7. Temperature initial conditions used in the simulations

Additionally, RMA10 must be supplied with initial conditions for the derivatives of the velocity components. Those are readily derived from the analytical solution. The initial water surface elevation displacement was set to zero. The velocity amplitude is 10 cm/s , which, together with the buoyancy frequency, determines an expected frequency for the baroclinic oscillations of 58.9 hours. The background stratification, together with the size of the time step (Δt) and the horizontal grid resolution (Δx), determines the internal wave gravity Courant number, IWCN. Table 3.13 shows the range of IWCN in the simulations, obtained by modifying the size of the time steps.

Time step (seconds)	IWCN
360	0.68
720	1.37
1080	2.05

Table 3.13. Time steps and internal wave gravity Courant number.

E_{xx} (Pa · sec)	8.0 x 10 ⁴
E_z (Pa · sec)	1.0 x 10 ⁻¹⁵
Manning n	1.0 x 10 ⁻¹⁵
D_{xx} (m²/s)	1.0 x 10 ⁻⁷
D_z (m²/s)	1.0 x 10 ⁻¹⁵

Table 3.14. Values of transport coefficients used in RMA10.

With RMA10 no stable solution was obtained unless the scalar and momentum exchange coefficients were set to a value of $O(10^{-15})$ Pa · s or higher. The set of coefficients used in the simulations is shown in Table 3.14. The high value of the horizontal momentum transport coefficient that is needed to obtain a smooth solution with RMA10 is remarkable. Other combinations produced either non-smooth or unstable solutions. The value of the parameter α in RMA10 that determines the type of time stepping scheme (see Eq. 3.2) was fixed to 2.0 in order to avoid dissipation associated with the time discretization algorithm. In SI3D all attempts to perform the simulations with a single trapezoidal iteration with temperature as the active scalar were unsuccessful, even with reduced time steps. The solution yielded by SI3D was only stable when higher numbers of iterations (at least 3) were used. Five trapezoidal iterations were used in the simulations shown below. This result illustrates the stabilizing effect of the trapezoidal iterations.

3.12. Error measures in the internal wave problem

Table 3.15 lists error measures for the longitudinal velocity and density solutions obtained with RMA10 and SI3D. Those measures (described in Table 3.7) are obtained by comparing analytical and numerical time series for each variable at two different nodes within the test basin: the shallowest node at the center of the rectangular basin, and the mid-depth node at one end. The accuracy of the velocity solution is evaluated at the center node, while the error in the density solution is calculated at the end node. The location of those nodes was selected so that the amplitude of the longitudinal velocity and density oscillations was the maximum predicted by the analytical solution, guaranteeing in this form a maximum signal to noise ratio that facilitates the identification of phase lag and dissipation factors by the algorithms described in section 3.4.

Table 3.15 shows that there exists a fundamental difference between the two codes: the finite element code is more dissipative but represents the phases of the waves better than the finite difference code. The dissipation introduced by RMA10 is considerable even though it was attempted to minimize this dissipation as much as possible by reducing the values of the friction and viscosity/diffusivities (see Table 3.14) and arranging the code to use a centered time-stepping algorithm. The reason for the numerical dissipation is either in the need to include high values of horizontal momentum transfer coefficients to keep the code stable, or in the fact that density is modeled linearly in the code even though the active scalars are interpolated quadratically. This treatment of density might introduce effectively a dissipative term in the governing equations. Both codes introduce a phase lead in the numerical solution relative to the analytical time series at these levels of

spatial resolution. The difference in behavior of the codes, revealed in Table 3.15, makes the error measures very sensitive to the length of the simulation. The measures due to a phase error will oscillate with a period, depending on the celerity ratio. The dissipation error will produce error measures that will grow with the simulation period and will not be subject to oscillations. Hence, the results given in Table 3.15 should be interpreted cautiously.

		Longitudinal velocity		Density	
		SI3D	RMA10	SI3D	RMA10
$\Delta t=360$ s	A	0.9999	0.9994	1.0000	0.9996
	Q	1.0275	1.0172	1.0274	1.0149
	D	0.9837	0.9441	0.9828	0.9713
	l_1	0.2077	0.3500	0.2299	0.2617
	l_2	0.2424	0.3873	0.2677	0.2988
	l_∞	0.4392	0.5757	0.4061	0.4390
	RMSE	1.7173	2.7673	0.0766	0.0861
$\Delta t=720$ s	A	0.9999	0.9988	1.0001	0.9991
	Q	1.0274	1.0170	1.0274	1.0149
	D	0.9837	0.9441	0.9828	0.9713
	l_1	0.2076	0.3502	0.2297	0.2612
	l_2	0.2424	0.3872	0.2675	0.2988
	l_∞	0.4391	0.5774	0.4068	0.4398
	RMSE	1.7171	2.7669	0.0766	0.0861
$\Delta t=1080$ s	A	0.9998	0.9983	1.0001	0.9987
	Q	1.0274	1.0168	1.0273	1.0147
	D	0.9837	0.9446	0.9829	0.9712
	l_1	0.2073	0.3489	0.2296	0.2613
	l_2	0.2420	0.3859	0.2672	0.2992
	l_∞	0.4369	0.5737	0.4061	0.4393
	RMSE	1.7146	2.7574	0.0764	0.0862

Table 3.15. Error measures for $L/\Delta x = 128$. A = dissipation factor; Q = celerity ratio, D = Index of agreement, l_1 , l_2 and l_∞ = error norms, and $RMSE$ = root means squared error.

They indicate that for the length of the simulation the error measures for SI3D were smaller than those for RMA10. Figures 3.8 and 3.9 show the time series from the codes plotted against their analytical counterparts for simulations with $IWCN = 1.37$. The high-frequency oscillations observed in the velocity time series are of barotropic origin. They are caused by longitudinal differences in the weight of the water column that result from horizontal density variations.

The effect of a lower spatial resolution on the numerical results was evaluated by using the same vertical stratification in basins with modified dimensions (basin lengths of 4096, 8192 and 12288 m). The distance between nodes remained unchanged ($\Delta x = 256$ m), and the same range of time steps was used in these experiments. The values of the horizontal non-dimensional wavelength ($L/\Delta x$) for those basins are 32, 64 and 96 respectively. Table 3.16 presents the results of the error analysis for $L/\Delta x = 64$. They indicate a slight deterioration of the phase and amplitude error in SI3D primarily resulting in slightly higher error measures. In RMA10 the solution is degraded by increased dissipation levels; the phase error decreased slightly when coarser grids were used. Figures 3.10 and 3.11 show the time series of density and longitudinal velocities obtained with SI3D and RMA10 in the 8192 m basin, using $\Delta t = 720$ seconds. High frequency oscillations in the velocity time series are observed, which, as it was the case of Figs. 3.8 and 3.9, have a barotropic origin. The increased dissipation rate is obvious by comparing Figs. 3.10 and 3.8. Even though Fig. 3.8 was obtained with the same time step and nodal spacing, it shows lower values of dissipation than Fig. 3.10. The same trends observed in Table 3.15 for the error measures in response to the IWCN appear in Table 3.16.

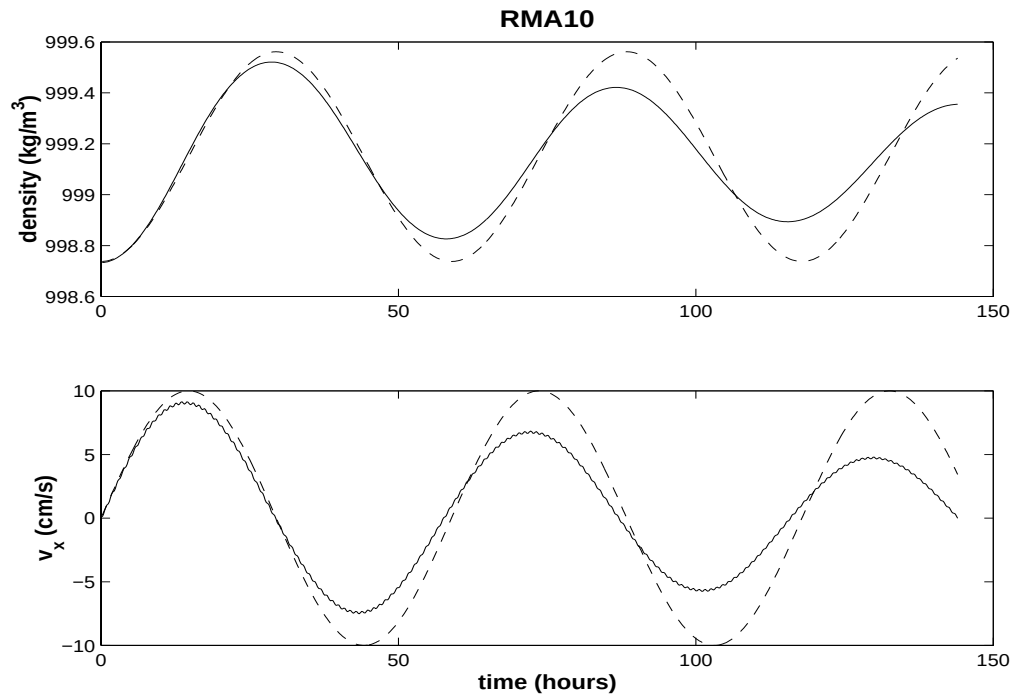


Figure 3.8. Time series of density and longitudinal velocity obtained with RMA10 (continuous line) and from analytical solution (dashed line) using $L/\Delta x = 128$.

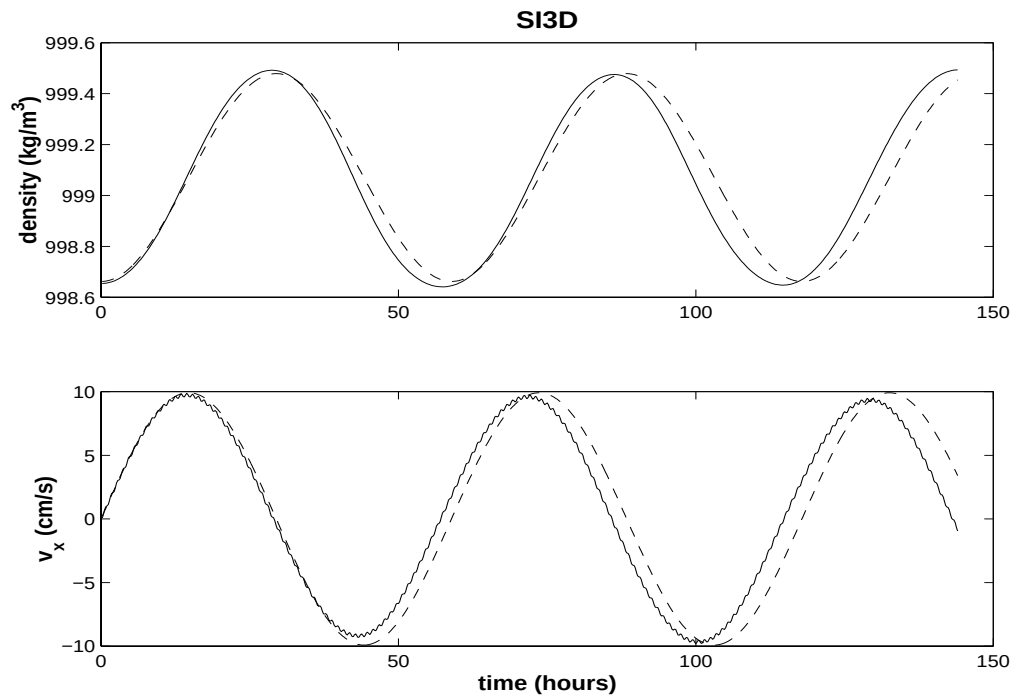


Figure 3.9. Time series of density and longitudinal velocity obtained with SI3D (continuous line) and from analytical solution (dashed line) using $L/\Delta x = 128$.

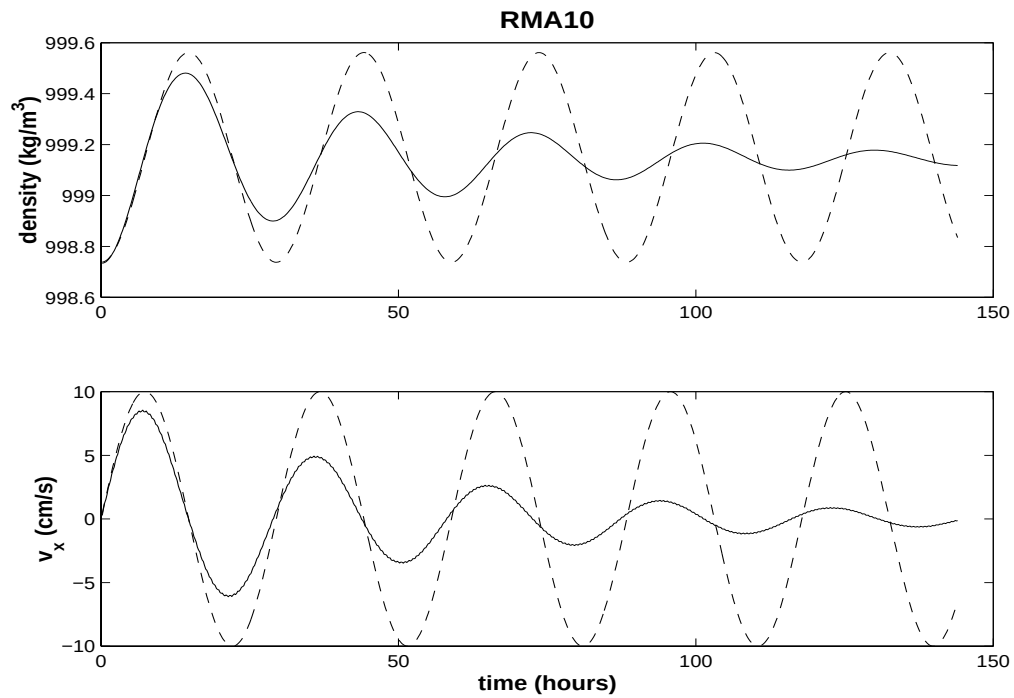


Figure 3.10. Time series of density and longitudinal velocity obtained with RMA10 (continuous line) and from analytical solution (dashed line) using $L/\Delta x = 64$.

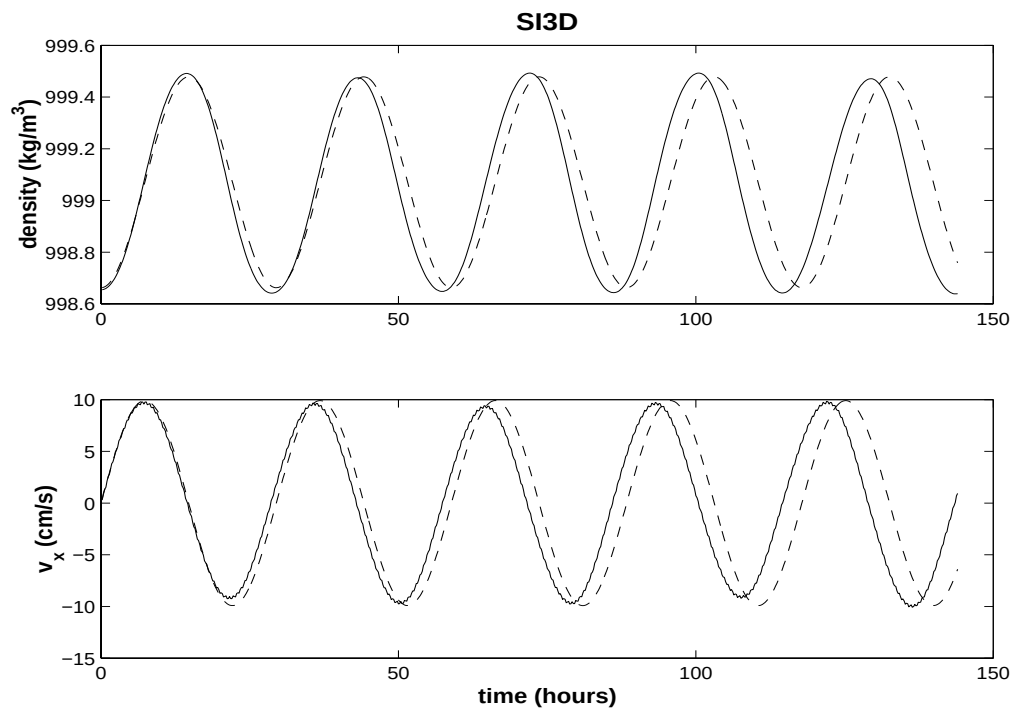


Figure 3.11. Time series of density and longitudinal velocity obtained with SI3D (continuous line) and from analytical solution (dashed line) using $L/\Delta x = 64$.

		Longitudinal velocity		Density	
		SI3D	RMA10	SI3D	RMA10
$\Delta t=360s$	A	1.0000	0.9979	1.0000	0.9982
	Q	1.0272	1.0149	1.0276	1.0142
	D	0.9416	0.6959	0.9361	0.7426
	l_1	0.4001	0.6909	0.4364	0.6483
	l_2	0.4613	0.7351	0.5065	0.7025
	l_∞	0.7540	0.9503	0.8700	0.9440
	RMSE	3.2600	5.2403	0.1457	0.2030
$\Delta t=720s$	A	0.9999	0.9959	1.0001	0.9963
	Q	1.0271	1.0150	1.0275	1.0146
	D	0.9420	0.6957	0.9365	0.7425
	l_1	0.3985	0.6910	0.4351	0.6484
	l_2	0.4595	0.7354	0.5051	0.7024
	l_∞	0.7475	0.9401	0.8676	0.9432
	RMSE	3.2479	5.2428	0.1452	0.2029
$\Delta t=1080s$	A	0.9999	0.9938	1.0001	0.9945
	Q	1.0268	1.0160	1.0273	1.0154
	D	0.9428	0.6966	0.9373	0.7428
	l_1	0.3958	0.6902	0.4324	0.6478
	l_2	0.4565	0.7348	0.5023	0.7022
	l_∞	0.7426	0.9508	0.8644	0.9424
	RMSE	3.2273	5.2393	0.1444	0.2028

Table 3.16. Error measures for $L/\Delta x = 64$. Symbols as in Table 3.15.

For the tests performed with the smallest of the basins ($L/\Delta x = 32$) no stable solution was obtained with SI3D with time steps greater than 90 seconds. RMA10 yielded a stable solution but this was extraordinarily dampened and the oscillations had disappeared after the first two oscillations.

3.13. Mass and energy conservation in representing stratified flows

The total kinetic energy, KE , total potential energy, PE , and total mass, M , in the system at a given time step are calculated as

$$KE = \iint_{x,y} \int_{z=0}^{z=H+\zeta} \frac{1}{2} \rho_0 (u^2 + v^2 + w^2) dz dx dy \quad (3.12.a)$$

$$PE = \iint_{x,y} \int_{z=0}^{z=H+\zeta} \rho g z dz dx dy \quad (3.12.b)$$

$$M = \iint_{x,y} \int_{z=0}^{z=H+\zeta} \rho dz dx dy \quad (3.12.c)$$

where ρ is density at location (x, y, z) , ρ_0 is a reference density, and (u, v, w) are the velocity components. The vertical location of the water surface is $H + \zeta$, measured from a datum taken at the base of the flat basin, and H is its equilibrium position (undisturbed state). The sum of KE and PE is the total energy E in the system and should remain constant in an enclosed basin that is not subject to frictional effects or any other sources or sinks of energy; hence this computation constitutes a good test to check for numerical dissipation or distortion of the temperature field. Similarly, the total mass in a closed system should remain constant. The test cases here are closed, and no sources of energy or mass exist. Hence, any variation in total energy or total mass in this test case is a numerical artifact and should be quantified.

The calculation of M , KE and PE in the finite element model needed the implementation of slight modifications in the source code, as RMA10 only supplies the value of water surface elevation, velocity and active scalars at nodal points. Integration, however, occurs naturally in the course of the computation of the element contributions to the stiffness matrix. Those contributions are nothing but the numerical integrals of the residuals within the domain of a given element. The integral over an individual element of a variable is calculated numerically by a weighted average of the value of the variables at the Gauss

points. The calculation of the M , KE and PE , involving the evaluation of the integrals in Eqs. 3.12.a-c, is then performed within the code and during the sweep through the elements in the computational domain that is repeated every time step. In the case of potential energy the integral calculation is performed as follows:

$$PE = \sum_{i=1}^{N_elements} PE_i \quad (3.13)$$

$$PE_i = \sum_{Gauss_point\ s} \rho'(x', y', z') \cdot g \cdot z' \cdot \left(\frac{h}{(b-a)} \right)^2 \cdot J \cdot w_{Gauss_point}$$

The primes in the latter equation denote the mapped variables in the computational coordinate system. In RMA10 the 3-D unsteady physical domain is mapped into a constant computational domain that does not suffer from deformations due to changes in the water surface elevation (King 1985). That coordinate mapping obeys the expression

$$\begin{aligned} \frac{z' - a}{b - a} &= \frac{z - a}{h} \\ x' &= x \\ y' &= y \end{aligned} \quad (3.14)$$

where h ($= H + \zeta$) is the thickness of the water column, b is the location of the undisturbed water surface elevation above a specified datum and a is the location of the bottom. $\rho'(x', y', z') = \rho(x, y, z)$ in Eq. 3.13. The factor $(h/(b-a))^2$ in equation 3.13 accounts for the coordinate transformation, w_{Gauss_point} is the weight given to the Gauss points, and J is the determinant of the Jacobian of the geometric transformation that allows one to map an irregular domain into the parent element with corners (in the case of a quadrilateral element) located at $(-1,-1)$, $(-1,1)$, $(1,-1)$ and $(1,1)$. For the calculation of KE

and M , a similar approach is followed. Here only the expressions for the KE are given, as the calculation of M follows immediately,

$$KE = \sum_{i=1}^{N_elements} KE_i \quad (3.15)$$

$$KE_i = \sum_{Gauss_points} \frac{1}{2} \cdot \rho_0 \cdot (u^2 + v^2 + w^2) \cdot J \cdot w_{gauss_point}$$

In the case of SI3D, the calculation of KE and PE can be done from the solution at any given time step. PE is calculated after all the computations of the hydrodynamic variables have been performed and is a straightforward adaptation of the "trapezoidal" rule for the evaluation of integrals. For the potential energy it reads

$$PE = \sum_i \sum_j \sum_k \rho(i, j, k) \cdot g \cdot z \cdot \Delta x \cdot \Delta y \cdot \Delta z \quad (3.16)$$

where $(\Delta x, \Delta y, \Delta z)$ are the fixed dimensions of the grid cell and equal to each other for all grid cells but for the surface and bottom cells. The bottom cells have as well a constant thickness in the case of a flat basin, and in this case only the surface cells are spatially and temporally variable.

In Table 3.17 is presented the fractional change in mass and total energy experienced in the numerical solutions during the 144 hours of simulations for the range of IWCN shown in Table 3.13 and $L/\Delta x = 128$. That change is calculated as the difference of mass or energy between its numerical estimate at the end of the simulation and that at the

beginning, normalized on the mass or energy content calculated from the initial conditions. Table 3.17 reveals that SI3D conserves both energy and mass better than RMA10 for all IWCN and for the degree of spatial resolution considered. The fractional change in energy shown by RMA10 for all IWCN is $O(10^{-5})$ and negative, confirming the dissipative character of the model. The use of larger time steps helps conserve energy, though. No trend in mass conservation was observed in the results obtained with RMA10. The solution calculated by SI3D, on the other hand, experiences changes in energy that are two orders of magnitude smaller than in RMA10, $O(10^{-7})$. The gain of mass or energy in SI3D during the simulations diminishes with higher values of IWCN, and eventually, as shown for values of $IWCN = 2.05$, mass and energy are lost.

	SI3D		RMA10	
IWCN	Mass change	Energy change	Mass change	Energy change
0.68	$+1.35716 \cdot 10^{-7}$	$+1.34155 \cdot 10^{-7}$	$+2.4435 \cdot 10^{-7}$	$-1.5042 \cdot 10^{-5}$
1.37	$+6.57203 \cdot 10^{-8}$	$+5.57237 \cdot 10^{-8}$	$+4.0725 \cdot 10^{-7}$	$-1.4364 \cdot 10^{-5}$
2.05	$-4.47415 \cdot 10^{-8}$	$-7.34791 \cdot 10^{-8}$	$+3.2580 \cdot 10^{-7}$	$-1.3459 \cdot 10^{-5}$

Table 3.17. Fractional change of mass and energy for the experiment with $L/\Delta x = 128$.

The fractional change shown in Table 3.17, however, does not reflect random or periodic changes suffered by the total mass and energy content during the course of simulation. In Table 3.18 will be found the ratio of the difference between the maximum value of the mass or energy and its minimum, to the average value of mass or energy during the simulation period. Table 3.18 indicates that the random changes in mass in RMA10 are higher than the changes in the 144-hour simulation period shown in Table 3.17. The

random changes in mass experienced by the solution computed with RMA10 is roughly one magnitude of order greater than by the solution calculated with SI3D.

	SI3D		RMA10	
IWCN	Mass change	Energy change	Mass change	Energy change
0.68	$1.3572 \cdot 10^{-7}$	$1.5774 \cdot 10^{-7}$	$1.7104 \cdot 10^{-6}$	$1.6794 \cdot 10^{-5}$
1.37	$6.5720 \cdot 10^{-8}$	$8.6549 \cdot 10^{-8}$	$1.7919 \cdot 10^{-6}$	$1.5376 \cdot 10^{-5}$
2.05	$4.5636 \cdot 10^{-8}$	$1.0457 \cdot 10^{-7}$	$1.3032 \cdot 10^{-6}$	$1.6231 \cdot 10^{-5}$

Table 3.18. Oscillations in conserved quantities vs. mean values during the course of simulations with $L/\Delta x = 128$.

Tables 3.17 and 3.18 show the amount of energy variation in the system, expressed as a fraction of the initial energy level, introduced by the numerical algorithms without referring to the mechanisms causing the error. The total potential energy has two subcomponents: the available potential energy, PE_a , that is actually transformed periodically into kinetic energy, and what Winters et al. (Winters *et al.* 1995) call background potential energy, PE_b , i.e. the potential energy in the undisturbed state. In the case of isothermal fluids the background potential energy is defined as the potential energy in the absence of any deviation of the water surface elevation from its equilibrium level. Any variations in the background potential energy are entirely due to mass loss in these circumstances. In stratified basins, however, diapycnal mixing and the non-conservation of active scalars may lead to changes in the background potential energy. Numerical diffusion is the main source of diapycnal mixing in purely advective simulations, which can induce variations in the background potential energy.

A transport scheme that either does not conserve total mass or that introduces numerical diffusion will not conserve the background potential energy. In order to separate the two effects, the total potential energy was estimated as a sum of two different components

$$\begin{aligned}
 PE &= PE_{mean} + PE' = \iint_{x,y} \int_{z=0}^{z=H+\eta} (\rho_{mean} + \rho') gz \, dz \, dx \, dy \\
 &= \rho_{mean} g \iint_{x,y} \int_{z=0}^{z=H+\eta} z \, dz \, dx \, dy + \iint_{x,y} \int_{z=0}^{z=H+\eta} \rho' gz \, dz \, dx \, dy
 \end{aligned} \tag{3.17}$$

in which ρ_{mean} stands for the mean density in the 3-D domain, and ρ' is the deviation from the mean in each water column. Changes in the first component PE_{mean} will represent changes in PE_b due to loss of mass in the whole domain. The second component PE' will be negative for stable stratification. It will capture the low-frequency changes in PE_b caused by diapycnal mixing or distortion of the density field, and it will also incorporate PE_a subject to oscillations with frequency equal to that of the baroclinic seiche. Diapycnal mixing will in general increase the magnitude of PE' , which eventually will become zero when fully mixed conditions are attained. The changes in PE_{mean} account for 98% of the changes in total energy in SI3D. In RMA10, however, PE_{mean} increases with time, while the PE and KE decrease.

Figures 3.12 and 3.13 show the time series of an estimate of PE_a and the calculated KE for both codes when they are applied to the solution of the test problem with $L/\Delta x = 128$, $\Delta t = 720$ seconds and a 1 m vertical resolution. The estimate of PE_a is obtained by subtracting its minimum value from the component PE' . The oscillatory part of PE' is the

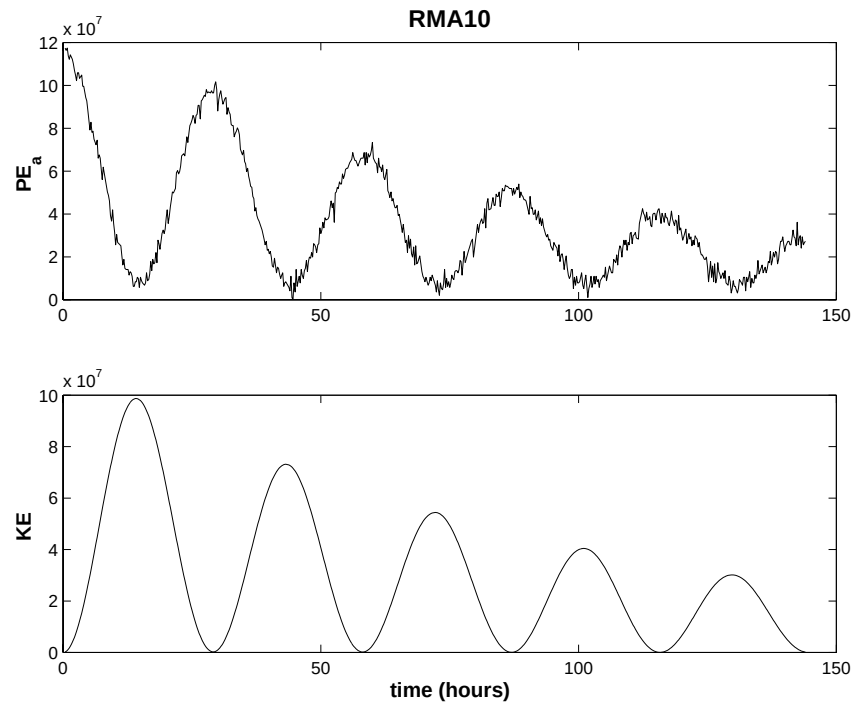


Figure 3.12. Evolution of the estimates of PE_a and KE in the simulations with RMA10 using $\Delta t = 720s$ and $L/\Delta x = 128$.

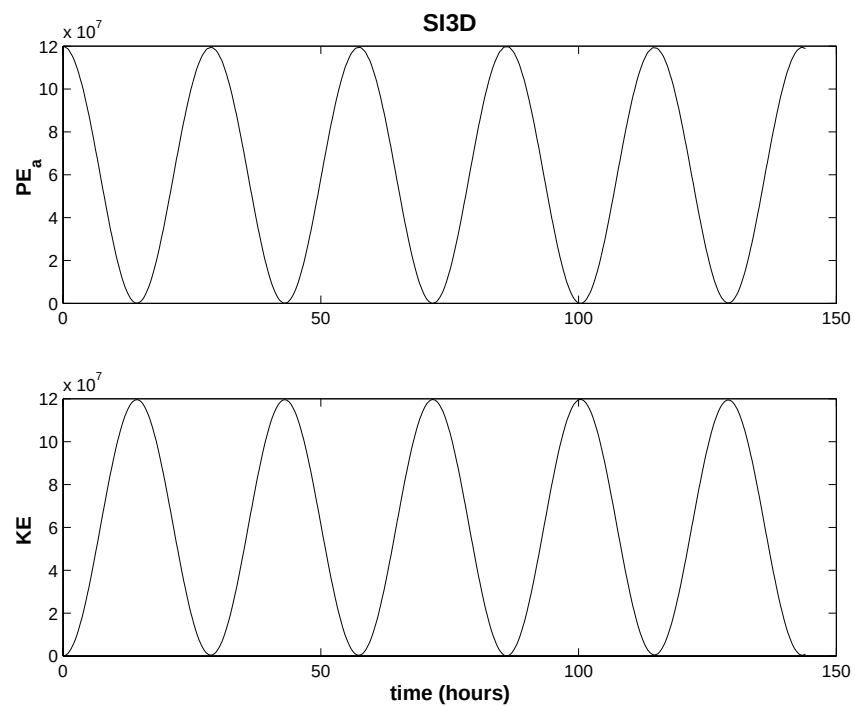


Figure 3.13. Evolution of the estimates of PE_a and KE in the simulations with SI3D using $\Delta t = 720s$ and $L/\Delta x = 128$.

PE_a and the non-oscillatory part represent the part of PE_b due to changes in density structure about the mean density of the whole system. The non-oscillatory component in Figs. 3.12 and 3.13 remains roughly constant in both cases, indicating that neither of the two codes introduces significant distortion to the background density field. The behavior of the oscillatory component is different in the two codes. The amplitude decays in RMA10, while it remains unchanged with time in the solution calculated by SI3D.

3.14. Computational time for stratified flows

The codes were timed for the solution of the problem of simulating internal waves in a rectangular flat-bottom basin to estimate differences in computational costs. The dimensions of the basin are 8192 x 512 x 12 m, and the computational nodes are separated 256 m in the horizontal and 1 m apart in the vertical. Given the differences in the arrangement of nodes between the two codes, the total number of computational points is 785 in RMA10 and 768 in SI3D. The time step is 720 seconds. A total of 5 trapezoidal iterations is performed in SI3D, and RMA10 may perform up to 5 iterations on the hydrodynamic components and 3 on the scalar transport equations. The codes are provided with adequate initial conditions. The experiments were run on an SGI Octane with a 175 MHz IP30 processor and 386 Mbytes of memory. An SGI MIPSpro f90 compiler version 7.3 with optimization level 2 was used. The time involved in the solution of the problem with RMA10 is 4603 seconds. The same problem is solved with SI3D in 50 seconds. The ratio of computational costs per node is roughly 90:1.

3.15. Summary

A series of benchmark problems have been presented that allow one to quantify the ability of 3-D codes to represent relevant phenomena in lake hydrodynamics. The tests consist of the simulation of linear phenomena with analytical solutions (Table 3.1 and section 3.11). The list of tests is not by any means a complete list of cases for which analytical solutions exist. Only those considered relevant to test the elements of the codes involved in the simulation of the barotropic and density-driven motions in natural lakes were included. Attention is focused on the representation of the frequency of barotropic seiches, the effects introduced through the approximation of curved boundaries (shores and bottom) on that representation, the ability to capture the effects of wind stress over the surface of a well-mixed and closed basin, and finally on the representation of internal waves or baroclinic seiches. The investigation included a detailed look at phase errors introduced by the schemes. The phase errors may not be important when dealing with individual components, as natural dissipation eventually eliminates oscillations before the lag becomes of considerable magnitude. However, in natural lakes, oscillations do not exist in isolation, and the interaction between the different wave components critically depends on the phase of the individual components.

The tests have been used in the comparison of two codes, selected as plausible candidates to provide the skeleton for a 3-D model of lake environments: SI3D, a finite difference code, and RMA10, a finite element model. RMA10 gave fairly accurate results when the trapezoidal time-stepping scheme is used and the grid is "sufficiently" refined. However, to obtain stable solutions in 3-D, artificial dissipation must be added by means of a non-

centered scheme, horizontal diffusion terms in the momentum equations or bottom friction. These sources of dissipation do not obey natural laws and are primarily included for stability reasons. The non-centered scheme is particularly dissipative, especially when dealing with waves that are not well resolved. This feature negates the introduction of higher-order turbulence schemes that try to model the dissipation due to non-resolved high frequency phenomena in the transport of momentum. Another of the serious pitfalls of RMA10 is the computational cost. It is very high per computational node, compared with SI3D, especially in 3-D. The simulations in 3-D, with more than one element in the vertical, are at least five times more expensive per computational node than those same simulations carried with only one element in the vertical or 2-D. Even though it is computationally expensive, the code is able to represent well a complex boundary with fewer number of nodes in the grid and compensate in this manner for the high cost per node. Grids with fewer nodes may seem to be an advantage in terms of computational time, but they are a source of phase error when dealing with higher modes of oscillation, as illustrated in the circular basin when simulating oscillations with a nodal circle.

SI3D uses a semi-implicit approach in the solution of the governing equations on a structured grid. The algorithm is extraordinarily fast, which makes the code at least 30 times faster than RMA10 when dealing with a similar number of nodes. The performance of the code is equally good in 2-D and 3-D. In 3-D simulations, RMA10 can be 150 times slower than SI3D when solving the equations with a similar number of nodes. Given the considerable difference in speed, one could afford, for 2-D runs, 5 times the resolution in the horizontal, and for 3-D, 10 times the resolution in the horizontal and the same

resolution in the vertical as in RMA10. This increase in the number of nodes is needed in order to represent curved boundaries with "reasonable" accuracy. The deficient definition of the shape of the boundaries introduces a considerable phase error in the solution. It is the phase error in basins with curved shorelines that makes the performance of the code worse than RMA10. The performance in rectangular basins is quite similar. The use of a large number of nodes offers a spatial resolution that allows SI3D to represent higher modes of oscillations better than RMA10 (ND-simulations vs. NC-simulations).

In the internal wave tests the ratio of speed per computational time is roughly 90:1, SI3D being faster than RMA10. SI3D outperformed RMA10 in all aspects except in the phase representation of internal waves in simple rectangular basins. The finite element code shows a considerable amount of dissipation that is built into the code and that does not depend on the time-stepping scheme used. This results in the damping and incorrect representation of internal waves, a feature that some users of RMA10 have noticed in their applications.

The tests are aimed at comparing the numerical algorithms, leaving aside a number of phenomena that may compromise the performance of the codes in applications to natural lakes. First of all, non-linear effects have not been considered and might be a source of instabilities. Second, and probably most important, it was assumed that the geometry of the basin and the boundary conditions are perfectly known. In the simulation of natural systems, however, one's knowledge of the boundary conditions is very approximate. The actual geometry of natural water bodies is never known as well as the geometry of the

circular basin. They are fractal objects, and even with the use of finite element meshes one introduces considerable simplification, as one strives to keep the computational cost within reasonable limits. And last, but not the least, the effects of turbulent phenomena in the large-scale motion can be resolved are not completely understood. The sources of error in applications to natural water bodies are considerable, and the numerical algorithms are only one of them. These need to be quantified so one can try to define the limits of one's confidence in numerical simulations of those systems.

CHAPTER 4

THREE-DIMENSIONAL HYDRODYNAMIC LAKE MODEL

4.1. Introduction

Given the lack of numerical dissipation and the high computational efficiency shown by SI3D in the solution of the benchmark problems in the previous chapter, it was adopted as the skeleton for a 3-D model of lake hydrodynamics. The disagreement between the analytical and numerical solution obtained with SI3D was mainly due to phase error. This error, however, can be partly compensated for by the increased spatial resolution needed to obtain a reasonable representation of curved boundaries.

In this chapter details are presented on the strategy used in SI3D for the solution of the 3-D hydrodynamic equations. For a thorough presentation of the leapfrog-trapezoidal method the reader is referred to Smith and Larock 1993. Special emphasis is placed in section 4.3 on the representation of the wind stress at the free surface, probably the main energizing mechanism in large lakes with large residence times, though not very well understood. This chapter does not discuss either the modeling of turbulent phenomena or the numerical solution of the transport equation. These features are presented with detail in chapter 5.

4.2 Numerical method

The model SI3D (Smith and Larock 1993, Smith 1997, Smith and Larock 1997) uses a semi-implicit leapfrog-trapezoidal finite difference scheme on a staggered grid to solve the layer-averaged form of the governing equations. The grid system is composed of horizontal layers, and the governing equations are integrated over the height of each layer. One advantage of the layer-averaged form is that the depth-integrated continuity equation (Eq.4.1) is effectively linear. Another benefit of integrating the governing equations is that the equations for a single-layer system reduce to a conservative form of the 2-D vertically averaged equations; thus in areas of shallow water where only one layer is required, the 3-D model becomes automatically a 2-D model without any special modifications to the model coding. The integrated form of the governing equations for layer k is given by (Smith 1997, p. 86).

Continuity equations

$$\frac{\partial \zeta}{\partial t} + \frac{\partial}{\partial x} \left[\sum_{k=1}^{km} U_k \right] + \frac{\partial}{\partial y} \left[\sum_{k=1}^{km} V_k \right] = 0 \quad (4.1)$$

$$w|_{k+1/2}^{k-1/2} = -\frac{\partial U_k}{\partial x} - \frac{\partial V_k}{\partial y} \quad (4.2)$$

Momentum equations

$$\begin{aligned} \frac{\partial U_k}{\partial t} + \frac{\partial (Uu)_k}{\partial x} + \frac{\partial (Vu)_k}{\partial y} + uw|_{k+1/2}^{k-1/2} - fV_k = \\ \frac{h_k}{\rho_k} g \rho_1 \frac{\partial \zeta}{\partial x} - \frac{h_k}{\rho_k} \left[\frac{gh_1}{2} \frac{\partial \rho_1}{\partial x} + \sum_{m=2}^k \left(\frac{gh_{m-1}}{2} \frac{\partial \rho_{m-1}}{\partial x} + \frac{gh_m}{2} \frac{\partial \rho_m}{\partial x} \right) \right] + \\ \frac{\partial}{\partial x} \left(K_H h \frac{\partial u}{\partial x} \right)_k + \frac{\partial}{\partial y} \left(K_H h \frac{\partial u}{\partial y} \right)_k + \left(K_V \frac{\partial U}{\partial z} \right)_{k-1/2} - \left(K_V \frac{\partial U}{\partial z} \right)_{k+1/2} \end{aligned} \quad (4.3)$$

$$\begin{aligned}
& \frac{\partial V_k}{\partial t} + \frac{\partial(Uv)_k}{\partial x} + \frac{\partial(Vv)_k}{\partial y} + vw|_{k+1/2}^{k-1/2} + fU_k = \\
& \frac{h_k}{\rho_k} g \rho_1 \frac{\partial \zeta}{\partial y} - \frac{h_k}{\rho_k} \left[\frac{gh_1}{2} \frac{\partial \rho_1}{\partial y} + \sum_{m=2}^k \left(\frac{gh_{m-1}}{2} \frac{\partial \rho_{m-1}}{\partial y} + \frac{gh_m}{2} \frac{\partial \rho_m}{\partial y} \right) \right] + \\
& \frac{\partial}{\partial x} \left(K_H h \frac{\partial V}{\partial x} \right)_k + \frac{\partial}{\partial y} \left(K_H h \frac{\partial V}{\partial y} \right)_k + \left(K_V \frac{\partial V}{\partial z} \right)_{k-1/2} - \left(K_V \frac{\partial V}{\partial z} \right)_{k+1/2}
\end{aligned} \tag{4.4}$$

In Eqs. 4.1 through 4.4, the layer-integrated volume transports (U , V) replace the velocity components (u , v) as the primary dependent variables. All other variables identified by using the notation already introduced in chapter 2.

The arrangement of variables in the 3-D domain follows the C-Arakawa staggered scheme (Smith 1997). The C-Arakawa arrangement has the advantage that difference quotients are readily centered in space to obtain second-order accuracy. Figure 4.1 displays the horizontal arrangement of variables on the numerical grid. The horizontal dimensions of each cell are Δx and Δy , which are taken as constant and equal. The vertical locations of the variables are pictured in Fig. 4.2 for the multi-level grid. The center of the 3-D cell is numbered with indices i , j , k . Pressure ($p_{i,j,k}$), active scalar concentration ($s_{i,j,k}$) and density ($\rho_{i,j,k}$) are defined at center. The x-direction velocity component ($u_{i+1/2,j,k}$) and volumetric transport ($U_{i+1/2,j,k}$) are located at half-integers of i and whole integers of j and k . The y-direction velocity component ($v_{i,j+1/2,k}$) and volumetric transport ($V_{i,j+1/2,k}$) are located at half-integers of j and whole integers of i and k . The vertical velocity component ($w_{i,j+1/2,k}$) is calculated at each layer interface between cell centers. The water surface elevation ($\zeta_{i,j}$) is a two-dimensional variable defined at the

center of each horizontal grid cell with integer values of i and j . The layer height ($h_{i,j,k}$) is defined at each cell center. The height of the surface layer and the bottom layer can vary in space, and in the surface layer it can also vary in time.

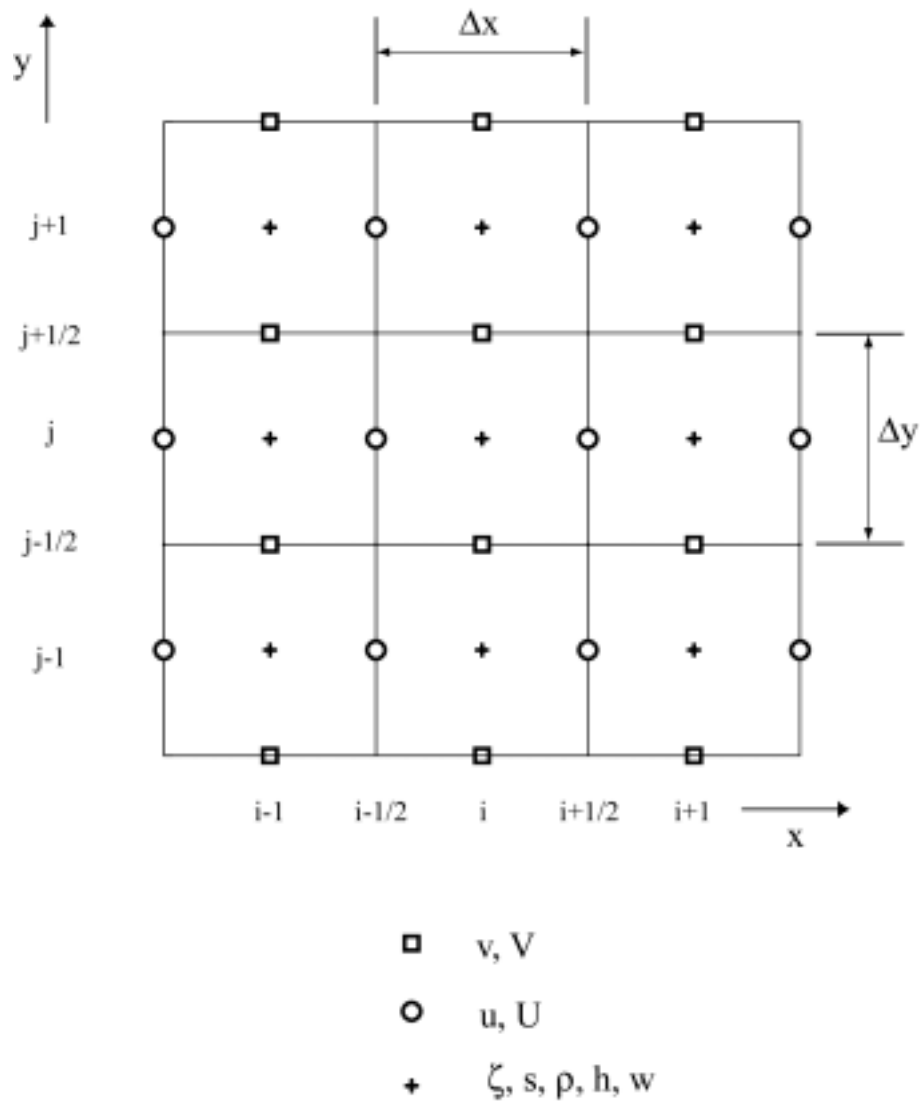


Figure 4.1. The space-staggered horizontal grid used in the 3-D model showing the location of variables (adapted from Smith 1997).

The semi-implicit method used in SI3D is based on treating the gravity wave and vertical diffusion terms implicitly while treating all other terms, including advection, explicitly. In this manner the method is guaranteed to be stable without the time-step limitation imposed by the gravity wave CFL condition. The finite-difference form of the governing equations for the leapfrog semi-implicit scheme can be written as

$$\begin{aligned} \zeta_{i,j}^{n+1} = & \zeta_{i,j}^{n-1} - \frac{\Delta t}{\Delta x} \left[\sum_{k=1}^{km} (U_{i+1/2,j,k}^{n+1} - U_{i-1/2,j,k}^{n+1} + U_{i+1/2,j,k}^{n-1} - U_{i-1/2,j,k}^{n-1}) \right] \\ & - \frac{\Delta t}{\Delta y} \left[\sum_{k=1}^{km} (V_{i,j+1/2,k}^{n+1} - V_{i,j-1/2,k}^{n+1} + V_{i,j+1/2,k}^{n-1} - V_{i,j-1/2,k}^{n-1}) \right] \end{aligned} \quad (4.5)$$

$$\begin{aligned} U_{i+1/2,j,k}^{n+1} = & \hat{U}_{i+1/2,j,k} - g \frac{\Delta t}{\Delta x} \bar{h}_{i+1/2,j,k}^n \left(\frac{\bar{\rho}_{i+1/2,j,1}^{-n}}{\bar{\rho}_{i+1/2,j,k}^{-n+1}} \right) (\zeta_{i+1,j}^{n+1} - \zeta_{i,j}^{n+1} + \zeta_{i+1,j}^{n-1} - \zeta_{i,j}^{n-1}) \\ & + \Delta t \left(K_{V_{i+1/2,j,k-1/2}}^n \cdot \left(\frac{(U/\bar{h})_{i+1/2,j,k-1}^{n+1} - (U/\bar{h})_{i+1/2,j,k}^{n+1}}{\bar{h}_{i+1/2,j,k-1/2}^{n+1}} + \frac{(u)_{i+1/2,j,k-1}^{n-1} - (u)_{i+1/2,j,k}^{n-1}}{\bar{h}_{i+1/2,j,k-1/2}^{n-1}} \right) \right) \\ & - \Delta t \left(K_{V_{i+1/2,j,k+1/2}}^n \cdot \left(\frac{(U/\bar{h})_{i+1/2,j,k}^{n+1} - (U/\bar{h})_{i+1/2,j,k+1}^{n+1}}{\bar{h}_{i+1/2,j,k+1/2}^{n+1}} + \frac{(u)_{i+1/2,j,k}^{n-1} - (u)_{i+1/2,j,k+1}^{n-1}}{\bar{h}_{i+1/2,j,k+1/2}^{n-1}} \right) \right) \end{aligned} \quad (4.6)$$

$$\begin{aligned} V_{i,j+1/2,k}^{n+1} = & \hat{V}_{i,j+1/2,k} - g \frac{\Delta t}{\Delta y} \bar{h}_{i,j+1/2,k}^n \left(\frac{\bar{\rho}_{i,j+1/2,1}^{-n}}{\bar{\rho}_{i,j+1/2,k}^{-n+1}} \right) (\zeta_{i,j+1}^{n+1} - \zeta_{i,j}^{n+1} + \zeta_{i,j+1}^{n-1} - \zeta_{i,j}^{n-1}) \\ & + \Delta t \left(K_{V_{i,j+1/2,k-1/2}}^n \cdot \left(\frac{(V/\bar{h})_{i,j+1/2,k-1}^{n+1} - (V/\bar{h})_{i,j+1/2,k}^{n+1}}{\bar{h}_{i,j+1/2,k-1/2}^{n+1}} + \frac{(v)_{i,j+1/2,k-1}^{n-1} - (v)_{i,j+1/2,k}^{n-1}}{\bar{h}_{i,j+1/2,k-1/2}^{n-1}} \right) \right) \\ & - \Delta t \left(K_{V_{i,j+1/2,k+1/2}}^n \cdot \left(\frac{(V/\bar{h})_{i,j+1/2,k}^{n+1} - (V/\bar{h})_{i,j+1/2,k+1}^{n+1}}{\bar{h}_{i,j+1/2,k+1/2}^{n+1}} + \frac{(v)_{i,j+1/2,k}^{n-1} - (v)_{i,j+1/2,k+1}^{n-1}}{\bar{h}_{i,j+1/2,k+1/2}^{n-1}} \right) \right) \end{aligned} \quad (4.7)$$

The subscripts denote the spatial location in the computational grid, and the superscripts the time level at which the variable is evaluated. The symbols k_1 and k_m in Eq. 4.1 denote the first (shallowest) and last (deepest) layer in a water column. In Eq. 4.6 the overbar ($\bar{}$) on a layer height or density variable is used to represent a spatial average in the x-direction between adjacent values. For example,

$$\bar{h}_{i+1/2,j,k} = (\bar{h}_{i,j,k} + \bar{h}_{i+1,j,k}) / 2 \quad (4.8)$$

Also, $\bar{\bar{h}}_{i+1/2,j,k-1/2}$ is defined to be the average of layer heights $\bar{h}_{i+1/2,j,k-1}$ and $\bar{h}_{i+1/2,j,k}$. The symbol ($\hat{}$) denotes a solution for the layer volumetric transport that includes only the contribution from the advection, Coriolis, baroclinic pressure and horizontal diffusion terms, treated explicitly in the semi-implicit scheme (see Appendix D in Smith 1997 for a description of the expanded finite-difference forms of these terms). All the explicit terms, with the exception of the horizontal diffusion, are centered in time, i.e. at time level n . The horizontal diffusion is treated, for stability purposes, at time level $n-1$. In Eq. 4.7 the overbar is again used to represent a spatial average between adjacent values, but this time in the y-direction,

$$\bar{h}_{i,j+1/2,k} = (\bar{h}_{i,j,k} + \bar{h}_{i,j+1,k}) / 2 \quad (4.9)$$

and $\bar{\bar{h}}_{i+1/2,j,k-1/2}$ is defined to be the average of layer heights $\bar{h}_{i+1/2,j,k-1}$ and $\bar{h}_{i+1/2,j,k}$. Similarly, as it was for the x-momentum equation, the symbol ($\hat{}$) in Eq. 4.7 represents the contribution of the explicit terms.

The wind stress is specified as a forcing function at the free surface by substituting

$$K_{V_{i+1/2,j,1/2}}^n \cdot \left(\frac{(U/\bar{h})_{i+1/2,j,0}^{n+1} - (U/\bar{h})_{i+1/2,j,1}^{n+1}}{h_{i+1/2,j,1/2}^{n+1}} + \frac{(u)_{i+1/2,j,0}^{n-1} - (u)_{i+1/2,j,1}^{n-1}}{h_{i+1/2,j,1/2}^{n-1}} \right) = \frac{(\tau_{xs})_{i+1/2,j,1/2}^n}{\rho_{i+1/2,j,1}^n} \quad (4.10)$$

$$K_{V_{i,j+1/2,1/2}}^n \cdot \left(\frac{(V/\bar{h})_{i,j+1/2,0}^{n+1} - (V/\bar{h})_{i,j+1/2,1}^{n+1}}{h_{i,j+1/2,1/2}^{n+1}} + \frac{(v)_{i,j+1/2,0}^{n-1} - (v)_{i,j+1/2,1}^{n-1}}{h_{i,j+1/2,1/2}^{n-1}} \right) = \frac{(\tau_{ys})_{i,j+1/2,1/2}^n}{\rho_{i,j+1/2,1}^n} \quad (4.11)$$

The symbols τ_{xs} and τ_{ys} in Eqs. 4.10 and 4.11 are the x- and y-components of the wind shear stress, and their subscripts define the spatial location where it is evaluated. In the model the wind stress is assumed, in general, to be spatially variable, and it is calculated as a quadratic function of the wind speed (see Eq.4.17 and Eq.4.36). The frictional stress, on the other hand, is specified at the bottom by substituting

$$K_{V_{i+1/2,j,km+1/2}}^n \cdot \left(\frac{(U/\bar{h})_{i+1/2,j,km}^{n+1} - (U/\bar{h})_{i+1/2,j,km+1}^{n+1}}{h_{i+1/2,j,km+1/2}^{n+1}} + \frac{(u)_{i+1/2,j,km}^{n-1} - (u)_{i+1/2,j,km+1}^{n-1}}{h_{i+1/2,j,km+1/2}^{n-1}} \right) = \frac{(\tau_{xb})_{i+1/2,j,km+1/2}^n}{\rho_{i+1/2,j,km}^n} \quad (4.12)$$

$$K_{V_{i,j+1/2,km+1/2}}^n \cdot \left(\frac{(V/\bar{h})_{i,j+1/2,km}^{n+1} - (V/\bar{h})_{i,j+1/2,km+1}^{n+1}}{h_{i,j+1/2,km+1/2}^{n+1}} + \frac{(v)_{i,j+1/2,km}^{n-1} - (v)_{i,j+1/2,km+1}^{n-1}}{h_{i,j+1/2,km+1/2}^{n-1}} \right) = \frac{(\tau_{yb})_{i,j+1/2,km+1/2}^n}{\rho_{i,j+1/2,km}^n} \quad (4.13)$$

The subscript k_{m+1} refers to a fictitious layer. The frictional stresses in the x- and y-directions, τ_{xb} and τ_{yb} , are calculated using the quadratic law 2.15 in terms of the horizontal volumetric transports computed in the bottom layer and a drag coefficient that is assumed spatially uniform and constant in time (see Eqs. 4.29 and 4.30 in Smith 1997).

The new water surface elevation is calculated by substituting Eqs. 4.6 and 4.7 into Eq. 4.5, resulting in a penta-diagonal matrix which is symmetric and positive definite. The system is then solved using a preconditioned non-symmetric conjugate gradient method (Golub and Loan 1996, Kincaid *et al.* 1989). The modified incomplete Cholesky method is used as preconditioner. After the new free surface elevation is computed, the horizontal velocity components are calculated using 4.6 and 4.7. The vertical velocity is then updated using the continuity equation (Eq. 2.1), which in finite difference form is

$$w_{i,j,k-1/2}^{n+1} = w_{i,j,k+1/2}^{n+1} - \frac{(U_{i+1/2,j,k}^{n+1} - U_{i-1/2,j,k}^{n+1})}{\Delta x} - \frac{(V_{i,j+1/2,k}^{n+1} - V_{i,j-1/2,k}^{n+1})}{\Delta y} \quad k = 2,3,...,km \quad (4.14)$$

This equation is solved explicitly, starting from the bottom layer in which $w_{i,j,km+1/2}^{n+1} = 0$.

The finite difference equations for the semi-implicit trapezoidal scheme are nearly identical to those of the semi-implicit leapfrog scheme, except that the time interval over which the scheme is applied is halved to Δt from $2\Delta t$. The integration procedure is centered at time level $(n+1/2)\Delta t$ in the trapezoidal method and no longer involves the time level $(n-1)\Delta t$. The equations for the trapezoidal method can be easily derived from Eqs. 4.1-4.10 by making the changes in the subscripts indicated in Table 4.1 and replacing Δt by $\Delta t/2$ (see Smith 1997 for a detailed presentation). The variables needed in the scheme at time level $(n+1/2)\Delta t$ are determined by averaging the value of the variable at time level $n\Delta t$ and the estimate of the variable at time $(n+1)\Delta t$ obtained in a previous leapfrog or trapezoidal iteration.

<i>Semi-implicit leapfrog scheme</i>	<i>Semi-implicit trapezoidal scheme</i>
$(n+1) \Delta t$	$(n+1) \Delta t$
$n \Delta t$	$(n+1/2) \Delta t$
$(n-1) \Delta t$	$n \Delta t$

Table 4.1. Simple guidelines to derive the finite-difference equations for the semi-implicit trapezoidal scheme from the discretized form of the equations using the semi-implicit leapfrog method (Eq.4.1-Eq.4.10). The subscripts in the equations for the leapfrog method, on the left column, should be replaced by those shown in the right column to obtain the finite-difference equations for the semi-implicit trapezoidal method.

4.3. Wind stress boundary conditions for the momentum equations

Wind forcing is the main driving mechanism in natural lakes with long residence times (so that the influence of inflows and outflows can be neglected). Two types of circulation can be driven by wind forcing in enclosed basins, referred to in the literature as direct and residual circulations (Cheng and Casulli 1982, Strub and Powell 1986). The first is the result of the repeated occurrence of internal wave cycles in the basin. The instantaneous velocities associated with the waves fluctuate, causing water parcels to move in an oscillatory fashion that results in a small net displacement after each oscillation (Strub and Powell 1986). The mean velocity field that results from averaging the instantaneous velocities over several wave periods is not zero²⁰, and it characterizes the residual circulation. In the direct circulation the parcels of water will move continuously in large gyres and the mean velocities are similar to the instantaneous velocities.

Strub and Powell (Strub and Powell 1986) studied the sources and sinks of circulation of the surface layer using the linear, baroclinic, transport equation for vorticity. For the surface layer the linear transport vorticity equation of a layered system is

²⁰ The mean velocities are much less than the instantaneous values

$$\frac{\partial \omega}{\partial t} = \underbrace{-gJ(h_1, \eta)}_{(1)} - \underbrace{f \left[\frac{\partial U}{\partial x} + \frac{\partial V}{\partial y} \right]}_{(2)} + \underbrace{\frac{1}{\rho} \bar{k} \cdot \text{curl}(\tau_s)}_{(3)} - \underbrace{\frac{1}{\rho} \bar{k} \cdot \text{curl}(\tau_b)}_{(4)} - \underbrace{K_h \nabla_h^2 \omega}_{(5)} \quad (4.15)$$

where ω is the vorticity of the layer transport, g is the acceleration of gravity, h_1 is the thickness of the surface layer, η is the water surface elevation, f is the Coriolis parameter, U and V are the vertically-integrated x- and y- transports in the layer, ρ is the density, τ_s is wind stress, τ_b is the stress at the bottom of the surface layer, K_h is the horizontal eddy viscosity, ∇_h is the horizontal gradient operator and J is the Jacobian operator,

$$J(h_1, \eta) = \left(\frac{\partial h_1}{\partial x} \frac{\partial \eta}{\partial y} - \frac{\partial h_1}{\partial y} \frac{\partial \eta}{\partial x} \right) \quad (4.16)$$

The five terms on the right side represent the sources or sinks of vorticity, and the generation of cyclonic or anti-cyclonic vorticity depends on the balance of those terms. They represent:

- (1) the generation of vorticity by baroclinic pressure forces arising from internal surfaces tilting at different angles with changes in depth;
- (2) the conversion of planetary to local vorticity by vortex stretching, resulting primarily in a stratified fluid from internal motions;
- (3) the flux of vorticity at the top boundary caused by the curl of the wind stress;
- (4) the flux of vorticity at the bottom boundary caused by the curl of the bottom stress;
- and
- (5) the horizontal diffusion of vorticity.

The first two terms have a baroclinic origin, and together with the wind stress they are sources of vorticity. The curl of the bottom stress and the horizontal diffusion terms are sinks of vorticity in the surface layer. The circulation induced by the baroclinic sources is of cyclonic nature (Strub and Powell 1986). The curl of the wind has been used to explain both the anti-cyclonic (Strub and Powell 1986) and cyclonic (Lemmin and D'Adamo 1996) nature of the circulation in lakes.

The variability of the wind stress over a lake surface may also induce differences in the deepening rates of the mixed layer²¹. Parker and Imberger (Parker and Imberger 1986) illustrated those effects with data collected in Wellington Reservoir (Australia). The spatial variations of the depth of the mixed layer will produce horizontal exchanges of fluid, as shown experimentally by Maxworthy and Monismith (Maxworthy and Monismith 1988), and hence will drive large-scale motions in a lake.

It is clear from Eq. 4.11, and from the differential deepening effect discussed in the previous paragraph, that the wind-stress field should be accurately characterized if one is to carry out 'realistic' numerical simulations in enclosed basins. The wind stress is often calculated in hydrodynamic models by using a bulk aerodynamic formulation (Fischer *et al.* 1979)

$$(\tau_s) = C_D \rho_A u_A^2 \quad (4.17)$$

where C_D is the wind drag coefficient, ρ_A is the air density, and u_A is the wind speed measured at a reference height above the free surface. Actually the wind field acting on

²¹ In deriving equation 4.11 it is assumed that there is no mixing across layers.

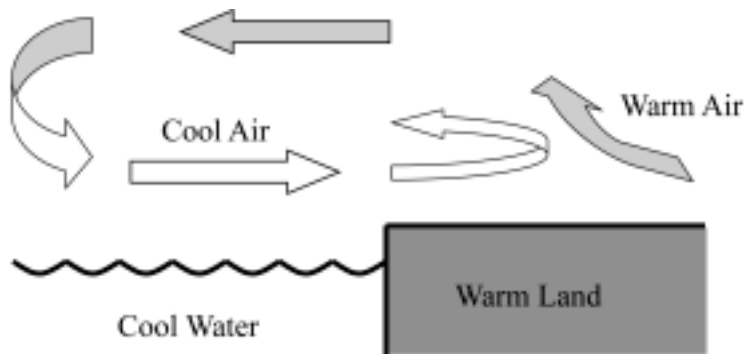
the surface of a lake is fairly difficult to characterize, as it is rarely uniform. Mesoscale wind variability causes a variation over large lakes, and wind sheltering by the surrounding terrain is the major factor affecting smaller lakes (Imberger and Patterson 1990). The characteristics of the terrain that will cause ‘small’ scale spatial variations in the wind field is reviewed in section 4.3.1, together with the models proposed in the literature to generate wind fields that capture those spatial variations. The routine used in this work to generate spatially variable wind fields is described in detail in section 4.3.2. The wind stress depends additionally (Eq. 4.17) on the specified magnitude of the drag coefficient C_D , which depends on the height of measurement, the wave state and the atmospheric stratification above the water surface (Geernaert 1990). Those dependencies are reviewed in section 4.3.3 and the method used to determine the drag coefficient in the model is also described. The resulting wind stress field will be provided to the hydrodynamic model as boundary conditions for the momentum equations at the free surface.

4.3.1 Driving mechanisms and representation of the wind field

The part of the atmosphere where the weather occurs, the troposphere, extends from the ground to an average altitude of 11 km. Only the lower few kilometers of the troposphere, however, are directly modified by the underlying surface. This is the atmospheric boundary layer (ABL). Over both land and ocean the general nature of the ABL is to be thinner in high-pressure regions than in low-pressure regions. The subsidence associated with the high-pressure regions moves boundary layer air from the high towards the lower pressure regions; this general trend is modulated by the earth’s rotation (Coriolis force) that will deflect the wind and align it with the isobars. The

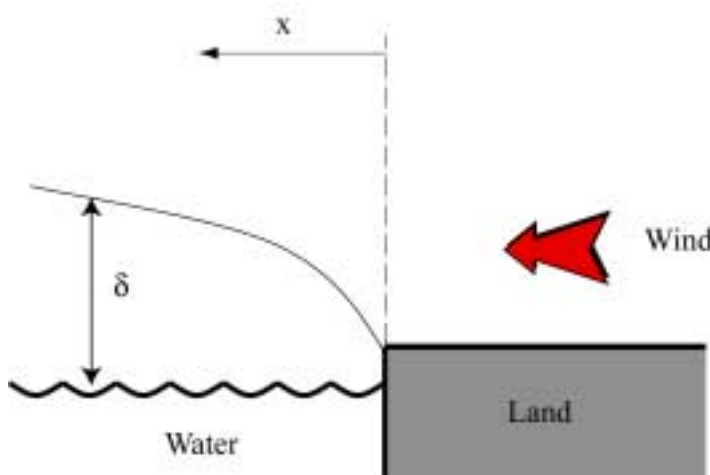
pressure gradient force and the Coriolis force, together with the centrifugal force that results from the change in direction of the wind, constitute the main driving mechanisms of wind on a *synoptic* scale (Gordon *et al.* 1998). On a *local* scale surface thermal and/or moisture properties, spatial variations of surface roughness, and surface elevation will modify and even generate boundary layer flows (Stull 1994).

The change of thermal properties is particularly pronounced at the shoreline. The large heat capacity of lakes and oceans reduces the water surface temperature change to near-zero values during the diurnal thermal cycle. The land surface, however, warms and cools more dramatically because the small molecular conductivity and heat capacity of soils prevents the diurnal temperature signal from propagating rapidly away from the surface (Stull 1994). As a result the air over land will be warmer during the day and cooler at night. These temperature differences will drive local winds or *breezes*. The *land breeze* occurs at night. The air over the water is warmer then, and the cool air from the land flows out to the lake or ocean at the lower levels. During mid-morning air rises over the warm land near the shoreline, and cooler air from the water flows in at the surface to replace it (Fig. 4.3). This is known as a *sea breeze* (or a *lake breeze*). In the absence of a background synoptic flow, the sea breeze can easily extend 20 to 50 km inland (Stull 1994).



*Figure 4.2
Idealized sea or
lake-breeze
circulation.*

Shorelines similarly define boundaries where surface roughness undergoes an abrupt transition²². Figure 4.4 is a schematic representation of a shoreline that is normal to the flow. It is assumed that the boundary layer upwind of the shoreline is in equilibrium with the forcing associated with the land surface. When the air crosses the shoreline and flows over the water surface, the bottom boundary layer will be modified by the new surface features. The thickness of this modified boundary layer will increase with distance downwind of the shoreline and is referred to as an internal boundary layer or IBL (Taylor and Lee 1984, Stull 1994).



*Figure 4.4.
Development
of an internal
boundary
layer away
from a
shoreline.*

²² The roughness lengths (z_0) above orchard and forest, for example, are on the order of 0.5 to 6 m, while above water surfaces z_0 could be three orders of magnitude smaller (Taylor and Lee 1984)

In the case of an IBL that develops in response to a change in roughness, its depth, δ , is parameterized as some power of the distance from the shoreline x , referred to as the fetch, such that

$$\frac{\delta}{z_o} = a_{IBL} \left(\frac{x}{z_o} \right)^{b_{IBL}} \quad (4.18)$$

Here z_o is the aerodynamic roughness length downwind of the shoreline. The exponent b_{IBL} is equal to about 0.8 for statically neutral conditions, but it is slightly smaller in statically stable conditions and larger for unstable conditions. The parameter a_{IBL} is in the range of 0.2 to 0.8, being large for unstable conditions and small for stable conditions (Stull 1994). Downwind of the shoreline the wind will speed up. The change in velocity, Δu , experienced at a point within the IBL located at height Δz_m and some distance x from the shoreline, relative to the velocity at a point at the same height but at $x = 0$ (i.e. at the shoreline) can be calculated as (Taylor and Lee 1984) as

$$\Delta u = \frac{\ln \frac{\Delta z_m}{z_o} \ln \frac{\delta}{z_{o1}}}{\ln \frac{\delta}{z_o} \ln \frac{\Delta z_m}{z_{o1}}} u_o(\Delta z_m) \quad (4.19)$$

The roughness lengths used in the derivation of Eq. 4.19 are assumed to be constant upwind and downwind of the shoreline. This assumption is not true, however, for water surfaces where the roughness lengths are generally regarded to be functions of a variable wave state (Geernaert 1990), which in turn will vary with fetch. Hence, Eq. 4.19 can only be considered to be approximate.

The topography is the third characteristic of the terrain that can modify and/or generate the boundary layer flow. Changes in surface elevation can induce horizontal differences in the air temperature that can drive complex 3-D ‘local’ air circulations such as valley or slope winds (Stull 1994). The most significant effect of the topography, however, is the aerodynamic modification of ambient synoptic winds. The simplest example is the speed-up observed as the wind blows over 2-D or 3-D hills. This case has been well characterized in the literature (see Taylor and Lee 1984 and references therein) because of its practical application in the location of wind turbines. The generally 3-D topography will additionally cause the ambient wind to change direction (deflection effect), and it will also create on the leeward side areas of momentum deficit or wakes (sheltering effect). These two latter effects are particularly relevant for lake applications, as they will induce a spatial variability in the wind field over the lower levels of the regional topography where lakes are usually located. Baines (Baines 1995) compiles several observational studies of flow past three-dimensional obstacles mounted on a surface. The observations of Hunt and Snyder (see Baines 1995) of flow past a “polynomial hill” are particularly descriptive. They show that, in general, the flow field past 3-D obstacles is characterized by the presence of a lee-side separation region and an associated wake, whose extent will depend on the strength of the stratification²³. The wake contains two distinct counter-rotating horseshoe vortices. The near-surface streamlines tend to diverge and follow the horizontal topographic contours over the lower part of the obstacles on the upstream side. On the downstream side the streamlines converge and become aligned

²³ The flow regimes around the “polynomial hill” are classified according to the magnitude of the Froude number = U/Nh where N is the buoyancy frequency, h is the height of the obstacle and U is the ambient velocity. In the “polynomial hill” the ratio of the maximum height to width of the obstacle is ~ 0.5 .

with the flow direction. Figure 4.5 is a sketch of flow past a 3-D isolated surface mounted obstacle.

The effect of a stable atmospheric stratification is to encourage air to flow around an obstacle rather than over it. The *critical* or *dividing stream-surface* (Baines 1995) identifies the surface ($z = H_{\text{CRIT}}$) that separates the flow into a lower part ($z < H_{\text{CRIT}}$) that mostly flows around the obstacles and an upper part that mostly flows upward and over it ($z > H_{\text{CRIT}}$). The criterion proposed by Sheppard (Sheppard 1956) allows a simple definition of H_{CRIT} (see Baines 1995 or Taylor and Lee 1984)

$$H_{\text{CRIT}} \approx h \left(1 - \frac{U}{Nh} \right) \quad (4.20)$$

This criterion is a consequence of equating the kinetic energy of the ambient fluid upstream of the obstacle ($= 0.5 \rho_0 U_A^2$) to the difference in potential energy due to the mean stratification between the height z_t , far upstream of the streamline that extends to the top of the obstacle, and the top of the obstacle h_m ($= 0.5 N^2 (h_m - z_t)^2$). The ‘Sheppard criterion’, though it is not physically correct, provides a good description of observations for a variety of obstacle shapes and density profiles (Baines 1995).

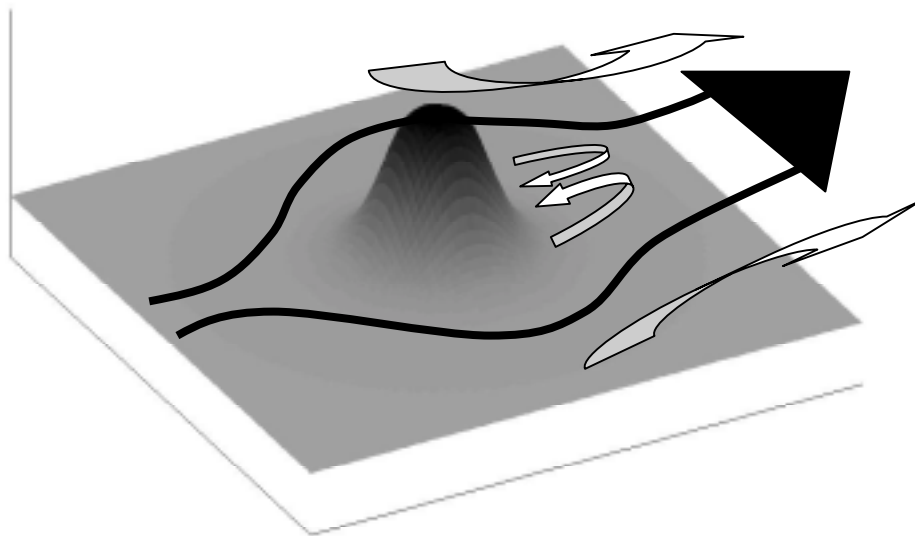


Figure 4.5. Effect of an isolated 3-D obstacle in a boundary layer flow. The spatial extent of the vortex pair will depend on the magnitude of the change in elevation, the atmospheric stability and the ambient wind speed (Baines 1995, pages 400-406).

Several modeling approaches have been used to represent wind fields over complex terrain. These generally fall into two categories (Cox *et al.* 1998). The first category is that of the prognostic models that use a set of unsteady 3-D hydrodynamic equations to predict the mean wind and turbulence fields. The second is that of diagnostic wind generation models. These models, also referred to in the literature as *objective analysis methods*, are based on wind measurements²⁴ and the interpolation of those measurements onto a grid. Most models adjust the initial interpolation to satisfy the mass conservation equation (see e.g. Erasmus 1986 or Montero *et al.* 1998), and some additionally

²⁴ Measurements are needed at several land or surface stations and also along vertical profiles so that an interpolated 3-D wind interpolated field can be generated.

incorporate the concept of a critical or dividing streamline concept²⁵ (Ludwig *et al.* 1991). The objective models generally lack predictive ability but involve much less computational cost than the prognostic models, which makes them the preferred method for practical applications (Cox *et al.* 1998). The quality of the 3-D wind field obtained by objective analysis of wind measurements depends primarily on the quality, amount and location of the wind measurements and the quality of the initial interpolation (Pielke 1985). For local analysis problems the most popular interpolation methods are those proposed by Barnes (Barnes 1973), Cressman (Cressman 1959), and the *multiquadric* scheme developed by Hardy for geophysical applications (Hardy 1971). The latter is receiving increasing attention in the meteorological literature, given its superior accuracy when compared with the Barnes or Cressman interpolation methods (Nuss and Titley 1994).

Diagnostic models are in general very successful in representing the *deflection* effect of the topography in the ambient wind (see results in Montero *et al.* 1998). However, they fail to represent the deficit of momentum (*sheltering* effect) or the presence of vortices in the wakes behind topographic obstacles unless wind measurements are taken in those areas. Under neutrally stratified conditions, the deficit of momentum can be described with an expression of the form (Taylor and Salmon 1993)

$$\frac{u}{U_0(h)} = \left(\frac{x}{h} \right)^c F(\eta) G(\zeta) \quad (4.21)$$

²⁵ The use of the critical or dividing streamline concept in diagnostic models requires not only velocity measurements but additionally requires information on the atmospheric stability.

where $\eta=(y/h)(x/h)^{-b}$ and $\zeta=(z/h)(x/h)^{-a}$. Here h is the maximum height of the 3-D object, $U_0(h)$ is the wind speed measured at the top of the obstacle, x denotes the distance downwind from the obstacle, z is the vertical axis, y is the horizontal distance normal to x , and u is the velocity at the location (x,y,z) . The function F represents the lateral velocity deficit variation in the wake and has a Gaussian form. The function G has an exponential form and characterizes the deficit variation in the direction of the flow. The parameters a , b and c in the model are $c = -1.6$ and $a = b = 0.5$ (see Taylor and Salmon 1993). The details of the flow within the separation wake are highly case-specific and rather complex (Taylor and Salmon 1993, Baines 1995).

4.3.2. Multiquadric interpolation method for wind field generation

In the 3-D hydrodynamic model the wind field is generated using wind observations that are interpolated to the wet nodes of the computational grid with a straightforward 2-D *multiquadric* method (Nuss and Titley 1994). The interpolation method is described below, and it was selected on the basis of its superior accuracy in relation to other schemes (Nuss and Titley 1994). In the applications of the model presented in this work the number of surface wind stations was increased so that the details of the wind field could be represented without the recursive procedure used in diagnostic models to account for the deflection effects of topography. The incorporation of the topographic effects in the objective analysis method requires wind speed data along vertical profiles, which is typically not available.

The foundation of the multiquadric interpolation method lies in the theory of interpolation using radial functions. The interpolation equation using radial functions is (Nuss and Titley 1994)

$$H(X) = \sum_{i=1}^N \alpha_i Q_i(X - X_i) \quad (4.22)$$

where $H(X)$ is the spatially varying wind field, $Q_i(X - X_i)$ is a radial basis function, and N denotes the number of observations. The argument of Q_i is the vector between an observation point X_i and any other point X in the n -dimensional space domain. In the multiquadric method the radial basis function is a hyperboloid function of the form

$$Q_i(X - X_i) = - \left(\frac{\|X - X_i\|^2}{c^2} + 1.0 \right)^{1/2} \quad (4.23)$$

where c is an arbitrary, and typically small, constant referred to as the multiquadric parameter. To determine the coefficients α_i , the interpolation equation (Eq. 4.22) is applied to every observation point (x_j, y_j) , resulting in the following set of equations

$$H(x_j, y_j) = \sum_{i=1}^N \alpha_i Q_i(x_j, y_j) \quad (4.24)$$

where

$$Q_i(x_j, y_j) = - \left(\frac{|x_j - x_i|^2 + |y_j - y_i|^2}{c^2} + 1.0 \right)^{1/2} \quad (4.25)$$

The system of equations in 4.24 ($\mathbf{H}_j = \mathbf{Q}_{ij}\boldsymbol{\alpha}_i$ in Cartesian tensor subscript form) is solved for the coefficients α_i using the LU decomposition method (Press *et al.* 1992). Once the weighting coefficients are calculated at the beginning of the procedure, the interpolation of the solution to any desired uniform grid $\mathbf{H}_g$, represented by grid points (x_g, y_g) , proceeds as follows:

$$\mathbf{H}_g = \mathbf{Q}_{gi}\boldsymbol{\alpha}_i = \mathbf{Q}_{gi}(\mathbf{Q}_{ij})^{-1}\mathbf{H}_i \quad (4.26)$$

Here each element in the matrix $\mathbf{Q}_{gi}$ is

$$Q_i(x_j, y_j) = - \left(\frac{|x_j - x_i|^2 + |y_j - y_i|^2}{c^2} + 1.0 \right)^{1/2} \quad (4.27)$$

The resolved scales and accuracy of the approximated function are determined by the number and spacing of the observations that are used to determine the coefficients. Errors and incomplete sampling of small-scale features may produce unrealistic results, however, which makes it desirable to introduce some smoothing. Thus the interpolation equation is modified slightly, resulting in

$$\mathbf{H}_g = [\mathbf{Q}_{gi} + (N\lambda\boldsymbol{\sigma}_i^2\delta_{ij})]\boldsymbol{\alpha}_i \quad (4.28)$$

where N is the number of observations, $\boldsymbol{\sigma}_i^2$ is the mean squared observation error, λ is a smoothing parameter, and δ_{ij} is the Kronecker delta. The parameter λ governs the degree

of smoothing in the approximated function and can be related to the half-power point of a spectral filter (Nuss and Titley 1994)²⁶.

4.3.3. Wind drag

In the hydrodynamic model the stress exerted by the wind on the water surface (wind drag) is calculated by using a bulk formulation (Fischer *et al.* 1979) in which the stress is related to the wind speed through a drag coefficient C_D . The drag coefficient is not a constant but depends on the environmental conditions such as the atmospheric stratification or the wave state. Those dependencies are revealed through a brief examination of the theory of the atmospheric surface layer.

4.3.3.1. Atmospheric surface layer velocity profiles

The region at the bottom of the atmospheric boundary layer where turbulent fluxes and stresses vary by less than 10% of their magnitude is referred to in the literature as the atmospheric surface layer or ASL (Stull 1994). Within the ASL and under neutral stratification the wind speed gradient can be scaled to the surface wind stress (τ_s) and a mixing length l_m , according to (Geernaert 1990, Tennekes and Lumley 1972)

$$\frac{\partial U}{\partial z} = |\tau_s / \rho|^{1/2} l_m \quad (4.29)$$

The mixing length, l_m , depends on the height above the interface, z , as

$$l_m = \kappa(z + z_0) \quad (4.30)$$

²⁶ It can be determined by using a technique called generalized cross-validation, though in this application it remains a free parameter.

where κ is the von Kármán constant ($= 0.4$), and z_0 is the aerodynamic roughness length that has a value on the order of 10^{-4} m for water. Integration of Eq. 4.29 results in the well-known logarithmic velocity profile²⁷

$$U_z = u_* \kappa^{-1} \ln(z / z_0) \quad (4.31)$$

where U_z is the wind speed at height z and u_* ($= |\tau / \rho|^{1/2}$) is the friction velocity. Under unstable stratification conditions turbulence is enhanced, and the mixing length becomes larger than κz (Geernaert 1990). For stable stratifications, on the other hand, the magnitude of the mixing length decreases below κz . The effect of stratification on the velocity profile is accounted for by introducing in Eq. 4.29 a stability function (Φ_M)

$$\frac{\partial U}{\partial z} = \frac{u_*}{\kappa z} \Phi_M \quad (4.32)$$

Upon integrating Eq. 4.32, one obtains the surface layer velocity profile²⁸ as

$$U_z = u_* \kappa^{-1} [\ln(z / z_0) - \psi_m] \quad (4.33)$$

The Businger-Dyer relationships (Stull 1994) provide the functional form of the stability functions Φ_M and ψ_m in terms of the non-dimensional group z/L , where L is the Monin-

²⁷ Integration is performed from z_0 up to height $z + z_0$. Given that $z \gg z_0$, it is assumed for practical purposes that $z + z_0 \sim z$. Similarly the velocity U at z_0 is negligible compared with U at z .

²⁸ The same integration limits and assumptions as in the derivation of equation 4.31 are used here.

Obukov length²⁹. In particular, the functional dependence of ψ_m on the parameter z/L is given for stable conditions ($z/L > 0$) by

$$\psi_m\left(\frac{z}{L}\right) = 4.7 \frac{z}{L} \quad (4.34)$$

and for unstable conditions ($z/L < 0$) by

$$\psi_m\left(\frac{z}{L}\right) = -2 \ln\left[\frac{1+x}{2}\right] - \ln\left[\frac{(1+x^2)}{2}\right] + 2 \tan^{-1}(x) - \frac{\pi}{2} \quad (4.35)$$

in which $x = [1 - (15z/L)]^{1/4}$.

4.3.3.2. Bulk aerodynamic formulation and wind drag coefficient

Equation 4.33 can be written as

$$u_*^2 = [\kappa / \ln(z/z_0) - \psi_m]^2 \quad U_z^2 = C_D \quad U_z^2 \quad (4.36)$$

where C_D , the drag coefficient, is defined as

$$C_D = [\kappa / \ln(z/z_0) - \psi_m]^2 \quad (4.37)$$

Equation 4.36 constitutes the bulk formulation of the surface friction velocity and relates u_* to the wind speed at a given height through the drag coefficient. The definition of the drag coefficient in Eq. 4.37 reveals that it is not a constant parameter, as is assumed in

²⁹ The Monin-Obukov length is defined as $L = \frac{u_*^3}{-\kappa g \overline{w' \rho'}}$, in terms of the friction velocity u_* and the vertical density flux $\overline{w' \rho'}$.

most hydrodynamic models, but depends on the height of measurement z , the aerodynamic roughness of the free surface³⁰ z_0 , and the atmospheric stratification parameter ψ_m .

An empirical formula relating z_0 and u_* was suggested in 1955 by Charnock (Charnock 1955) purely on dimensional grounds. His expression was subsequently rationalized as an equation of state characterizing the equilibrium between wind and waves, with gravity waves acting as roughness elements (Wu 1985). The Charnock relation, as it is referred to in the literature, can be written as

$$z_0 = a \frac{u_*^2}{g} \quad (4.38)$$

where g is the acceleration of gravity and a is the Charnock's coefficient, originally assumed to be constant³¹ (~ 0.015). Several authors, however, have challenged the assumption of a constant Charnock coefficient. Amorocho and DeVries (Amorocho and DeVries 1980) analyzed numerous data sets and concluded that a is not constant in the range of wind velocities from zero to 40 m/s. Kitaigorodskii and Zaslabskii (Kitaigorodskii and Zaslabskii 1974) indicated that the Charnock relation should be modified to include the wave age ($= C_0/u_*$),

$$\frac{z_0 g}{u_*^2} = a \left(\frac{C_0}{u_*} \right) \quad (4.39)$$

³⁰ The roughness lengths z_0 are regarded as functions of a variable wave state (Geernaert 1990).

³¹ Experimentally determined values of “ a ” generally fall between 0.01 and 0.02.

In the definition of the wave age, C_0 represents the phase velocity of the dominant wave (wave number at the maximum energy). Large values of C_0/u_* represent decaying waves or large swell, and small values indicate a growing sea (Geernaert 1990). Geernaert et al. (Geernaert *et al.* 1987) derived a relationship between the wave age and wave steepness, so that the Charnock coefficient can be expressed as

$$a = \frac{H_{1/3}}{L_w} \quad (4.40)$$

where $H_{1/3}$ is the significant wave height and L_w is the wave length at the peak of the energy spectrum. Other models (see Geernaert 1990) relate the roughness coefficient to the spectral characteristics of the surface waves, and it can be reduced, with appropriate assumptions, to the Charnock relationship with a variable coefficient a .

Donelan and others (reference in Geernaert 1990 or Imberger and Patterson 1990) proposed in 1984 a model where the total drag exerted by the wind on the free surface is partitioned between *form drag* induced by the longer waves and *frictional drag* exerted by the short, steep gravity and capillary waves, i.e.

$$C_D = C_{LOW} + C_{HIGH} \quad (4.41)$$

C_{LOW} is based on a roughness length described by the wave energy denoted primarily by the dominant wave of the equilibrium range, while the roughness length contribution to C_{HIGH} is based on the wave energy corresponding to the waves having at least twice the spectral peak frequency. Evidence suggests that the form and friction drag mechanisms both must play an important role in transferring momentum to the wave surface. Banner

and Peirson (Banner and Peirson 1998) conducted a detailed laboratory experiment using particle image velocimetry techniques to examine the contributions of the tangential stress (or frictional drag) and the form drag to the total stress exerted by the wind at the free surface. They challenge the results previously obtained that point towards the tangential stress as the dominant component of the wind stress. According to their observations, however, the tangential stress contributes the entire wind stress prior to the formation of wind waves, but as the wave field develops the form drag provides the major proportion of the wind stress.

The modeling approaches described above to represent the wave-state dependence of the drag coefficient require information on the wave climate at the free surface. These data are not usually available. Alternatively, they could be estimated through the solution of some kind of wave model. A simpler approach is that proposed by Amorocho and DeVries (Amorocho and DeVries 1980) where the dependence of the drag coefficient on the wave-state is treated implicitly. They proposed the following empirical equation to express the normalized drag coefficient³² as a function of the wind speed (U_{10}) at 10 m above the free surface,

$$C_{DN} = 0.0015 \left[1 + \exp \left(- \frac{U_{10} - 12.5}{1.56} \right) \right]^{-1} + 0.00104 \quad (4.42)$$

The model of Amorocho and DeVries shows three distinct regions in the development of the wind shear stress marked by different states in the wave climate: (1) a lower region in

³² Drag coefficient measured at 10 m above the free water surface under neutrally stable atmospheric stratification.

which the wind waves have not begun to break, and for which C_{DN} is approximately constant; (2) a transitional region, after the onset of breakers, for which C_{DN} varies non-linearly with U_{10} ; and (3) a limiting region for which C_{DN} tends again toward a constant value, and corresponds to a condition of breaker saturation (Amorocho and DeVries 1980). Given its simplicity, this has been the model adopted to represent the normalized drag coefficient in the hydrodynamic model. This model further eliminates a source of subjectivity in the application of the hydrodynamic code to real systems.

The influence of the atmospheric stratification on the magnitude of the drag coefficient is represented through the stability function ψ_m . Equations 4.34 and 4.35 give the Businger-Dyer formulation of the stability functions. Liu and Schwab (Liu and Schwab 1987) compared the stability functions of several models proposed in the literature. They conclude that any of the methods could be used effectively in practical applications. The argument of the stability functions is the non-dimensional variable (z/L) , where z is height above the free surface and L is the Monin-Obukov length scale. The calculation of L , however, requires the specification of the heat flux at the water surface, which depends on one of the prognostic variables (i.e. surface temperature). Hence the calculation of the wind drag cannot be performed separately from the code. Furthermore it will require some kind of iteration procedure (Liu and Schwab 1987) because of the dependence of L on u^* . This might increase the computational cost in the model applications. It was, hence, preferred not to incorporate the effects of stratification into the drag coefficients for the sake of the simplicity and efficiency of the hydrodynamic model, though it would be desirable in the future to test the consequences of this approach. In the applications of

the model, the strongest wind events take place during late afternoon and evening hours when the atmospheric stratification could be expected to approach neutral stability.

CHAPTER 5

THREE-DIMENSIONAL TEMPERATURE DISTRIBUTION MODEL

5.1. Introduction

Most lakes undergo periods of variable length when they stratify, alternating with others when the lake is nearly homogeneous and well mixed in what is called a stratification cycle. Under stratified conditions, water motions primarily result from the combined forcing of wind stress, the rotation of the Earth, and the density gradients that, in the particular case of fresh water lakes, are mainly determined by the spatial distribution of temperature. The correct simulation of lake hydrodynamics under those conditions requires the simultaneous solution and coupling of the 3-D equations for the circulation and the distribution of temperature, or more generally the distribution of the so-called active scalar components characterized by their ability to modify density and hence to interact with the velocity field. The temperature distribution in a stratified lake is the result of a subtle balance among many processes, including air-sea exchange, transport by currents and vertical turbulent mixing, that must be represented adequately if the distribution is to be realistic. The modules of the 3-D model representing those processes have been carefully developed to provide an accurate representation of the temperature field, and they are described in the next few sections. In section 5.2 the equations for the representation of the exchange of heat between the water and the atmosphere are introduced. The algorithms for the representation of the advective transport of temperature (and in general any scalar) in the computational domain are then described.

The advection scheme originally incorporated in SI3D, leapfrog-trapezoidal, is evaluated, and an alternative scheme is proposed that is intended to eliminate the problems associated with the original algorithm. Finally, in section 5.4, the model used to represent turbulent processes in the 3-D code is introduced and tested. Turbulent processes in a water column are primarily driven by surface stresses (shear or buoyancy fluxes), and their effects penetrate to a distance that defines the limit of a boundary layer or mixed layer, a region within which the water properties are almost uniform³³.

5.2. Heat budget and thermodynamics at a free surface

Heat exchange at a free surface is generally classified according to the transfer mechanisms: radiation, convection, conduction and evaporation. For numerical modeling of water surface heat fluxes, however, it is convenient to classify the heat transfer in terms of its ability to penetrate the water surface. Evaporation or latent heat flux, Q_E , conduction or sensible heat, Q_H , and long wave (or back) radiation, Q_B , are all *surface* heat transfer effects that occur only at the surface of the water. Short-wave radiation, Q_S , on the other hand, is a *penetrative* effect that distributes heat throughout a significant range of the water column. In general, more than half of the short-wave solar radiation (solar insolation or solar irradiance) that enters the water column through the surface is absorbed within the top half meter (Rosati and Miyakoda 1988). The remaining fraction penetrates and is absorbed in deeper layers. The distribution of the solar irradiance within the water column will depend on the water clarity properties and is site specific. The water clarity may undergo strong seasonal and even diurnal oscillations in response to

³³ The boundary layer that develops adjacent to the free surface is called the surface boundary layer to differentiate it from the bottom or benthic boundary layer that develops in the vicinity of the bottom.

biological cycles, and it will be affected by wind-driven sediment re-suspension in shallow lakes.

The classification of the surface and penetrative components of the surface heat flux is not only a conceptual matter but has its mathematical consequences. The temperature transport equation may be expressed in vector notation as

$$\frac{\partial T}{\partial t} = \mathbf{v} \cdot \nabla T + \frac{\partial}{\partial z} \left(D_V \frac{\partial T}{\partial z} \right) + \frac{1}{\rho_0 c_p} \frac{\partial I}{\partial z} \quad (5.1)$$

Here T represents water temperature, ∇ is the gradient operator, $\mathbf{v}$ is the velocity vector, I is the downward irradiance, c_p is specific heat of water, D_V is the vertical eddy diffusivity, and ρ_0 represents a reference density. Equation 5.1 includes the divergence of the downward irradiance I , that represents the absorption of the short-wave radiation and acts as a source term for each infinitesimal layer. The remaining components of the heat flux provide the necessary boundary condition at the free surface for the temperature transport equation (Eq. 5.1).

In the computation of the heat fluxes at a free surface, the downward fluxes that increase the thermal energy of the water column are considered positive while those fluxes (upwards) that result in a loss of thermal energy are negative. This sign convention is consistent with a thermodynamic definition of a system (i.e. the surface is the system boundary and transfer out of the system is a loss). The incoming solar short-wave radiation will, accordingly, always be a positive flux.

In the next sections the formulation used in the transport model for each of the components of the surface heat flux is presented.

5.2.1. Short-wave radiation

The downward flux of solar radiation penetrating the free surface, Q_s , is given by the amount of insolation reaching the water surface, Q_G , reduced by the albedo A_s of the water surface,

$$Q_s = Q_G \cdot (1 - A_s) \quad (5.2)$$

The insolation reaching the top of the atmosphere at a given site is a function of its geographical coordinates, time of day and other astronomical data (Tennessee Valley Authority (T.V.A) 1972). That function must, however, be corrected for atmospheric attenuation and for cloud cover to obtain the short-wave radiation that reaches the water surface. These corrections are usually site specific and not well known. Alternatively, many weather stations record short-wave radiation at the earth surface (after attenuation and cloud cover), observations that can be directly used in Eq. 5.2.

The albedo or shortwave reflectivity is, in general, a function of solar zenith angle and the state of the surface (Henderson-Sellers 1986). For a mean daily parameterization, Henderson-Sellers suggests a value of 0.06, based on Lake Hefner studies. However, for a prognostic model it is recommended that one incorporate temporal variations. An algebraic equation that allows for those effects is that attributed to Pivovarov (referenced in Henderson-Sellers 1986)

$$A_s = \frac{a_0}{a_0 + \sin \alpha^*} \quad (5.3)$$

where α^* is the solar altitude, a_0 is a function of fractional cloud cover C_1 , where $C_1 = 1$ represents total cloud cover, and Julian day, J ,

$$a_0 = 0.02 + 0.01 \cdot (0.5 - C_1) \cdot \{1 - \sin[\pi \cdot (J - 81)/183]\} \quad (5.4)$$

The solar altitude α^* is determined by astronomical data and also geographical latitude and time of day. The expression that relates solar altitude to latitude, θ , solar declination, δ , and hour angle is given by

$$\sin \alpha^* = \sin \theta \cdot \sin \delta + \cos \theta \cdot \cos \delta \cdot \cos h \quad (5.5)$$

The hour angle h is zero at local noon and increases in magnitude by $\pi/12$ for each hour before and after noon. The solar declination on the other hand will vary for each Julian day, J , and for the northern hemisphere is given by the expression

$$\delta = 0.4093 \cdot \sin[2\pi \cdot (J - 79.75)/365] \quad (5.6)$$

5.2.2. Penetration of solar irradiance in a water column

The short-wave solar radiation that reaches a depth z , $I(z)$, is given by the solar irradiance that penetrates the surface, Q_s , multiplied by an attenuation function, $\Phi_A(z)$, that accounts for the attenuation (a combination of absorption and scattering) of radiation in the upper layers. The attenuation of solar radiation with depth, $\Phi_A(z)$, can be parameterized in several ways. Perhaps the simplest model for the solar attenuation is a modified form of the Beer-Lambert law, given by (Henderson-Sellers 1986) as

$$\Phi_A(z) = (1 - \beta) \exp[-\eta(z - z_A)] \quad (5.7)$$

The coefficient β accounts for the portion (40 to 50 %) of the incoming solar radiation that is absorbed in a thin layer at the top of the water column, and z_A defines the thickness of that surface absorption layer ($z_A \sim 0.6$ m). The extinction or attenuation coefficient ranges from $\eta \approx 0.05$ for clear waters to $\eta > 1.0$, and it has been related to the Secchi depth (Henderson-Sellers 1984 page 42), probably the most extended form of “clarity” index available,

$$\eta \approx \frac{5.6}{\text{Secchi depth}} \quad (5.8)$$

An alternative approach is to parameterize the solar attenuation with a double exponential depth dependence (Rosati and Miyakoda 1988),

$$\Phi_A(z) = \left[R \exp\left(-\frac{z}{L_1}\right) + (1 - R) \exp\left(-\frac{z}{L_2}\right) \right] \quad (5.9)$$

where L_1 and L_2 are attenuation lengths and R is an empirical constant. For clear waters $R = 0.58$, $L_1 = 0.35$ m and $L_2 = 23$ m are values suggested in the literature (Rosati and Miyakoda 1988). For other types of waters, the reader is referred to (Jerlov 1968) where five types of water are defined, varying from clear (Type I) to increasingly turbid (Type V). The double exponential dependence of Eq. 5.9 includes the Beer-Lambert law as a particular case. For example, it will suffice to set $R = 1$ and $L_1 = 1/\eta$, or $R = 0$ and $L_2 =$

$1/\eta$ in Eq.5.10 to obtain the Beer-Lambert attenuation law. Equation 5.9, hence, has been the preferred formulation of the solar attenuation factor in the transport model.

5.2.3. Long-wave radiation

The radiant energy emission (long-wave or infrared radiation) by a black body is (Holman 1990) described by

$$Q_{LW} = \sigma T_{(K)}^4 \quad (5.10)$$

where Q_{LW} is the long-wave energy transfer per unit surface area (W/m^2), σ is the Stephan-Boltzman constant ($5.669 \times 10^{-8} \text{ Wm}^{-2}\text{K}^{-4}$), and $T_{(K)}$ is the absolute temperature in degrees Kelvin. The computation of radiant energy exchanges between real bodies requires (1) emissivities ε that account for the difference in behavior between real and black bodies, (2) shape factors that take into account the shape of the objects, (3) the absorption of radiation in a gas such as air, and (4) the relative temperature difference (see Holman 1990). For the particular case of atmospheric-water surface interaction (1) emissivity, (2) cloud cover, and (3) water surface reflectivity are used. In the heat exchange module of the transport model, the long-wave radiation component Q_B is decomposed into emitted long-wave radiation Q_B^{ro} and incident long-wave radiation (or atmospheric radiation) after reflection Q_B^{ra} and modeled as

$$\begin{aligned} Q_B &= Q_B^{ro} + Q_B^{ra} \\ Q_B^{ro} &= \varepsilon_w \sigma \cdot T_s^4 \\ Q_B^{ra} &= Q_B^{ri} (1 - A_L) = \varepsilon_a \sigma \cdot T_A^4 (1 - A_L) \end{aligned} \quad (5.11)$$

Here ε_w and ε_a are the emissivities of water and air, and A_L is the long-wave reflectivity (usually assumed constant at 0.03). The factor ε_w is essentially constant and equal to 0.972; T_S is the water surface temperature and T_A is the atmospheric temperature, both in degrees Kelvin. The emissivity of air, on the other hand, depends on variables such as cloud cover C_1 , the air temperature T_A , and its vapor pressure e_A in Pascals. Here a physically based expression is adopted for ε_a under clear conditions suggested by (Brutsaert 1975), corrected by a multiplicative factor of the form $(1+k_1C_1^2)$ to account for the influence of cloud cover. The resulting expression is

$$\varepsilon_a = 0.642 \cdot \left(\frac{e_A}{T_A} \right)^{1/7} \cdot (1 + 0.17 \cdot C_1^2) \quad (5.12)$$

5.2.4. Latent heat flux

The evaporative energy flux Q_E is calculated in terms of the difference between the saturated vapor pressure at the surface water temperature (e_S^{sat}) and the actual vapor pressure at the temperature of the air (e_A), following the expression (Henderson-Sellers 1986)

$$Q_E = \rho_w \cdot L_v \cdot 1.15 \cdot 10^{-8} \cdot c_D \cdot u_2 \cdot (1 + a_3 \cdot c_R) (e_S^{sat} - e_A) = C_E \cdot (e_S^{sat} - e_A) \quad (5.13)$$

in which L_v is the latent heat of vaporization (a function of temperature), c_D is the aerodynamic drag coefficient (set to a constant 1.3×10^{-3} for the purposes of the calculation of heat fluxes), and c_R and a_3 are coefficients that depend on the stability of the atmosphere above the water mass, parameterized by the difference between the

surface water temperature (T_S) and the air temperature (T_A). Here we give the expressions used for L_v , c_R and a_3 in Eq. 5.13

$$L_v = 1.91846 \cdot 10^{-6} \cdot [T_S / (T_S - 33.91)]^2 \quad (5.14)$$

$$c_R = \frac{0.0017 \cdot (T_S - T_A)^{1/3}}{c_D \cdot u_2}, \quad T_S > T_A \quad (5.15)$$

$$c_R = 0, \quad T_S > T_A$$

$$a_3 = 0.73 \cdot c_R \quad c_R \leq 1.37 \quad (5.16)$$

$$a_3 = 1 \quad c_R > 1.37$$

The following formulation (SI units) is adopted to represent the dependence of the saturation vapor pressure on temperature (Henderson-Sellers 1986),

$$e^{sat}(T_A) = 2.1718 \cdot 10^{10} \exp[-4157 / (T_A - 33.1)] \quad (5.17)$$

The actual vapor pressure e_A at the air temperature is related to the saturated vapor pressure at that temperature $e_A^{sat}(= e^{sat}(T_A))$, via the relative humidity R_h

$$e_A = R_h e_A^{sat} \quad (5.18)$$

5.2.5. Sensible heat flux

The sensible heat flux is related to the evaporative energy loss by the Bowen ratio (Q_H/Q_E). In the heat exchange module of the transport model the Bowen ratio is assumed to be of the form

$$Q_H / Q_E = 0.61 \cdot 10^{-3} \cdot p \cdot \frac{T_S - T_A}{e_S^{sat} - e_A} \quad (5.19)$$

where p is the atmospheric pressure (Pa). Expression (5.19) is independent of the stability of the atmosphere (Henderson-Sellers 1986); together with Eq. 5.13 for the latent heat flux it gives the following expression for the sensible heat flux:

$$Q_H = 0.61 \cdot 10^{-3} \cdot C_E \cdot p \cdot (T_S - T_A) \quad (5.20)$$

5.2.6. Numerical implementation

The transport model solves the layer-averaged form of the advection-diffusion equation, which for layer k obeys the expression

$$\frac{\partial(hT)_k}{\partial t} + \frac{\partial(UT)_k}{\partial x} + \frac{\partial(UT)_k}{\partial y} + (wT)_{k+1/2}^{k-1/2} = \left(D_v \frac{\partial T}{\partial z} \right)_{k+1/2}^{k-1/2} + \frac{1}{\rho_0 c_p} (I)_{k+1/2}^{k-1/2} \quad (5.21)$$

where the notation $(\dots)_{k+1/2}^{k-1/2}$ is used to represent the difference between the upper and lower interface values for layer k .

The last term on the right side is a source term; it represents the short-wave radiation absorbed in layer k . It is calculated as the short-wave radiation that penetrates the free surface multiplied by the fraction of irradiance that is absorbed at layer k , such that

$$\frac{1}{\rho_0 c_p} (I) \Big|_{k+1/2}^{k-1/2} = \frac{Q_s}{\rho_0 c_p} [\Phi \Big|_{k-1/2} - \Phi \Big|_{k+1/2}] \quad (5.22)$$

In the present version of the model the fraction of the incoming radiation that is absorbed at layer k is calculated once at the beginning of the simulation with the water clarity parameters defined in the input file, and each such fraction remains constant thereafter. It would, however, be possible to compute those values at every time step to account for transient variations of water clarity.

In the particular case that the fraction of incoming irradiance that is absorbed in the water column is not 1, the remaining energy is distributed uniformly among the layers. This could occur, for example, in shallow, clear waters. Heat gains or losses to the ground are not allowed in the present version of the model.

The surface boundary conditions for the transport equation are

$$(wT) \Big|_{1/2} = 0 \quad (\text{zero advective flux}) \quad (5.23)$$

$$\left(D_v \frac{\partial T}{\partial z} \right) \Big|_{1/2} = (Q_E + Q_B + Q_H) \quad (5.24)$$

The different components of the surface heat flux are calculated in the initial iteration, using the prognostic variables (water surface temperature) at time n . In successive iterations as new estimates of the water surface temperature at time $n+1$ are obtained, the heat components are calculated by using the value of the prognostic variable at time $n+1/2$. The meteorological variables are assumed to be constant during the span of a time step, and their values are equal to that at time n . In Table 5.1 is listed the different

meteorological variables that the user must define in the input file, as well as the geographic information that the heat module needs to perform the calculations of the heat exchange with the atmosphere.

a. Meteorologic data - Dry bulb air temperature ($^{\circ}\text{C}$), T_A - Relative humidity (%), RH - Atmospheric pressure (Pa), p - Wind velocity (m/s), u (assumed uniform over the whole lake for the purpose of heat balance computation) - Cloudiness as a fraction (C_1) - Precipitation (not presently used) b. Geographic data - Longitude of standard meridian of the location (Pacific Time: GMT – 8.00 h) - Longitude of the local meridian - Latitude

Table 5.1. Input data for the heat module of SI3D. Precipitation is neglected in the present version of the model.

5.3. Advection algorithm

The leapfrog-trapezoidal scheme, used in SI3D for the advection of active scalars, was successfully tested in chapter 3 under idealized conditions, with a smooth temperature field and the simplest possible 3-D domain (a rectangular basin). However, in more realistic geophysical flows there exist “irregular” or non-smooth fields that may compromise the performance of the model. The following two cases illustrate scenarios where non-smooth fields will develop as a consequence of a *scale contraction*.

The first case of *scale contraction* occurs during episodes of upwelling of hypolimnetic waters in a stratified lake subject to strong winds. The ratio of characteristic horizontal to

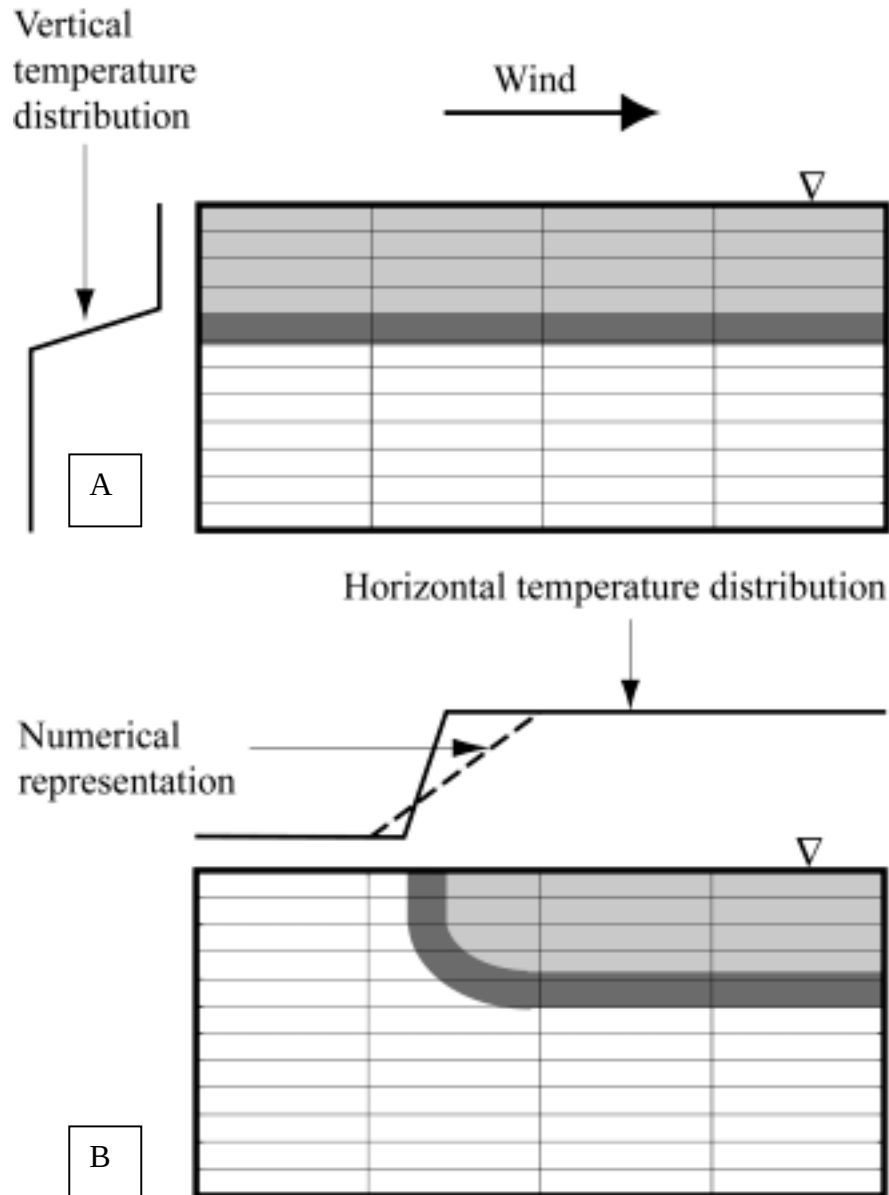


Figure 5.1(a). Examples of scale contraction in geophysical flows. An upwelling event in a stratified lake caused by wind. The figure represents a reservoir with a three-layer structure: homogeneous hypolimnion on the bottom, homogeneous epilimnion at the top and a region of strong gradients in between. A grid has been overlaid with higher resolution in the vertical than in the horizontal, as is usually the case in geophysical applications. The vertical gradients of temperature are well represented when the temperature gradients occur mainly in the vertical direction (A). Owing to the different resolution of the grid in the vertical and the horizontal directions, those gradients will be under-resolved and misrepresented when reaching the surface after episodes of strong winds (B).

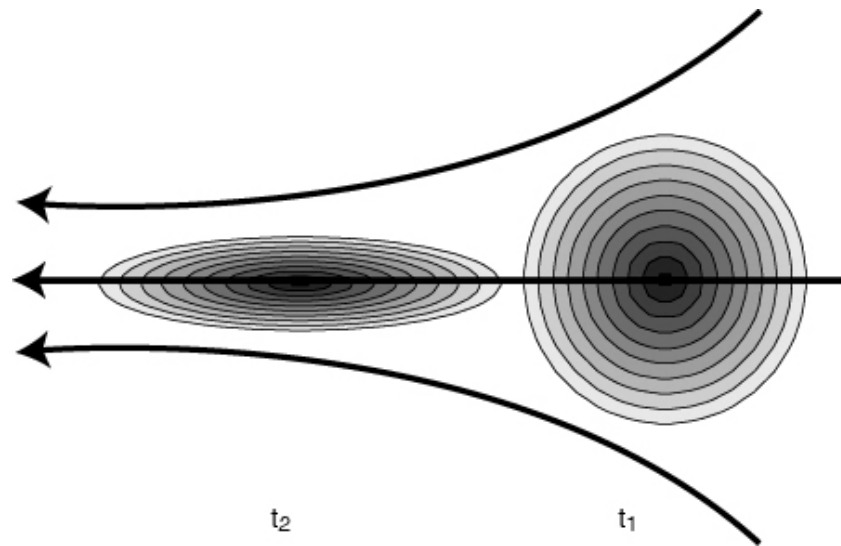


Figure 5.1.(b). Examples of scale contraction in geophysical flows. Deformation from a circular pattern at t_1 to an ellipse at t_2 of a scalar field in a convergent flow field.

the vertical length scales in natural water bodies is $O(10^2\text{-}10^3)$. This remarkable difference in length scales is reflected in the computational grid where the values of Δz are typically two orders of magnitude smaller than the horizontal spacing between nodes. Vertical gradients that may be well resolved when the isotherms remain in the (x, y) plane will be poorly resolved when those gradients reach the surface and the isotherms become vertical, as occurs during upwelling events. Figure 5.1(a) shows this process schematically.

The second case of *scale contraction* is the result of a stretching and shearing deformation by the velocity field caused by a sudden contraction in the geometry of a lake. The effects of flow deformation on an initially circular distribution of a passive tracer are illustrated in Fig. 5.1(b). As the scale of the tracer distribution shrinks in the direction perpendicular to the axis of dilatation, the concentration field will eventually become difficult to resolve adequately on a fixed numerical mesh.

The numerical solution of the advection equation with the leapfrog-trapezoidal scheme³⁴ (LT) in conditions of inadequate spatio-temporal resolution (non-smooth scalar fields) is characterized by the development of spurious oscillations in the vicinity of discontinuities and poorly resolved gradients³⁵. Monotone schemes (upwind), on the other hand, yield numerical solutions that are free from spurious ripples but are only first-order accurate and highly diffusive. The strategy behind flux-limited methods is to approximate a conservation law with a high-order scheme in regions where the solution is smooth while using a low-order monotone scheme where the solution is poorly resolved or discontinuous (Durrán 1999). The solution obtained in this manner is free from spurious ripples near poorly-resolved gradients while fairly accurate in regions where the field is well resolved. The need for flux limiters is most pronounced in problems where undershoots and overshoots in the scalar concentration field can couple with other physical processes to trigger spurious behavior. This is the case for ‘active’ components such as temperature, salinity or suspended sediment concentration. Oscillations in the temperature field, for example, may induce high values of mixing coefficients and hence lead to an over-estimation of mixing rates for other tracers.

In section 5.3.1 the leapfrog-trapezoidal scheme and a two-level flux-limited schemes are briefly described and applied to the solution of the advection equation in 1-D. The flux-limited method uses the monotone upstream differencing scheme and the Lax-Wendroff

³⁴ Spurious oscillations develop, in general, if space-centered finite difference algorithms are used to discretize the advection operator

³⁵ The velocity field is less prone to develop those kinds of discontinuities, at least in the relatively low energetic environments of lakes, and hence it was not considered necessary to introduce alternative schemes for the treatment of the advection terms in the momentum equations.

second-order method. Hereafter it will be referred to as the FL method. The numerical algorithms are compared in a 1-D system with constant velocity and a variety of scalar fields. The implementation of 1-D algorithms in higher dimensions is discussed next, and numerical experiments are presented to illustrate the performance of the methods in 2-D. Consistency with continuity (Gross 1997) is a critical feature in the design of a 2-D or 3-D advection scheme in shallow water equation models, and it is the focus of section 5.3.3. Then the implementation of the 2-level FL-method in the 3-level architecture of SI3D is further discussed. The simulation of baroclinic seiches in a rectangular basin is used as a final test for the 3-D FL advection scheme.

5.3.1. Advection schemes in 1-D

Many of the partial differential equations arising in fluid dynamics can be expressed as systems of *conservation laws* of the form

$$\frac{\partial \mathbf{c}}{\partial t} + \sum_j \frac{\partial}{\partial x_j} \mathbf{f}_j(\mathbf{c}) = \mathbf{0} \quad (5.25)$$

This equation states that the rate of change of $\mathbf{c}$ at each point is determined by the convergence of the fluxes $\mathbf{f}_j$ at that point. The conservation law governing the advection of a scalar in a 1-D system, in particular, obeys the expression

$$\frac{\partial S}{\partial t} + \frac{\partial}{\partial x} f(S) = \frac{\partial S}{\partial t} + \frac{\partial}{\partial x} (uS) = 0 \quad (5.26)$$

where S is the concentration of the scalar being transported and u is the velocity of the fluid. The leapfrog-trapezoidal method when formulated in conservation form and applied in 1-D, results in the following expressions

Leapfrog step

$$\frac{S_j^{n+1} - S_j^{n-1}}{2\Delta t} + \left[\frac{F_{j+1/2}^n - F_{j-1/2}^n}{\Delta x} \right] = 0 \quad \text{with} \quad F_{j+1/2}^n = u_{j+1/2}^n \left(\frac{S_{j+1}^n + S_j^n}{2} \right) \quad (5.27)$$

Trapezoidal step

$$\frac{S_j^{n+1} - S_j^n}{\Delta t} + \left[\frac{F_{i+1/2}^{n+1/2} - F_{i-1/2}^{n+1/2}}{\Delta x} \right] = 0 \quad \text{with} \\ F_{j+1/2}^{n+1/2} = \left(\frac{\tilde{u}_{j+1/2}^{n+1} + u_{j+1/2}^n}{2} \right) \left(\frac{(\tilde{S}_{j+1}^{n+1} + S_{j+1}^n) + (\tilde{S}_j^{n+1} + S_j^n)}{4} \right) \quad (5.28)$$

Here the tilde identifies variables estimated in previous leapfrog or trapezoidal iterations. When referring to the LT method, it will be assumed, unless otherwise noted, a unique trapezoidal iteration after the initial leapfrog step.

The FL-method approximates Eq. 5.26 in conservation form as

$$\frac{S_j^{n+1} - S_j^n}{\Delta t} + \left[\frac{F_{j+1/2}^n - F_{j-1/2}^n}{\Delta x} \right] = 0 \quad (5.29)$$

The algorithm is a two-level scheme, as it uses only information from the previous step in order to obtain the solution at the present time. The corrected fluxes (F) are evaluated by using information from the previous time step and are constructed as

$$F_{i+1/2} = F_{i+1/2}^L + C_{i+1/2} (F_{i+1/2}^H - F_{i+1/2}^L) \quad (5.30)$$

F^L and F^H denote the fluxes obtained by using monotone and high-order schemes, and $C_{i+1/2}$ is a multiplicative limiter. The performance of the scheme is highly dependent on the algorithm for specifying $C_{i+1/2}$. As mentioned in the introduction, the monotone upstream differencing scheme and the Lax-Wendroff second-order method are used here. The monotonized centered, or “MC”, limiter was selected as it is the best choice for “general” applications (Durrant 1999). The following steps are taken to calculate the corrected fluxes ($F_{i+1/2}$):

$$r_{i+1/2}^n = \frac{S_{i+\gamma+1}^n - S_{i+\gamma}^n}{S_{i+1}^n - S_i^n} \quad \text{where } \gamma = -\text{sgn}(u_{i+1/2}^{n+1/2}) \quad (5.31)$$

$$C_{i+1/2}^n = \max \left[0, \min \left(2r_{i+1/2}^n, \frac{1+r_{i+1/2}^n}{2}, 2 \right) \right] \quad (5.32)$$

$$F_{i+1/2}^n(u^{n+1/2}, S^n) = u_{i+1/2}^{n+1/2} \frac{S_{i+1}^n + S_i^n}{2} - \left[(1 - C_{i+1/2}^n) |u_{i+1/2}^{n+1/2}| + (u_{i+1/2}^{n+1/2})^2 \frac{\Delta t}{\Delta x} C_{i+1/2}^n \right] \left(\frac{S_{i+1}^n - S_i^n}{2} \right) \quad (5.33)$$

The resulting scheme is TVD or ‘total variation diminishing’ (Durrant 1999, p.252; Hirsch 1990). Since flux limiter methods revert to first-order schemes in the vicinity of minima and maxima, they do not give fully second order approximations. The effective order of accuracy can be estimated empirically, and for the particular FL method adopted in this work, it has been shown to be 1.9 (Durrant 1999, p. 270).

Figures 5.2 and 5.3 present the results obtained with the two algorithms when they are applied to three benchmark test problems consisting of the advection with constant speed of three different scalar profiles. The first is a unit step change in S over one mesh width. The second is an isolated sine-squared wave of width $20\Delta x$,

$$S(t=0) = \begin{cases} \sin^2\left(\frac{\pi x}{20\Delta x}\right), & 0 \leq x \leq 20\Delta x \\ 0, & \text{otherwise} \end{cases} \quad (5.34)$$

The third test profile consists of a semi-ellipse of width $2i_w\Delta x$, initially centered at i_c ,

$$S(t=0) = \begin{cases} \sqrt{1 - (i - i_c)^2 / i_w^2}, & \text{if } |i - i_c| < i_w \\ 0, & \text{otherwise} \end{cases} \quad (5.35)$$

In the exact solutions the profiles advance by 45 mesh-widths. Two Courant numbers were used: $CN = 0.05$ (Fig. 5.2) and $CN = 0.5$ (Fig. 5.3). In each case the exact solution is shown by the solid line.

It is clear from these plots that the original LT method introduces significant oscillations in the solution, especially at low c and in the regions where the gradients are not well resolved. This is not the case, however, for the FL method that eliminates undershoot and overshoot from the solution. The flux-limiters introduce some distortion in the solution around the peaks in the scalar field.

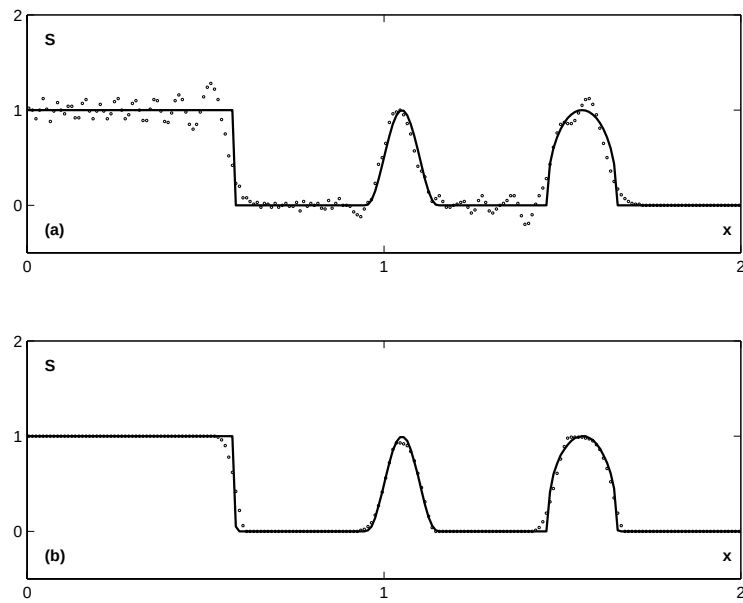


Figure 5.2. Solution of 1-D test problem with $CN = 0.05$. (a) LT method, and (b) FL method.

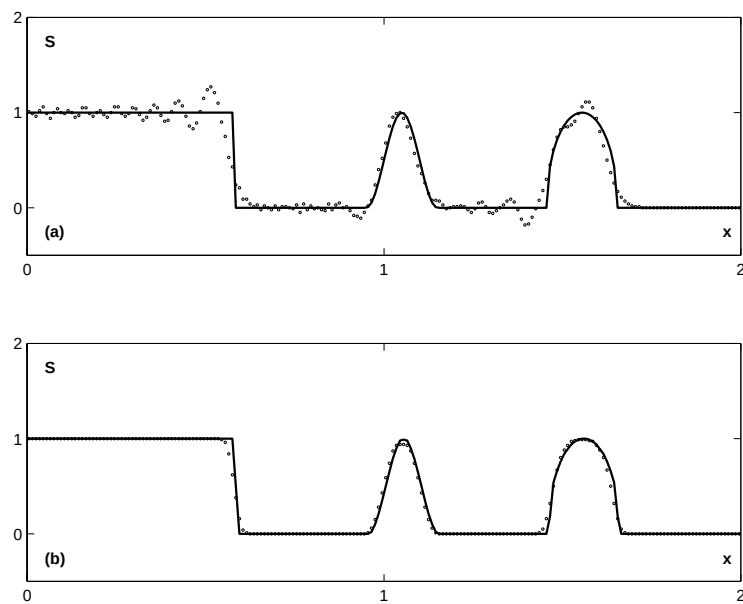


Figure 5.3. Solution of 1-D test problem with $CN = 0.5$. (a) LT method, and (b) FL method.

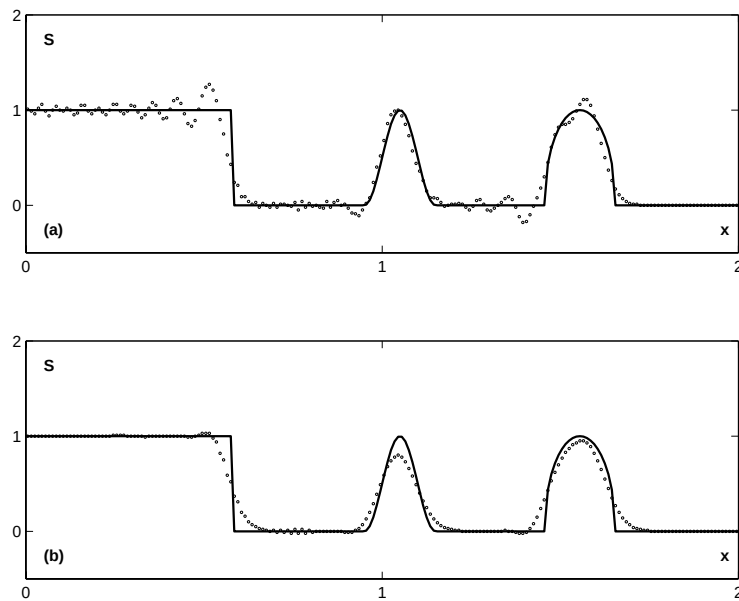


Figure 5.4. Solution of 1-D test problem with $CN = 0.5$. (a) LT method with 1 trapezoidal iteration, and (b) LT method and two trapezoidal iterations.

The effect of introducing additional trapezoidal iterations in the LT method is illustrated in Fig. 5.4. Figure 5.4.a presents the results obtained for $CN = 0.5$ and 1 iteration, while 5.4.b displays the results obtained by using 2 iterations. The effect of additional iterations is to increase the diffusion in the solution. This numerical diffusion will almost remove the oscillations in the poorly resolved areas, but it also significantly decreases the magnitude of the peaks.

5.3.2. Extension to higher dimensions: directional splitting

The extension to higher dimensions of the LT method applied to the advection equation is straightforward and is presented next for the sake of completeness for the 2-D case:

Leapfrog step

$$\begin{aligned} \frac{S_{i,j}^{n+1} - S_{i,j}^{n-1}}{2\Delta t} + \left[\frac{F_{i+1/2,j}^n - F_{i-1/2,j}^n}{\Delta x} + \frac{F_{i,j+1/2}^n - F_{i,j-1/2}^n}{\Delta y} \right] &= 0 \quad \text{with} \\ F_{i,j+1/2}^n &= v_{i,j+1/2}^n \left(\frac{S_{i,j+1}^n + S_{i,j}^n}{2} \right) \quad \text{and} \quad F_{i+1/2,j}^n = u_{i+1/2,j}^n \left(\frac{S_{i+1,j}^n + S_{i,j}^n}{2} \right) \end{aligned} \quad (5.36)$$

Trapezoidal step

$$\begin{aligned} \frac{S_{i,j}^{n+1} - S_{i,j}^n}{\Delta t} + \left[\frac{F_{i+1/2,j}^{n+1/2} - F_{i-1/2,j}^{n+1/2}}{\Delta x} + \frac{F_{i,j+1/2}^{n+1/2} - F_{i,j-1/2}^{n+1/2}}{\Delta y} \right] &= 0 \quad \text{with} \\ F_{i+1/2,j}^{n+1/2} &= \left(\frac{\tilde{u}_{i+1/2,j}^{n+1} + u_{i+1/2,j}^n}{2} \right) \left(\frac{(\tilde{S}_{i+1,j}^{n+1} + S_{i+1,j}^n) + (\tilde{S}_{i,j}^{n+1} + S_{i,j}^n)}{4} \right) \quad \text{and} \\ F_{i,j+1/2}^{n+1/2} &= \left(\frac{\tilde{v}_{i,j+1/2}^{n+1} + v_{i,j+1/2}^n}{2} \right) \left(\frac{(\tilde{S}_{i,j+1}^{n+1} + S_{i,j+1}^n) + (\tilde{S}_{i,j}^{n+1} + S_{i,j}^n)}{4} \right) \end{aligned} \quad (5.37)$$

The extension of the 1-D FL method for advection to higher dimensions follows a directional-splitting approach. Non-splitting or multidimensional³⁶ advection algorithms that produce virtually no numerical oscillations can also be found in the literature, including those of Zalesak (Zalesak 1979), Smolarkiewicz (Smolarkiewicz 1984), Leonard and Niknafs (referenced in Lin and Falconer 1997), and LeVeque (LeVeque 1996). The application of these methods to 2-D and 3-D flows that are characterized by spatially and temporally variable water depth is subject, though, to very significant difficulties and could also be prone to numerical oscillations. Gross (Gross 1997), for example, proved that the multidimensional scheme of Smolarkiewicz applied to

³⁶ The word “multidimensional” refers to those methods that avoid operator-splitting procedures to discretize differential equations in multiple dimensions.

simulation of advection in estuarine flows violated a condition that he referred to as “consistency with continuity” and could, hence, produce mass conservation errors. Lin and Falconer (Lin and Falconer 1997) compared the implementation of the ULTIMATE QUICKEST advection algorithm (Leonard 1991) in multiple dimensions using both splitting and non-splitting methods and showed that the split algorithms were more effective in controlling numerical oscillations.

In the directional-splitting approach the advective fluxes in each Cartesian direction are calculated sequentially. Consider now the 2-D advection equation written in operator form,

$$\frac{\partial S}{\partial t} + \left(u \frac{\partial}{\partial x} + v \frac{\partial}{\partial y} \right) S = L(S) = L_1(S) + L_2(S) \quad (5.38)$$

where L is that linear operator formed by the sum of two time-independent linear operators L_1 and L_2 , in this case the operators involving derivatives parallel to each spatial coordinate. If one uses the forward Euler approximation to the time derivative term, the operator-splitting approach to the discretization of the 2-D advection equation consists of the following two steps:

$$\tilde{S}^n = S^n + \Delta t L_1(S^n) = (I + \Delta t L_1) S^n = F_1(S^n) \quad (5.39)$$

$$S^{n+1} = \tilde{S}^n + \Delta t L_2(\tilde{S}^n) = (I + \Delta t L_2) \tilde{S}^n = F_2(\tilde{S}^n) \quad (5.40)$$

The fluxes L_1 and L_2 are calculated following the steps already described in section 5.3.1 for 1-D systems. The resulting scheme is only $O(\Delta t)$ accurate, even if one uses second order accurate approximations instead of F_1 and F_2 , unless the operators L_1 and L_2 commute. Strang splitting could be used to retain $O(\Delta t)^2$ accuracy, but in general this step requires 50% more work per time step than does conventional splitting (Durran 1999).

Two benchmark problems are selected to illustrate the performance of the LT method and the FL method in the solution of the 2-D advection equation. The first consists of the simulation of the solid body rotation of a cone, which has been frequently used as a test case for advection schemes. The computations were done on a domain $[0,1] \times [0,1]$ with $\Delta x = 0.01$, $\Delta t = 0.01$ and angular velocity = 1 radian/second. The initial conditions are shown in Fig. 5.5.a. The solutions obtained with leapfrog-trapezoidal and FL methods after 10 cycles are shown in Figs. 5.5.b and 5.5.c respectively. The error in the solution was calculated using the ℓ_1 norm defined previously in Table 3.7. The computational time was 23 seconds for the leapfrog-trapezoidal method and 170 seconds for the FL method. The results show the development of spurious oscillations when the solution is calculated with the leapfrog-trapezoidal scheme. The ℓ_1 error norm for this case was 1.73. The FL method suppresses overshoots and undershoots and considerably improves the accuracy of the solution ($\ell_1 = 0.19$). However, the higher accuracy is achieved at the cost of a higher computational time.

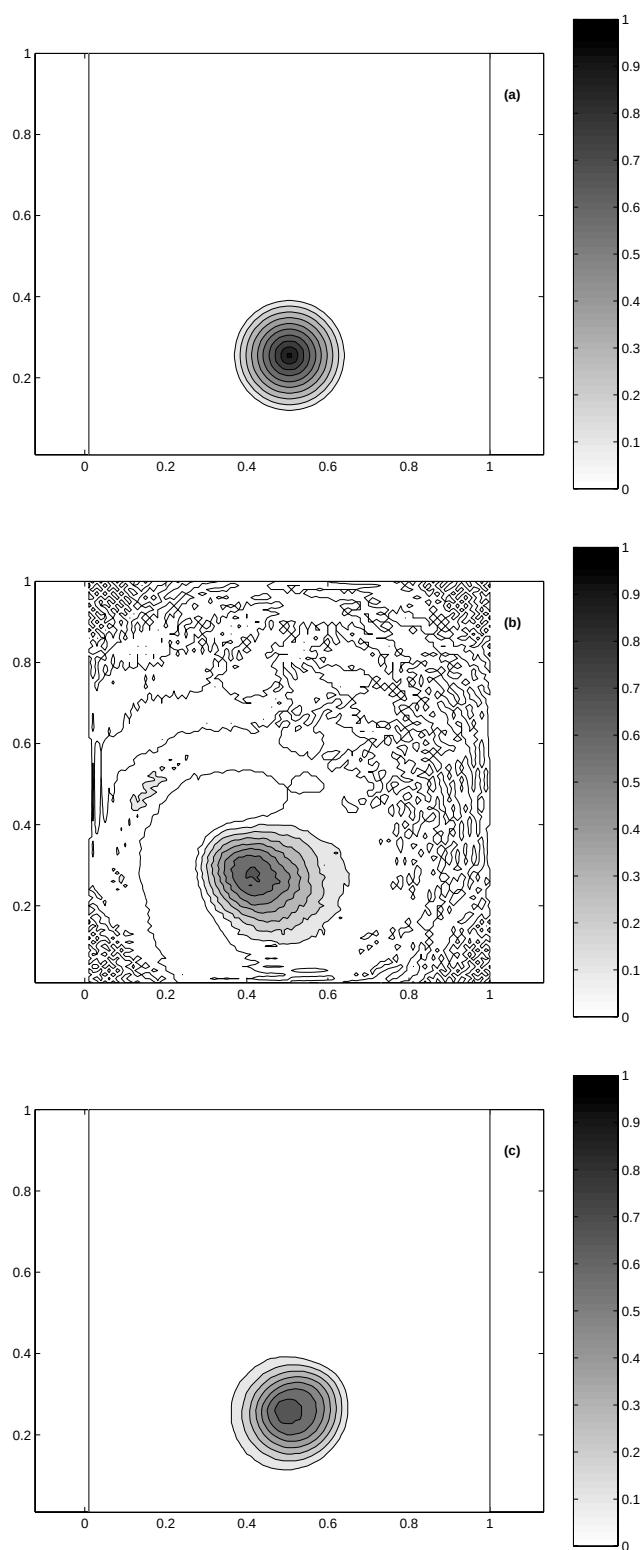


Figure 5.5. Advection of a rotating cone. (a) Initial conditions; (b) Solution after 10 cycles obtained with the LT method; and (c) the FL method. The units in the gray scale are normalized on the maximum concentration at the center of cone.

The second problem is probably a more severe test and uses a swirling deformation flow of the form (LeVeque 1996)

$$u(x, y, t) = \sin^2(\pi x) \sin(2\pi y) \cos(\pi \frac{t}{T}), \quad v(x, y, t) = -\sin^2(\pi y) \sin(2\pi x) \cos(\pi \frac{t}{T}) \quad (5.41)$$

This flow satisfies $u = v = 0$ on the boundaries of the unit square. It slows down and reverses direction in such a way that the initial data should ideally be recovered at time T .

The initial condition is given by a smooth hump of the form

$$S(x, y, 0) = \frac{1}{2}(1 + \cos(\pi r(x, y))), \quad \text{where } r(x, y) = \min(\sqrt{(x - x_0)^2 + (y - y_0)^2}, r_0) \quad (5.42)$$

The computations were done on a domain $[0, 1] \times [0, 1]$, with $x_0 = y_0 = r_0 = 0.25$ as illustrated in Fig. 5.6.

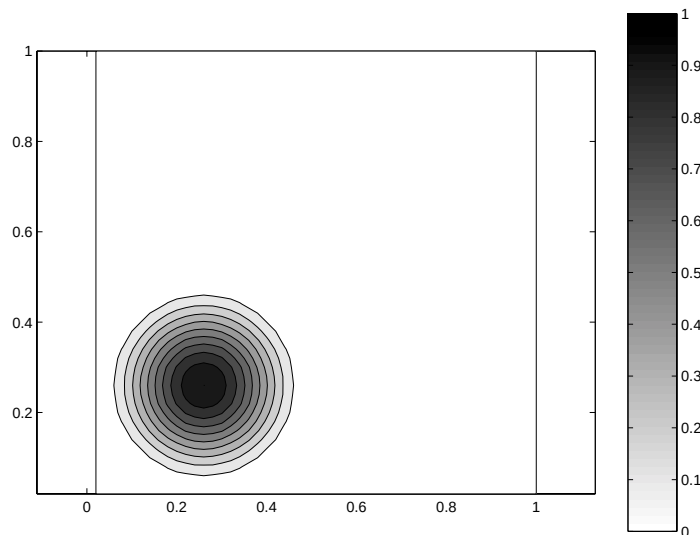


Figure 5.6. Initial conditions for test case of 2-D advection of a smooth hump in a swirling deformation flow.

The effect of the spatial and temporal discretizations was evaluated using a high-resolution ($\Delta x = 0.01$, $\Delta t = 0.005$) and a low-resolution test ($\Delta x = 0.02$, $\Delta t = 0.01$). The test case with low resolution was additionally solved with the leapfrog-trapezoidal method using 5 trapezoidal iterations per time step (Fig. 5.9). The results are shown in Figs. 5.7, 5.8 and 5.9 for the low-resolution case and in Figs. 5.10 and 5.11 for the high-resolution case. In those figures the solution is presented both at $T/2$, the time of maximum deformation, and after a complete cycle T , when one should recover the initial conditions. The leapfrog-trapezoidal solution at the time of maximum deformation $T/2$ is quite oscillatory, especially in cases of low resolution. After one period T the LT scheme recovers the original scalar field fairly well, both in the low-resolution ($\ell_1 = 0.05$) and the high-resolution ($\ell_1 = 0.028$) cases. The computed scalar field with the FL method again does not show any oscillations (even in the low-resolution case) at $T/2$. However, the FL method introduces some deformation in the scalar field that is especially noticeable in the low-resolution case and at time T . The deformation did not show up in the LT method with one trapezoidal iteration, but it appears if two or more trapezoidal iterations are performed. This fact suggests that the deformation in the field is the result of either the non-linearity of the schemes or, most probably, the introduction of non-reversible diffusion through the upwind scheme used in the FL method or the additional trapezoidal iterations. The increased number of iterations smoothed the oscillations in the scalar field at $T/2$.

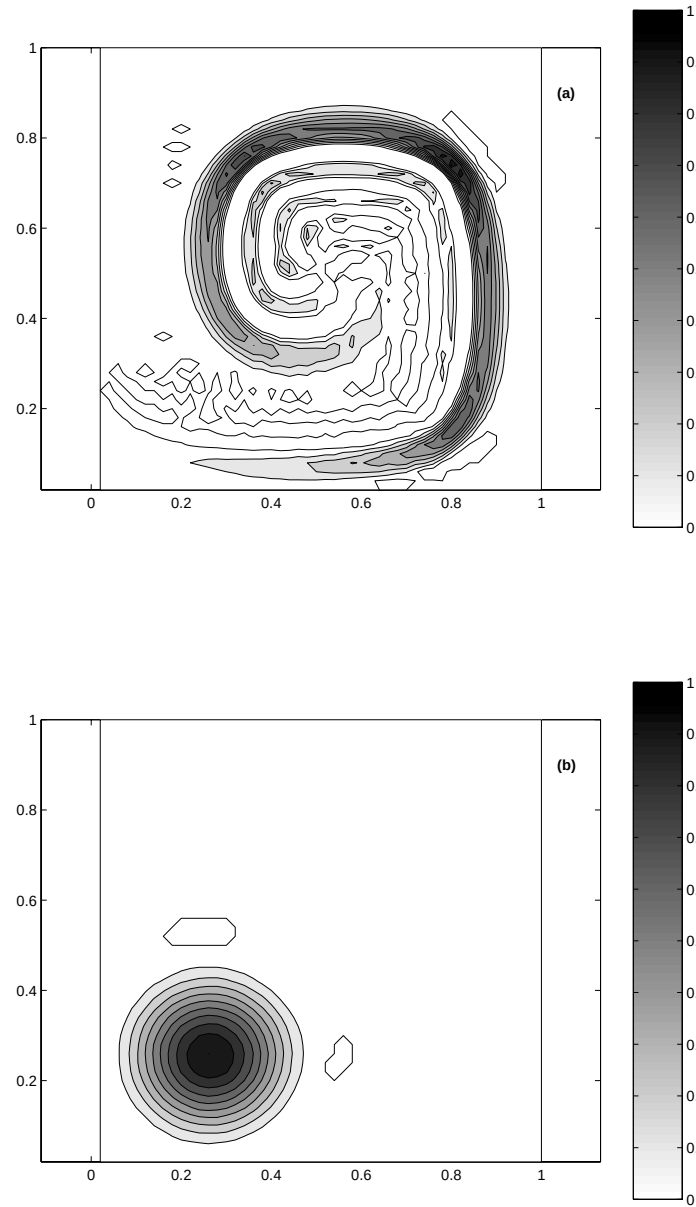


Figure 5.7. Advection of a smooth hump in a swirling deformation flow. Solution obtained with the LT method, using 1 trapezoidal iteration and a low-resolution discretization (a) at $T/2$ (b) at T .

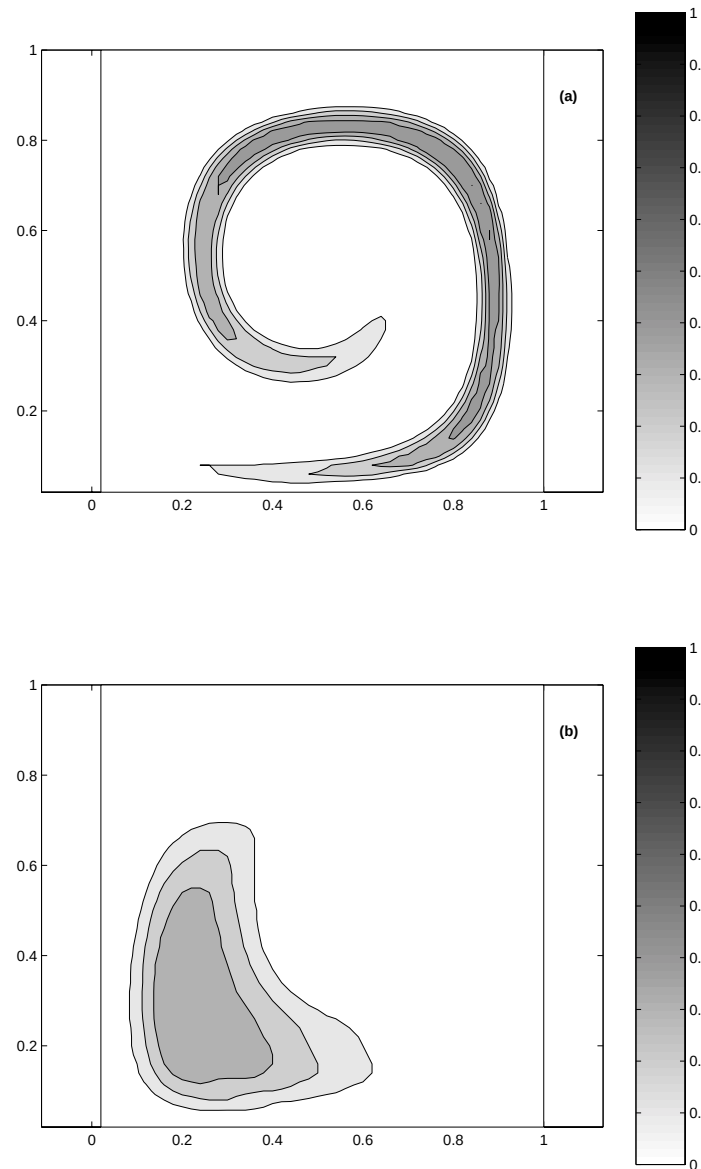


Figure 5.8. Advection of a smooth hump in a swirling deformation flow. Solution obtained with the FL method and a low resolution discretization (a) at $T/2$ (b) at T .

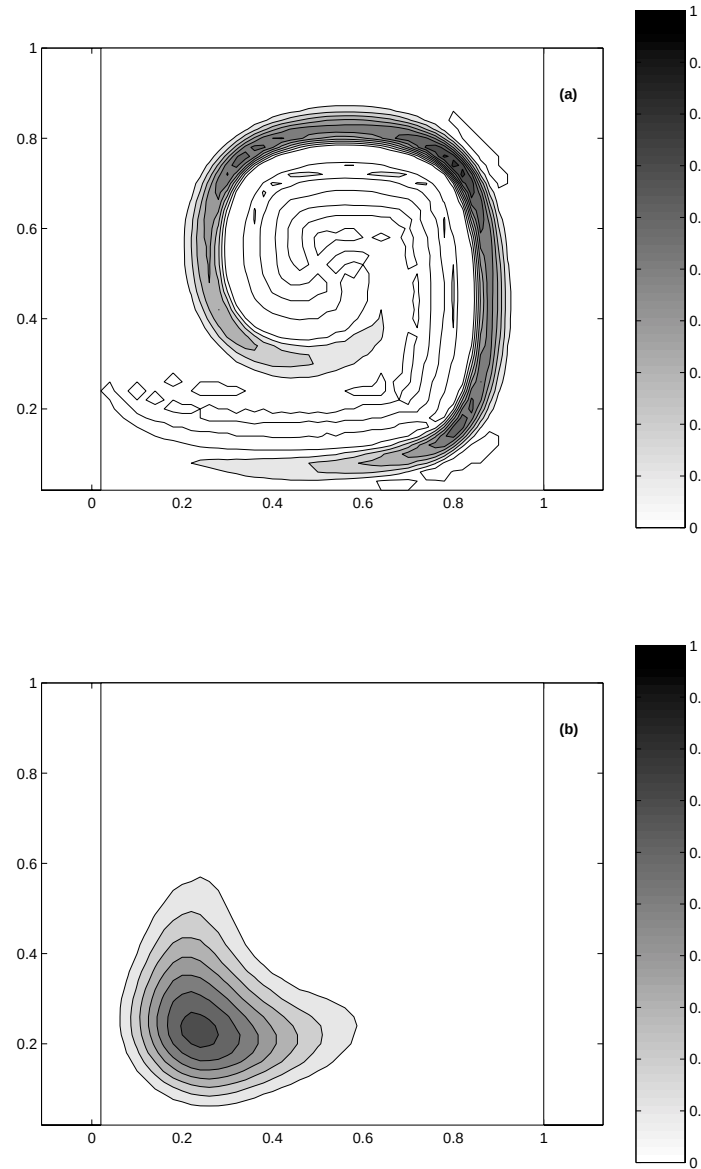


Figure 5.9. Advection of a smooth hump in a swirling deformation flow. Solution obtained with the LT method (5 trapezoidal iterations) and low-resolution discretization (a) at $T/2$ (b) at T .

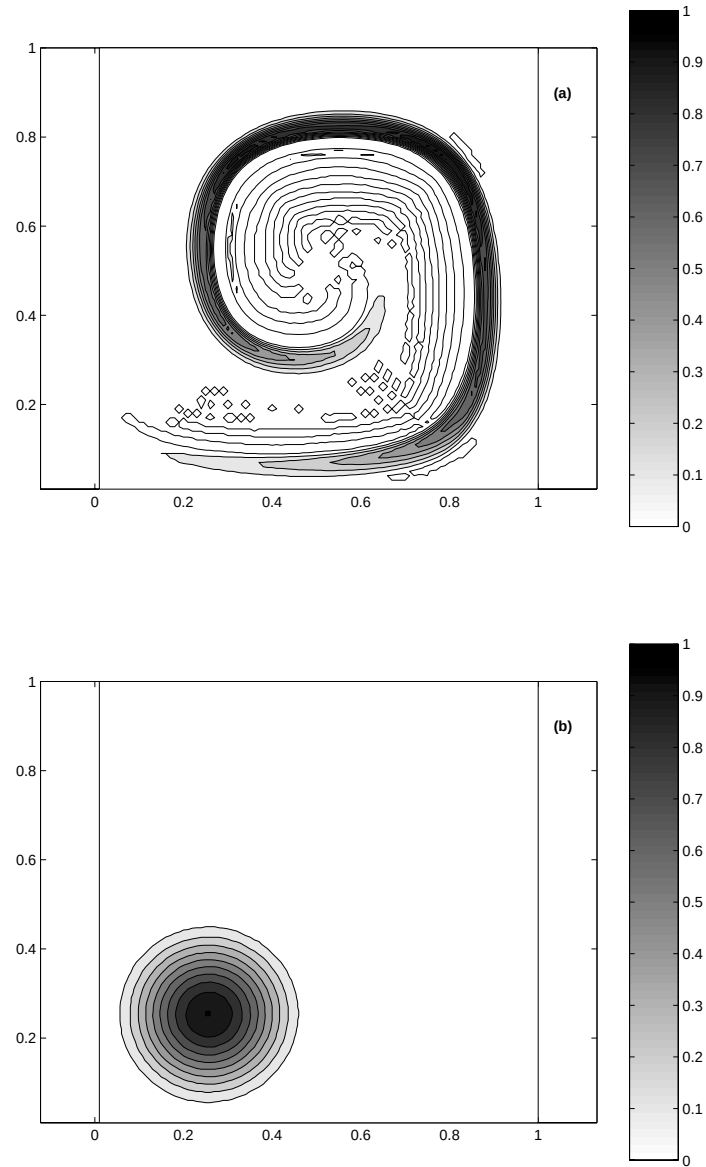


Figure 5.10. Advection of a smooth hump in a swirling deformation flow. Solution obtained with the LT method (1 trapezoidal iteration) and high-resolution discretization (a) at $T/2$ (b) at T .

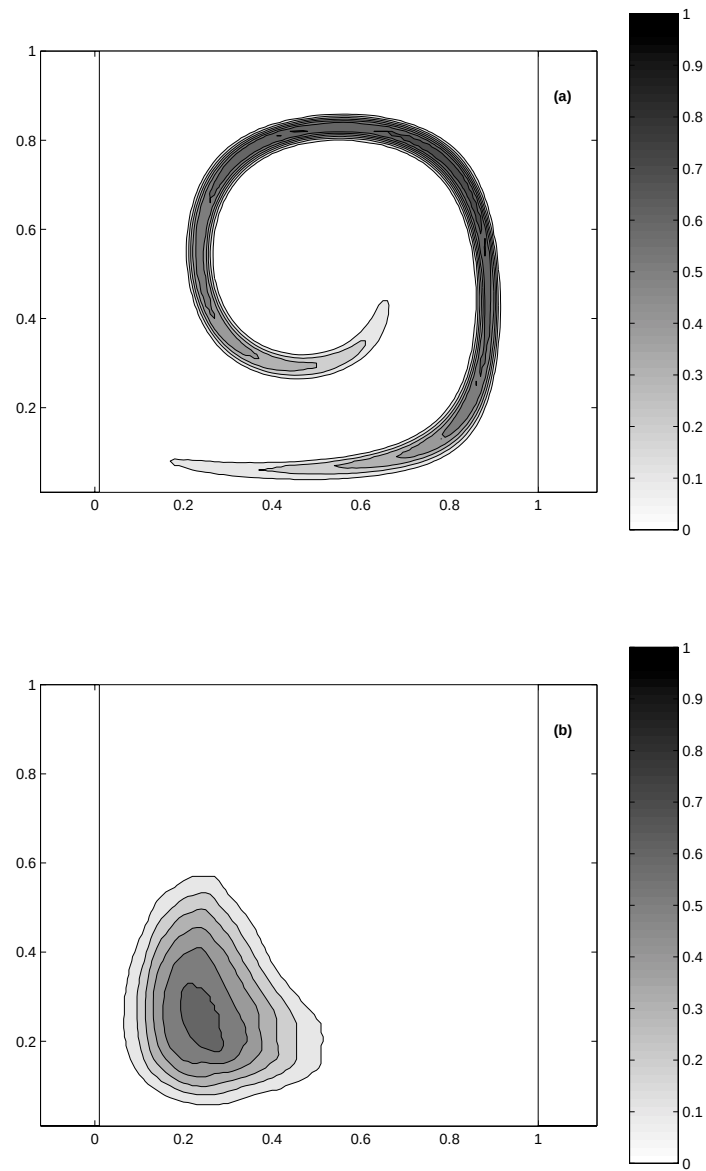


Figure 5.11. Advection of a smooth hump in a swirling deformation flow. Solution obtained with the FL method and high-resolution discretization (a) at $T/2$ (b) at T .

5.3.3. 3-D advection scheme in SI3D

The leapfrog-trapezoidal advection scheme in the 3-D hydrodynamic model SI3D is now applied to the layer-averaged form of the scalar transport equation,

$$\frac{\partial(hS)_k}{\partial t} + \frac{\partial(US)_k}{\partial x} + \frac{\partial(US)_k}{\partial y} + ws|_{k-1/2} - ws|_{k+1/2} = 0 \quad (5.43)$$

Here k identifies a given layer, h is the thickness of the layer, U and V are horizontal transports, and w is the vertical velocity. The implementation of the LT scheme does not use operator splitting and is a straightforward generalization of Eq. 5.36 and Eq. 5.37 for 2-D. Only the velocities (u, v) need to be replaced by depth-averaged transport (U, V) in the horizontal direction, while in the vertical direction the difference in the fluxes ($w s$) evaluated at the upper ($k-1/2$) and lower ($k+1/2$) interfaces should be used instead of the gradient. This scheme is “consistent with continuity” (Gross 1997). The consistency will guarantee that mass is conserved and in a uniform scalar field means that the scalar transport scheme will reproduce the numerical solution of the continuity or mass conservation equation.

The two-level flux-limited method was implemented in 3-D using operator splitting. The resulting scheme is only first order in time, though of 1.9 order in space (Durrant 1999). In order to achieve “consistency with continuity”, intermediate values of the scalar quantities as well as layer thickness must be calculated. These intermediate values are calculated as follows:

Transport in the E-W direction

$$\tilde{H}_{i,j,k}^n = H_{i,j,k}^n - \frac{\Delta t}{\Delta x} (U_{i+1/2,j,k}^{n+1/2} - U_{i-1/2,j,k}^{n+1/2}) \quad (5.44)$$

$$\tilde{S}_{i,j,k}^n \tilde{H}_{i,j,k}^n = S_{i,j,k}^n H_{i,j,k}^n - \frac{\Delta t}{\Delta x} (F_{i+1/2,j,k} (U_{i+1/2,j,k}^{n+1/2}, S^n) - F_{i-1/2,j,k} (U_{i-1/2,j,k}^{n+1/2}, S^n)) \quad (5.45)$$

Transport in the N-S direction

$$\tilde{H}_{i,j,k}^n = \tilde{H}_{i,j,k}^n - \frac{\Delta t}{\Delta y} (V_{i,j+1/2,k}^{n+1/2} - V_{i,j-1/2,k}^{n+1/2}) \quad (5.46)$$

$$\tilde{S}_{i,j,k}^n \tilde{H}_{i,j,k}^n = \tilde{S}_{i,j,k}^n \tilde{H}_{i,j,k}^n - \frac{\Delta t}{\Delta y} (F_{i,j+1/2,k} (V_{i,j+1/2,k}^{n+1/2}, \tilde{S}^n) - F_{i,j-1/2,k} (V_{i,j-1/2,k}^{n+1/2}, \tilde{S}^n)) \quad (5.47)$$

Transport in vertical direction

$$S_{i,j,k}^{n+1} H_{i,j,k}^{n+1} = \tilde{S}_{i,j,k} \tilde{H}_{i,j,k}^n - \Delta t (F_{i,j,k-1/2} (w_{i,j,k-1/2}^n, \tilde{S}^n) - F_{i,j,k+1/2} (w_{i,j,k+1/2}^n, \tilde{S}^n)) \quad (5.48)$$

The corrected fluxes F in Eqs. 5.44-5.48 are constructed for each direction with the monotone upstream differencing scheme and the Lax-Wendroff second-order method, as shown in Eqs. 5.30-5.33. At flux faces next to the boundaries, the calculation of the multiplicative MC limiter might involve scalar values from outside the wet computational domain. In those cases the Lax-Wendroff method was utilized to calculate fluxes. The calculation of fluxes with the upwind scheme will introduce diffusion and it will produce a significant deformation of the scalar field. The effects are particularly deleterious in the vertical where the stratification and hence the speed of propagation of internal waves can be modified. The discretization of the diffusion terms in the vertical direction follows the Crank-Nicholson or trapezoidal method. No lateral diffusion or turbulent transfer coefficient is used with the LT method in SI3D.

In the original formulation of SI3D, the leapfrog method was applied to both the hydrodynamic and scalar transport equations in the first step, which provided an initial estimate of the prognostic variables for successive (if any) trapezoidal iterations. This sequence is slightly modified to use the FL method for the solution of the transport equation. In this case the momentum and continuity equations are solved using a leapfrog step and a single trapezoidal iteration before solving the transport equation. In this manner the velocity field used for the advection of the scalar is free from the computational mode associated with the leapfrog method and has better phase-propagation characteristics. No additional iterations are needed to avoid oscillations and smooth the results, unless an internal gravity wave number time-step limitation must be overcome.

The original transport scheme in SI3D was used in chapter 3 to simulate the evolution of a baroclinic seiche excited by conveniently chosen initial conditions. That same benchmark problem was used to test the performance of the FL method for the transport equation within SI3D and compare it with the leapfrog-trapezoidal method. The simulation of a baroclinic seiche is a problem that is particularly sensitive to errors in mass conservation, and hence it is an apt test of “consistency with continuity”. The domain was 16384 m long, 512 m wide and 12 m deep. It was discretized with cells 256 m x 256 m in the horizontal ($\Delta x = \Delta y = 256$ m) and 1 m in the vertical ($\Delta z = 1$ m). A total of 144 hours were simulated using $\Delta t = 720$ seconds (internal gravity wave courant number = 1.37). A single iteration was performed after the solution of the scalar transport equation was obtained in each time step. Hence, in each time step the momentum and

continuity equations were solved 3 times, while the scalar transport was solved only twice. The initial conditions are those shown in Fig. 3.7.

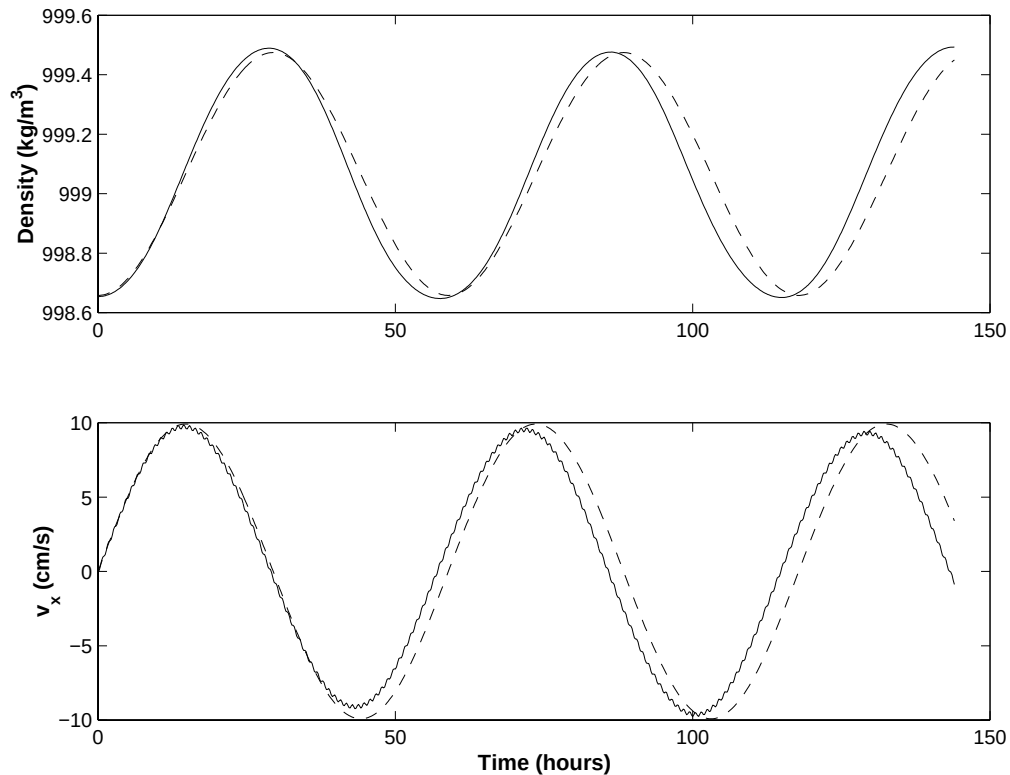


Figure 5.12. Time series of density and longitudinal velocity obtained with the FL advection scheme in SI3D (continuous line) and from the analytical solution (dashed line) using $L/\Delta x = 128$. The high frequency oscillations observed in the velocity time series has a barotropic origin as explained in section 3.12.

The results of the simulation at points where the amplitude of the velocity and the density signals reach their maxima are shown in Fig. 5.12. The error measures of the density and longitudinal velocity solution obtained with the FL and LT methods at those same points are shown in Table 5.2. The definition of the error measures is given in Table 3.7. Two trapezoidal iterations per time step were used in the LT method. The results indicate a

	Longitudinal velocity		Density	
	FL method	LT method	FL method	LT method
A	1.0001	1.0001	0.9999	0.9999
Q	1.0237	1.0232	1.0278	1.0292
D	0.9834	0.9838	0.9830	0.9829
ℓ_1	0.2087	0.2075	0.2274	0.2296
ℓ_2	0.2435	0.2417	0.2656	0.2672
ℓ_∞	0.4310	0.4379	0.4040	0.4052
RMSE	1.7252	1.7123	0.0759	0.0763

Table 5.2. Error measures for $L/\Delta x = 128$. A = dissipation error; Q = celerity ratio; D = Index of agreement; ℓ_1 , ℓ_2 , ℓ_∞ = error norms, and RMSE = root mean square error.

very similar accuracy of the Leapfrog-Trapezoidal and the FL method for this problem. The computational time for the simulation of 288 hours in this problem is 72 seconds on a PC system having 128MB of memory and a Pentium-III processor at 450MHz. The same problem was solved using the leapfrog-trapezoidal method (two trapezoidal iterations per time step) in 68 seconds.

5.4. Vertical mixing algorithm

5.4.1. Review

The characteristic length scale of the turbulent processes responsible for vertical mixing is such that they cannot be resolved in other than very limited domains and thus must be parameterized. The schemes in the literature for the parameterization of turbulent phenomena can be grouped in two broad categories: bulk and diffusion models (Kantha and Clayson 1994). *Bulk or mixed-layer models* exploit the fact that within the boundary layers water properties are well mixed and almost uniform. The conservation laws for those properties (momentum, temperature, etc.) are integrated over the depth of the mixed layer, and the extent of the well-mixed region is parameterized. The resulting

model consists of a system of ordinary differential equations for the evolution of the uniform water properties within the mixed-layer and simulates the advance or retreat of the mixed layer in response to surface stresses and the shear stress within the water column. For a detailed discussion of the mixed-layer models the reader is referred to Martin 1985, Large *et al.* 1994, Imberger and Patterson 1990 and Sherman *et al.* 1978. The use of mixed-layer models in 1-D oceanic studies is relatively common (e.g. Martin 1985, McCormick and Meadows 1988), and only recently have they been incorporated into 3-D oceanic general circulation and lake models (Hodges *et al.* 2000 and references therein) to model the evolution of the surface wind driven mixed layer. The bottom boundary layer in these applications is simply neglected. Very few attempts exist to extend the mixed-layer models to bottom boundary layers (e.g. Monismith and Fong 1996), and even those have been developed only for estuarine environments. In lake environments the benthic boundary layer may make a very significant contribution to scalar mixing (Imberger 1998) and hence should be explicitly represented. In shallow lakes, however, there does not exist a clear-cut separation between the surface boundary layer, the bottom boundary layer and the lake interior, as is the case for deep lake environments. The interaction between the bottom and surface boundary layers when they overlap is not yet resolved and might pose considerable difficulties in the design of an efficient algorithm.

Diffusion models parameterize turbulent mixing as a diffusion process, and they probably are the oldest and most popular method of representing turbulent phenomena. The parameterization may be based on abundant theoretical and observational knowledge of

the surface layers and hence contains empirical or semi-empirical formulations for turbulent diffusion (semi-empirical models), or it can be based on actual modeling of the turbulence quantities by appealing to turbulence closure at the second or higher moment levles (closure schemes). In the *semi-empirical models* the turbulent fluxes of momentum (u , v) and scalar (θ) are represented using the well-known *Boussinesq's eddy viscosity concept*

$$\langle u'w', v'w' \rangle = K_v \left(\frac{\partial u}{\partial z}, \frac{\partial v}{\partial z} \right) \quad (5.49)$$

$$\langle \theta'w' \rangle = D_v \left(\frac{\partial \theta}{\partial z} \right) \quad (5.50)$$

where K_v and D_v represent the eddy viscosity and eddy diffusivity respectively. Many relationships based upon experimental, theoretical and numerical studies have been suggested for the eddy coefficients in terms of the gradient Richardson number

$$Ri = -\frac{g}{\rho_0} \frac{\partial \rho}{\partial z} \bigg/ \left| \frac{\partial v}{\partial z} \right|^2 \quad (5.51)$$

and the turbulent Prandtl number ($Pr = K_v/D_v$). These relationships assume that the eddy coefficient can be expressed as a neutral value ($Ri = 0$), denoted below with a zero subscript, multiplied by a function of the stability parameter, Ri , and can be written as

$$K_v = K_{v0} \Phi_M(Ri) \quad (5.52)$$

$$D_v = D_{v0} \Phi_H(Ri) \quad (5.53)$$

The eddy coefficients under neutral conditions (K_{v0} and D_{v0}) usually appeal to the Prandtl mixing-length hypothesis that defines the characteristics turbulent velocity as

$$u' \sim w' \sim \ell \frac{\partial u}{\partial z} \quad (5.54)$$

At present there is no sound physical basis for stipulating ℓ as a function of depth, but several possibilities exist. One of those formulations for the mixing length was proposed by Keulegan (reference in Smith 1997) and is given by

$$\ell = \kappa z \left(1 - \frac{z'}{H} \right) \quad (5.55)$$

and can be derived by assuming a logarithmic profile. In Eq. 5.55, κ is the von Karman constant (~ 0.4), z' is the vertical distance measured from the bottom (for the benthic boundary layer) or the water surface (for the surface layer), and H is the total water depth. Even under non-stratified conditions the velocity profiles in unsteady lake flows are often non-logarithmic owing to the effects of accelerations and secondary currents that are induced by complex geometry and density gradients. For these reasons Eq. 5.55 must be considered only approximate for lake applications.

In proposing a theory of the thermocline, Munk and Anderson (Munk and Anderson 1948) generalized the stability functions of a number of previous investigations and suggested

$$\Phi_M = (1 + aRi)^{-n} \quad (5.56)$$

$$\Phi_H = (1 + bRi)^{-m} \quad (5.57)$$

The values proposed in the literature for a , b , m and n vary widely, and should be verified for each particular application (Blumberg 1986). Other forms of stability functions have been proposed in the literature (e.g. Henderson-Sellers 1982), none of which, however, has enjoyed as widespread usage as the Munk-Anderson formulation. In general, these expressions are valid only in the range of stable conditions ($Ri > 0$). Whenever the water column becomes unstable, as in cooling periods, free convection or extremely rapid mixing must be accommodated. The convection adjustment process of Bryan and Cox (Bryan and Cox 1967) is usually employed in these cases. Whenever a portion of a water column is found to be unstable, an adjustment is made to bring the column back to neutral stability.

The *turbulence closure models* are based on the Reynolds averaged equations for momentum and heat formulated in terms of mean variables and higher moments of the fluctuations. In second order closure models, one solves conservation equations for the mean variables and the second order moments (Reynolds stresses, heat fluxes and scalar variances). The higher-order correlations arising in the formulation of the equations for the second moments (third-moment turbulence quantities and pressure strain rate and pressure-scalar gradient covariance terms) are modeled by appealing to physical intuition and/or observational evidence. The introduction of additional simplifications results in a hierarchy of models, of which the most useful in geophysical applications is the model that consists of one conservation equation for the turbulent kinetic energy (TKE) and a

set of algebraic equations for turbulence second-order moments. This set is supplemented either by auxiliary information on the turbulence length scale such as the Blackadar formula or by utilizing an equation for a quantity that includes the turbulence length scale such as dissipation (k - ε models) or, as in the *Mellor-Yamada models* the product of the turbulence length scale and twice the TKE. The so-called 2.5 Level of the Mellor-Yamada hierarchy of models (MY2.5) has been widely used in geophysical applications and was the preferred scheme for the vertical mixing parameterization. As originally formulated (Mellor and Yamada 1982; Mellor and Yamada 1974), the model has been shown to underpredict the depth of the surface mixed layer in oceanic applications (e.g. Martin 1985). However, further developments of the model have appeared in the literature (Galperin *et al.* 1988 and Kantha and Clayson 1994) that correct for those deficiencies. This work follows Kantha and Clayson (Kantha and Clayson 1994) in the formulation of the turbulence closure scheme for vertical mixing in the 3-D lake model. The governing equations are presented in section 5.4.2, followed by the numerical algorithm used for their solution. The vertical mixing parameterization is then tested using two benchmark problems in section 5.4.4.

5.4.2. The Mellor-Yamada turbulence closure model level 2.5

The model used to specify the Reynolds stresses and turbulent fluxes in the governing equations was proposed by Kantha and Clayson (Kantha and Clayson 1994) and belongs to the Mellor-Yamada hierarchy. It uses the quasi-equilibrium turbulence model (QETE) of Galperin *et al.* (Galperin *et al.* 1988) with readjusted parameters, and it further incorporates a parameterization of shear-instability-induced mixing in the strongly stratified region below the surface mixed layer. The model uses two partial differential

equations to calculate the turbulent kinetic energy, $TKE = q^2/2$, and a turbulent macroscale, l . It is assumed here that the balance of TKE and the length scale is dominated by local production, local dissipation and vertical turbulent diffusion in the water column (i.e. horizontal advection is neglected). The equation for turbulent kinetic energy is

$$\frac{\partial q^2}{\partial t} - \frac{\partial}{\partial z} \left(l q S_q \frac{\partial q^2}{\partial z} \right) = 2[P_s + P_b - \varepsilon] \quad (5.58)$$

where P_s , P_b and ε stand for shear production, buoyant production and the dissipation of turbulent kinetic energy. They are described further by the expressions

$$P_s = K_v \left(\frac{\partial u}{\partial z} \right)^2 + K_v \left(\frac{\partial u}{\partial z} \right)^2; \quad P_b = \frac{g}{\rho_0} D_v \frac{\partial \rho}{\partial z}; \quad \varepsilon = \frac{q^3}{B_1 l} \quad (5.59)$$

in which B_1 is an empirical constant. The equation governing the turbulent macroscale is

$$\frac{\partial (q^2 l)}{\partial t} - \frac{\partial}{\partial z} \left(l q S_q \frac{\partial (q^2 l)}{\partial z} \right) = l E_1 [P_s + P_b] - \frac{q^3}{B_1} \left[1 + E_2 \left(\frac{l}{\kappa z} \right)^2 + E_3 \left(\frac{l}{\kappa (H - z)} \right)^2 \right] \quad (5.60)$$

where $z = 0$ at the free surface, H is the depth of the water column, and κ is the von Karman constant. The last two terms are wall proximity terms and account of the effects of solid walls and free surfaces respectively on the length scales. The boundary conditions applied at the top and the bottom boundaries are (Blumberg 1986)

$$q^2 = B_1^{2/3} (u_*)^2, q^2 l = 0 \quad (5.61)$$

The turbulent fluxes of momentum (u , v) and scalar (θ) are represented using a diffusion model and the concepts of eddy viscosity and diffusivity, as in Eq. 5.49 and Eq. 5.50. The eddy viscosity and diffusivity are calculated from q and the turbulent macroscale as $K_v = S_m l q$ and $D_v = S_h l q$ in which the stability functions (S_m , S_h) are given by (Kantha and Clayson 1994)

$$S_h = \frac{A_2 \left(1 - \frac{6A_1}{B_1} \right)}{1 - 3A_2 G_h [6A_1 + B_2(1 - C_3)]} \quad (5.62)$$

$$S_m = A_1 \left\{ \frac{\left(1 - \frac{6A_1}{B_1} - 3C_1 \right) + 9[2A_1 + A_2(1 - C_2)] S_h G_h}{(1 - 9A_1 A_2 G_h)} \right\} \quad (5.63)$$

$$G_h = - \left(\frac{Nl}{q} \right)^2 \quad (5.64)$$

The values of the constants in the model are $(A_1, A_2, B_1, B_2, C_1, C_2, C_3, S_q, E_1, E_2) = (0.92, 0.74, 16.6, 10.1, 0.08, 0.7, 0.2, 0.2, 1.8, 1.33)$. The term G_h is bounded from above, so that the variance of the velocity fluctuations is positive definite, according to

$$G_h \leq \{A_2 [B_1 + 12A_1 + 3B_2(1 - C_3)]\}^{-1} \quad (5.65)$$

In the code the upper bound on G_h is set to 0.029 (Kantha and Clayson 1994). There is as well a lower bound on G_h which reflects that, under strong stable stratification conditions, there is a size limit beyond which eddies are incapable of overturning. This bound is given by the Ozmidov scale, $L_o = (\varepsilon/N^3)^{1/2}$. Imposing a bound on the turbulence length scale that is proportional to L_o leads to the following result:

$$\frac{N}{q}l \leq C_4 \quad (5.66)$$

Here N is the buoyancy frequency ($N^2 = - (g/\rho_0)\partial\rho/\partial z$) and C_4 is 0.53 (Galperin *et al.* 1988). This value of C_4 leads to a lower bound on G_h of -0.28 .

In the strongly stable but highly sheared region below the mixed layer, mixing is mainly caused by episodic Kelvin-Helmholtz instabilities. Additionally, internal waves in this region modulate the shear and periodically enhance the instabilities. These processes are difficult to model, and they are usually represented by using *ad hoc* parameterizations. The model of Kantha and Clayson (Kantha and Clayson 1994) parameterizes the shear-induced diffusivities as a strongly decreasing function of the gradient Richardson number Ri , and the internal-wave-induced diffusion as constant values (Table 5.2). The values of the eddy viscosity and diffusivity are specified as in Table 5.3 immediately below the fully turbulent mixed layer (where the values of q^2 drop below $10^{-6} \text{ m}^2/\text{s}^2$). The model has in addition a weak but constant background mixing coefficients ($= 10^{-5}$). The unit for all diffusivities in Table 5.3 is m^2/s .

Range of Ri	Eddy viscosity	Eddy Diffusivity
$Ri > 0.7$	$K_{MR} = 10^{-4}$	$K_{HR} = 5 \times 10^{-5}$
$0 < Ri < 0.7$	$K_{MB} = K_{MR} + 5 \cdot 10^{-3} \left[1 - \left(\frac{Ri}{0.7} \right)^2 \right]^3$	$K_{HB} = K_{HR} + 5 \cdot 10^{-3} \left[1 - \left(\frac{Ri}{0.7} \right)^2 \right]^3$
$Ri < 0$	$K_{MB} = K_{MR} + 5 \cdot 10^{-3}$	$K_{HB} = K_{HR} + 5 \cdot 10^{-3}$

Table 5.3. Parameterization of shear-induced and internal-wave-induced (K_{MR}) mixing in the turbulence model.

5.4.3. Numerical method for the turbulence closure model

The approach adopted for the solution of the governing equations for the TKE and the turbulent macroscale is a 3-level fully implicit algorithm (Fletcher 1991; Ferziger and Peric 1996). The method has a truncation error of $O(\Delta t^2, \Delta x^2)$, is A-stable and damps out spurious oscillations associated with stiff problems. The algorithm results in a tri-diagonal system of equations for each water column that is solved using the Thomas algorithm. The variables q^2 and q^2l are defined at vertical velocity points (interfaces between two cells aligned in the vertical direction). The discretized equations are written as

$$\begin{aligned}
& \frac{3}{2} \frac{(q^2)_{k-1/2}^{n+1} - (q^2)_{k-1/2}^n}{\Delta t} - \frac{1}{2} \frac{(q^2)_{k-1/2}^n - (q^2)_{k-1/2}^{n-1}}{\Delta t} \\
& - \frac{1}{\Delta z_{k+1/2}^{n+1}} \left[q_{k-1}^n l_{k-1}^n S_q \left(\frac{(q^2)_{k-3/2}^{n+1} - (q^2)_{k-1/2}^{n+1}}{\Delta z_{k-1}^{n+1}} \right) - q_k^n l_k^n S_q \left(\frac{(q^2)_{k-1/2}^{n+1} - (q^2)_{k+1/2}^{n+1}}{\Delta z_{k+1}^{n+1}} \right) \right] \\
& = 2A_{V_{k-1/2}}^n \left[\left(\frac{u_{k-1}^n - u_k^n}{\Delta z_{k-1}^{n+1}} \right)^2 + \left(\frac{v_{k-1}^n - v_k^n}{\Delta z_{k-1}^{n+1}} \right)^2 \right] + 2 \frac{g}{\rho_0} D_{V_{k-1/2}}^n \left(\frac{\rho_{k-1}^n - \rho_k^n}{\Delta z_{k-1}^{n+1}} \right) - 2 \frac{(q^2)_{k-1/2}^{n+1} q_{k-1/2}^n}{B_1 l_{k-1/2}^n}
\end{aligned} \tag{5.67}$$

$$\begin{aligned}
& \frac{3}{2} \frac{(q^2 l)_{k-1/2}^{n+1} - (q^2 l)_{k-1/2}^n}{\Delta t} - \frac{1}{2} \frac{(q^2 l)_{k-1/2}^n - (q^2 l)_{k-1/2}^{n-1}}{\Delta t} \\
& - \frac{1}{\Delta z_{k+1/2}^{n+1}} \left[q_{k-1}^n l_{k-1}^n S_q \left(\frac{(q^2 l)_{k-3/2}^{n+1} - (q^2 l)_{k-1/2}^{n+1}}{\Delta z_{k-1}^{n+1}} \right) - q_k^n l_k^n S_q \left(\frac{(q^2 l)_{k-1/2}^{n+1} - (q^2 l)_{k+1/2}^{n+1}}{\Delta z_{k+1}^{n+1}} \right) \right] \\
& = E_1 l_{k-1/2}^n A_{V_{k-1/2}}^n \left[\left(\frac{u_{k-1}^n - u_k^n}{\Delta z_{k-1}^{n+1}} \right)^2 + \left(\frac{v_{k-1}^n - v_k^n}{\Delta z_{k-1}^{n+1}} \right)^2 \right] + E_1 l_{k-1/2}^n \frac{g}{\rho_0} D_{V_{k-1/2}}^n \left(\frac{\rho_{k-1}^n - \rho_k^n}{\Delta z_{k-1}^{n+1}} \right) \\
& - \frac{(q^2 l)_{k-1/2}^{n+1} q_{k-1/2}^n}{B_1 l_{k-1/2}^n} \left[1 + E_2 \left(\frac{l_{k-1/2}^n}{\kappa z_{k-1/2}^n} \right)^2 + E_3 \left(\frac{l_{k-1/2}^n}{\kappa (H^n - z_{k-1/2}^n)} \right)^2 \right]
\end{aligned} \tag{5.68}$$

In the leapfrog solution of the momentum equation the eddy coefficients are centered in time, i.e. the eddy viscosity and diffusivity are those calculated for time n . Hence the solution for q^2 and $q^2 l$ at time $n+1$ is not needed for the calculation of the velocity solution during the initial leapfrog step. The solution for q^2 and $q^2 l$ is obtained after the solution of the momentum and continuity equations with the leapfrog step, and no further iterations are performed. Once q^2 and $q^2 l$ at $n+1$ have been calculated at the end of the leapfrog step, the eddy viscosity and diffusivity are calculated at $n+1$, and then they are averaged with the coefficients already calculated for the previous time n to obtain the eddy coefficients at time $n+1/2$ that are used in the trapezoidal iterations.

5.4.4. Test cases for the vertical mixing algorithm

The turbulence model and its numerical implementation were tested against laboratory scale observations of stratified turbulent flows published in the literature. The first test case consists of the simulation of wind-induced deepening of the turbulent surface mixed layer on an initially quiescent uniformly stratified medium. The solution with the Kantha and Clayson model (KC) is compared with that obtained with other models and with deepening laws published in the literature (Kato and Phillips 1969, Pollard *et al.* 1973, Kundu 1980, Sherman *et al.* 1978) and with the simulations from a 1-D vertical model, coded for this purpose, that uses the vertical mixing algorithm already incorporated in SI3D. The same 1-D vertical numerical algorithm is used in section 5.4.4.2 to test the representation with the turbulence model of the deepening of the surface-mixed layer driven by convective cooling.

5.4.4.1. Deepening on a uniformly stratified medium

Since its publication in 1967, the experiment of Kato and Phillips (Kato and Phillips 1969) has become a benchmark problem for the test of turbulence models (e.g. **General Ocean Turbulence Model** (GOTM) in <http://www.gotm.net>). It consists of a constant stress applied to the surface of an initially quiescent tank of fluid with a uniform density gradient. Kato and Phillips postulated that the depth of the mixed-layer (h_m) would evolve according to the expression

$$h_m = u_* \left(\frac{15t}{N_0^2} \right)^{1/3} \propto t^{1/3} \quad (5.70)$$

where N_0 is the initially constant buoyancy frequency. The 1/3-power law time dependence ($h_m \propto t^{1/3}$), however, was later subject to discussion in the literature. Pollard et al. (Pollard *et al.* 1973) conducted one of the most fundamental theoretical works on the dynamics of the oceanic mixed-surface layer; it showed that the depth of the surface mixed layer should follow a 1/2-power law dependence on time ($h_m \propto t^{1/2}$). They show that their theory (PRT) provides a fairly good representation of the laboratory data from Kato and Phillips, attributing the slight discrepancy to the differences in the magnitude of the dissipation from sidewalls between the laboratory and ocean scales. The depth of the mixed layer in an initially quiescent fluid with uniform density gradient subject to a sudden wind stress in a non-rotating framework will in the PRT model obey the expression

$$h_m = (2Ri_0)^{1/4} u_* \left(\frac{t}{N_0} \right)^{1/2} \quad (5.71)$$

The criterion that in their model specifies the deepening rate is

$$Ri_o = \frac{gh\Delta\rho}{\rho(\Delta U^2 + \Delta V^2)} = \text{constant } (=1) \text{ during deepening} \quad (5.72)$$

In Eq. 5.72 (ΔU , ΔV) and $\Delta\rho$ are the discontinuities in the velocity and density at the bottom of the surface mixed layer. As noted by Niiler (Niiler 1975), the PRT model is equivalent to equating the potential energy increase to the shear production by the entrainment stress at the mixed layer base. Subsequent laboratory and field data suggest that the PRT model gives answers of the right order if Ri_o is set to a smaller value in the range from 0.35-0.8 (Price 1979). This result was later confirmed and tested (Kundu 1980) using an algebraic stress model for the vertical mixing.

In another fundamental work, Kraus and Turner (Kraus and Turner 1967) presented a model where the rate of production of turbulent kinetic energy by wind waves was assumed to be equal to the rate of increase of potential energy due to deepening. It has been shown later (Niiler 1975) that the PRT and KT models are two asymptotic limits of the general balance of the turbulent kinetic energy equation. In fact they represent two different mechanisms of deepening of the surface mixed-layer (Spigel and Imberger 1980): surface stirring (KT model) and shear entrainment (PRT model). In a 1978 article, reviewing mixing in stratified fluids, Sherman and others (Sherman *et al.* 1978) published a complete energy balance for entrainment in mixed-layers that incorporates both

deepening mechanisms. This mixing model was later incorporated in the hydrodynamic component of DYRESM, one of the most successful 1-D vertical models for water quality in lakes. The energy balance in the Sherman et al. model (from here on denoted as SIC) can be written as

$$\frac{1}{2} \left(C_T q^2 + \frac{\Delta \rho}{\rho} gh \right) \Delta h = \frac{C_K}{2} q^3 \Delta t + C_S (\Delta U)^2 \Delta h - \Lambda_L \Delta t \quad (5.73)$$

(1) (2) (3) (4) (5)

Here C_T , C_K , C_S are constant parameters, h is the depth of the mixed-layer. A turbulent intensity velocity scale q is defined that combines the effects of wind stirring and surface cooling

$$q^3 = u_f^3 + \eta^3 u_*^3 \quad (5.74)$$

where

$$u_f^3 = -\frac{\alpha gh H_N}{c_p \rho} \quad (5.75)$$

H_N is the heat loss at the surface during cooling periods, α is the coefficient of thermal expansion and c_p is the specific heat of water at constant pressure. The left side in Eq. 5.73 represents the energy required to mix a layer of thickness Δh in the time Δt . The right side gives the energy available for mixing. Term 1 accounts for the rate of change of TKE within the water column. Term 2 is the buoyancy term and represents the rate of change of potential energy in the water column. Terms 3 and 4 represent sources of TKE. Term 4 is the shear production and term 3 represents the surface stirring mechanisms of

KT. The efficiency of the stirring process is represented by the constant η in expression 5.74. The last term on the right side, term 5, accounts for losses of TKE from the mixed layer via generation of internal waves in the hypolimnion. This term was not deemed of primary importance in Spigel and Imberger 1980 for the description of mixed layers in lakes and reservoirs, in comparison with terms 3 and 4. If one neglects that term, Eq. 5.73 can be rewritten as

$$\frac{dh}{dt} = \frac{C_K \eta^3 u_*^3}{C_T \eta^2 + g' h / u_*^2 - C_S \Delta U^2 / u_*^2} \quad (5.76)$$

where g' is the modified acceleration of gravity ($= g\Delta\rho/\rho$). Also $\Delta\rho$ and ΔU represent the jump in density and velocity at the base of the mixed layer, which must be specified. The SIC model is completed with a 1-D momentum and scalar transport equation for such purpose. In the particular case of a water-column subject to wind stress and no heat exchange with the atmosphere, the discretized equations for ΔT and ΔU at a given moment t_0 in time are

$$\Delta U(t = t_0) = \frac{u_*^2 t_0}{h(t = t_0)} \quad (5.77)$$

$$\Delta T = T_{ml}(t = t_0) - T(z = h(t_0), t = 0); \quad T_{ml}(t = t_0) = \int_{z=-h(t_0)}^{z=0} T(z, t = 0) dz \quad (5.78)$$

where T_{ml} represents the uniform temperature at the mixed layer. The numerical implementation of the SIC model employs a simple fully explicit scheme.

The values of the constant parameters in the SIC model are selected to match a wide range of experimental results. They are chosen to be the following (Spigel 1978):

- C_K measures the stirring efficiency of convective overturn ($C_K = 0.125$)
- $C_K \eta$ is a coefficient measuring stirring efficiency of the wind ($\eta = 1.23$)
- C_S represents the efficiency of shear production for entrainment ($C_S \sim 0.2-0.5$).
Spigel (Spigel 1978) suggests 0.5 as a good estimate from most experimental results.
- C_T is associated with temporal, or unsteady, non-equilibrium effects due to changes in surface wind stress or surface cooling ($C_T \sim 0.5$)

A comparison was made of the simulations from a 1-D vertical model that uses the same vertical mixing algorithm already incorporated in SI3D, against the deepening law given by Eq. 5.71 and the results from the SIC model. The 1-D model solves the layer-averaged form of the momentum equation

$$\frac{\partial U_k}{\partial t} = K_v \left. \frac{\partial U}{\partial z} \right|_{k-1/2} - K_v \left. \frac{\partial U}{\partial z} \right|_{k+1/2} \quad (5.79)$$

subject to the boundary condition

$$K_v \left. \frac{\partial U}{\partial z} \right|_{k_1-1/2} = \frac{\tau_{sx}}{\rho} \quad (5.80)$$

Here U represents the layer-averaged transport, defined at the center of each cell, and τ_{sx} is the wind stress applied at the surface. The model additionally solves the 1-D heat equation for the temperature

$$\frac{\partial(h_k T_k)}{\partial t} = D_v \left. \frac{\partial T}{\partial z} \right|_{k-1/2} - D_v \left. \frac{\partial T}{\partial z} \right|_{k+1/2} + \frac{I}{\rho} \left. \right|_{k-1/2} - \frac{I}{\rho} \left. \right|_{k+1/2} \quad (5.81)$$

subject to the heat flux boundary condition Eq. 5.24. Here h_k is the thickness of layer k , T_k is its temperature, and I is solar irradiance. For the problems used to test the turbulence closure, the solar irradiance and surface heat flux were set to zero. The values of the eddy viscosity and diffusivity (K_v and D_v) are calculated at the interfaces between consecutive cells using the model of Kantha and Clayson, KC (Kantha and Clayson 1994). The same numerical scheme used in SI3D was implemented in the 1-D model.

The depth of the mixed layer in the simulations with the 1-D model was calculated as the depth of the interface between consecutive cells that shows the maximum density difference. Owing to the definition of the depth of the mixed-layer the simulated results will have jumps, the jumps being of the same magnitude as the spatial vertical discretization (Δz). For the tests a 50 m deep water column was used. A wind of 5 m/s was imposed in the x direction on the surface, and the evolution of the system was tracked for 48 hours. The water column was discretized using $\Delta z = 1$ m, and the time step was set to $\Delta t = 30$ seconds. The surface temperature was set to 25 °C and the bottom temperature to 5 °C. For the comparison with the PRT model, the initial temperature distribution was such that a linear density profile was obtained when using the non-linear

equation of state from SI3D. A linear temperature profile was used, however, to compare the KC turbulence closure model against the SIC model, so as to avoid numerical integration errors from the calculation of the temperature of the mixed layer in Eq. 5.78.

The evolution of the wind mixed-layer depth simulated with the KC model is compared in Fig. 5.15 against the estimated mixed-layer depth given by the PRT model (Eq. 5.71). In Figure 5.16 the results from the KC model are compared with those of the SIC model using two different values of C_S that define the lower ($C_S = 0.2$) and upper limits ($C_S = 0.5$) of the range of values suggested in the literature for this constant parameter (Imberger 1985). Figures 5.15 and 5.16 show that the KC model compares favorably with

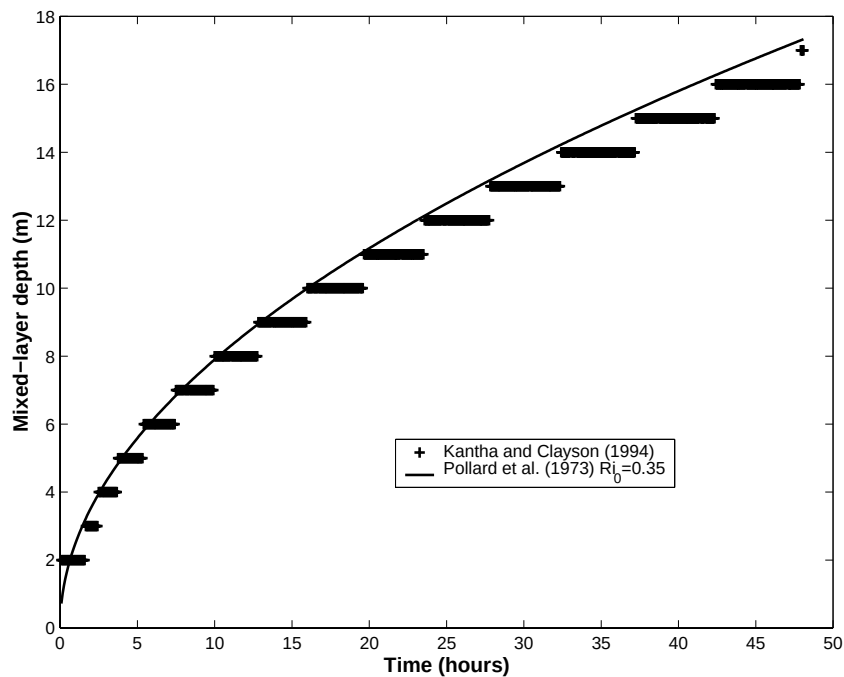


Figure 5.15. Evolution of the depth of the mixed layer in an initially quiescent tank of fluid with a uniform density gradient subject to a constant stress, predicted by the model of Kantha and Clayson (1994, ref. Kantha and Clayson 1994) and by the deepening law given by Eq. 5.71.

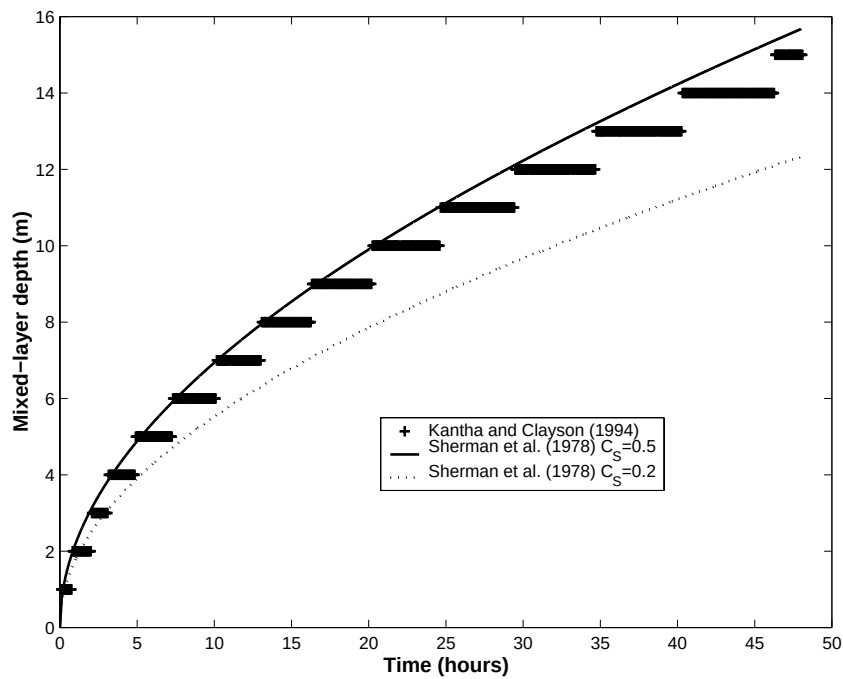


Figure 5.16. Evolution of the depth of the mixed layer in an initially quiescent tank of fluid with a uniform density gradient subject to a constant stress, predicted by the model of Kantha and Clayson (1994, ref. Kantha and Clayson 1994) and Sherman et al. (1978, ref. Sherman et al. 1978).

with previously published results and mixed-layer models and supports the suitability of the turbulence closure to represent wind-driven mixed layers in 3-D models.

5.4.4.2. Convective cooling

During cooling periods the vertical mixing in a water column is primarily driven by convective turbulence. The thermal exchange at the surface is dominated by heat losses that generate density profiles characterized by the presence of colder and heavier water at the surface, overlaying a mass of warmer and lighter water. This density distribution is unstable and will drive convective motions which are able to mix the upper layers.

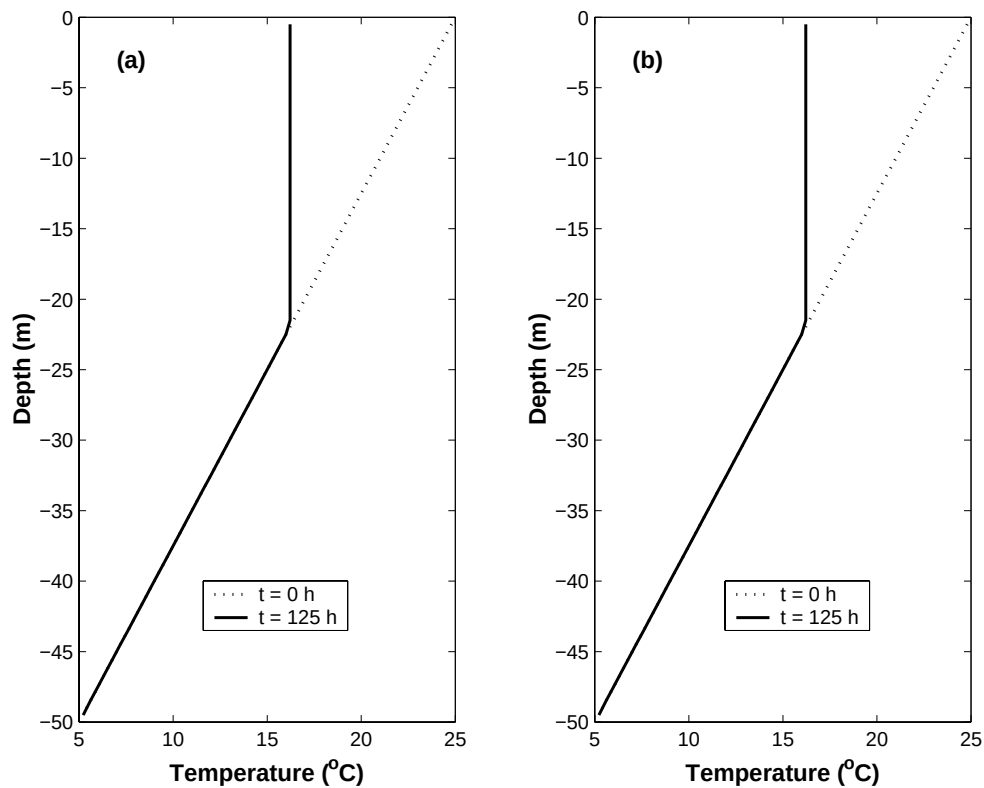


Figure 5.17. Initial and final temperature profiles obtained with (a) a convective adjustment process and (b) with the present implementation of the Kantha and Clayson (Kantha and Clayson 1994), applied to the simulation of mixing driven by convective cooling at the free surface.

The performance of the turbulence closure in SI3D-L is examined here when it is applied to the simulation of mixing processes in a water column during cooling periods. The 50 meter deep water column in the test problem is initially quiescent and has a uniform temperature gradient, with 25°C at the surface and 5°C at the bottom. It is then suddenly subject to heat losses at the surface of $H_N = -900 \text{ W/m}^2$. The evolution of the temperature profile is then simulated using $\Delta t = 300$ seconds for 125 hours. The vertical discretization is $\Delta z = 1\text{m}$. The reference profiles, used to check that the turbulence closure model is correctly designed and implemented, were obtained by using the convective adjustment

process of Bryan and Cox (Bryan and Cox 1967) described in section 5.4.1. The same 1-D momentum and scalar solver was used for the reference runs. Figure 5.17 shows on the left the temperature profiles at $t = 0$ and at $t = 125$ hours, simulated with the convective adjustment process. On the right, is shown the profiles at those same times, but obtained with the turbulence closure scheme of Kantha and Clayson (Kantha and Clayson 1994). The differences between profiles (a) and (b) in Fig. 5.17 are not apparent, which supports the correctness of the implementation of the turbulence closure scheme.

CHAPTER 6

PARTICLE TRACKING

6.1. Introduction and literature review

The use of three-dimensional hydrodynamic models in engineering applications is usually motivated by the need to give accurate predictions of the transport and dispersion of pollutants in natural systems. It is widely accepted that the advection-diffusion equation (also known as the transport equation) provides a mathematical description of the transport of pollutants in water. Numerical solutions of the transport equation may be obtained in two ways: (1) by *Eulerian* methods, or (2) by the simulation of the continuum of dispersed ‘contaminant’ by a cloud of discrete particles, also known as the *Lagrangian* or *particle tracking* method (Hunter *et al.* 1993). *Eulerian* methods are probably the more popular solution techniques, possibly because of their ease of integration into solutions of the hydrodynamic equations, which are, almost exclusively, solved using similar methods (Hunter 1987). However, they have some drawbacks. In particular,

- They are often not mass conserving (Heemink 1990);
- To avoid negative concentrations and oscillatory solutions near high concentration gradients, filter techniques must be used which will introduce numerical diffusion, e.g. Asselin filter (Gross *et al.* 1999), and hence will involve a loss of accuracy (Dunsbergen and Stelling 1993). Those problems are mainly associated with the representation of the advection terms³⁷.

³⁷ The treatment of diffusion and decay processes in Eulerian methods is quite satisfactory, even using very simple numerical schemes (Hunter 1987).

In the particle models, also referred to as *random walk*, *random flights* or *lagrangian stochastic* models in the literature, advection is simulated with a high degree of accuracy. The difficulties associated with the representation of the advection of high concentration gradients and the mass conservation problems are overcome. The representation of dispersion processes is done through stochastic methods, which requires the use of a large number of particles to obtain accurate results³⁸. The different formulations of random-walk models that have been proposed in the literature can be classified, according to (Naslund *et al.* 1994, Wilson and Sawford 1996), into two broad categories

1. Langevin Equation Models (LEM) or first-order random flights.
2. Random Displacement Models (RDM).

The Langevin Equation models (LEM) are described by the following set of stochastic differential equations (e.g. Thomson 1987):

$$du_i = a_i(\mathbf{x}, \mathbf{u}, t)dt + b_{ij}(\mathbf{x}, t) dW_j \quad (6.1)$$

$$dx_i = u_i dt \quad (6.2)$$

where $\mathbf{x}$ and $\mathbf{u}$ are the particle position (x_1, x_2, x_3) and velocity (u_1, u_2, u_3); a is a function of $\mathbf{x}$ and $\mathbf{u}$, and time t ; b is a function of $\mathbf{x}$ and t ; and dW_j are the increments of a vector-valued Wiener process with independent components (Gardiner 1990). These equations describe a Markov process. The formulation of a_i and b_{ij} will determine whether the model satisfies the ‘*well-mixed*’ criterion (Thomson 1987) and hence whether the model is consistent with the physics. This criterion states that if the particles are well mixed at a

³⁸ The accuracy increases as the square root of the number of computations (Hunter 1987)

given moment in time they should stay well mixed. Thomson (Thomson 1987) proposed a 3-D model that satisfies that criterion, where the formulas for a_i and b_{ij} are

$$a_i = - \left(\frac{C_0 \varepsilon}{2} \right) \delta_{ij} \Gamma_{jk} (u_k - \bar{u}_k) + \frac{\Phi_i}{g_a} \quad (6.3)$$

$$b_{ij} = - (C_0 \varepsilon)^{1/2} \delta_{ij} \quad (6.4)$$

where C_0 ($= 5.7$) is a universal constant, ε is the rate of dissipation of turbulent kinetic energy, δ_{ij} is the Kronecker delta, Γ_{jk} is the tensor inverse of the Reynolds stress tensor τ_{jk} , u_k is the k -component of the mean Eulerian velocity, and Φ_i/g_a obeys an expression that in expanded form contains 63 terms in each coordinate axis. The LEM have been recently applied to 3-D atmospheric pollutant dispersion problems (e.g. Naslund *et al.* 1994, Lee and Naslund 1998), and their 1-D counterparts have received considerable attention as well in the atmospheric literature (e.g. Legg and Raupach 1982, Sawford 1985, Hsieh *et al.* 1997).

In the Random Displacement Models (RDM) the ‘state’ of a particle is described with a Markov process (Gardiner 1990), as in the LEM models (Eqs. 6.1 and 6.2), but in this case the state of the particle is described only by its position, in contrast with the LEM models where both the position and the velocity describe the ‘state’ of a given particle. In general, the RDM are only valid when the particle velocity is no longer correlated with its initial value, i.e. for times much larger than the Lagrangian time scale ($t \gg T_L$), and RDM will overestimate the diffusion at short times (Naslund *et al.* 1994). The equation describing the particle position is

$$dx_i = a_i(\mathbf{x}, t) \cdot dt + \sum_{j=1}^3 b_{ij}(\mathbf{x}, t) \cdot dW_j(t) \quad (6.5)$$

(1)
(2)

where $W_j(t)$ is a Wiener or Brownian Motion process. In order to simulate a particle trajectory, determined by the above equation, that is consistent with the convection diffusion equation a_i and b_{ij} must satisfy the following equations (Dimou and Adams 1993, Dunsbergen and Stelling 1993, Tompson and Gelhar 1990, Hunter *et al.* 1993, Kitanidis 1994),

$$a_i = u_i + \sum_{j=1}^3 \frac{\partial D_{ij}}{\partial x_j} \quad (6.6.a)$$

$$\frac{1}{2} \sum_{k=1}^3 b_{ik} b_{jk} = D_{ij} \quad (6.6.b)$$

where the repeated index summation convention (Einstein notation) is not applied to the sub-indexes. The expressions in Eq. 6.6.b have the form in vector notation

$$\mathbf{A} \equiv \mathbf{u} + \nabla \cdot \mathbf{D} \quad ; \quad \mathbf{B} \cdot \mathbf{B}^T \equiv 2\mathbf{D} \quad (6.7)$$

in which $\mathbf{D}$ is the diffusion tensor (a 3x3 matrix in 3-D simulations). Durbin (Durbin 1984) showed that a 1-D version of Eqs. 6.1-6.3 with no mean velocity can be transformed to

$$dx_i = \frac{\partial}{\partial x_i} (\tau_{ii} T_L) dt + (2\tau_{ii} T_L)^{1/2} dW_i \quad (6.8)$$

where T_L is the Lagrangian time scale (Csanady 1980) and τ_{ii} represents the i -th diagonal element of the Reynolds stress tensor (no summation implied by repeated indexes). Equation 6.8 is consistent with Eq. 6.5 with an eddy diffusivity defined as

$$K_i = T_L \tau_{ii} \quad (6.9)$$

Given their simplicity and the direct relationship with the advection-diffusion equation, the RDM have been given considerable attention in the literature. Many of the applications are in groundwater hydrology (e.g. Kinzelbach 1988, Uffink 1988, Thompson and Gelhar 1990, Scheibe and Cole 1994, LaBolle *et al.* 1996), though the methodology has been also used in the field of water resources (Hunter 1987, Van den Boogaard *et al.* 1993, Dimou and Adams 1993, Heemink 1990, Riddle 1998). The modest amount of input (essentially the velocity field and the eddy diffusivity) required by the RDM allows the use of particle-tracking modules as a post-processing tool, and it is the main motivation for the current choice of the RDM as the method to apply in the present particle-tracking routine. The routine is called *ptrack*, and it will be described in the next few sections. The particles in the present version of *ptrack* are assumed to be neutrally buoyant. When those particles encounter the boundaries of the physical domain they are reflected and relocated in the water body, so that no mass is lost during the particle-tracking simulations.

6.2. Methodology

The equation describing the particle position is Eq. 6.5 where, for consistency with the advection diffusion equation a_i and b_{ij} must satisfy Eq. 6.6 or equivalently Eq. 6.7 (Dimou and Adams 1993, Dunsbergen and Stelling 1993, Thompson and Gelhar 1990,

Hunter *et al.* 1993, Kitanidis 1994). If one assumes isotropic horizontal diffusion, as it is in the hydrodynamic and transport model, and takes the vertical direction as one of the principal directions of the diffusion tensor, the matrix $\mathbf{D}$ in Eq. 6.7 can be written as

$$\mathbf{D} = \begin{pmatrix} D_H(\mathbf{x}, t) & 0 & 0 \\ 0 & D_H(\mathbf{x}, t) & 0 \\ 0 & 0 & D_V(\mathbf{x}, t) \end{pmatrix} \quad (6.10)$$

and $\mathbf{B}$ will similarly be diagonal with elements of the form

$$b_{ij} = \sqrt{2D_{ij}} \delta_{ij} \quad (6.11)$$

In the particle-tracking method the particle position is given by the summation of successive particle displacements over time. Each of those displacements is described by the stochastic differential Eq. 6.5 and consists of a **deterministic part** (term 1 on the right hand) and a **stochastic part** (term 2). The discretized version of Eq. 6.5 that represents the transition from the state (= particle position) at $n\Delta t$ to the state at $(n+1)\Delta t$ is written as

$$\mathbf{x}_i^{(n+1)} = \mathbf{x}_i^{(n)} + \mathbf{a}_i(\mathbf{x}^{(n)}, n\Delta t) \cdot \Delta t + \sum_{j=1}^3 b_{ij}(\mathbf{x}^{(n)}, n\Delta t) \cdot \Delta W_j(n\Delta t) \quad (6.12)$$

where the bold variables are vector quantities. This equation is consistent with the use of the Ito integration rule applied to Eq. 6.5 (Gardiner 1990, Kitanidis 1994). The factor $\Delta W_j(t)$ is simulated by a Gaussian number generator, with

$$E(\Delta W_j) = 0 \quad (6.13)$$

$$E(\Delta W_i \Delta W_j) = \delta_{ij} \Delta t \quad (6.14)$$

In vector notation the transition equation has the form

$$\mathbf{x}^{n+1} = \mathbf{x}^n + \mathbf{A}(\mathbf{x}^n, n\Delta t) \cdot \Delta t + \mathbf{B}(\mathbf{x}^n) \cdot \mathbf{z} \sqrt{\Delta t}$$

in which $\mathbf{A}$ is a deterministic forcing vector, $\mathbf{B}$ is a deterministic scaling matrix, and $\mathbf{z}$ is a vector of three independent random numbers with mean zero and variance one.

6.3. Deterministic displacement (drift)

The deterministic displacement of the particle in Cartesian coordinates contains the influence of the local velocity and the gradient of the turbulent diffusivity (Eq. 6.6.a). The latter has received particular attention in the literature as, when neglected (in the initial applications of the technique), it leads to a physically unrealistic gathering of mass in low-dispersion areas (Kinzelbach 1988, Uffink 1988). Its implementation is quite straightforward and does not pose any problems. Since the horizontal mixing processes are not totally understood in shallow water flow problems, an isotropic horizontal diffusion D_H is assumed, and hence the drift term associated with the spatial variation of the turbulent scalar transfer coefficient is zero in the horizontal directions. On the other hand, the vertical eddy diffusivities D_V will vary with depth and are calculated in the hydrodynamic model at the center of every vertical-flux face³⁹ of the computational domain. In the particle-tracking module an approximation is made by taking the vertical diffusivity to be constant at each of the vertical-flux faces and applying the usual first-order finite difference scheme to calculate a constant gradient of D_V within each cell.

³⁹ Vertical-flux faces refer here to the horizontal faces crossed by vertical fluxes. In those faces, it is where the 3-D model defines the vertical velocity components (see Fig. 4.2).

The advection algorithm requires special consideration. The position of the particles ($\mathbf{X}_p$) will not coincide, in general, with the discrete set of nodes where the velocities are defined in the 3-D hydrodynamic model. Hence, one must interpolate among the discrete set of velocity vectors defined at the nodes of the computational grid to obtain a 3-D continuous velocity field. Various methods can be used to obtain such a continuous field. Frequently the 3-D velocity field is continuously extended by tri-linear interpolation. This may not be a bad approach for visualization purposes. However, such an interpolation procedure does not guarantee that the continuity equation is satisfied in each point of the continuous space, which implies an inconsistency with the physics. An interpolation scheme that is consistent with continuity is adopted here and follows the algorithm originally proposed by Pollock (Pollock 1988) for groundwater applications.

The “mass conserving” interpolation scheme is briefly detailed below. Without loss of generality, one can assume that a grid cell is defined as a set of points $\mathbf{x} = (x, y, z)$ in $\mathbb{R}^3$ space such that $\{0 \leq x \leq \Delta x, 0 \leq y \leq \Delta y, 0 \leq z \leq \Delta z\}$. The continuous extension of the 3-D discrete flow field is written as

$$u_i^{ext}(\mathbf{x}, t) = \left(\frac{\partial u_i}{\partial x_i} \right)_{num} x_i + \left(u_i \Big|_{x_i=0} \right)_{num} = A_i \cdot x_i + B_i \quad (6.15)$$

The gradient term A_i corresponds to the numerical implementation of $\partial u_i / \partial x_i$ (no index summation convention) as it appears in the continuity equation of the hydrodynamic model. By using Eq. 6.15, the divergence of the continuous extension of our 3-D discrete velocity field obeys the equation

$$\sum_{i=1}^3 \frac{\partial u_i^{ext}}{\partial x_i} = \sum_{i=1}^3 \left(\frac{\partial u_i}{\partial x_i} \right)_{num} = 0 \quad (6.16)$$

Hence, a mass conserving hydrodynamic model, which gives the latter equality in Eq. 6.16, now implies a mass conserving advection step in the particle model. This method of interpolation guarantees that the particle will not be able to cross a cell boundary where the velocity component normal to this boundary is zero. For the particular case of the surface layer cells in the 3-D hydrodynamic model, where the upper boundary cell has a vertical velocity associated with the change in the water surface elevation, special care must be taken.

Within a cell ('host cell') the streamline through the present position of the particle ($\mathbf{x} = \mathbf{X}_p$) can be obtained analytically, by solving the linear ordinary differential equation

$$\frac{\partial x_i(t)}{\partial t} = u_i^{ext}(\mathbf{x}, t); \quad x_i(t_0) = (\mathbf{X}_p)_i \quad (6.17)$$

The analytical solution to Eq. 6.17 in 1-D problem reads

$$\begin{aligned} X(t) &= (X_p + B/A) \exp(A \Delta t) - B/A & \text{If } A \neq 0 \\ X(t) &= (X_p + B \Delta t) & \text{otherwise} \end{aligned} \quad (6.18)$$

The solution in Eq. 6.18 only applies within a computational cell. It can be used to determine the particle's exit point and the time to reach that point on the cell boundary, called the time to crossing, T_p .

Table 6.1 summarizes the algorithm implemented in the particle-tracking routine to determine T_p and the exit point in any given direction (adapted from Dunsbergen and Stelling 1993). In Dunsbergen and Stelling 1993 no mention is made of the case where $A (= \partial u / \partial x) < 0$ and the velocities in both flux faces

Table 6.1. Algorithm for the advection step in the Lagrangian particle tracking model.

CASES		TIME $T_p > 0$ TO REACH CELL BOUNDARY	POSITION OF PARTICLE AFTER T_p
$U_p = 0$		∞	X_p
$A > 0$	$X_p + C > 0$	$\frac{1}{A} \ln \frac{X_E + C}{X_p + C}$	X_E
	$X_p + C < 0$	$\frac{1}{A} \ln \frac{X_W + C}{X_p + C}$	X_W
$A < 0$	$X_W + C > 0$ & $X_E + C > 0$	$\frac{1}{A} \ln \frac{X_W + C}{X_p + C}$	X_W
	$X_E + C < 0$ & $X_W + C < 0$	$\frac{1}{A} \ln \frac{X_E + C}{X_p + C}$	X_E
	$X_E + C \leq 0$ & $X_W + C \geq 0$	∞	—
$A = 0$	$B > 0$	$\frac{X_E - X_p}{B}$	X_E
	$B < 0$	$\frac{X_W - X_p}{B}$	X_W

U_E = velocity at east boundary

U_W = velocity at west boundary

Δx = grid cell size

X_p = particle location

X_W = location of west boundary (= 0.0)

X_E = location of east boundary (= Δx)

$U_p = A X_p + B$ (Particle velocity)

$A = (U_E - U_W) / \Delta x$

$B = U_E$

$C = B/A$

have different directions (i.e. the flow converges into the computational cell). This case turns out to be particularly important in representing what happens at the boundaries of the physical domain. When the time step selected for the advection (Δt) is greater than T_p , the advection algorithm is split into sub-steps. In each sub-steps Eq. 6.17 is solved in different host cells. In multi-dimensional problems one can use the algorithm in Table 6.1 in each of the three Cartesian directions. The time to crossing T_p is taken as the minimum of the T_p 's calculated for each direction, and the exit face will be given by the algorithm in Table 1 applied to the direction with minimum T_p . The new host cell will be the one sharing the exit face with the present host cell.

6.4. Stochastic or random displacement

The stochastic contribution of the particle displacement in a free surface flow, with an isotropic horizontal diffusivity and the vertical direction as one of the principal directions of the dispersion or diffusion tensor, is given by

$$dx_i = \begin{cases} \sqrt{2 \cdot D_H} \cdot \Delta W_i & i = 1, 2 \\ \sqrt{2 \cdot D_V} \cdot \Delta W_i & i = 3 \end{cases} \quad (6.19)$$

A first-order approximation is adopted in the particle-tracking model by taking the vertical diffusivity constant at each of the vertical-flux faces. The vertical diffusion coefficient is approximated at the particle position with a straightforward linear interpolation technique from the values of D_V at the vertical faces of the cell that hosts the particle at the beginning of the time step.

The random Wiener increment can be realized by using a normal random number generator with mean zero and variance Δt . However, the computational cost can be significantly reduced by approximating $\sqrt{2 \cdot D \cdot \Delta t} \cdot W$ by $\sqrt{6 \cdot D \cdot \Delta t} \cdot R$, where R represents a random number from a uniform distribution in the interval $[-1, 1]$ (Dunsbergen and Stelling 1993, Hathhorn 1997).

6.5. Reflection algorithm

One of the properties of the particle tracking approach to the solution of the transport equation is that of being mass conservative. However, if one wants to guarantee that the method strictly conserves mass, then special attention must be given to the behavior of particles close to the boundaries of the computational domain. Particles should not be allowed to leave the computational domain through the free surface or through solid boundaries (bottom or lateral). At open boundaries, however, particles should be allowed to leave the computational domain. In this case they should then be marked as inactive particles. Finally, in the course of the computations, particles might attempt to leave the computational domain through the free surface or through a solid boundary. In those cases, a reflection algorithm is called.

The particles in the particle-tracking model are assumed to follow rectilinear trajectories within one time step, i.e. from the initial point to the final location in one time step the trajectory described by the particle is a line. This line will be a straight line if no reflection occurs, or a broken line if reflection occurs at the boundaries. The velocity is assumed to be uniform and equal to $(\mathbf{X}_{\mathbf{p_final}} - \mathbf{X}_{\mathbf{p_initial}})/\Delta t$. If the particle leaves the computational domain it is because it crossed at least a cell interface (that defines the

limits of the wet computational domain). The reflection algorithm is then applied; hence, there is a need to track the particle location from its initial position to each of the interfaces that the particle might cross in a given time step. Given the initial *host* cell and the velocity components, the exit face is identified as indicated in Table 6.1. Once the particle has arrived at a face, a check is made. If the next *host* cell is dry, then the velocity of the particle perpendicular to that face has its sign changed, and the algorithm continues tracking the particle with velocities changed. Otherwise the algorithm continues with the velocity unchanged. For example, if the particle reaches a vertical-flux face (either the bottom or the free surface) and the next cell to visit is dry, the algorithm at the next sub-step would take the location of the particle at the face as the initial position and a velocity equal to $(u_1, u_2, -u_3)$.

The algorithm was tested in complex geometries and employed particles that were seeded very close to the free surface or the bottom and gave fairly satisfactory results. The number of occurrences, however, of a particle leaving the domain was almost negligible.

CHAPTER 7

STUDYING THE HYDRODYNAMICS OF CLEAR LAKE

7.1. Motivation

Clear Lake, in the Central Coastal Range of Northern California, is the largest natural freshwater body lying entirely within the borders of California, though probably it is better known for being the site of one of the largest mercury mines in U.S history, the Sulphur Bank Mercury Mine (Fig. 7.1). The mine site, on the eastern shore of the lake, is presently abandoned, but a huge open pit filled with highly acidic water, the Herman Pit, and large quantities of mine tailings along the shoreline remain as evidence of its former size and productivity. From 1873 to 1957 the mining operations removed between 4400 and 7000 metric tons of mercury (Hg) from the site and resulted in about 100 metric tons of Hg being deposited into the aquatic ecosystem of Clear Lake (Chamberlin *et al.* 1990). The lake, as a result, is now one of the most Hg-contaminated systems in the world (Suchanek *et al.* 1998). Routine analyses for total Hg in Clear Lake, carried since the detection of elevated levels in the early 70s, reveal that the Hg levels in fish often exceed U.S. Food and Drug Administration and National Academy of Sciences guidelines for human consumption. This prompted in 1986 and subsequent years the issuing of public health advisories by the California Department of Health Services that recommended restrictions on consumption of Clear Lake fish. In 1991 the Mine, identified as the main source of mercury in the lake, was declared a U.S. EPA Superfund site.

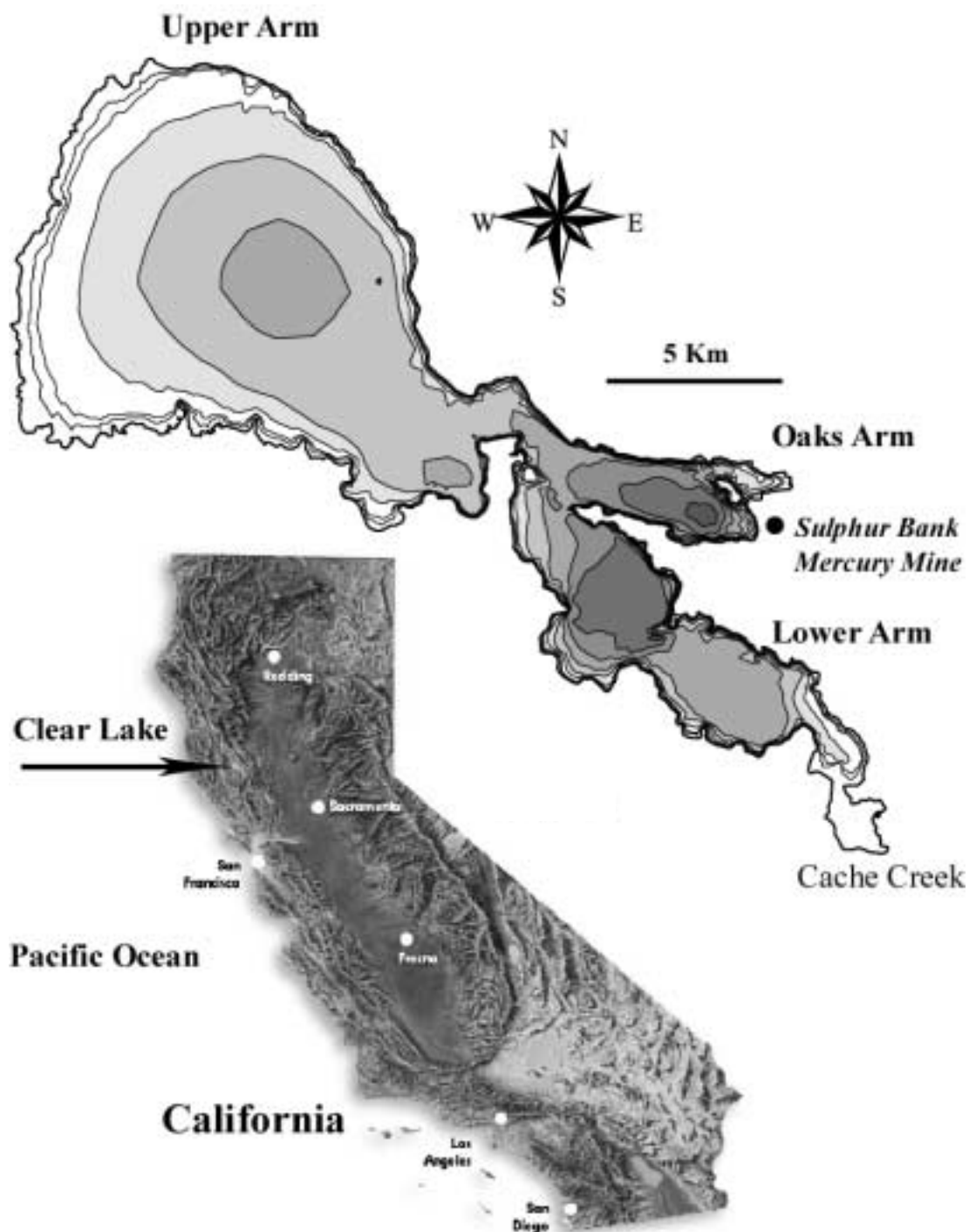


Figure 7.1. Geographic setting of Clear Lake, in the Coastal Range of northern California (bottom left), with the Pacific Ocean to the west and Sacramento Valley to the east. A bathymetric map of the lake (top right) with contours shown every 2 meters.

Two likely independent sources of mercury have been identified in the Clear Lake system (Suchanek *et al.* 1997). First, inorganic mercury⁴⁰ (mostly from the erosion of mercury-contaminated waste rock and tailings piles) has contaminated Clear Lake for many decades. Second, methyl mercury (a highly toxic and readily bioavailable species of mercury) appears to form effectively as a secondary product of bacterial metabolic activity on flocculent material formed from acid mine drainage⁴¹ into the lake and in sediments throughout the lake. Once formed, the floc becomes transformed over several months to a medium which is rich in methyl mercury, which in turn is prone to re-suspension and is easily transported by currents.

The first application of the 3-D model (SI3D-L) seeks to identify the driving mechanisms and circulation patterns in Clear Lake that may provide a better understanding of the distribution of mercury in this ecosystem. Particular attention is focused on one of the three sub-basins of Clear Lake (the Oaks arm) that houses the main source of mercury, the Sulphur Bank Mercury Mine, and addresses the key issues of exchange with the other arms and the Oaks Arm's internal circulation. The problem is addressed by using an integrated approach, where field observations and 3-D simulations with the hydrodynamic model described in chapters 4 and 5 are combined to reinforce the conclusions. Previous studies of the lake hydrodynamics have exclusively relied on either model simulations (Bale 1995) or field observations (e.g. Lynch 1996, Lynch and

⁴⁰ Bale's (Bale 1995) model, as well as other mercury speciation models, considers the following operational groups: reactive inorganic mercury (Hg^{++}), organic mercury whose prevalent form is methyl mercury (CH_3Hg^+), elemental and volatile mercury (Hg^0), and inert mercury.

⁴¹ The acid mine drainage was first discovered in 1995. Tracer release studies in the Herman Pit (Oton *et al.* 1998) indicate that mine water could be draining into the lake at rates in the range of 6000 to 8000 gallons per minute.

Schladow 1996, Lallatin 1975, Horne 1975), and their results were either inconclusive or fundamentally wrong in regards to the large-scale circulation in Clear Lake. Bale (Bale 1995), for example, carried out simulations of currents in Clear Lake, assuming that the hydrodynamic processes could be adequately described by using a 2-D depth-averaged model driven by a uniform wind field applied over the free surface of the lake. The assumptions and hence the results of Bale's work were later proven to be incorrect (e.g. Lynch 1996). Despite the large and shallow nature of Clear Lake, the water column undergoes periods of time of variable length when it is not well mixed. It is during those periods that stratification exerts a considerable influence on the lake hydrodynamics, modifying the circulation patterns and intensifying the transport rate within the system. These are precisely the effects that this study seeks to document.

Special attention will be given to one of the circulation mechanisms in the Oaks Arm of Clear Lake under stratified conditions which has been termed baroclinic wind pumping (hereafter it is referred to as baroclinic pumping). It arises from the interaction of a strong quasi-periodic forcing, the wind field, with the sporadic stratification in the temperature cycle of the lake, modulated by the Earth's rotation. This mechanism produces a counterclockwise residual circulation and differs from internal Kelvin waves (Hodges *et al.* 2000) in that it is a forced response, and thus its periodicity is dictated by the wind forcing. A similar process, referred to as *tidal pumping*, has been described in the literature for large estuarine systems (Fischer *et al.* 1979). In that case the mechanism is the result of the stratification (caused by salt concentration) interacting with the periodic forcing of the tides under the influence of the Earth's rotation. Baroclinic pumping is in

all likelihood not exclusive to Clear Lake but probably exists in other large and relatively shallow lakes forced by strong and almost periodic wind events where basin-scale internal waves do not play a major role in the dynamics of the system.

Two field data collection campaigns were conducted as part of the experimental work, and they are briefly described in section 7.3, after an overview of the physical characteristics of Clear Lake (section 7.2). The experiments were part of a collaborative effort with UC Berkeley, UC Santa Barbara, and Stanford University. The field data are analyzed in sections 7.4 and 7.5, and a conceptual model of circulation in the Oaks Arm is then presented, based on the experimental evidence. In the last sections are presented the results of the modeling exercise, and they provide a complete description of the processes leading to the baroclinic pumping effect, identifying the circulation patterns and the driving mechanisms in Clear Lake. The model setup for the simulation of Clear Lake hydrodynamic processes is described in section 7.6.1. Only the events during the second experiment were simulated, when all variables needed to run the model were carefully collected. The transport of particles released in the immediate vicinity of the mine site is studied by using the particle tracking routines developed in section 7.7. In section 7.8 a finite-difference Eulerian solution algorithm of the advection equation (the same as described in chapter 5 for the advection of temperature) is applied to simulate the distribution of a passive tracer released at two locations in the lake and to illustrate lake circulation patterns.

7.2. Physical characteristics of Clear Lake

Clear Lake is a large (176 km²) and highly eutrophic natural lake located at an elevation of 404 m above sea level in Northern California (39°00'N, 122°45'W). The system is warm, relatively shallow⁴² and polymictic. It occupies a natural corridor in the Central Coastal Range, a small mountain range that separates the dry Central Valley to the east from the marine climate to the west. The drainage area is generally mountainous with several large valleys, typically adjacent to the lake, with a total area tributary to the lake of about 1200 km², excluding the lake surface (Bale 1995).

The basin has been shaped by faults, volcanism and erosion over the last 500,000 years and possibly longer yielding a fairly complex geometry. It is naturally divided into three distinct sub-basins or arms: the Upper Arm, the Lower Arm and the Oaks Arm (Fig. 7.1). The large shallow basin of the Upper Arm comprises almost 72% of the lake's surface and receives more than 90% of the watershed runoff. At the east end of the Upper Arm a narrow passage (the Narrows) connects it with the two other arms of the lake, both of which are considerably narrower and deeper than the Upper Arm. The Lower Arm is a conduit to the lake's only outlet, Cache Creek, at the extreme southeast end of the lake. With an average annual inflow estimated to be 514×10^6 m³/year (Lynch 1996), the corresponding lake residence time is two to three years, and advective flow through the lake is not considered to be a major factor in water movement. The far end of the Oaks Arm is the site of the Sulphur Bank Mercury Mine, and its circulation has received particular attention in this work as being the key to understanding the distribution of

⁴² The mean depth is about 8 meters.

mercury from its source to the other two basins. The photograph in Fig. 7.2 gives an aerial view of the Oaks Arm taken from the Narrows, showing the location of the mine at its easternmost end. A summary of the lake physical parameters is provided in Table 7.1.

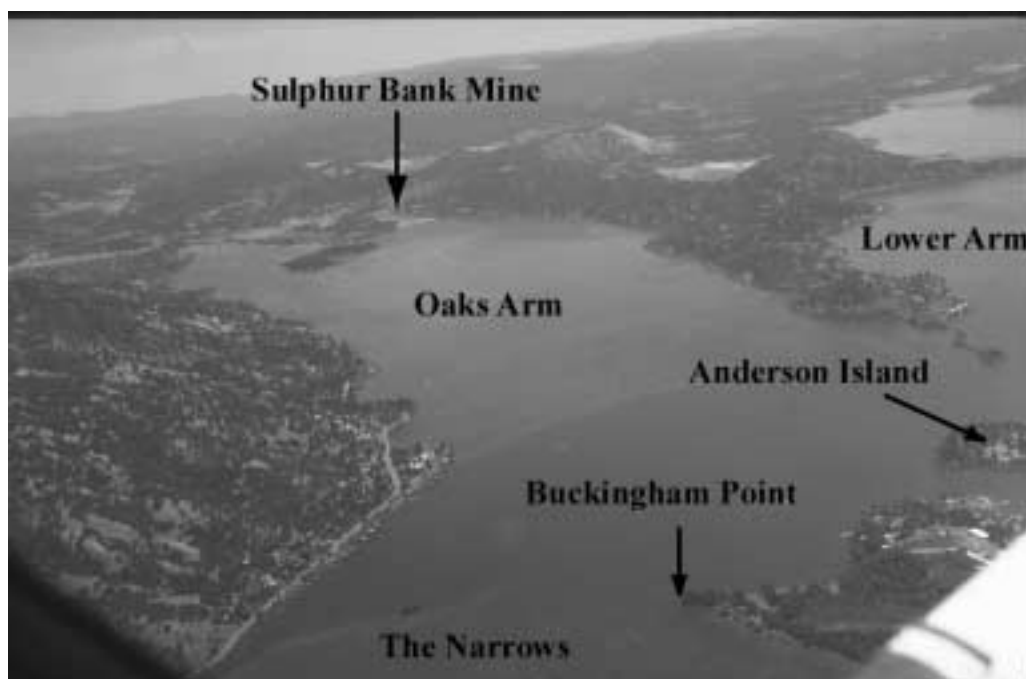


Figure 7.2. Aerial photograph of the Oaks Arm of Clear Lake, from the Narrows.

Basin	Area	Depth		Shoreline	Fetch		Width	Volume
		Mean	Max			Axis		
	(ha)	(m)	(m)	(km)	(km)		(km)	($10^6 m^3$)
Upper Arm	12700	7.1	12.2	56	16.4	SE-NW	12.2	904
Lower Arm	3270	10.3	18.4	39	13.4	SE-NW	4.3	384
Oaks Arm	1250	11.1	18.4	19	8.5	E-W	2.6	138
Overall	17760	8.1	18.4	114	--	--	--	1426

Table 7.1. Clear Lake Physical Parameters (after Horne 1975).

The location of Clear Lake, occupying a natural corridor between the dry Central Valley to the east and the marine climate to the west, imposes a characteristic wind regime over the system (see Fig. 7.1). Although episodic storms may be driven from the south, the

wind in Clear Lake basin is generally from the west-northwest and characterized as either nighttime and early morning drainage flow, later morning upslope flow and afternoon “pushes” of air from the maritime region to the west across Clear lake toward the Sacramento Valley (Giroux and Lawton 1980). A two-year study by Horne (Horne 1975) characterized the monthly variations in average wind direction and typical wind patterns for a winter and a summer day at Buckingham Point (Fig. 7.2), located on the Narrows near the geometric center of the lake. The study shows that the afternoon push described in Giroux and Lawton 1980 can, on a winter day, blow from the east (Sacramento Valley) westward towards the coast. On a typical winter day Buckingham Point endures mostly northwesterlies of 1-4 m/s interrupted by afternoon easterlies of 1-2 m/s. On a typical summer day the winds are primarily from the northwest, minimal between midnight and dawn and reaching a maximum about dusk.

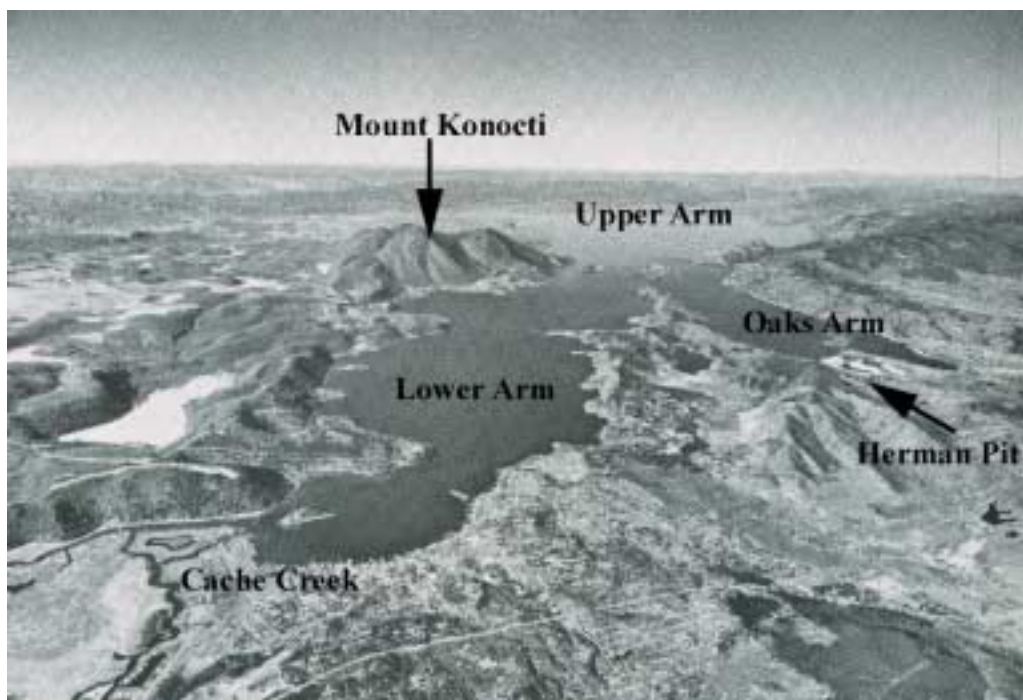


Figure 7.3. Aerial photograph of Clear Lake from its southeast corner, showing the irregular topography that surrounds the water body.

Given the large extent of Clear Lake and the irregular nature of its surrounding topography, the wind field acting on the water surface exhibits a considerable degree of spatial variability. Figure 7.3 shows an aerial photograph taken in 1971 from the southeast corner of Clear Lake that illustrates the complex landscape of the region, dominated by the presence of Mount Konocti, a dormant volcano rising 1000 m above the lake level. As noted by Lallatin (Lallatin 1975) in his report, Mount Konocti shelters the Lower Arm from the predominant northwesterly winds and funnels it into the Oaks Arm.

7.3. Material and methods for field experiments

7.3.1. August 1997: Experiment I

In the first of the field experiments a total of five mooring stations were deployed in the Oaks Arm of Clear Lake. Four of the mooring stations (stations SI.1 through SI.4 in Fig. 7.4) were equipped with acoustic Doppler velocity instruments (Acoustic Doppler Profilers, ADP, from Sontek, Inc.), while the fifth (identified as SI.5 in Fig. 7.4) consisted of two point acoustic Doppler velocity sensors (Argonauts, from Sontek, Inc.) measuring currents at 4.5 and 7.0 m depth. Both the ADPs and the Argonauts measure the velocity of water using the same Doppler shift principle in which, if a source of sound is moving relative to the receiver, the frequency of the sound at the receiver is shifted from the transmitted frequency by an amount that can be related to the velocity of the source (Sontek 1996). The instruments have three transducers, each used as a transmitter and a receiver. The transducer initially generates a short pulse of sound at a known frequency, which propagates through the water and is reflected in all directions by particulate matter

and bubbles. Some portion of the reflected energy travels back to the transducer, and, once received, the processing electronics measures the change in frequency. The location of measurements made by the instruments is a function of the time at which the return signal is sampled and the speed of propagation of sound in the medium. The velocity profiles measured by the ADP are obtained by measuring the return signal at different times following the initial pulse. The profile of water velocity is divided into range cells, where each cell represents the average of the return signal for a given period of time.

The Sontek Argonauts, used in the experiment, had a nominal acoustic frequency of 3 MHz, and the size of the sampling bin is approximately 1m. They were set to record the velocity on 15 minute intervals. Each record was obtained by averaging the velocity measurements obtained during 5 minute periods when the sensor was continuously repeating the cycle described in the previous paragraph. In the manufacturer's terminology, the averaging interval was set to 5 minutes and the profile interval to 15 minutes. The same sampling strategy was followed with the ADPs. Specifications of the ADPs and their setup for the experiment are given in Table 7.2.

<i>Station</i>	<i>Acoustic frequency (MHz)</i>	<i>Bin size (cm)</i>	<i>Profile interval (s)</i>	<i>Averaging interval (s)</i>
SI.1	3.0	25	900	300
SI.2	1.5	50	900	300
SI.3	1.5	50	900	300
SI.4	1.5	50	900	300

Table 7.2. Acoustic frequency, bin size, averaging and measurement interval of Sontek ADPs used in Clear Lake experiment.

The four mooring stations with the acoustic profilers (SI.1 through SI.4) were additionally equipped with self-contained 9311 temperature loggers from Oregon Environmental Instruments, Inc. (formerly Alpha Omega Computer Systems, Inc. and referred to as Alpha-Omega loggers). Three temperature loggers were deployed at each station at depths indicated in Table 7.3 and were set up to measure temperature every 60 seconds. They were calibrated to an accuracy of 0.003°C, and each had a resolution of 0.001°C.

<i>Station</i>	<i>Sensor 1</i>	<i>Sensor 2</i>	<i>Sensor 3</i>
SI.1	0.5	2.5	4
SI.2	2.5	6.5	11
SI.3	2.5	6.5	11.5
SI.4	2.5	6.5	11.5

Table 7.3. Depth (m) of temperature sensors at mooring stations (meters below water surface).

All of the instruments at the mooring stations commenced recording at 0h on August 2, 1997, and they continued for approximately 4 days. The locations were selected to highlight the flow patterns in the Oaks Arm and to monitor the exchange between the three basins. In addition to the mooring stations, several temperature profiles were taken with a Seabird SBE-25 CTD profiler on August 5, 1997, to characterize the temperature field during the experiment. The profiles were collected at the times and locations shown in Table 7.4, and the locations are shown in Fig. 7.4. Wind (speed and direction) and air temperature records for the four days of the experiment were obtained from the Sulphur Bank Mercury Mine site. The location of the meteorological station is shown in Fig. 7.5.

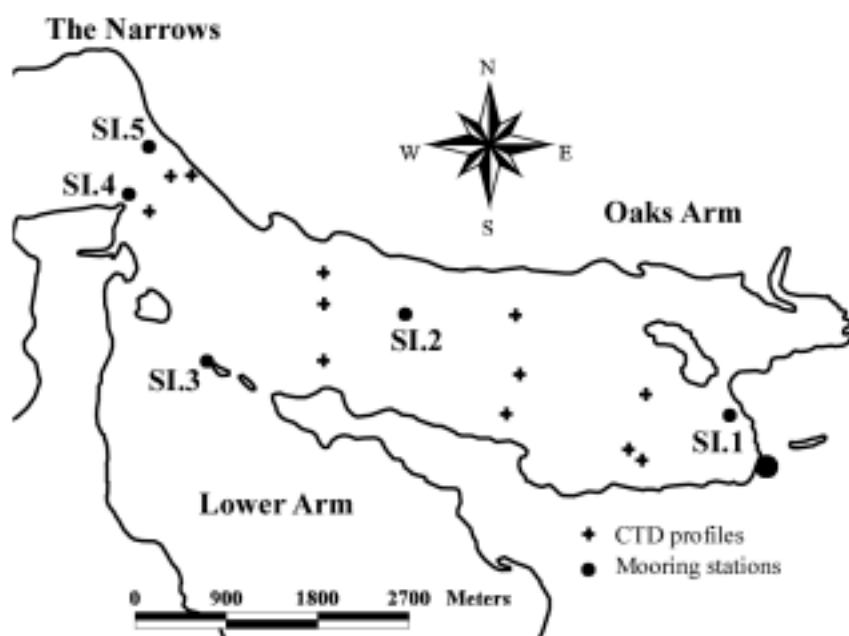


Figure 7.4. Experiment I. Layout

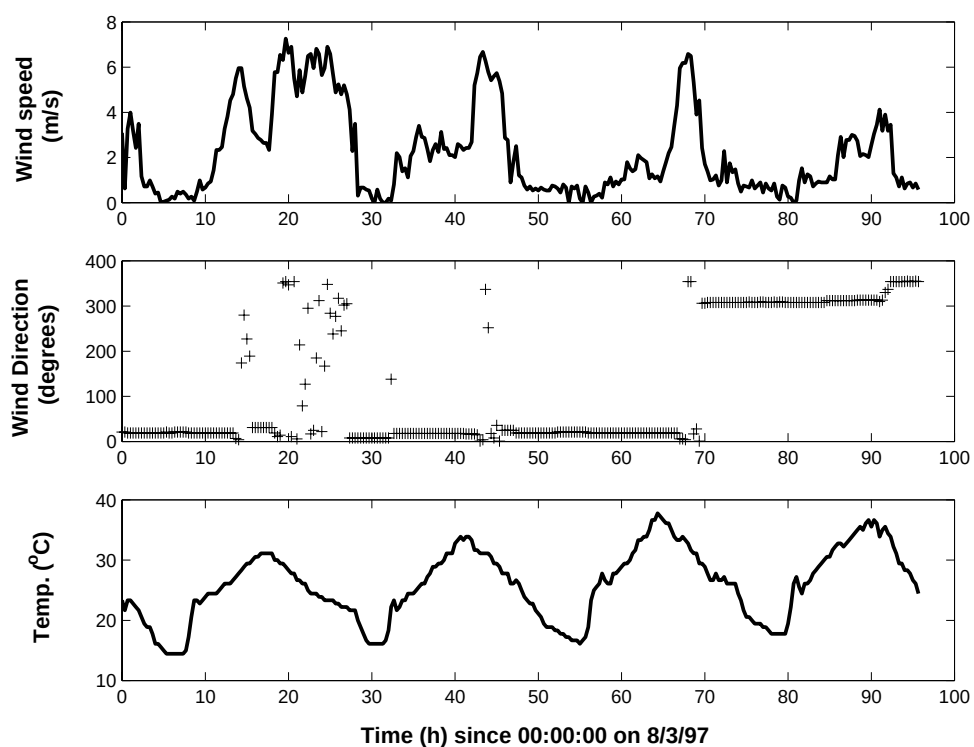


Figure 7.5. Experiment I. Meteorological information.

<i>Station</i>	<i>Date</i>	<i>Time</i>	<i>Longitude</i>	<i>Latitude</i>
8402	8/5/97	14:34:17	W 122.7440	N 39.0256
8403	8/5/97	14:42:37	W 122.7414	N 39.0291
8405	8/5/97	14:52:27	W 122.7388	N 39.0291
8406	8/5/97	15:01:48	W 122.7226	N 39.0197
8407	8/5/97	15:08:53	W 122.7226	N 39.0167
8408	8/5/97	15:17:54	W 122.7227	N 39.0113
8409	8/5/97	15:50:06	W 122.7002	N 39.0061
8410	8/5/97	16:00:40	W 122.6985	N 39.0099
8411	8/5/97	16:16:52	W 122.6991	N 39.0156
8412	8/5/97	16:30:28	W 122.6831	N 39.0080
8413	8/5/97	16:39:57	W 122.6835	N 39.0016

Table 7.4. Time and location of CTD profiles taken during 08/05/97.

7.3.2. May 1999: Experiment II

Thermistor chains were deployed in the Oaks Arm during two weeks in May 1999, at three locations (Fig. 7.6, top) selected to highlight longitudinal and lateral tilts of internal surfaces in the lake under stratified conditions. Each of the thermistor chains had nine Alpha-Omega temperature loggers arranged at about 1m intervals close to the surface, and further apart closer to the bottom. The instruments were calibrated to an accuracy of 0.003°C and had a resolution of 0.001°C. In addition to temperature records, water velocity profiles, with 0.5m bins, were collected every 15 minutes at one of the mooring stations (TII.3) with a 5 beam 600 kHz RDI Broad-band acoustic current velocity profiler (600kHz RDI BB-ADCP) and also along several velocity transects performed during Julian days 143 and 144 perpendicular to the longitudinal axis of Oaks Arm with a 600kHz RDI-Workhorse (identified with the symbol R in Fig. 7.6, top). At station TII.1 a 3Mhz Sontek ADP was deployed, but it unfortunately failed shortly after the deployment. The wind field was characterized by using an array of 14 stations, located as shown in Fig. 7.6, which recorded the mean wind speed and the dominant wind direction at 15



Figure 7.6. Layout of experiment II. The bottom portion of the figure shows a map of Clear Lake with the location of 14 stations (W1 through W14) where wind speed and direction were measured. The detailed view of the Oaks Arm area (top right frame) shows the location of thermistor chains (TII.1, TII.2 and TII.3) and the approximate boat track in the velocity transects shown in this work.

minute intervals. The accuracy was variable among the stations; in general the wind direction at seven of them (all but W5, W8, W11 and W14 in Fig. 7.6) was given using a 16-quadrant-description. The characteristics of the wind instruments used in the characterization of the wind field in this experiment are given in Table 7.5. Air temperature, relative humidity and atmospheric pressure information were obtained from the weather station at Kelseyville (Lake County), maintained by Lake County Air Quality Management District, and solar radiation was obtained from “Hopland FS” CIMIS station #85 in nearby Mendocino County. Figure 7.7 shows records of air temperature, solar radiation, relative humidity, and the wind speed and direction obtained at a station located close to the Narrows, for a period that includes the dates of the experiments.

Station ID	Manufacturer & Model	Wind speed Accuracy	Wind direction Accuracy
W1-W4, W7, W9, W10, W12, W13	Davis Instruments Corp. Weather Monitor Wizard III	$\pm 5\%$ at wind speeds up to 34 m/s	16-quadrant description ($\pm 12.5^\circ$)
W14	Handar UltraSonic wind sensor Model 425	± 0.135 m/s or $\pm 3\%$ whichever is greater, up to 50 m/s	$\pm 2\%$
W5, W8	MetOne uvw anemometer	$< \pm 3\%$ (EW shaft)	$< \pm 3\%$ (NS shaft)

Table 7.5. Characteristics of wind instrumentation used in May99 experiment. No information could be obtained for the instrumentation at station W11.

On Julian day 143 in 1999 four GPS-tracked Lagrangian current followers were released at different locations in the Oaks Arm to reveal large-scale circulation patterns. They were retrieved two days after the deployment, on Julian day 145. Each system has two

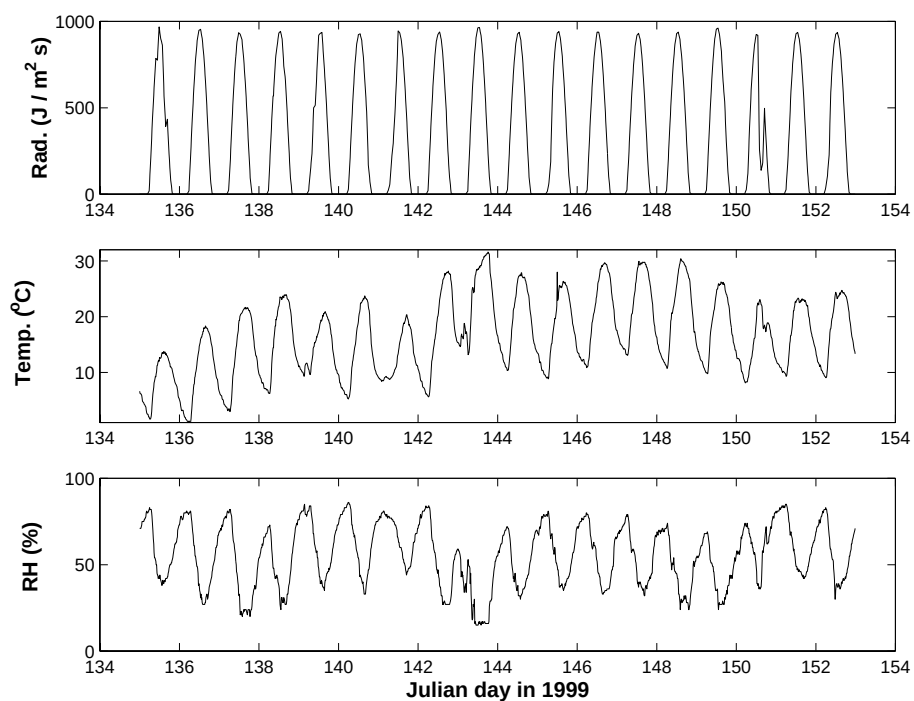


Figure 7.7.a. Experiment II. Meteorological information.

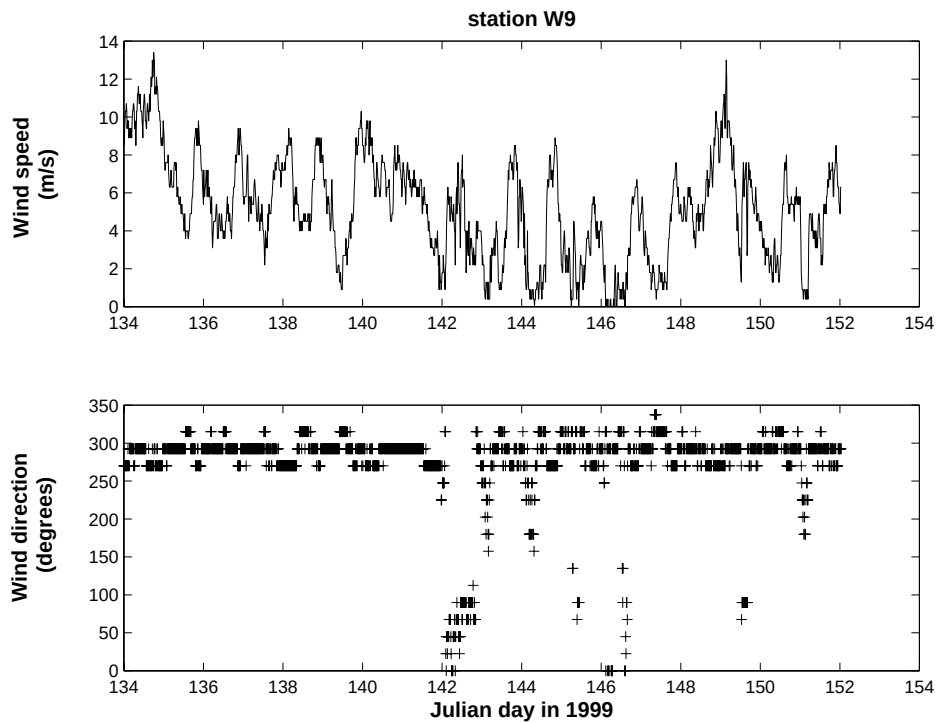


Figure 7.7.b. Experiment II. Wind records obtained at station W9 (Fig. 7.6).

components: a surface GPS tracking unit and a subsurface drogue at the depth of interest (Fig. 7.8). The GPS tracking unit (or TAG) is a small, waterproof, spar buoy. It contains a differential GPS board and antenna, a CPU board (with 512K flash RAM), a radio transmitter and antenna, and a controller board. The GPS board was programmed to awaken every 15 minutes and to collect a valid position within a 60 second period. The TAGs are equipped with radio transmitters that operate at unique frequencies while the CPU/GPS sleeps. A directional radio receiver is used to recover each unit. Upon recovery, data are downloaded from TAG via a serial port to a host PC and differentially corrected using Trimble Pathfinder software and data from the nearest Trimble base station.

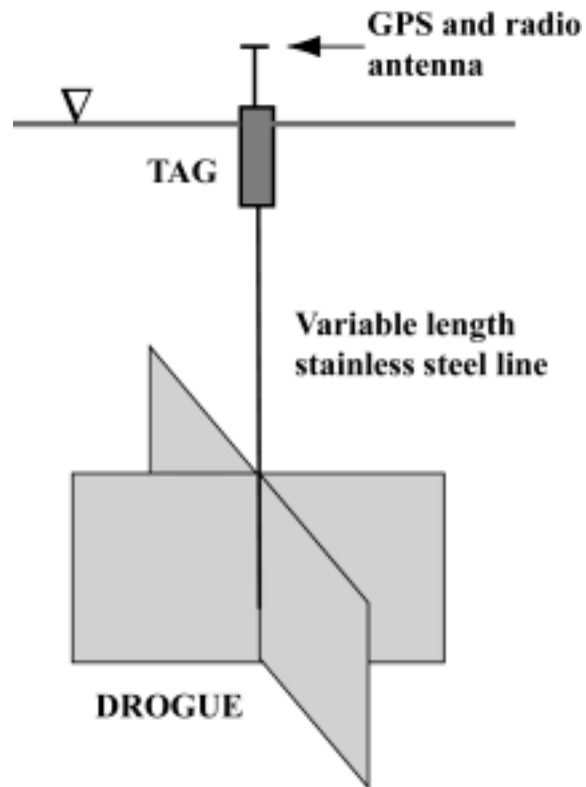


Figure 7.8. GPS-tracked drogue design. It comprises two parts: the GPS tracked unit (TAG) and a subsurface drogue at the depth of interest.

The drogue consists of 4 rip-stop nylon panels attached to a PVC frame, each approximately 1 square meter in area. The large size of the drogue is necessary to compensate for the effects of surface wind and wave drag on the TAG. The small mass/area ratio of the drogues makes them very sensitive to any changes in water currents, and the displacements of the drogues can reasonably describe the Lagrangian behavior of water particles. The drogue is connected to the TAG by stainless steel aircraft cable, the length of which can be varied to locate the drogue at any required depth. Three of the drogues were 2 m below the surface (surface drogues), and one was 7 m below the surface (deep drogue). When deployed, the panels are at right angles to each other (forming a cross in plan view).

7.4. Wind fields over Clear Lake

The records in Fig. 7.5 and Fig. 7.7 reveal that the wind followed, on most days and for both experiments, the pattern described in section 7.1 for a typical day in the summer period: primarily from the northwest, minimal between midnight and dawn and reaching a maximum about dusk. The diurnal periodicity is quite remarkable in the records. Only during Julian day 142 in 1999 did the lake experience the influence of relatively strong easterly winds.

The array of stations shown in Fig. 7.6 was used to characterize the spatial variability of the wind over Clear Lake during the experiment of May 1999. In particular, the vertical component of the vorticity in the wind field was analyzed, which, as discussed in chapter 4 (Eq. 4.11), is probably the feature that most strongly influences the circulation

patterns⁴³. From a generalization of Stokes' theorem, the vertical component of the vorticity can be computed either by differentiation or by integration (Schaefer and Doswell 1979), such that

$$\omega = \mathbf{k} \cdot \nabla \times \mathbf{v} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = \lim_{A \rightarrow 0} \frac{1}{A} \oint_{\Gamma} \mathbf{v} \cdot d\mathbf{r} \quad \text{with } A = \iint_S d\sigma \quad (7.1)$$

where A is the plan area of surface S , $\mathbf{k}$ is a unit vector normal to S , $d\sigma$ is a differential surface area, and $d\mathbf{r}$ is the differential of the position vector along the curve Γ which bounds S and lies in the horizontal plane. The integral definition can be applied over triangular areas whose vertices are located at the wind observation points (stations A, B and C in Fig. 7.9). If one further assumes that the wind velocity components vary linearly along each leg of the triangle, the evaluation of the line integral in Eq. 7.1 to provide an estimate of the mean vorticity (denoted with an overbar) over the triangular area is straightforward and follows:

$$\begin{aligned} \bar{\omega} = & \left[\frac{u_B + u_A}{2} (x_B - x_A) + \frac{v_B + v_A}{2} (y_B - y_A) \right] + \\ & \left[\frac{u_C + u_B}{2} (x_C - x_B) + \frac{v_C + v_B}{2} (y_C - y_B) \right] + \\ & \left[\frac{u_A + u_C}{2} (x_A - x_C) + \frac{v_A + v_C}{2} (y_A - y_C) \right] \end{aligned} \quad (7.2)$$

Here (u_j, v_j) and (x_j, y_j) represent the wind velocity components and the spatial coordinates at station j .

⁴³Strictly speaking, it is the vorticity of the wind stress field, calculated as the product of a drag coefficient times the square of the wind speed, the forcing term in Eq. 4.11. However, if one ignores the diabatic corrections to the drag coefficient and assume neutral stability conditions, the wind stress is directly dependent on the wind velocity.

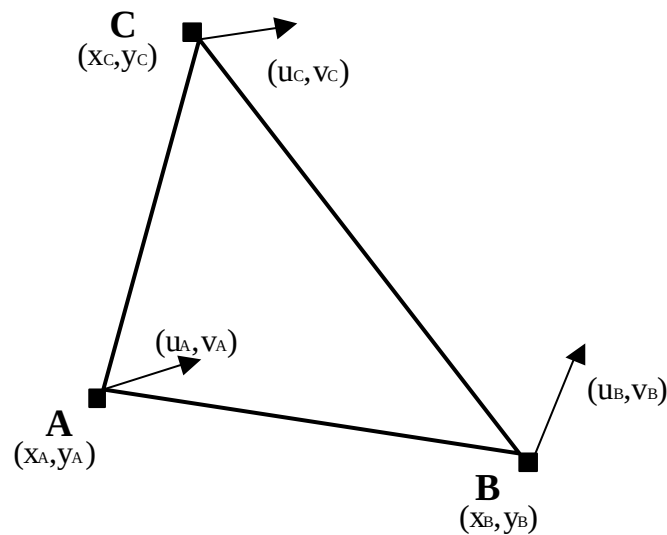


Figure 7.9. Triangular area defined by three wind observation points (A, B and C). The average vorticity over this area is estimated by using Eq. 7.2.

The same procedure was used to obtain vorticity estimates over triangular areas of the Oaks Arm and the Lower Arm with vertices at the observation points shown in Fig. 7.6. The focus is particularly on the evolution of the vorticity over the areas defined by stations W5, W8, and W14, on Lower Arm (Fig. 7.10.a), and on the triangle defined by stations W14, W9 and W13, over the westernmost region of the Oaks Arm (Fig. 7.10.b). In these two regions the northwesterly wind is expected to experience the effects of Mount Konocti, blocking and deflecting it and hence creating vorticity in the wind field, as discussed in chapter 4 and illustrated in Fig. 4.3. The vorticity time series, shown in Fig. 7.10, are calculated from the wind records every 15 minutes and smoothed using a 4-point moving average filter. For those days with predominant northwesterly winds, the time series over the Oaks Arm (Fig. 7.10.a) shows positive values of vorticity (counterclockwise rotation) with maxima during the evening hours for those days with northwesterly winds. The Lower Arm is located in the wake of Mount Konocti, and the

wind circulation for those same days over this region has negative vorticity (clockwise rotation), reaching the maximum absolute magnitudes at dusk, coinciding with or briefly after the peak in the wind speed. On day 142, when easterly winds dominated, the signs of the vorticity were the opposite of those shown during the days with northwesterly winds.

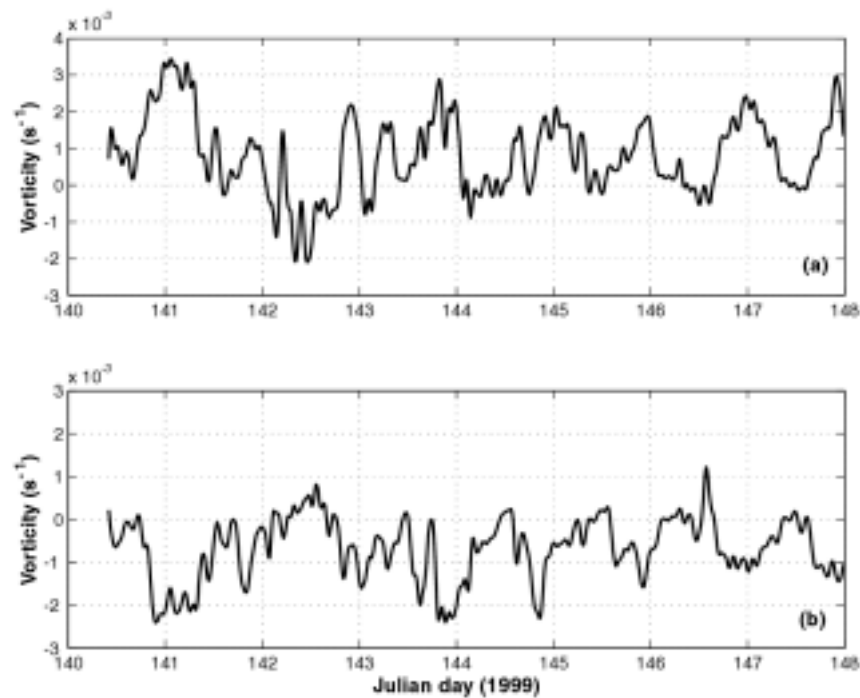


Figure 7.10. Evolution of average vorticity over: (a) the west portion of the Oaks Arm, triangular area defined by stations W14, W9 and W13; and (b) the north part of the Lower Arm, the region defined by stations W5, W8 and W14.

The differential definition of vorticity in Eq. 7.1 was used, in discrete form, to characterize the vorticity of the wind fields provided as boundary conditions to the 3-D model, which were obtained with a multiquadric algorithm (chapter 4). In Fig. 7.11 the results of the vorticity analysis at 1700h on Julian day 139, 1999, with predominant northwesterly winds, are presented. Probably the most remarkable characteristic of the vorticity field is the presence of two cells, one of positive vorticity over the Oaks Arm

and another of opposite sign over the Lower Arm. These features agree with the description of the effects of an isolated obstacle on an otherwise uniform boundary layer flow, already presented in section 4.4, and they suggest that the wind field over the more easterly basins of Clear Lake is shaped and dominated by the presence of Mount Konocti.

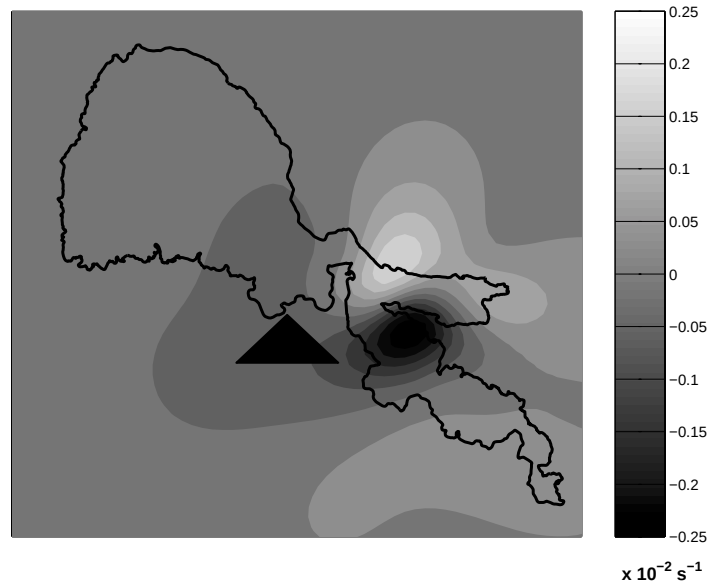


Figure 7.11. Differential estimate of the vorticity of a wind field constructed by means of multiquadric interpolation from wind records taken at 1700h on Julian day 139, 1999, at the locations shown in Fig. 7.6. The triangle represents the location of Mount Konocti.

The characteristics of this wind field impose a very distinctive signature on the observed circulations in the Oaks Arm and especially in the Lower Arm. Figure 7.12 shows depth-averaged currents in those two basins, measured by Lynch, Schladow and Stacey (Lynch and Schladow 1996) on November 10, 1995, under almost isothermal conditions⁴⁴. The most prominent features in the depth-averaged currents are a counterclockwise gyre

⁴⁴ The observations gathered in November 1995 indicate that there exists some degree, though very weak, of stratification. The current profiles measured in the Lower Arm suggest that the flow directions at the top and bottom portions of the water columns are similar, and they support a 2-D depth-averaged description of the currents. In the Oaks arm, that description is not, however, completely correct (see Lynch and Schladow 1996, p. 79).

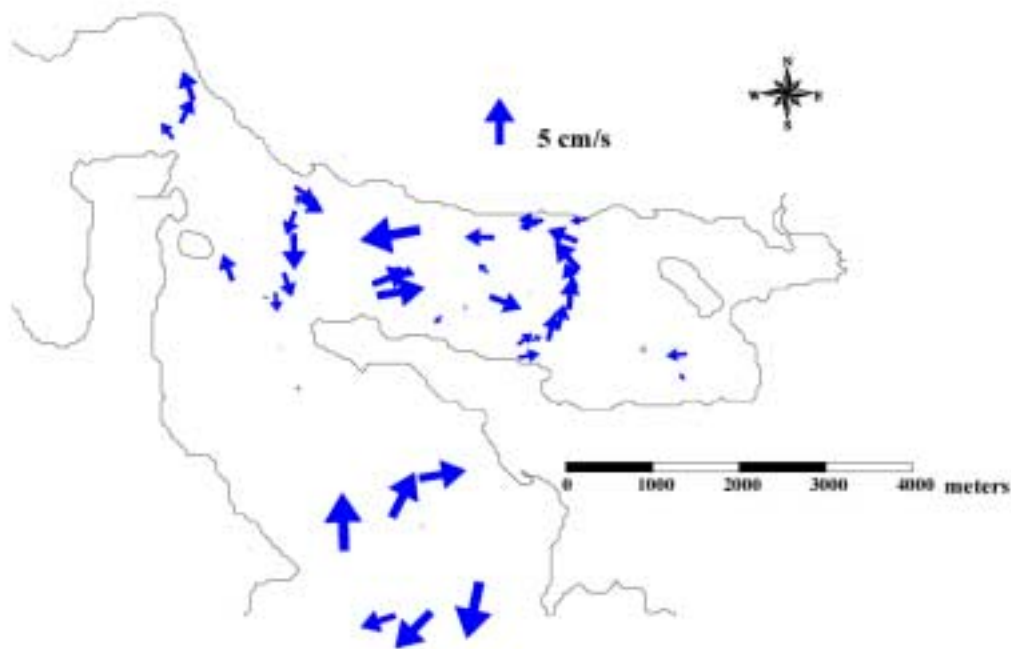


Figure 7.12. Depth-averaged currents in the Oaks and Lower Arms of Clear Lake, measured on November 10, 1995 (data from Lynch and Schladow 1996).

(positive vorticity) in the Oaks Arm and a gyre with the opposite sign of rotation (negative vorticity) in the Lower Arm, suggesting that the vorticity in the lake is shaped by the rotational character of the wind field. The linear equation for conservation of mean (depth-averaged) flow vorticity in homogeneous basins driven by horizontally varying wind field obeys the expression (Strub and Powell 1986)

$$\frac{\partial}{\partial t} \left(\frac{\partial \bar{v}}{\partial x} - \frac{\partial \bar{u}}{\partial y} \right) = \mathbf{k} \cdot \left[\nabla \times \left(\frac{\boldsymbol{\tau}_s}{H} \right) \right] \quad (7.3)$$

where the effects of bottom friction have been neglected, the overbar on the velocity (u , v) denotes depth-averaged components, $\boldsymbol{\tau}_s$ is the wind stress at the free surface, H is the water depth, and $\mathbf{k}$ is a vertical unit vector. Equation 7.3 demonstrates the dependence of

the depth-averaged circulation on the wind stress field and, together with Fig. 7.11 and Fig. 7.12, strongly suggests that the wind field and the effects of the surrounding topography must be carefully characterized to carry out realistic simulations⁴⁵. The 2-D depth-averaged hydrodynamic simulations of Clear Lake carried out by Bale (Bale 1995) were driven by northwesterly winds applied uniformly over the free surface of the lake. The simulated circulation in the Lower Arm had a positive vorticity, in contrast with the clockwise gyre (negative vorticity) that is evident in the observations.

7.5. Analysis of experimental results in the Oaks Arm

7.5.1. Evolution of temperature field: longitudinal response

The evolution of the temperature field along the main axis of the Oaks Arm in response to the diurnal northwesterly winds can be better understood with the aid of the simplest possible model of wind effects on a stratified reservoir, consisting of a two-layer fluid contained in a rectangular box and subject to an impulsively applied wind stress. In this model it is further assumed that rotational effects and wind stress variability are negligible. Spiegel and Imberger (Spiegel and Imberger 1980) showed that different types of flow regimes are possible in such a model, depending on the relative strength of mixing process and large-scale internal motions. The Wedderburn number

$$W = \frac{\alpha g \Delta T h^2}{u_*^2 L} \quad (7.4)$$

⁴⁵ Under stratified conditions, the equation governing the circulation is given in chapter 4, Eq. 4.11, and similarly reveals the critical role of the wind forcing.

can be used to define the flow regime. Herein, u_* is the water friction velocity due to wind stress, h is the depth of the upper layer, g is the acceleration of gravity, α is the coefficient of thermal expansion, L is the fetch length of the lake, and ΔT is the temperature difference between top and bottom layers. The Wedderburn number W gives the ratio of the thickness of the upper layer h to the maximum deflection of the single internal interface, which according to the linear, two-layer theory is

$$(Z_i)_{\max} = \frac{-L}{Ri^*} \quad (7.5)$$

In Eq. 7.5, the Richardson number (Ri^*) is given by

$$Ri^* = \frac{\alpha g \Delta T h}{u_*^2} \quad (7.6)$$

The interface just reaches the free surface for $W = 1$ leading to upwelling of the lower layer fluid (Monismith and Fong 1996). When $W < 1$ the interface surfaces, and significant horizontal variations in the temperature structure will result. As shown in laboratory and field experiments (Imberger and Patterson 1990), the response in weakly and continuously stratified system – that probably describes best the observed stratification in Clear Lake – is characterized by isotherms surfacing and stretching at the upwind end, while at the downwind end the isotherms will deepen and compress. In any case, the wind stress will cause the generation of significant temperature gradients in the direction of the wind.

To estimate W for the conditions existing during the first experiment, a two-layer profile was defined that best represented⁴⁶ the temperature profile calculated by averaging all of the CTD casts taken on August 5, 1997 (Fig. 7.13). The location of the temperature jump in the two-layer profile was taken as a surrogate for the depth of the thermocline. Using a wind speed of about 6 m/s, the Wedderburn number is $W = 0.03$, suggesting that the stratification in the Oaks Arm is very weak and cannot resist the overturning moment supplied at the surface by the wind.

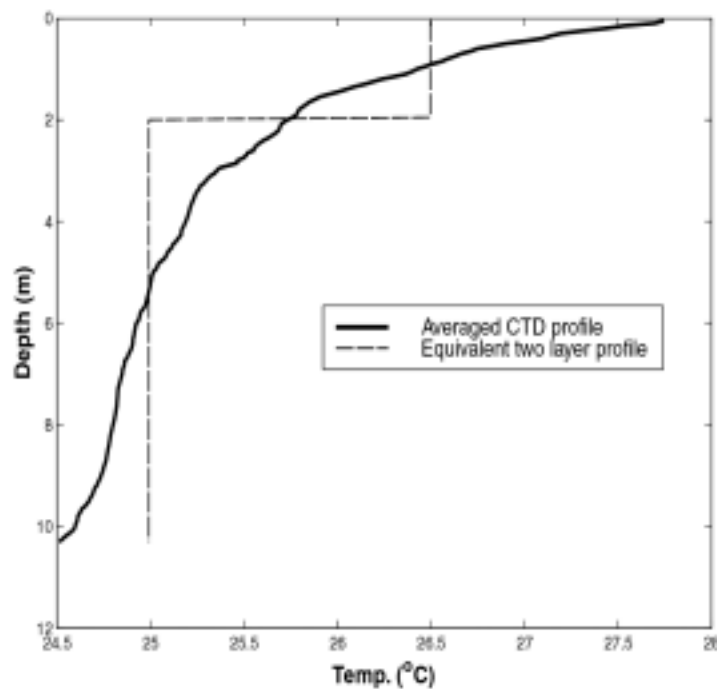


Figure 7.13. Average temperature profile for August 5, 1997, and its representation with a two-layer structure.

Similar calculations were completed for each of the first 14 days of the experiment in May 1999 (Table 7.7). For each day of the experiment a two-layer profile was used to represent hourly-averaged temperature profile from station TII.3 at 1400. This time was

⁴⁶ The “best” two-layer profile is taken here as that which minimizes the sum of squared errors between the model and the observed profile.

considered as the beginning of the wind event. The wind event was assumed to extend from 1400 to 2000 hours every day, and the value of the friction velocity was calculated from the mean wind speed during that period of time, evaluated at station W9 in Fig. 7.6. Table 7.7 shows in all cases that $W < 1$ or $W \sim 1$, and hence the diurnal wind events, with a strong component along the main axis of the Oaks Arm, will generate upwelling and significant horizontal temperature gradients in the East-West direction.

Julian Day, 1999	ΔT	h	u_*	W
138	0.61	6.5000	0.0079	0.100
139	0.26	5.0000	0.0062	0.040
140	0.23	4.5000	0.0084	0.016
141	0.27	4.5000	0.0053	0.048
142	0.71	6.5000	0.0043	0.399
143	0.95	5.7500	0.0092	0.091
144	1.11	8.7500	0.0088	0.269
145	1.67	8.5000	0.0054	1.013
146	2.72	5.0000	0.0045	0.822
147	3.44	5.7500	0.0049	1.160
148	4.08	8.0000	0.0093	0.740
149	2.27	10.0000	0.0087	0.735
150	2.37	7.0000	0.0085	0.394
151	2.88	9.5000	0.0089	0.804

Table 7.7. Estimates of Wedderburn number for each of the days of the experiment II.

The records obtained with the thermistor chains during the first of the field experiments revealed that significant longitudinal horizontal temperature gradients developed in the Oaks Arm, in response to the diurnal northwesterly winds (Fig. 7.14). The plots show the time series of wind speed and temperature records 2.5 m below the water surface obtained at stations TI.1 and TI.2 (Fig. 7.4), located at the east end and center of the Oaks Arm respectively, about 4 km apart. The temperature differences reach maxima of up to

4°C, coinciding with the strongest evening winds, and decline thereafter to disappear before the onset of the next wind event.

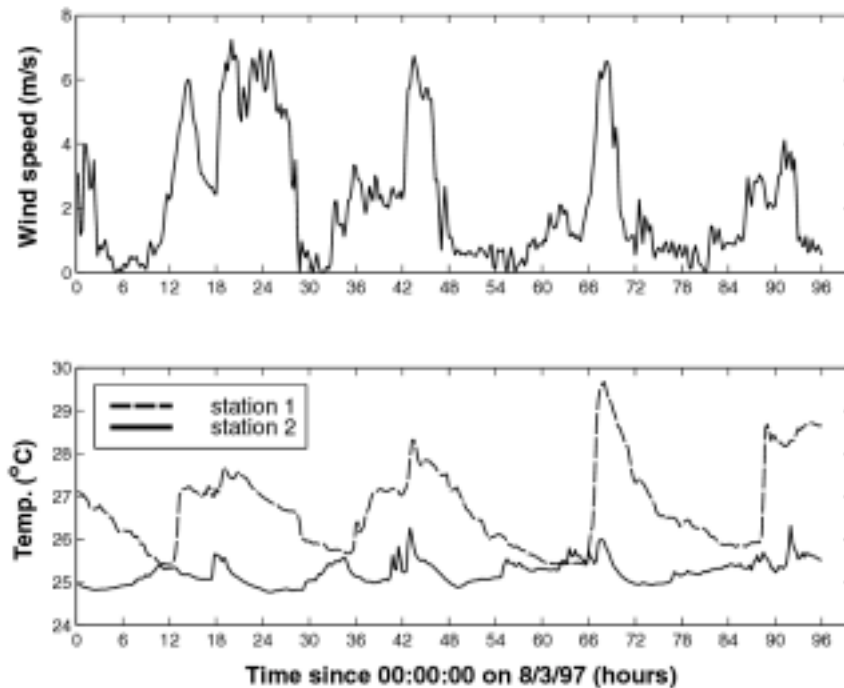


Figure 7.14. Wind forcing and the generation of longitudinal differences in temperature, during experiment I (August 1997).

7.5.2. A simple conceptual model for longitudinal circulation

The presence of horizontal temperature gradients is a key feature in the internal dynamics of the Oaks Arm, and of Clear Lake in general, as it induces water currents that tend to reestablish the equilibrium state with horizontal isotherms that existed prior to the onset of the wind. Thus the longitudinal circulation in the Oaks Arm may be characterized by cycles of setup and relaxation of horizontal temperature gradients being driven by a succession of windy and calm periods. Figure 7.15 presents a conceptual model of the setup and relaxation cycle. In that sketch the rectangular basin represents the Oaks Arm viewed from the south, with the rightmost end being the mine site. The dominant winds

are aligned with the main axis of the basin. The temperature is represented by the intensity of the gray scale, with the higher temperatures corresponding to darker colors. Also drawn is a line to represent the location of the thermocline in an equivalent two-layer system.

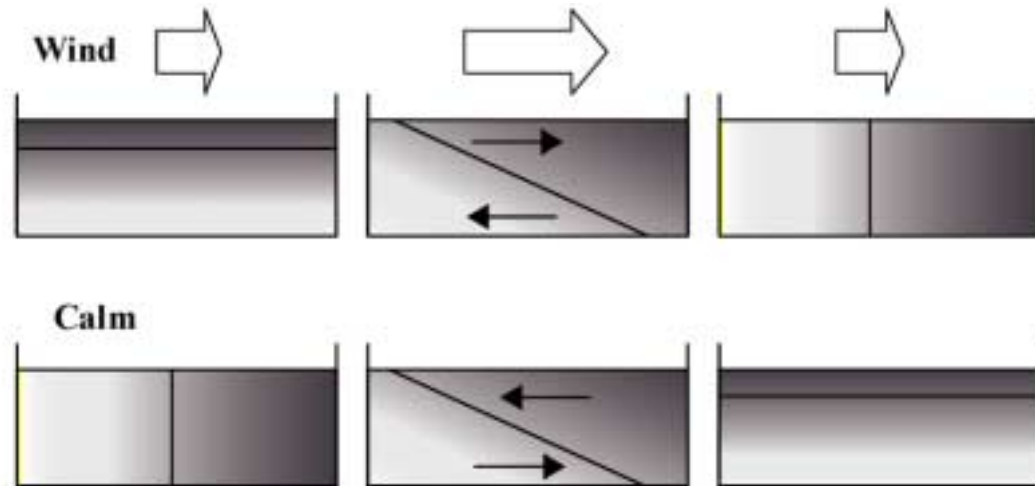


Figure 7.15. A conceptual model of the longitudinal circulation in the Oaks Arm driven by diurnal northwesterly winds.

The wind acts on this weakly stratified system to generate horizontal temperature gradients, as discussed in the previous section, and a strong wind event may cause a complete overturning of the temperature stratification (see Fig. 5.5, p. 33, in Lynch and Schladow 1996). The surface water will move downwind in response to northwesterly winds while currents in the lower layers will be forced to move upwind. After the wind dies down (the lower part of Fig. 7.15), the temperature gradients induce baroclinic pressure gradients that act to create momentum that forces the system to return to its original equilibrium state. As explained by Imberger and Patterson (Imberger and

Patterson 1990), the internal displacement will degenerate into damped motions with no internal oscillations. The pattern of currents in calm conditions is the opposite of that driven by the wind, i.e. westwards in the surface layer and eastwards at the lower layer.

The conceptual model in Fig. 7.15 is confirmed by experimental observations of velocity profiles obtained at station 2, near the center of the Oaks Arm, during the four days of the field campaign in 1997 and shown in Fig. 7.16. In the figure the magnitude of each velocity component, either parallel or perpendicular to the longitudinal axis of the arm (v_x and v_y , respectively), is represented by using the intensity of the gray scale, shown in a vertical bar next to the figure. Eastward (positive x or downwind) currents have a higher gray intensity, while westward (negative x or upwind) currents are represented with the lighter gray scale intensities. The horizontal axis represents time, and the vertical axis represents depth, with the surface at the top. The alternation of vertical re-circulation patterns exhibited in Fig. 7.16 is quite remarkable, and the pattern obeys the succession of windy (late evening) and calm (night and early morning) periods in the diurnal cycle.

During the night and early morning periods of calm, the baroclinic pressure gradients create the only driving force that is able to generate currents in the lake. Otherwise the circulation set up by the previous wind event would slowly disappear as a consequence of bottom friction and internal friction. The role of the baroclinic forcing in the generation of motions in the Oaks Arm of Clear Lake during calm periods is revealed in the following analysis of field observations, where a simple 1-D momentum balance that

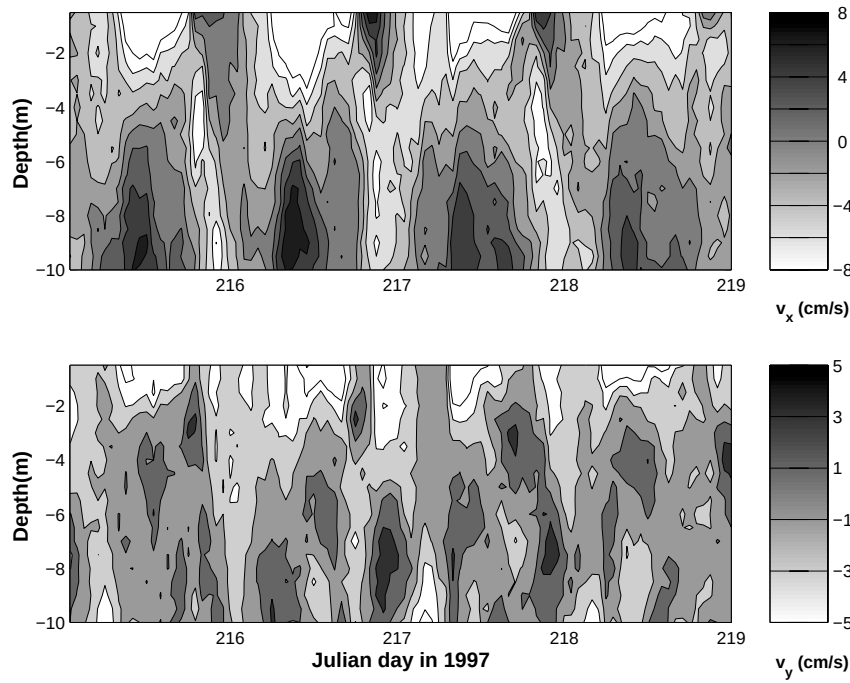


Figure 7.16. Experiment I. Time series of hourly average velocity profiles at station SI.2. The horizontal axis in the plots represents time and the vertical axis is depth measured from the surface, positive upwards. The intensity of the gray scale (on the right) represents the value of the velocity component. The x and y direction in the velocity components correspond to the longitudinal and transversal axis of the Oaks Arm. The contours are shown every 2 cm/s.

neglects Coriolis effects and non-linear terms is applied. The equation governing the momentum in the longitudinal direction is given by

$$\frac{\partial U}{\partial t} = -g \frac{\partial \zeta}{\partial x} - \frac{gH}{2\rho_0} \frac{\partial \rho}{\partial x} + \frac{\tau^w}{\rho_0} - \frac{\tau^b}{\rho_0} \quad (7.7)$$

(1) (2) (3) (4) (5)

where ζ is the water surface departure from the equilibrium elevation and U represents the integrated transport

$$U = \int_{-H}^{\zeta} u \, dz \quad (7.8)$$

While a 1-D momentum representation is an oversimplification, it will provide some insight on the role of horizontal temperature gradients in driving transport. The variations in water surface elevation (2) will be neglected in the discussion that follows. Term 2 is only relevant on time scales of the barotropic seiches (~ 3 hours in Clear Lake, see appendix B for a detailed analysis of barotropic seiches). The bottom stress term (5) always acts as a sink of momentum (decelerating the water column). Primary attention will be focused on the two remaining sources of momentum: the wind stress (4) and the baroclinic pressure gradient (3). The time series of velocity profiles collected near station TII.3 with an ADCP during experiment II (Fig. 7.6) will be used for the analysis. In Fig. 7.17 (lower frame) is represented the depth-averaged acceleration in the East-West direction derived from the ADCP records (term 1, Eq. 7.7) together with the East-West wind stress term (term 4). The latter was obtained by using a constant drag coefficient equal to 1.3×10^{-3} in Eq. 4.13. The velocity records do not include the first 1-2 meters from the bottom (accounting for the blanking distance and the height of the sensors above the lake bottom). However, one can assume that the velocity close to the bottom is negligible. In the top frame of Fig. 7.17 the unsteady term is plotted with the other source term, which is the baroclinic forcing. The baroclinic term was calculated by using only the information from 2 to 9 meters of water so as to avoid the influence of changing depth between stations TII.1 and TII.3. It is also assumed that stations TII.1 and TII.3 may be considered to be aligned with the longitudinal axis of the Oaks Arm.

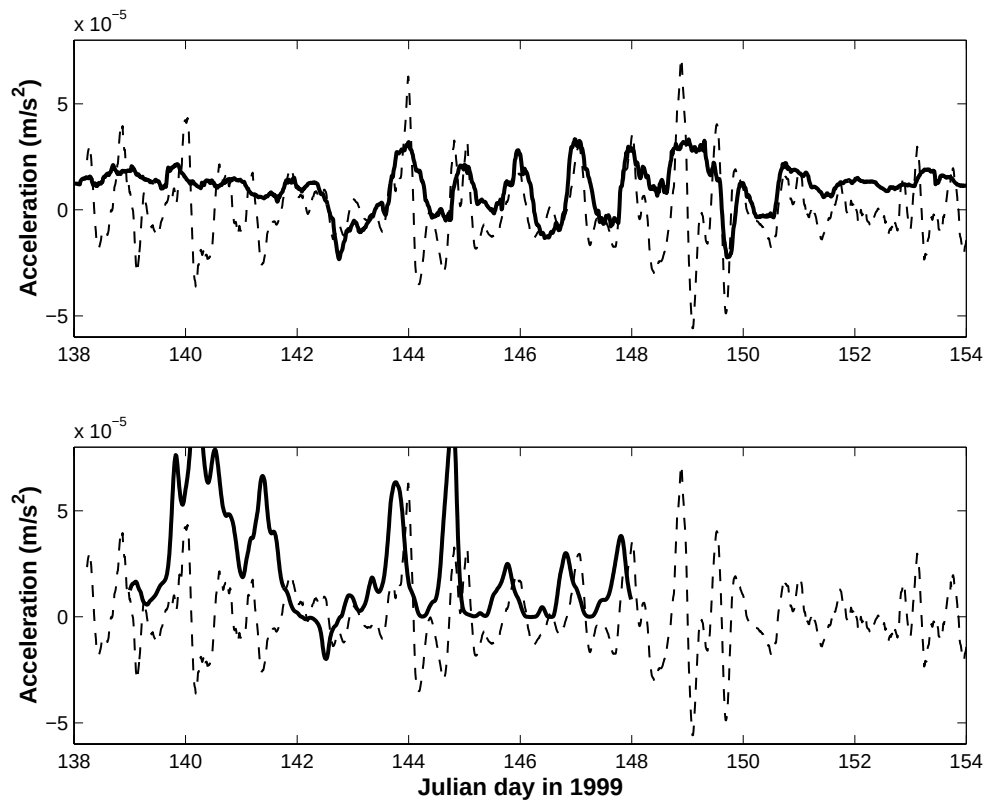


Figure 7.17. Time series of the unsteady term (term 1 in Eq. 7.7) (dashed line) and the forcing terms (solid lines) in Eq. 7.7: baroclinic pressure forcing (term 3) (top), and wind stress forcing (term 4) (bottom).

At the beginning of a wind event one can assume that the horizontal temperature gradients have relaxed and that the only source of momentum in Eq. 7.7 is the wind stress applied on the top of the column (τ^w). In the 1-D equation the wind would act to create downwind currents (i.e. towards the southeast in the case of Clear Lake). In 2-D depth-averaged systems, however, the wind acting uniformly on a lake with depth variations will create topographic gyres (Csanady 1975), where the currents are directed downwind in the shallower parts of the lake but upwind in the deeper portions. Since the ADCP was deployed in the deepest point in the Oaks Arm, one would expect the northwesterly winds to drive westward (negative) currents. This effect explains why the unsteady term

becomes negative during the wind events. As the wind peaks and decreases, the unsteady term starts to increase and eventually becomes positive, reflecting the buildup of baroclinic pressure gradients acting against the wind forcing. Once the wind forcing is negligible one is left with one unique source to drive water motions. Figure 7.17 (top) illustrates that the time series of depth-averaged acceleration follows closely that of the baroclinic forcing term, especially during the period of “stronger” stratification from day 144 to 148 (see section 7.5.4). The peaks in the unsteady term occur after those in the wind stress and at the same time as the peaks in the baroclinic forcing.

One of the velocity transects collected at the center of the Oaks Arm, in the second experiment of May 1999 at 1600h local time, during a strong wind event (Fig. 7.18) supports the conceptual model of the longitudinal response, but it also indicates the presence of significant lateral effects⁴⁷. The component of the velocity along the main axis of the Oaks Arm is shown on top, and that in the perpendicular direction on the bottom of the figure. The horizontal axis represents the distance from the north of the basin, the vertical is depth and the intensity of the gray represents the value of the velocity component (positive is eastward). Figure 7.18 shows a classical re-circulation pattern not only along the longitudinal axis of the Oaks Arm but across it, the latter being probably the result of rotational effects.

⁴⁷ Since the measurement of the velocity components in the direction of the ship track during a transecting exercise is biased (Monismith 1999), the information collected by the current meters working in transect mode should be interpreted very carefully. Thus, that information is used in this thesis only in a qualitative sense.

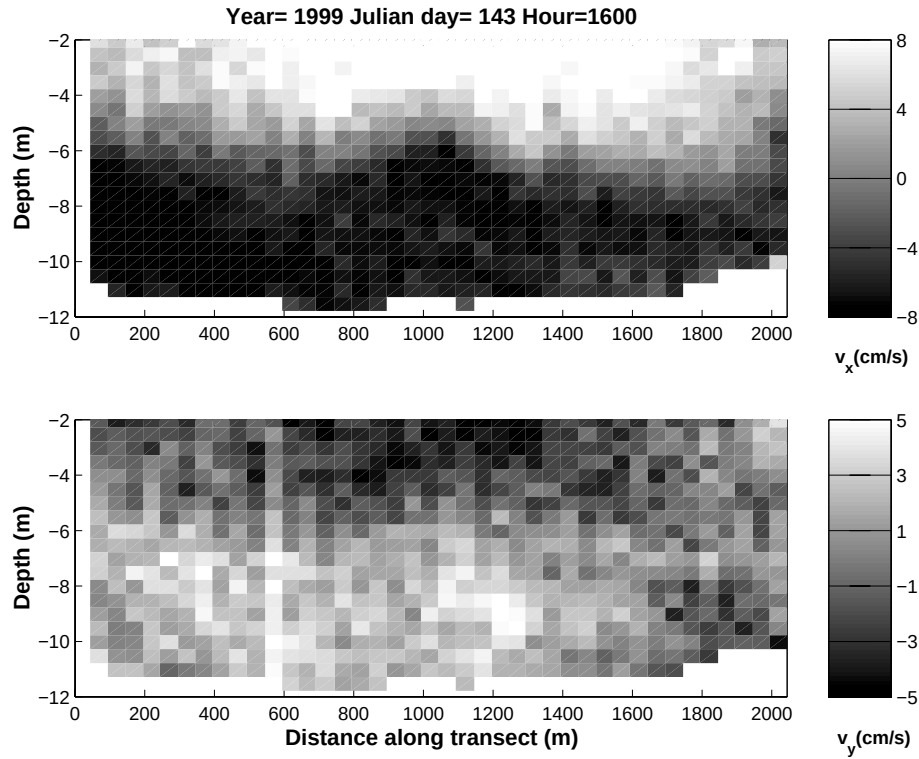


Figure 7.18. Experiment II. Transect taken at 1600 hours in Experiment 2 (identified as RII.3 in Fig. 7.6) showing re-circulation cells both along (top) and perpendicular to (bottom) the main axis of the Oaks Arm. Velocity is given in cm/s. The transect starts on the north shore and ends on the south shore of the Oaks Arm.

7.5.3. Influence of the Earth's rotation

The Earth's rotation can exert a significant influence on the internal motions of lakes on temporal scales of the order of the inertial period (~ 19 hours in Clear Lake) and on spatial scales of the same order as the Rossby radius of deformation. For stratified systems the internal Rossby radius of deformation, Λ , is defined as

$$\Lambda = c_1 / f \quad (7.9)$$

where f is the Coriolis parameter (inertial frequency) and c_1 is the speed of propagation of internal waves (only the first vertical mode is considered here). The speed c_1 was

calculated for each of the days of the experiment in May 1999 using an hourly-averaged temperature profile taken at 1400h from station TII.3 and a modal decomposition approach (Kundu 1990 and described in the appendix D). Then Λ was calculated; it is given in Table 7.8 for the first 14 days. A Rossby radius of deformation on the order of the width of the lake (~ 1500 - 2000 m for the Oaks Arm) suggests that the Earth's rotation will influence the internal dynamics of the lake.

Owing to rotational effects, the diurnal along-shore westerly winds may induce upwelling on the north shore of the Oaks Arm. In an initially quiescent two-layer system of depth H , subject to an impulsively applied wind stress, the equation that describes the magnitude of the interface displacement is (Cushman-Roisin 1994)

$$Z_i = A \exp\left(-\frac{y}{\Lambda}\right) \quad (7.10)$$

in which A is the amplitude of the upwelled state and is related to the wind impulse ($I = u_*^2 t$). In the case where the interface does not intersect the surface $A \sim I (H - h)/Hc_1$. The interface may intersect the water surface – complete upwelling – when (Csanady 1982) the amplitude exceeds the depth of the upper layer h , i.e.

$$I \geq \frac{hH}{H - h} c_1 \quad (7.11)$$

The two-layer stratification model and the diurnal wind events, defined as in section 7.5.1, was used to estimate the magnitude of the internal displacements induced by a long-shore wind, and hence the occurrence of upwelling on the north shore of the Oaks

Arm resulting from Coriolis effects. The ratio of the wind impulse to the right side of inequality 7.11 is listed in Table 7.8. During those days of maximum stratification (mainly Julian days 145 through 151) the condition of total upwelling was not achieved but considerable interface displacement, of O(4-7 m), would occur across the Oaks Arm as a consequence of the Coriolis effect. In any case, and because of Coriolis effect, the northwesterly winds affecting the Oaks Arm basin will induce upwelling on the north shore and will hence result in significant lateral temperature gradients. Figure 7.19 shows a simple conceptual model of the transverse response of the Oaks Arm to the diurnal longitudinal wind event. In the sketch the rectangular area represents a cross section of the Oaks Arm viewed from the west, with the right edge being the south shore. The wind acts into the page.

<i>Julian Day, 1999</i>	<i>Speed of Propagation (m/s)</i>	<i>Rossby Radius (m)</i>	<i>$I(H - h)/hHc_1$</i>
138	0.053	576	0.63
139	0.033	357	0.42
140	0.031	343	0.18
141	0.034	368	0.48
142	0.059	646	2.30
143	0.066	724	0.46
144	0.073	802	1.25
145	0.093	1012	3.82
146	0.116	1264	2.55
147	0.137	1501	3.11
148	0.154	1679	1.77
149	0.105	1143	2.50
150	0.116	1271	1.24
151	0.127	1382	2.37

Table 7.8. Rossby radius of deformation and conditions for long-shore wind-induced upwelling in the Oaks Arm of Clear Lake, calculated for 14 days of the experiment in May 1999.

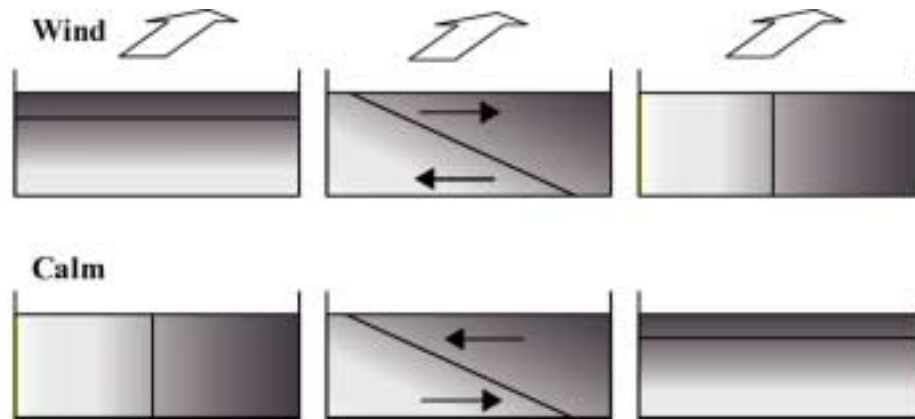


Figure 7.19. A conceptual model of the transverse circulation in the Oaks Arm of Clear Lake, driven by diurnal northwesterly winds and Coriolis effects.

7.5.4. Evolution of the temperature field in 3-D

The arrangement of thermistor chains in the Oaks Arm during experiment II (Fig. 7.6) was intended to reveal the evolution of the temperature field in the arm in 3-D. Figure 7.20 shows the time series of temperature at depths from 2 to 9 meters obtained at stations TII.1, TII.2 and TII.3 in May 1999. The experiment started with the lake fairly well mixed at a temperature of $\sim 15\text{-}16^{\circ}\text{C}$. The lake then underwent a warming period, stratified, and cooled down with the air temperature, to end two weeks later well mixed again at $\sim 19^{\circ}\text{C}$. The records show very significant differences in the phase of the temperature records between station TII.1 and the other two stations (TII.2 and TII.3). Temperatures in the upper layers increased at stations TII.2 and TII.3 in evening hours coinciding with the diurnal wind event, and it decreased afterwards. At station TII.1 the temperatures initially increased in response to the onset of wind, but they decreased immediately after, even before the end of the wind event.

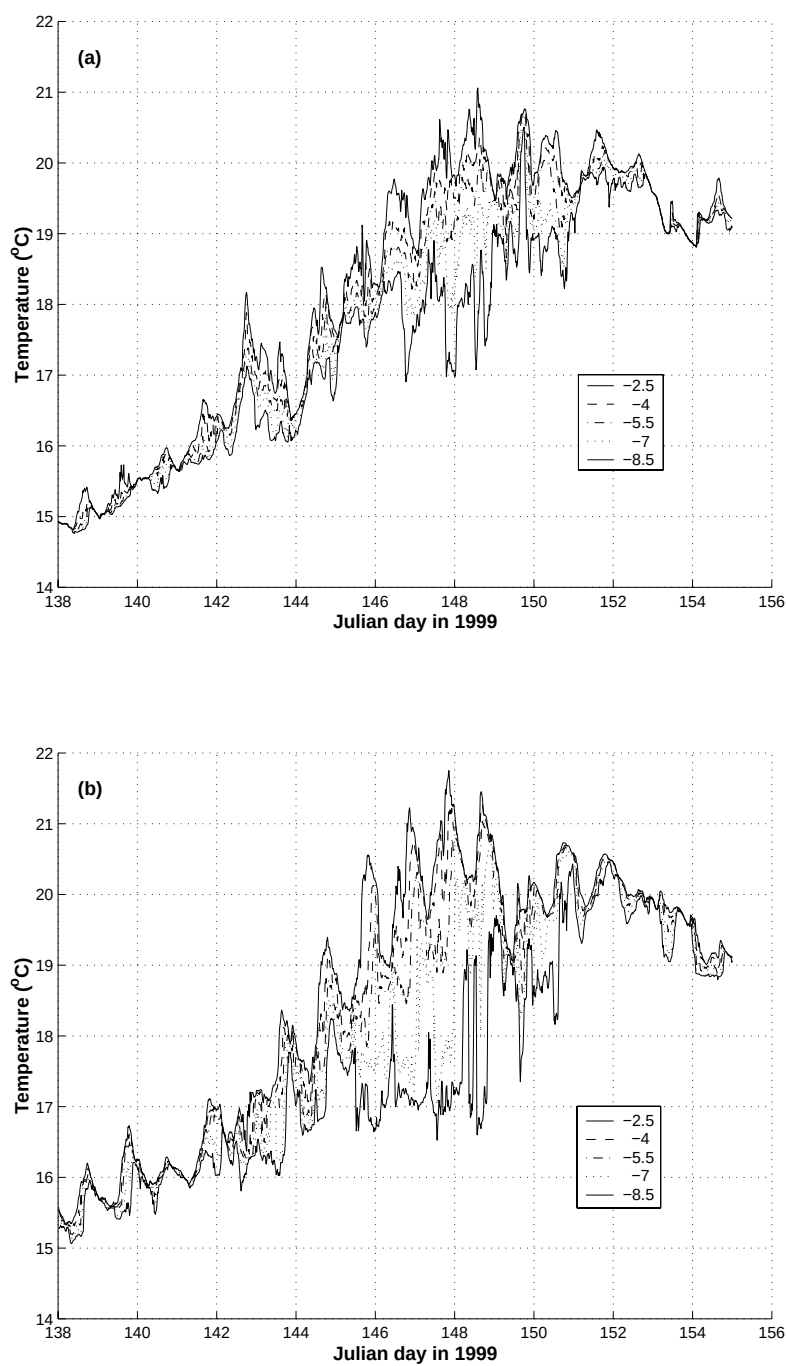


Figure 7.20. Experiment II. Time series of temperature from selected depths (shown in legends) at stations TII.1 (a), TII.2 (b) and TII.3 (c)

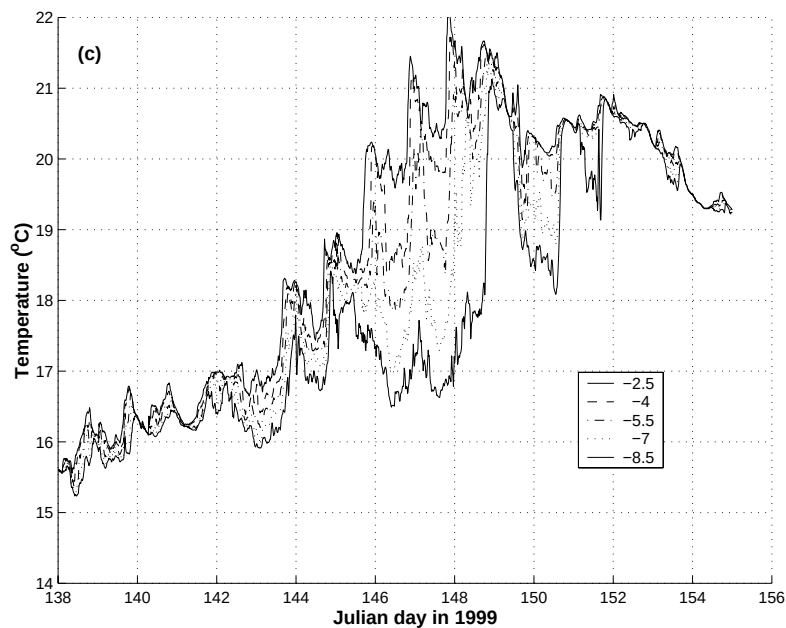


Figure 7.20 (continued).

A six-day portion of the experiment which represented the evolution of the isotherms at stations TII.1 and TII.2 (Fig. 7.21), located at the north and south ends of the Oaks Arm respectively, was selected for the study. At station TII.1 cold water accumulates at the bottom of the water column and isotherms stretch apart in response to the wind event. At station TII.2, on the other hand, isotherms seem to be compressed in response to wind forcing, and warmer water accumulates at the surface. The re-circulation cells, both along and across the longitudinal axis of the Oaks Arm, excited by northwesterly winds and revealed in the velocity transect of Fig. 7.18, provide the driving mechanism for such an evolution of the temperature field. The north-west (NW) currents at the lower levels will carry colder hypolimnetic waters to station TII.1, located at the NW end of the Oaks Arm, while the south-east (SE) currents at the surface will transport warmer water to stations TII.2 and TII.3. Figures 7.20 and 7.21 indicate that significant temperature gradients develop across the Oaks Arm in response to the action of the wind.

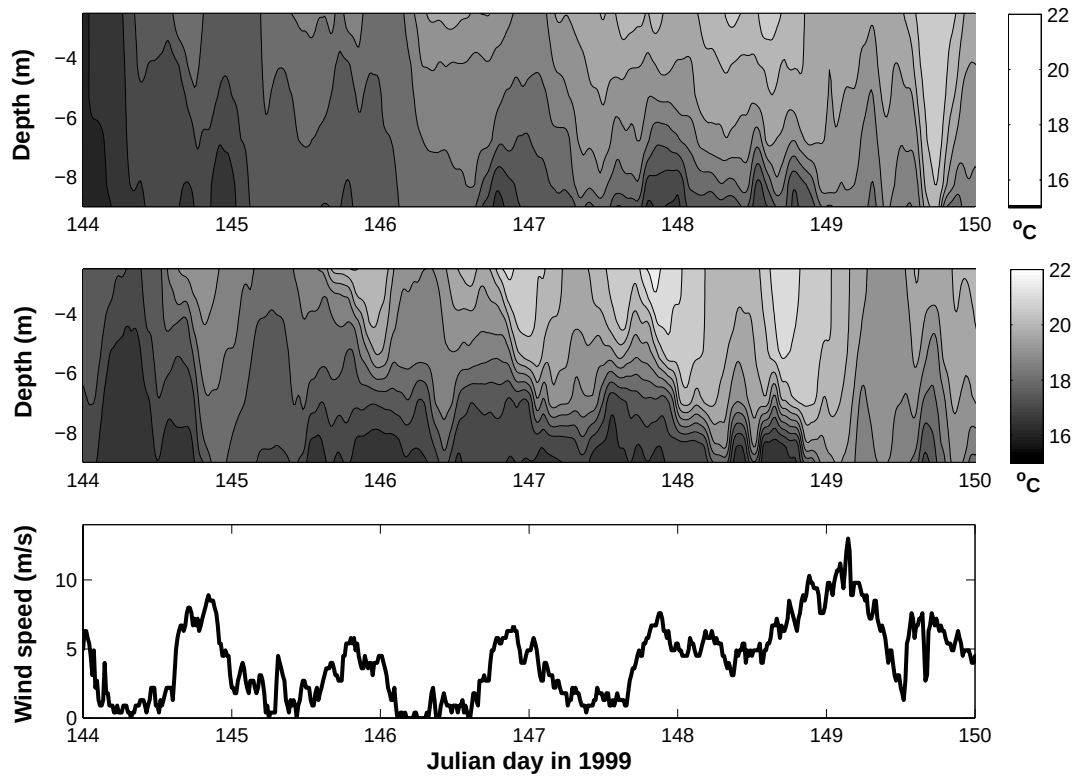


Figure 7.21. Experiment II. Isotherm displacements at stations TII.1 (top) and TII.2 (middle) are caused by diurnal northwesterly wind events during six days of the experiment, starting on May 23, 1999. The time series of wind speed recorded at station W9 is shown in the bottom frame for reference.

A clearer picture arises if the time series of integrated potential energy is used to describe the evolution of a temperature profile. The integrated potential energy (IPE) is calculated from a temperature profile as

$$IPE(t) = \int_{z_0}^{z_1} \rho(z, t) g z dz \quad (7.12)$$

where ρ is density, z is the vertical coordinate positive upwards and measured from the lake surface, and z_0 and z_1 denote two depths in the water column. The analysis of the internal dynamics with IPE does not involve the uncertainty of choosing a representative

isotherm, and the analysis is particularly suitable for temporal scales long enough that significant changes in the water temperature are encountered. Those temporal scales are seasonal scales in large and deep monomictic lakes, but in Clear Lake and other shallow polymictic lakes the time scales are weeks or even days. The heating and cooling of a lake on those scales will appear as low frequency oscillations, which can be removed by filtering or de-trending. Herein IPE has been estimated using $z_0 = -9\text{m}$ and $z_1 = -2\text{m}$ in Eq. 7.12, and it will be negative because of the coordinate system. Neglected are the variations in the total depth due to expansion or contraction of water driven by heating or cooling, or caused by releases at the outlet of the lake, that could influence the estimations of IPE.

The IPE describes the stratification in a given water column and can be used to trace the advection of cold or warm water or even the occurrence of mixing. For example, the advection of cold water into the water column used in calculations of IPE will increase the density and hence will cause a decrease in the IPE. The IPE will increase, on the other hand, in response to the advection of warm water. The time series of IPE was calculated for all stations, de-trended and band-pass filtered to isolate the 24-hour period signal, the characteristic frequency of the wind events. In Fig. 7.22 the time series of the de-trended and filtered IPE at stations TII.1, TII.2 and TII.3 are presented. They show during the period from Julian day 142 to 150, when the lake was stratified, that station TII.1 received cold water in response to daily wind events, reflected in a decrease in the IPE. In stations TII.2 and TII.3, however, the IPE increased after the onset of the wind event and reveals wind mixing and warmer water being transported to the southernmost and

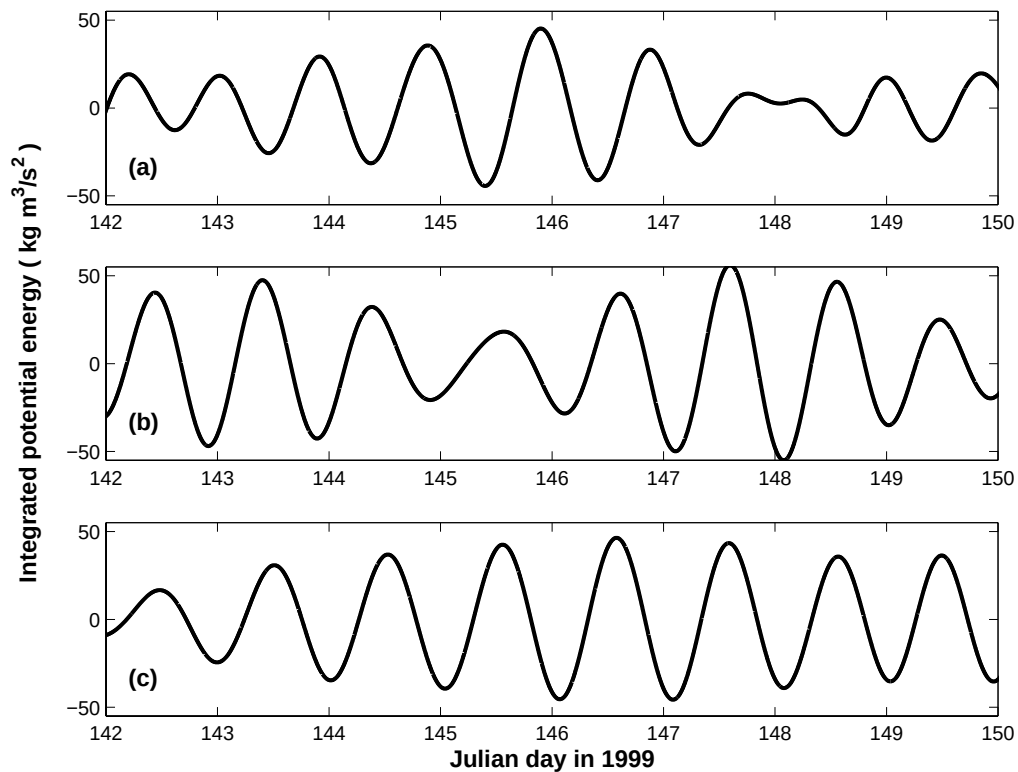


Figure 7.22. Experiment II. Filtered and de-trended time series of Integrated Potential Energy (IPE) at stations (a) TII.1, (b) TII.2 and (c) TII.3 during 8 days of the experiment, when the lake was stratified. The filter isolates the 24h period component

easternmost ends of the Oaks Arm. During calm periods, station TII.1 experiences an increase in IPE, whereas stations TII.2 and TII.3 the IPE decreases, probably indicating the advection of warm water towards the northwest end of the Oaks Arm and the transport of cold water to the southernmost and easternmost ends of the arm. The opposition in phase of the IPE time series is quite remarkable, and it suggests that the warm pool of water accumulated by the wind at the southeast end of the arm moves back to the west after the wind event by following the north shore, probably due to Coriolis effects. Experimental evidence was obtained in one of the velocity transects taken across the Oaks Arm at 1400 hours on May 24, 1999, showing that the return flow that brings

warmer water to the west does follow the north shore of the basin (Fig. 7.23). It is this cycle of setup and relaxation of the temperature gradients modulated by the Earth's rotation, that is termed here baroclinic pumping, and as it will be seen in sections 7.7 and 7.8 may cause a considerable increase in the rate of transport of dissolved and particulate matter within and out of the Oaks Arm.

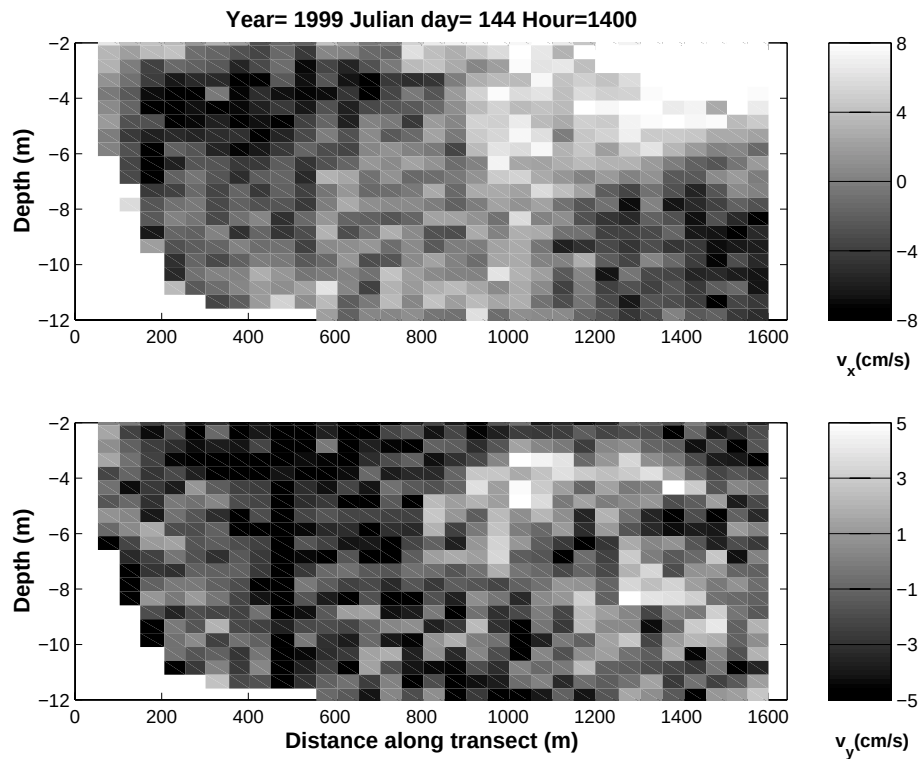


Figure 7.23. Experiment II. Transect taken at 1400h on Julian day 144, 1999. The boat track is identified as RII.4 in Fig. 7.6. The top frame shows the velocity component along the main axis of the Oaks Arm (positive is eastward), and the bottom frame represents the component perpendicular to it. Velocity is given in cm/s. The transect starts on the north shore and ends on the south end of the arm.

7.5.5. Exchange with the other basins

For the analysis of the migration paths of particulate and dissolved species from the mine site to the rest of the lake, the internal circulation within the Oaks Arm was first

considered, followed by an analysis of the exchange of water masses between the Oaks Arm and the other two arms of the multi-basin Clear Lake. To address the second aspect, two ADPs were deployed during experiment I in 1997 (see Fig.7.4) in the channels connecting the Oaks Arm with the Lower Arm (station SI.3) and the Upper Arm (station SI.4). The data collected at the junctions between the basins follow the patterns observed by Lynch and Schladow (Lynch and Schladow 1996). However, in that work the data did not provide any light into the origin of the water masses crossing the limits of the Oaks Arm basin, and the question may need to be addressed using tracers. Attention shall be focused on that question later using numerical tracer release simulations.

The time series of velocity profiles obtained at stations SI.3 and SI.4 are shown as gray scale plots in Figs. 7.24 and 7.25 respectively. The velocity records from station SI.3 show a consistent southward velocity in the surface layers for the duration of experiment I, and a changing flow direction in the lower layers. The water column flows are driven by northwesterly winds to the south towards the Lower Arm during the late afternoon and evening hours, while during calm periods the lower layers in the velocity profiles flow northwards, opposing the surface currents. The east-west component of the velocity vector is positive (eastward) during evening windy conditions but negative (westward) in periods of calm (night and early morning). These observations suggest that water flows either from the Upper Arm or the Oaks Arm, depending on the wind. The uniform

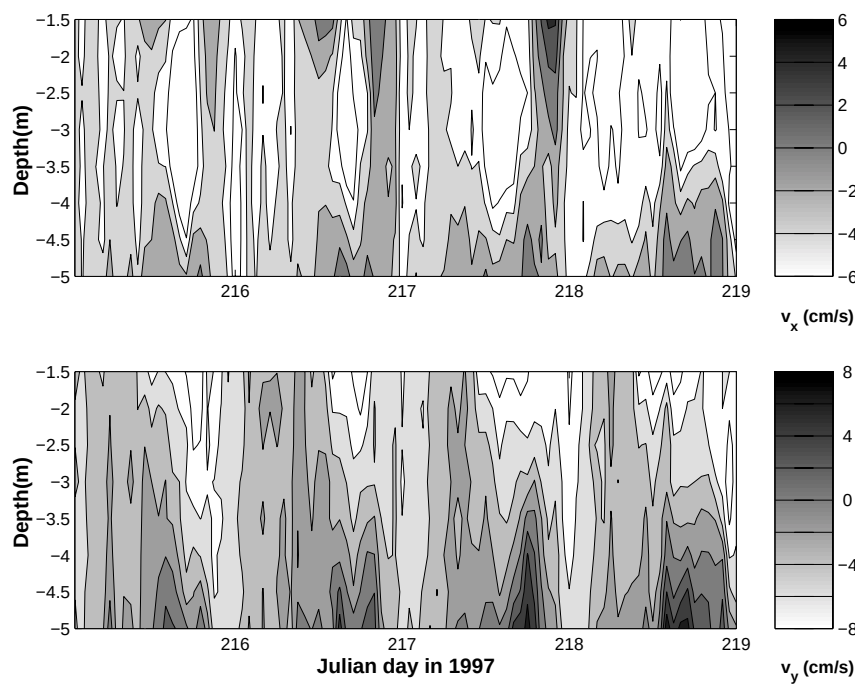


Figure 7.24. Experiment I. Time series of velocity profiles at station SI.3.

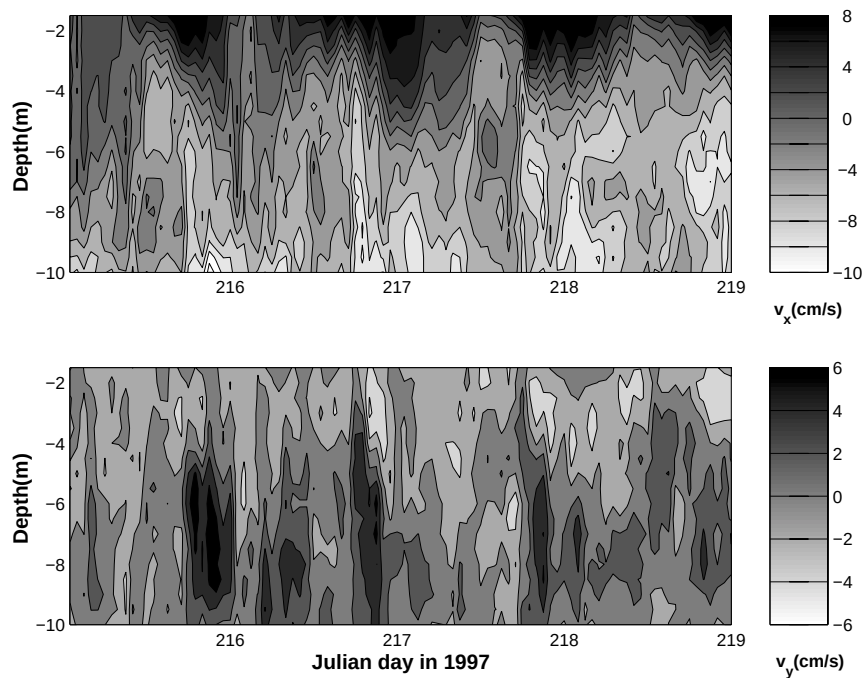


Figure 7.25. Experiment I. Time series of velocity profiles at station SI.4. The classical wind re-circulation pattern is very evident in the east-west direction (i.e. across the Narrows).

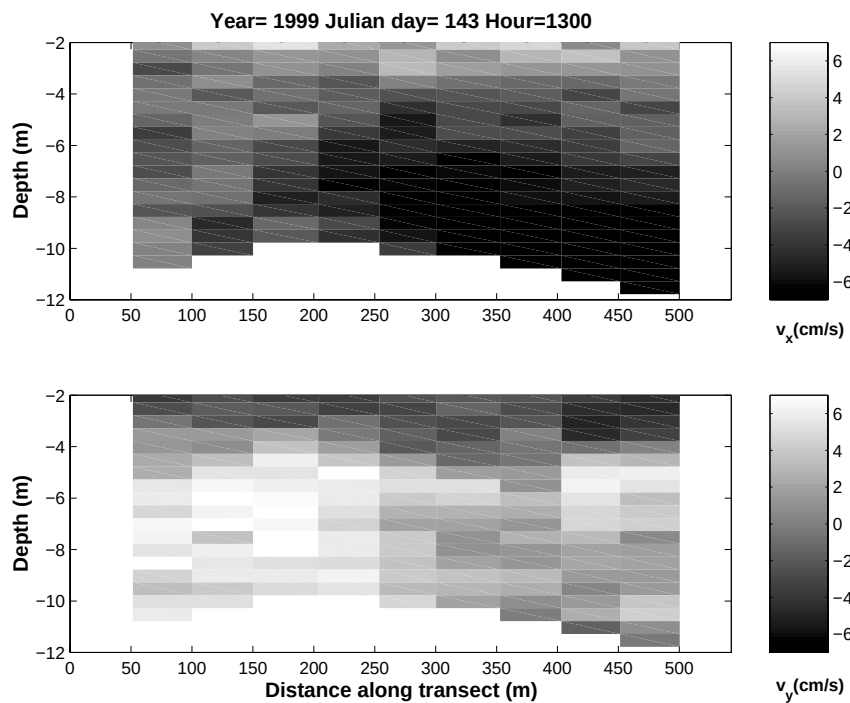


Figure 7.26. Velocity transect across the channel connecting the Oaks Arm and the Lower Arm. The starting point is at Anderson Island, the largest island in that area, and the x-axis represents distance covered by the boat along its trajectory towards the east.

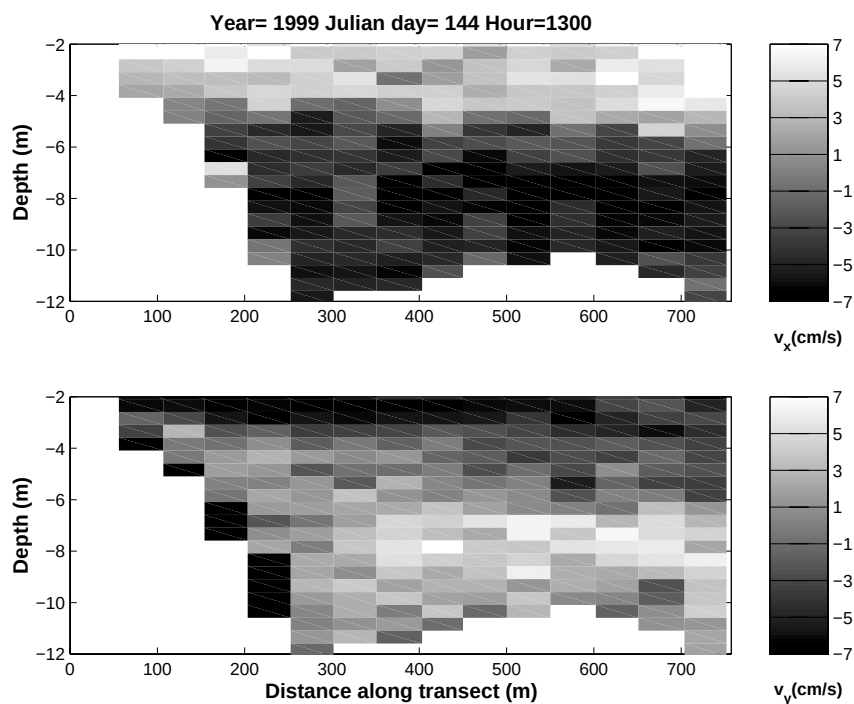


Figure 7.27. Velocity transect across the Narrows. The velocity components v_x and v_y are along, and perpendicular to, the main axis of the Oaks Arm.

southward velocity in the water column at station SI.3 during windy periods of experiment I is probably due to the shallow location of the instrument, as it was kept out of the main channel to avoid heavy boat traffic. At deeper locations in the channel the vertical velocity profile would probably show the classical wind re-circulation pattern, with the thickness of the southward surface currents varying according to the magnitude of the wind speed. In the velocity transect taken across the channel connecting the Oaks Arm and the Lower Arm at about 1300h on Julian day 144 during a windy period in experiment II, the re-circulation pattern was evident all across the channel (Fig. 7.26).

In the Narrows the velocity sensors recorded a classical wind exchange pattern, with windward motion at the surface layer and return flow in the deeper positions. The depth of the surface layer varied on a diurnal basis in response to the intensity of the wind stress, with larger magnitudes in the evening hours and smaller ones during the night and early morning. The velocity transects in the field (Fig. 7.27) showed a generalized re-circulation pattern all across the Narrows with very small variations in the horizontal. In addition to the diurnal response, the Fourier analysis of the velocity signal of all bins showed a characteristic period of approximately 3-4 hours, representing the first mode barotropic seiche in Clear Lake (see appendix B).

7.5.6. Drogues

The drogue tracks from experiment II are shown in Fig. 7.28. The release location of each drogue is shown in the figures by a dot, and the path is described with a continuous line beginning at the release point. The elapsed time between sequential positions of the

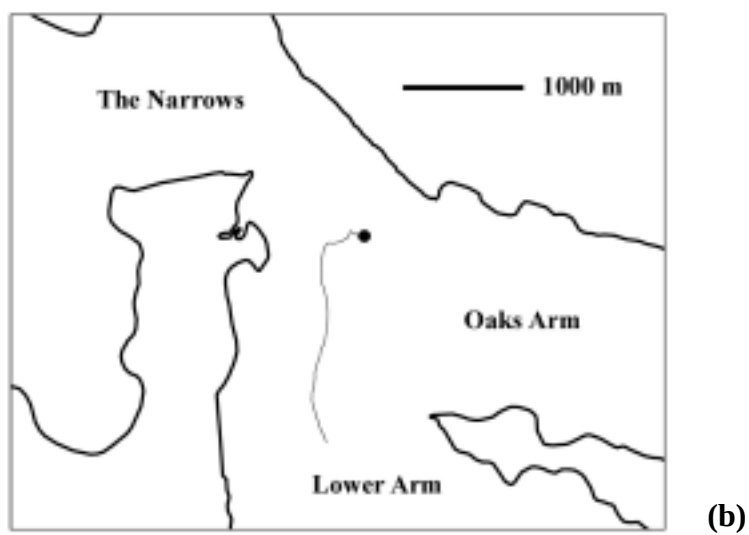
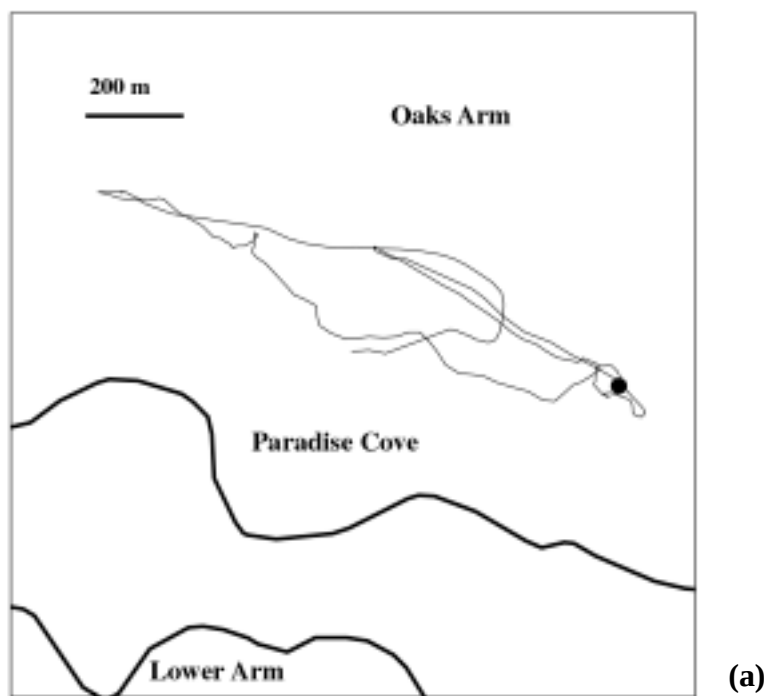
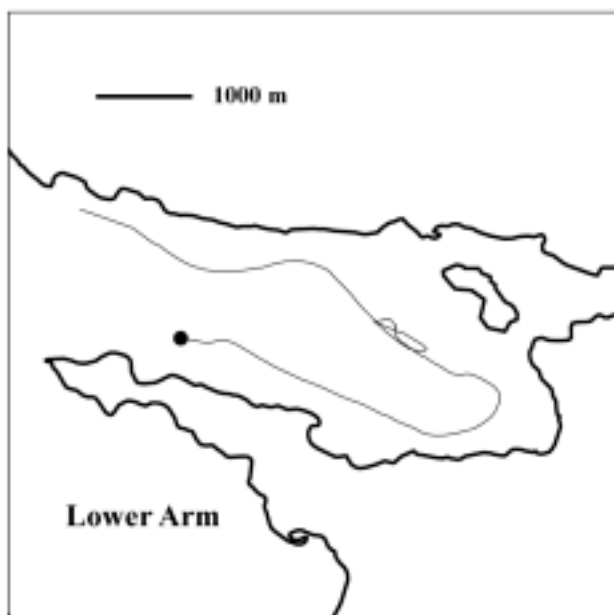
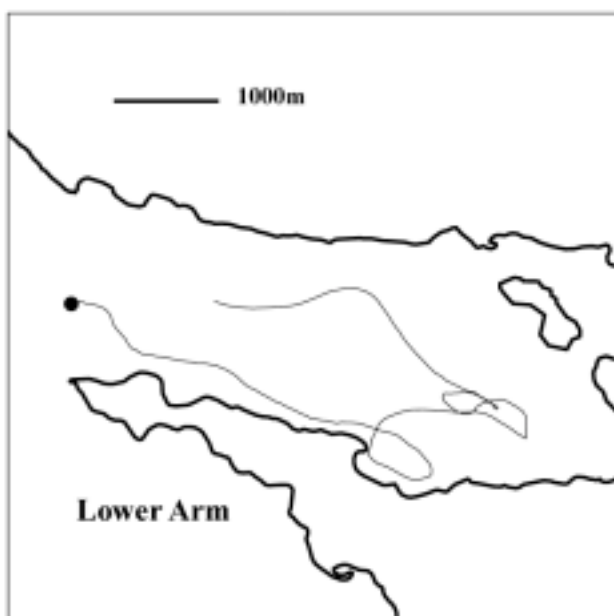


Figure 7.28. Trajectories followed by the drogues during experiment II: (a) drogue 1, (b) drogue 2, (c) drogue 3, and (d) drogue 4.



(c)



(d)

Figure 7.28 (continued).

drogues is fixed and equal to 15 minutes, and from those records it is possible to calculate the Lagrangian velocity magnitudes and directions and compare them with the wind records (Fig. 7.7b). The time series of the Lagrangian velocity components in the east-west (positive toward the east) and north-south (positive towards the north) directions are shown in Fig. 7.29 for three of the four drogues (1, 3 and 4) released in the experiment.

One of the surface (2 m deep) drogues (drogue 2 shown in Fig. 2.28b) was deployed late in the evening of Julian day 142, and it moved in a southward direction and into the Lower Arm where it was retrieved early the next day. This result agrees with the current meter data shown in Fig. 7.24, which suggests a very active exchange between the basins, and that water in the region connecting the three basins to the east of the Narrows flows with a strong southward component towards the Lower Arm or the southern part of the Oaks Arm. The trajectories followed by the other two surface drogues (Fig. 7.28 c and d) reveal a basin-scale counterclockwise gyre in the Oaks Arm that dominated their behavior. Orbital motions on spatial scales smaller than the basin were also present, as seen in the trajectory of drogue 4, which reveals a separation vortex in the embayment at the south end of the Oaks Arm. These vortices are caused by shoreline irregularities in the lake, such as side embayments, which can induce enhanced rates of longitudinal dispersion by “trapping” and later releasing particles that in general follow the large-scale circulation. Fischer et al. (Fischer *et al.* 1979, p. 241) describe a similar mechanism in estuarine environments, which they refer to as tidal “trapping”. In that case the entrapment of particles could be caused by phase differences in the velocity field between the main channel and side embayments or small branching channels.

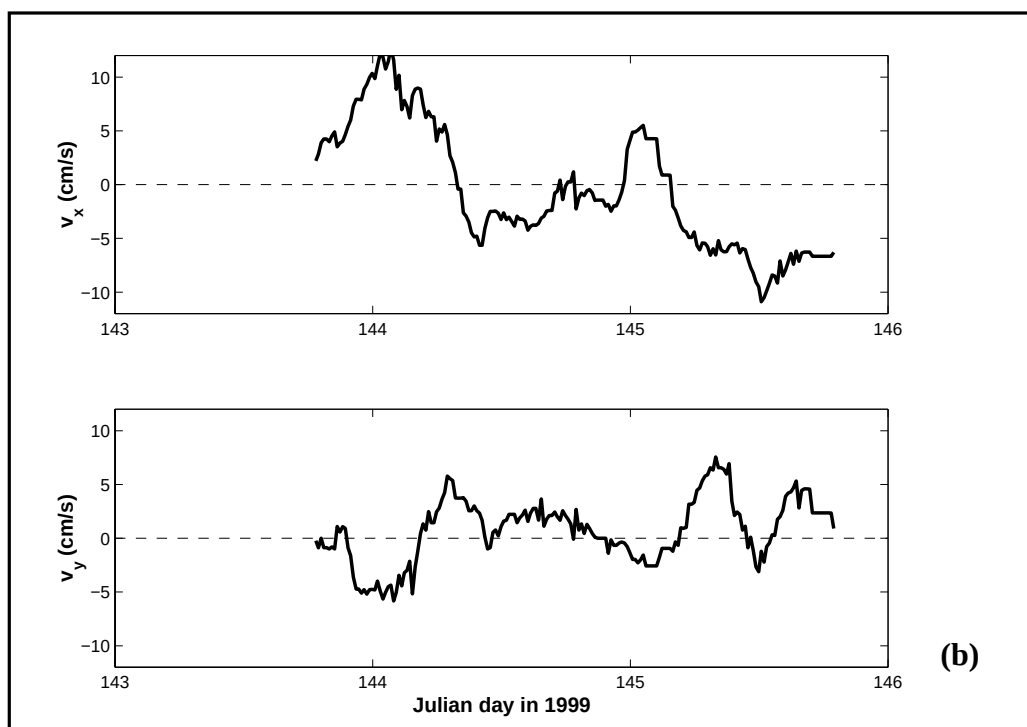
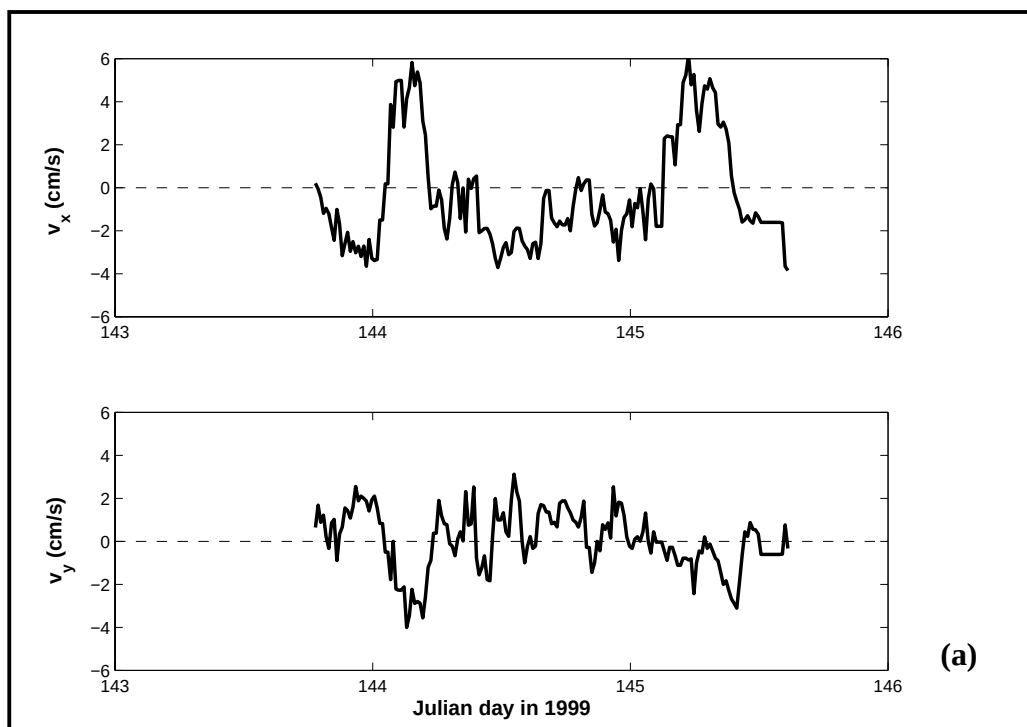


Figure 7.29. Time series of Lagrangian velocity components: (a) drogue 1, (b) drogue 3, and (c) drogue 4

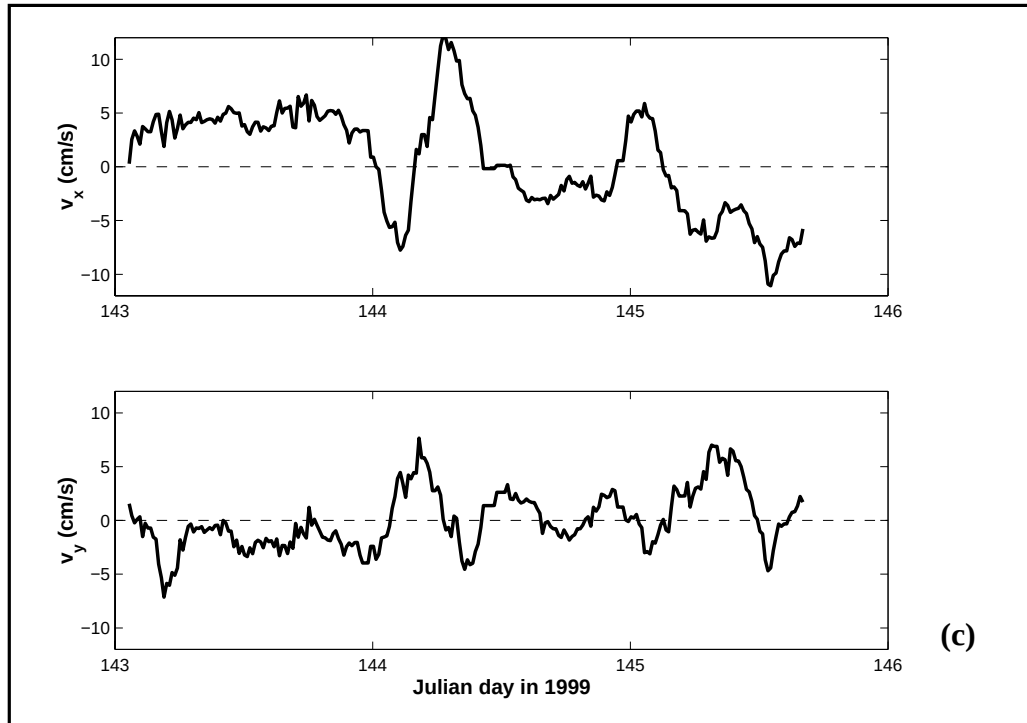


Figure 7.29 (continued).

Okubo (Okubo 1973) developed a simplified model of the entrapment mechanism that is independent of the hydrodynamic features causing the phenomenon. In that model the effective longitudinal diffusivity B , in a uniform flow of velocity u_0 caused by identical and uniformly distributed “traps” obeys the expression

$$B = \left\{ \frac{1}{1 + \varepsilon} + \frac{\varepsilon}{(1 + \varepsilon)^3} \frac{u^2}{Ak} \right\} A \quad (7.13)$$

where ε is the ratio of trap volume to channel volume, k^{-1} is the characteristic exchange time between traps and the main flow, and A is the constant eddy diffusivity in the main channel in the absence of traps. For the dispersion to be enhanced, there must be a mean flow passing the trap with a velocity larger than $(1 + \varepsilon)(Ak)^{1/2}$. If one assumes that the

gyre in the Oaks Arm is a steady feature and applies Okubo's model to one of its branches, the width of the channel could be roughly taken as half of the width of the basin, i.e. approximately 1 km. A reasonable value for the width of the embayments is about 100 m, giving a ratio, ε , of approximately 0.1. The time lapsed since drogue 4 reversed its eastward motion and moved west to the time when the drogue's velocity changed sign again after describing a clockwise loop in the embayment in the south of the Oaks Arm is approximately 5 hours, which will be taken as a characteristic exchange time. Lynch and Schladow (Lynch 1996, p. 53) estimated, from tracer release experiments, a value of the diffusivity of approximately $10 \text{ m}^2/\text{s}$ for non-stratified conditions. If one uses the values of ε , k^{-1} and A indicated above for the Oaks Arm, the velocity in the gyre (taken as approximately 10 cm/s) is enough for the trapping mechanism to enhance the dispersion. A simple calculation indicates that the dispersion coefficient could be at least twice its value in the mean flow, due to lateral embayments, and suggests that the effects of shoreline topographic features should be resolved, by using small cell sizes, or at least parameterized in any hydrodynamic model.

The Lagrangian east-west velocity component of drogue 3 typically shows two peaks within a 24h period (Fig. 7.29b): one of negative sign (westward) at about noon after the wind has reached its minimum, and the second, this time of positive sign after midnight, following the peak in wind speed. This oscillatory behavior reveals the basin's longitudinal baroclinic response to the evening westerly winds in Clear Lake. The westerly wind pushes warm surface water towards the easternmost end of the Oaks Arm, causing the drogues to move towards the mine site (Fig. 7.28c) during the evening hours

and setting up horizontal temperature gradients in the water. The relaxation phase of the cycle described in Fig. 7.15 causes the surface current to flow westward and hence causes the drogues to move away from the mine. The eastward motion of drogue 3 tended to be associated with southward displacements. As a drogue moving to the east decelerated, it drifted north, revealing the lateral circulation sketched in Fig. 7.19. A similar behavior was observed in drogue 4, though in this case it was complicated by the trapping mechanism discussed above. Both drogues 3 and 4 experienced eastward velocities during Julian day 144 during the whole night and early morning hours until approximately 800h, with peak values at about 200h for drogue 3 and at about 600h for drogue 4. The lag in time of the positive peak in the drogues' velocity (at night hours) relative to the maximum wind speed (late in the evening) could be due either to the inertia of the water currents, or most probably to baroclinic circulation. These features of the flow field, revealed in the drogue trajectories, will be further analyzed in the results from SI3D-L.

It is interesting to notice that the surface eastward motions are stronger (up to +12 cm/s) while the surface drogues are located in the southern part of the Oaks Arm, while the westward (negative) Lagrangian velocities reach higher magnitudes (up to -10 cm/s) when the drogues have traveled around the basin describing a counterclockwise trajectory and are sampling the velocity field on the northern part of the Oaks Arm. The drogue motions are weaker in general in the easternmost end of the Oaks Arm, suggesting that wind and baroclinic circulations are weaker in this end of the basin and stronger at its center, as would be expected given that the proximity of a boundary reduces the

magnitude of the longitudinal horizontal velocities. The asymmetric behavior of the drogues in the north and south portions of the basin, on the other hand, indicates that the baroclinic circulation is stronger in the north while the circulation in the south is dominated by the wind. The asymmetry in the large-scale circulation of the Oaks Arm will be further studied in section 7.6 when analyzing the numerical results.

The magnitude of the deeper currents revealed in the track of drogue 1 (Fig. 7.29a) was typically less than half those of the surface currents. Over the two days of the deployment, the drogue oscillated back and forth, parallel to the longitudinal axis of the Oaks Arm with peak velocities of up to 6 cm/s. The motions tended to be 180 degrees out of phase with the surface currents, as seen by comparing Fig. 7.28a and Fig. 7.28c showing the trajectories of drogues 1 and 3, both released at the same time and horizontal location but at different depths (7 and 2 meters respectively). This behavior indicates that the oscillatory motions of drogue 1, at the bottom of the lake, were also part of the basin's longitudinal baroclinic response to the wind, illustrated in Fig. 7.15. During the evening periods of both days 144 and 145, when the lake was been subjected to westerly winds, the drogue moved upwind (to the west), changing direction and moving to the east after the wind decreases in the night and early morning hours (Fig. 7.29a). The bottom drogue tended to respond to changes in the wind faster than the surface drogues, due to the smaller velocity magnitudes, and hence less inertia. It is very interesting to note that the magnitude of the eastward velocities, though driven by baroclinic forces, was larger than the westward wind-driven motions.

7.6. Modeling the hydrodynamics of Clear Lake: the Oaks Arm

Here and in the next few sections are presented the results of simulations carried out with the 3-D hydrodynamic model described in chapters 4 and 5 (i.e. the extended version of SI3D for lake environments, or SI3D-L). The general picture outlined in section 7.5, and revealed by the experimental information, is supported by the simulation results and emerges as a very plausible model of the circulation in the Oaks Arm.

7.6.1. Model setup

The domain of Clear Lake was discretized by constructing a grid with 100 m resolution in the horizontal and 1 m in the vertical direction. The geometry of the lake was previously rotated counterclockwise 27.5 degrees so as to reduce the number of grid cells and the ratio of wet to dry cells. The computational grid is a prismatic box having $282 \times 115 \times 16$ (= 518,880) cells, of which only a small fraction, about 20% (100731 cells), are active (wet). Figure 7.30 shows the discretized version of the bathymetry of Clear Lake used in the model, obtained by linearly interpolating the bathymetric information, available at irregularly spaced locations, to the spatial locations of the grid points. The time step was set to $\Delta t = 180$ s. In each time step, two trapezoidal iterations were performed after the initial leapfrog solution, which was the operational sequence that gave best results in the initial experiments with the code for stratified basins, shown in chapter 5. No instabilities were observed in the course the of simulation, indicating that the level of spatio-temporal discretization was appropriate.

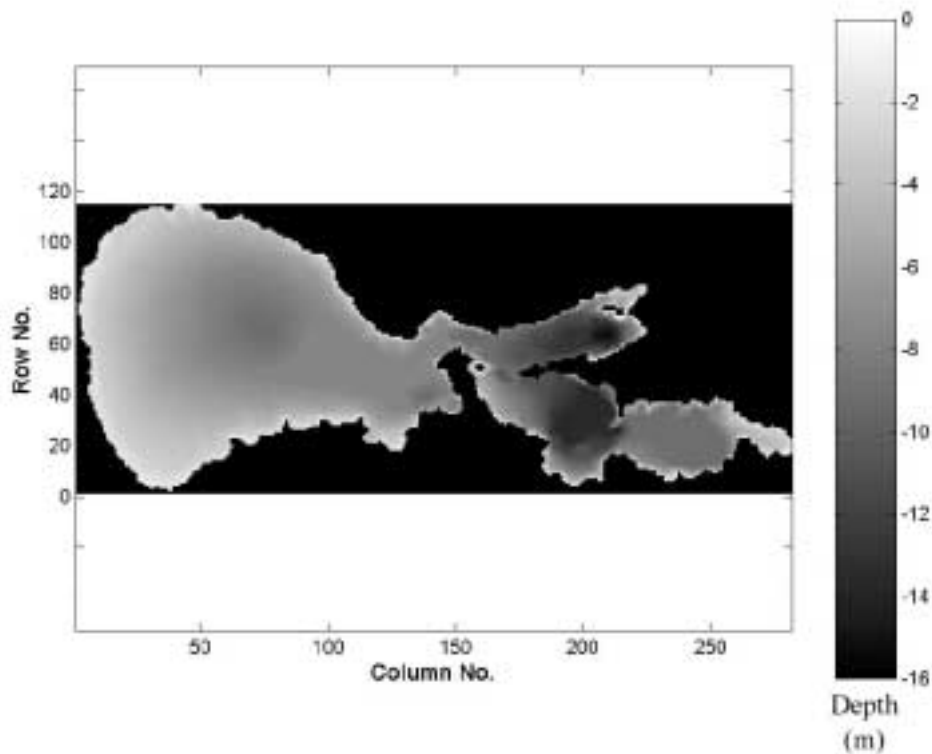


Figure 7.30. Bathymetry of Clear Lake supplied to the 3-D model. The geometry is rotated 27.5 degrees counterclockwise to reduce the dimensions of the computational grid and increase the fraction of wet grid cells.

It was assumed in the model that the incoming solar irradiance was absorbed in the first 1-m layer. This is a good approximation in Clear Lake where Secchi depth measurements during the past years (1-2 meters) suggest attenuation coefficients of about $0.25\text{--}0.5\text{ m}^{-1}$. The code was driven by meteorological data collected as indicated in section 7.3.2. There was assumed to be no cloud cover.

The wind records were used to construct wind fields over Clear Lake with a multiquadric interpolation scheme (Nuss and Titley 1994, see chapter 4). The wind records from stations located on the lake where instruments were deployed for the length of the experiment II (marked as W5, W8, W9, W13 and W14 in Fig. 7.6 over the Oaks and

Lower Arms) were previously corrected for height above the water surface using an expression given in Amorocho and DeVries 1980 for neutral conditions. The wind records from other stations were not corrected, as little information was available on the physical characteristics of the sites. The interpolated wind fields were constructed every 15 minutes and supplied to the 3-D model as a boundary condition. The model interpolates linearly in time between consecutive wind fields to obtain the wind forcing at the time of computations. The wind drag coefficient was set to follow the law proposed in Amorocho and DeVries 1980, which is given in Eq. 4.38 as a function of wind speed, to account for the effects of the free surface roughness. No adjustments to the drag coefficient were made to allow for the influence of diabatic conditions in the wind drag. The bottom drag coefficient was selected as 0.002, which corresponds to a roughness length $z_0 = 0.02$ cm, appropriate for a mud bottom (Smith 1997, p.52). The hydrographic survey of Clear Lake carried out in 1948 by the U.S. Coast & Geodetic Survey (Survey 1948-1949) states that “a thin layer of ooze overlies most of the lake bottom reaching a thickness of about 3 feet”. This, together with direct and casual observations by other researchers involved in the study of Clear Lake (Nelson 2000), supports the choice of a mud bottom as the correct bottom boundary condition for the model. Given the fairly good agreement obtained between model results and observations, no sensitivity analysis of model results with respect to the value of the bottom drag coefficient was attempted. The horizontal eddy viscosity was set to a constant value $A_H = 1 \text{ m}^2/\text{s}$.

The model was initialized with a uniform temperature of 15.5°C , and the period of the simulation started on day 139 at 000 hours and extended for eight days. Field

observations (section 7.5) support the selection of a well-mixed water column throughout the lake as the initial condition. Missing data from several wind stations after day 147 made it unfeasible to extend the simulation period with a spatially variable wind field until the end of the experiment on day 154, and hence the simulated period was ended on day 147. The model was run on an SGI Octane with a 175 MHz IP30 processor and 642 Mbytes of memory. Each run of the 3-D hydrodynamic code (3840 time steps) took about 18 hours to completion, giving a ratio of computational to real time of about 1:10.

7.6.2. Three-dimensional temperature simulations

7.6.2.1. Validation of model results

The temperature data collected at stations TII.1, TII.2 and TII.3 in May 1999 were used for model validation⁴⁸ purposes. Only results from the last six days of simulation were used in the validation exercise. The first two days were not included in the exercise to allow the model to spin-up. The 1-minute observed temperature records were smoothed by using a moving average filter with 15 points, so as to remove high-frequency oscillations and create 15-minute averages. The smoothed temperature records were then resampled to give a time series with data points every quarter of an hour, to replicate the output frequency from the 3-D model, also set to 15 minutes. The evolution of measured and simulated profiles at stations TII.1, TII.2 and TII.3 for days 142 through 147 is shown in Figs. 7.31(a), 7.31(b) and 7.31(c). Figure 7.32 presents simulated versus observed temperatures during 5 days (days 142 to 146) at 6 different depths at the three validation stations.

⁴⁸ The word validation is used here instead of calibration, since no attempt was made to improve the model results by adjusting parameters. In fact, the model is free from adjustable parameters other than the bottom drag coefficient, which in the present application was taken from the literature.

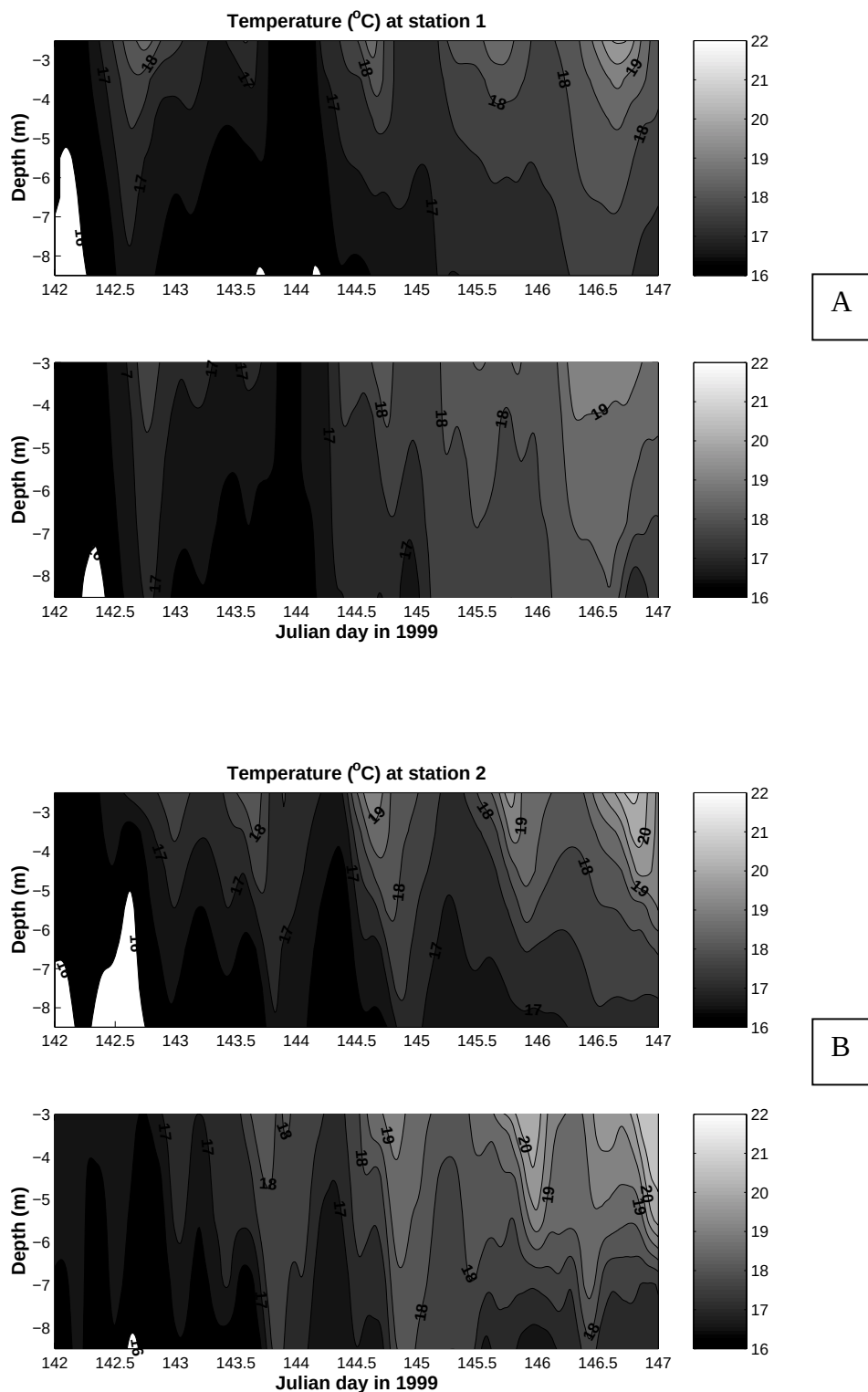


Figure 7.31. Observed (top) and simulated (bottom) evolution of the temperature field during 5 days of experiment II at station (a) TII.1, (b) TII.2, and (c) TII.3.

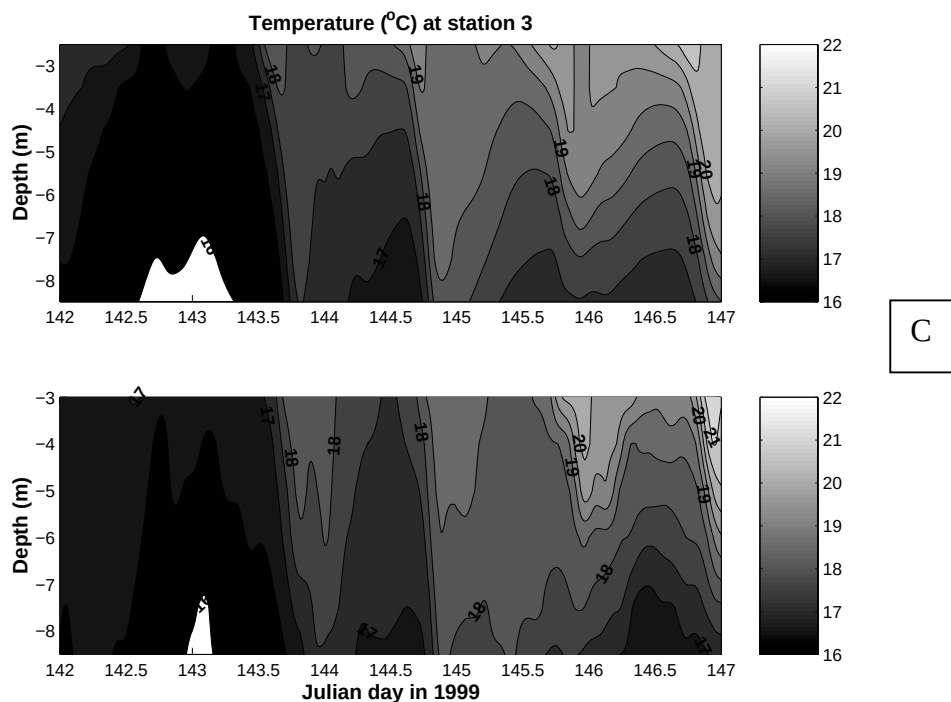


Figure 7.31 (continued)

The agreement between observed and simulated water temperatures was quantified by using the ℓ_1 , ℓ_2 , ℓ_∞ error norms and the root mean square error (Table 3.7⁴⁹). Table 7.8 presents the error norms at TII.1, TII.2 and TII.3 that result from comparing observed and simulated records during the period of validation at all depths. The agreement is rather satisfactory when compared with the results of other 3-D modeling studies published in the literature. Jin et al. (Jin *et al.* 2000), for example, in an application of a 3-D hydrodynamic model to Lake Okeechobee, found a minimum RMSE of 0.7, with a minimum data range of 6 °C. The maximum RMSE encountered in the comparison was 0.6 in station TII.2, where the temperature values had a range of approximately 5.4 °C. The Lake Okeechobee simulation was for a total of 30 days, and in this comparison

⁴⁹ For the purpose of comparison of modeled and observed records, the observations are assumed to provide the reference or “analytical” solution in the definition of the error measures in Table 3.7.

exercise only the records at one given depth were used. Lake Okeechobee is a larger and shallower system than Clear Lake, though its geometry is much simpler. Another example of 3-D lake modeling studies is that of Hodges et al. (Hodges *et al.* 2000). A 3-D semi-implicit model (ELCOM) was used in that application to simulate the internal dynamics of Lake Kinneret. The lake is ovoid in shape, the north-south length is 20 km, east-west width is 12 km, and the maximum depth is 42 m. The ℓ_1 and ℓ_2 norms encountered in the comparison of the simulated and the observed temperatures in Lake Kinneret during 8 days were one order of magnitude larger than those shown in Table 7.8. The average ℓ_1 and ℓ_2 norms from 8 temperature stations were $\ell_1 = 0.125$ and $\ell_2 = 0.164$). The high values of the ℓ -norms were attributed, in that work, to phase errors. In another application of a semi-implicit model (Gross 1997, Gross *et al.* 1999), this case to San Francisco Bay, Gross compared simulated and observed time series of salinity and found a ℓ_1 norm of 0.019, on the same order as that found in this work. The San Francisco Bay simulation was for a period of 10 days, and the salinity values observed during that period ranged from 20 to 25 ppt.

	ℓ_1	ℓ_2	ℓ_∞	RMSE	Data range
TII.1	0.020	0.024	0.072	0.414	5.52
TII.2	0.028	0.035	0.078	0.608	5.36
TII.3	0.018	0.024	0.082	0.415	4.13

Table 7.8. Error norms of the three-dimensional temperature simulations. The results and the data used are given in $^{\circ}\text{C}$.

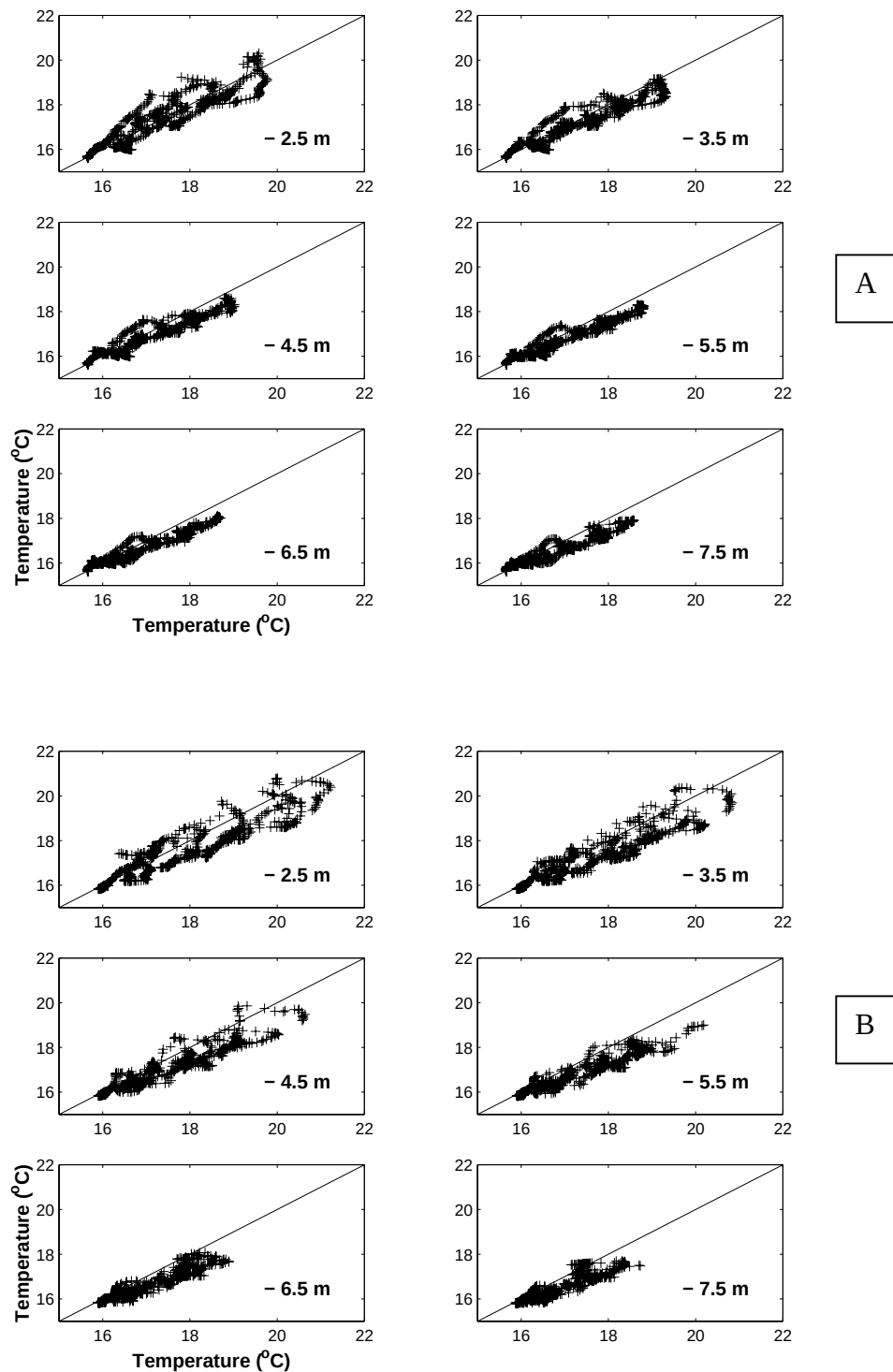


Figure 7.32. Simulated (y-axis) versus observed (x-axis) temperatures at 6 depths in stations (a) TII.1, (b) TII.2 and (c) TII.3. The continuous line has a 1:1 slope and represents complete agreement. The time period represented by the plotted points extends from Julian day 142 to 147.

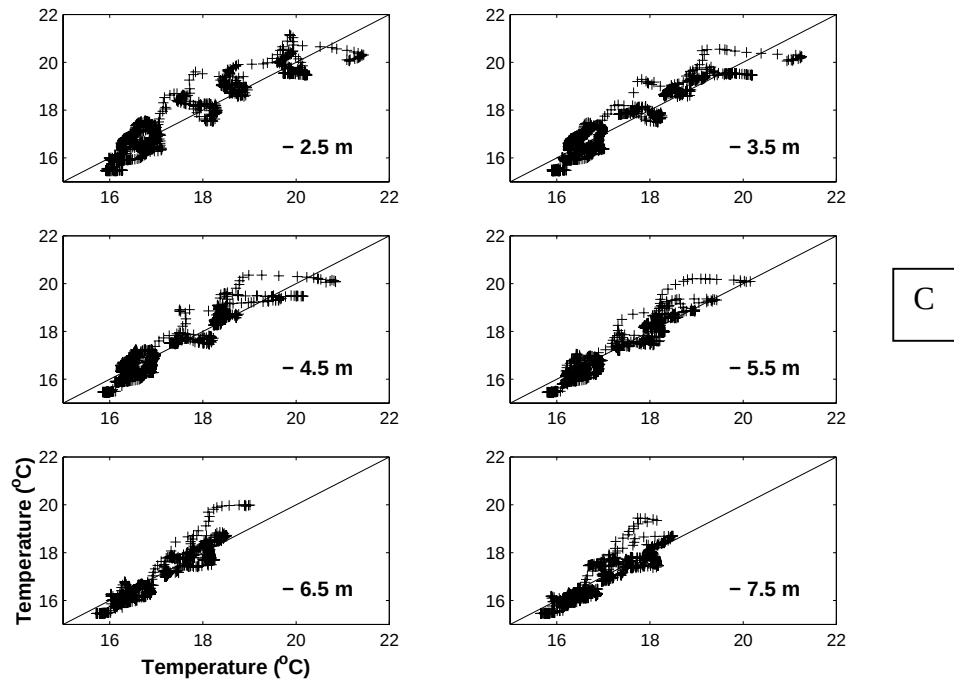


Figure 7.32 (continued).

A statistical analysis of the model performance was also conducted using observations and simulations at all depths at station TII.3 for the validation period. The analysis consisted of fitting, using the least-square method, a regression line of the form $y = \beta_1 x + \beta_2$ through the points in a plot of observed versus simulated temperature records, and studying the goodness of the fit. In Fig. 7.33 the simulated temperatures at station TII.3 at all depths are plotted against their observed counterparts. The discontinuous lines define the confidence band at the 95% level of the regression linear model, which includes the 1:1 slope line representing complete agreement. The confidence limits should be interpreted as follows. If one were to predict observations from SI3D-L simulations, the observed temperatures should be in 95% of the cases within the confidence intervals. The confidence band at the 95% level includes a range of temperatures of approximately 1.5

°C centered at the regression line. The regression line fitted to the data had a slope of 1.02 and an intercept of -0.4 . In the next section is analyzed the possible causes of the model error. Even though the model contains error, the agreement in phase was good as revealed in Fig. 7.31, which supports the use of the model in the study of the hydrodynamics of Clear Lake.

7.6.2.2. Sources of error

The comparison between model and field temperatures indicated good agreement, according to different error measures and a statistical analysis. Errors exist, though, that cause the model results to be below the observations, especially at stations TII.1 and TII.2. In this section the possible causes are reviewed. Given the good results obtained with the model in reproducing analytical solutions and the results of controlled laboratory experiments, one must conclude that the errors are most probably caused by a somewhat deficient representation of the initial and boundary conditions over the lake.

In recent modeling applications (Cook 2000) it has been pointed out that the spin-up period has a considerable influence on the agreement between observations and simulations, suggesting that natural aquatic systems have a “long” memory, and the effects of events, such as storms, may shape and influence the hydrodynamics of those systems for periods of up to weeks. In the present application, unfortunately, complete meteorological information was only available for a limited period of time prior to the start of the experiment, so that it was not possible to test the influence of the spin-up period in the results. It is hence expected that any error in the representation of the lake

initial conditions might affect the results. In modeling the internal waves of Lake Kinneret, Hodges et al. (Hodges *et al.* 2000) only allowed 24 hours as spin-up time, to minimize the effect of the initial conditions. Though the errors were attributed to a flawed representation of wave phases (Hodges *et al.* 2000), part of the disagreement could have been caused by the inadequacy of the initial conditions. The memory in Lake Kinneret, two or three times deeper than Clear Lake, could be more prolonged because of its higher inertia.

In the application of SI3D-L to Clear Lake it was assumed that the temperature field at the beginning of the simulation period was uniform and equal to 15.5 °C. A careful analysis of the observations, though, shows that the temperature field was not uniform on day 139. There existed a certain amount, though weak, of stratification in the water column, and the degree of stratification was different, depending on the station. Furthermore, the mean temperatures at stations TII.1, TII.2 and TII.2 differed, thus also revealing differences in the horizontal directions. The mean temperatures at stations TII.1, TII.2 and TII.3 were 15.017, 15.742, and 16.005, respectively. It is difficult, though, to construct a realistic temperature and velocity field to be provided to the model as initial conditions, and this is the reason why in most modeling applications the simplest approach, i.e. separating the validation period as much as possible from the initiation of the simulation, is adopted.

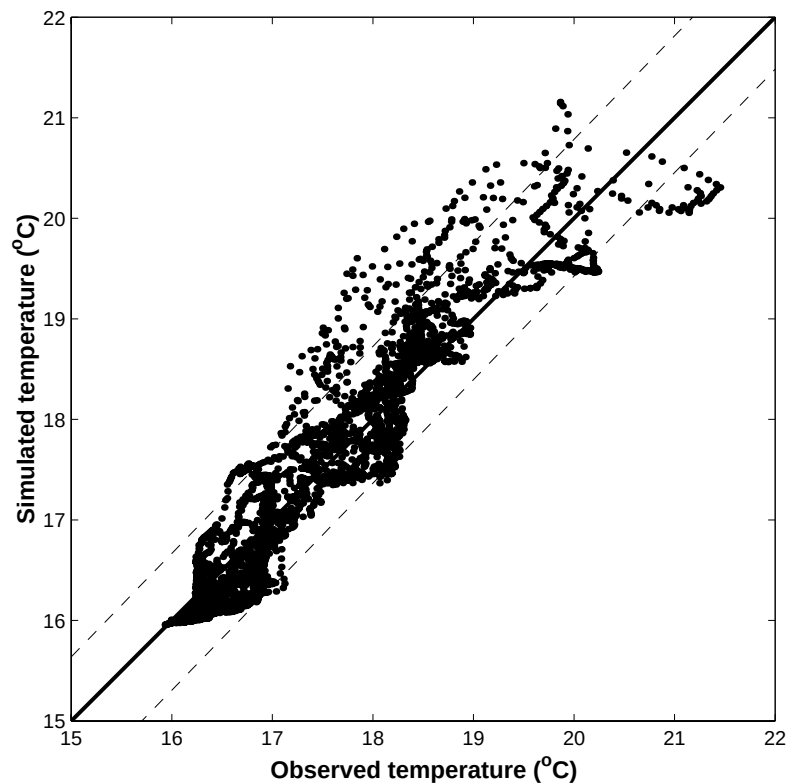


Figure 7.33. Simulated (y-axis) versus observed (x-axis) temperatures at station TII.3 at all depths in experiment II. The solid line has a 1:1 slope and represents complete agreement between model and observations. The dashed define the limits of the 95% confidence band for the regression line. The time period represented by the plotted points extends from Julian day 141 to 147.

It is probably not an overstatement to say that an important part of the success of a modeling exercise lies in the characterization of the boundary conditions. One of the boundary conditions needed in the model is the wind stress field, calculated by using the wind speed and directional information. The wind field on the free surface was constructed by spatially interpolating wind records from 14 different stations. Those records, in the first place, contain measurement errors associated with equipment limitations or the location of the meteorological stations. Some of the wind stations were located in urban areas surrounding the lake, where the wind probably differs from that

existing over the lake. Some of the sensors only provided a coarse description of wind direction. Even if the interpolation routines were perfect, the measurement errors would propagate spatially and would give an inaccurate description of the wind field. The interpolation method is not, though, flawless, and one could generate an imperfect wind field from exact wind records at several locations. The interpolation error has not been studied in this work, though it would be desirable to carry out that exercise, possibly using non-parametric statistical techniques, such as cross-validation (Efron and Gong 1983). Other meteorological variables used in the construction of the boundary conditions for the temperature transport equations were collected in the city of Kelseyville, located approximately 3-4 km from Clear Lake. The atmospheric conditions at this location are probably not representative of the conditions existing over the lake surface, which are needed for an accurate calculation of the heat fluxes. This is especially true for variables such as air temperature or relative humidity, whose values can have a profound impact in the calculations of the heat fluxes.

7.6.2.3. Validation of the conceptual model

Figures 7.34 and 7.35 show the simulated temperature fields 2.5 meters below the surface at 2200 on Julian day 145, immediately following the wind event, and 14 hours later (at 1200 hours on day 146). They reveal the presence of significant temperature gradients that evolve, the evolution being dictated by the wind conditions. The northwesterly winds set up horizontal temperature variations of about 2.5–3 °C in the Oaks Arm, with higher temperatures to the south and east within the basin. The warm water mass returns during the night and early morning calm periods to the northwest by following the north shore, and this movement will result in higher temperatures at the north than in the south of the

Oaks Arm during early morning hours. Lines of foam and suspended material that were aligned with the isotherms in Fig. 7.35 were observed on the mornings during the second experiment.

To evaluate the influence of the earth's rotation on the setup and evolution of the temperature gradients, SI3D was run with the Coriolis parameter set to zero. Figures 7.36 and 7.37 show the results of these simulations at 2200 hours on day 145 and 1200 hours on day 146, as in Fig. 7.34 and Fig. 7.35. The horizontal temperature gradients resulting from the northwesterly winds are aligned with the longitudinal axes of the Oaks Arm, and lateral temperature variations are not evident 2.5 m below the surface. Another simulation, using a spatially uniform wind field as measured at station W9 over the Oaks Arm to test the effect of the variability of the wind field on the results. Figures 7.38 and 7.39 show the simulated temperature field 2.5 m below the surface, obtained at 2200 hours on day 145 and 1200 hours on day 146. The Coriolis term is operational in these two simulations. No major differences can be observed between the temperature fields in the Oaks Arm obtained with a variable or a uniform wind field, which strongly suggests that the northwesterly wind field and Coriolis effects jointly control the evolution of the temperature field in the Oaks Arm. The temperature fields of the Upper and the Lower Arms, however, do display significant differences between the uniform- and the variable-wind cases.



Figure 7.34. Simulated temperature field ($^{\circ}\text{C}$) at 2.5 m below the surface immediately after a wind event at 2200 hours on day 145, 1999.



Figure 7.35. Simulated temperature field ($^{\circ}\text{C}$) at 2.5 m below the surface at noon on Julian day 146 in 1999, after the relaxation of horizontal temperature gradients.



Figure 7.36. Simulated temperature field ($^{\circ}\text{C}$) without rotational effects at 2.5 m below the surface immediately after a wind event at 2200h on day 145, 1999. The temperature gradients across the Oaks Arm do not appear.



Figure 7.37. Simulated temperature field ($^{\circ}\text{C}$) without rotational effects at 2.5 m below the surface after relaxation of density gradients, at 1200 hours on Julian day 145, 1999.

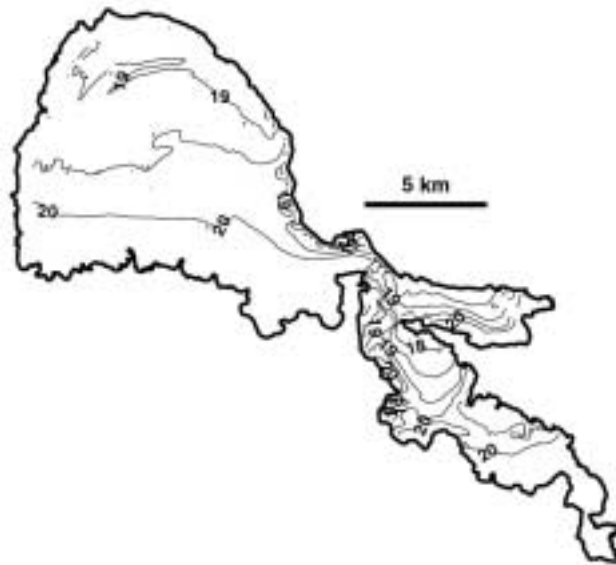


Figure 7.38. Simulated temperature field ($^{\circ}\text{C}$) with a uniform surface wind field at 2.5 m below the surface immediately after a wind event at 2200 h on Julian day 145, 1999.



Figure 7.39. Simulated temperature field ($^{\circ}\text{C}$) with a uniform surface wind field at 2.5 m below the surface after relaxation of density gradients, at 1200 h on Julian day 146, 1999.

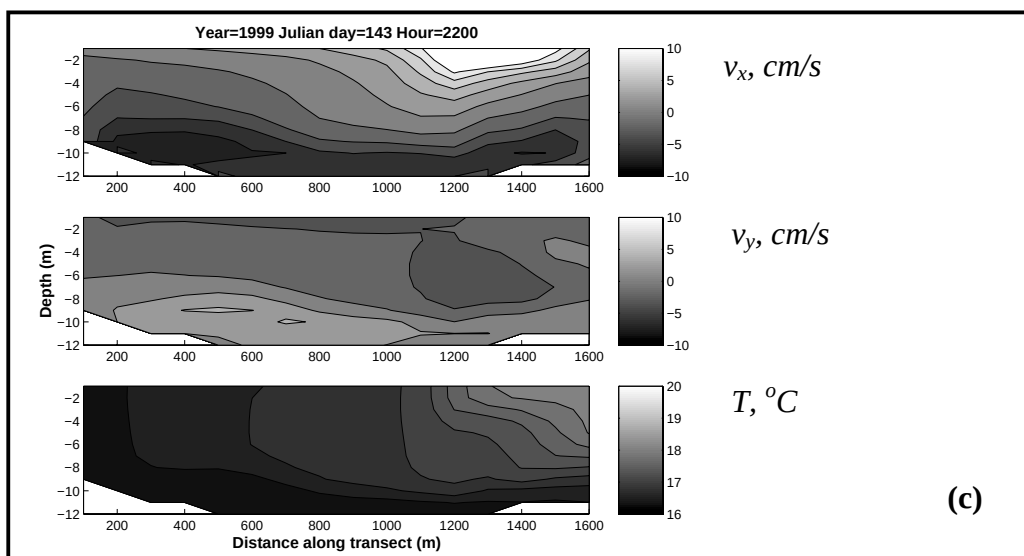
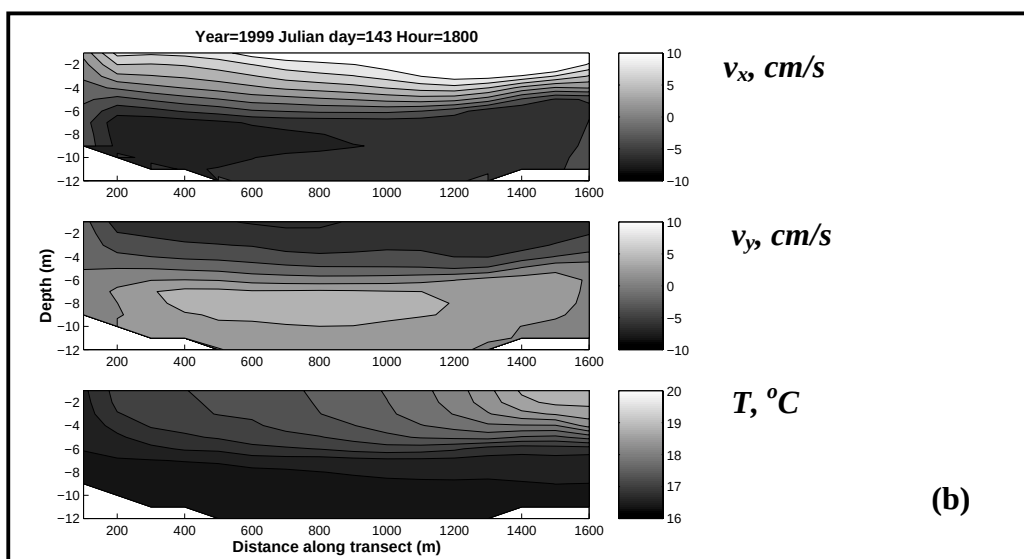
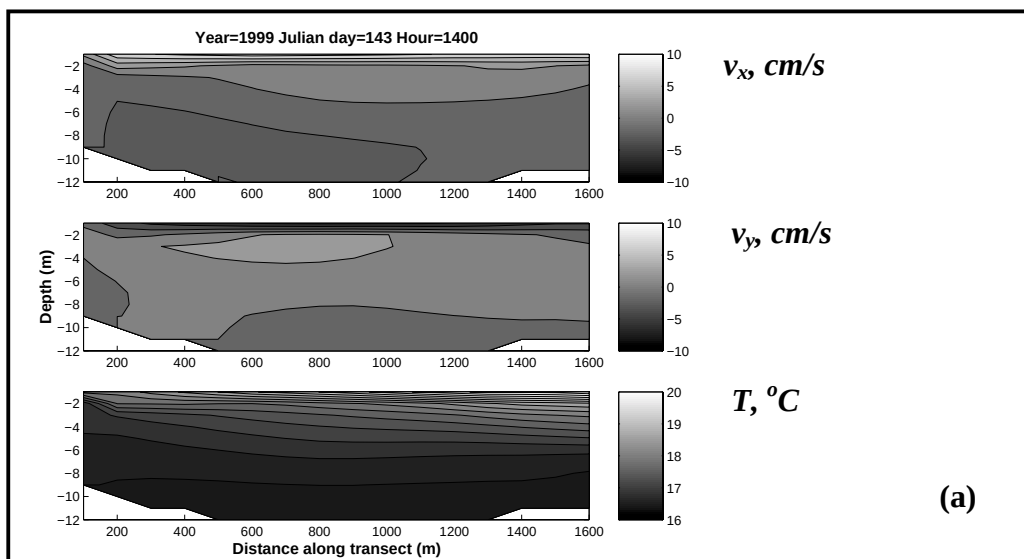
7.6.3. Simulations in a cross-section of the Oaks Arm

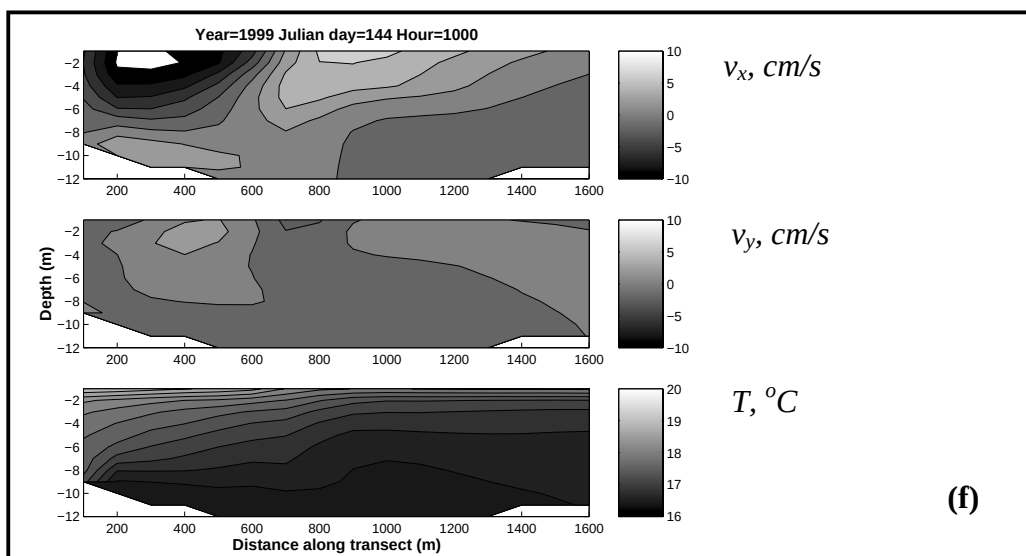
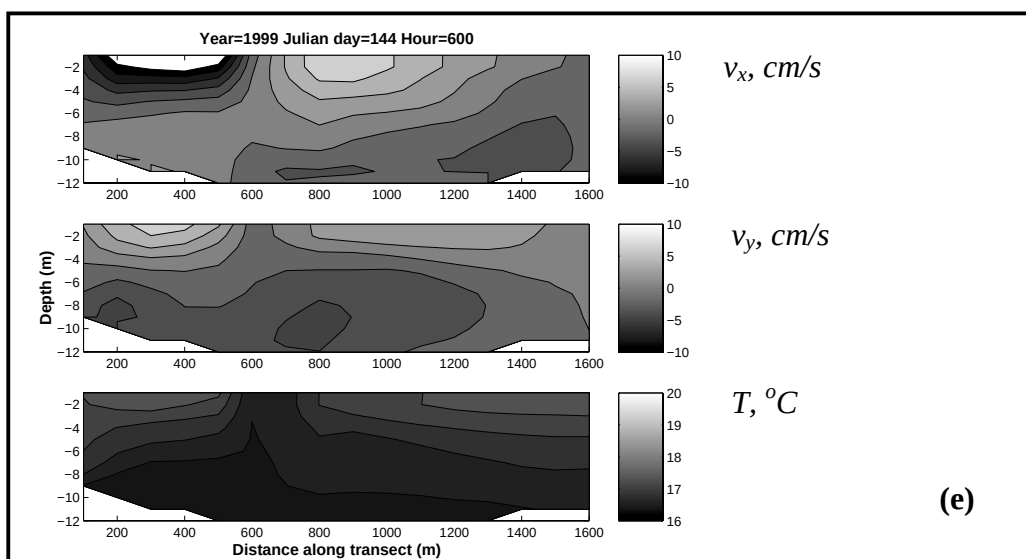
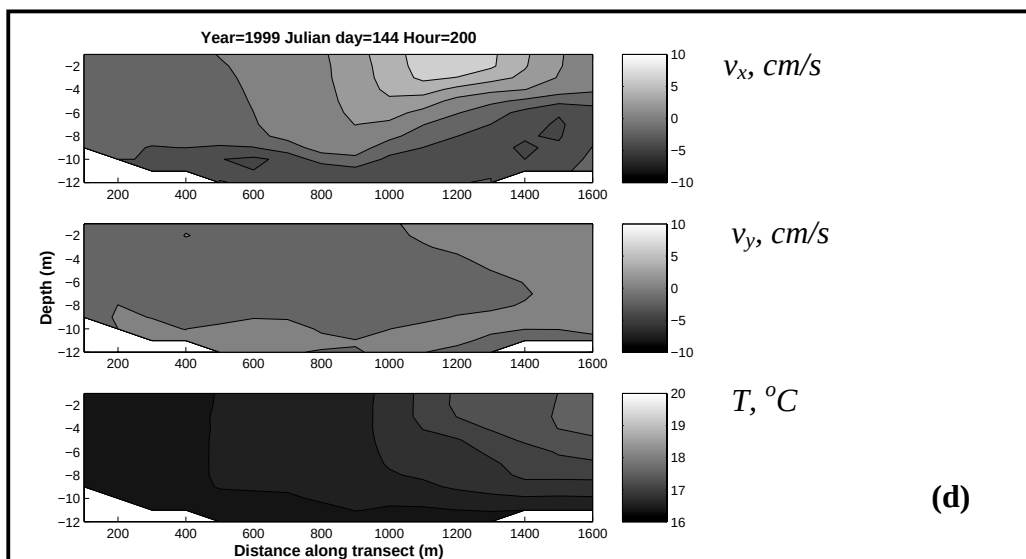
Velocity transects obtained during the field experiments (Fig. 7.23 and Fig. 7.18) with acoustic profilers (ADCP) have provided considerable insight on the changing structure of the velocity field in the Oaks Arm. In Fig. 7.40 one sees simulated transects across the Oaks Arm 4 hours apart during a cycle of 24 hours that shows a remarkable parallelism with the observed velocity transects. They illustrate the asymmetry in the longitudinal circulation, mostly caused by the Earth's rotation. The cross-sections are presented with the north side on the left of the figure, that is, as if one looks from the Narrows towards the mine site at the east end of the Oaks Arm. The plot at the bottom of each set represents the temperature field, and the other two represent the east-west velocity component on top and north-south component in the center. The simulated transects illustrate the evolution of the temperature and velocity fields in response to a diurnal northwesterly wind event. The results in Fig. 7.40.a and Fig. 7.40.b exhibit the presence of wind-driven recirculation cells both along and, because of Coriolis effects, perpendicular to the longitudinal axis of the Oaks Arm: the water at the top moves downwind and to right of the wind at the surface, while it returns to the NW through the lower layers, as observed in the field (see velocity transect in Fig. 7.18). The isotherms stretch and tilt upwards at the north side of the Oaks Arm, and they will compress and deepen at its southernmost and westernmost ends. The resulting temperature field shows significant horizontal gradients across the arm, as illustrated in Figs. 7.40.c and 7.40.d, and along the Oaks basin. The baroclinic pressure gradients induced by those temperature differences are strong enough to drive currents of $O(10-15 \text{ cm/s})$ at the surface, carrying warm water back to the northwest (Fig. 7.18). At 600h on day 144 (Fig. 7.40.e) the

simulated transect sees the arrival of warm water carried by this “return” flow. Owing to the Coriolis effect and the distribution of temperatures (Fig. 7.34), that return flow is concentrated along the north shore. The westward currents in the north persist during the morning hours (Fig. 7.40.f). Figure 7.40.g shows the onset of the next wind event (1400h on day 144) which conveys warm surface water to the southern end of the Oaks Arm, with the return westerly flow still persisting along the north shore. The agreement of the longitudinal velocity field in Fig. 7.40g with the observations of Fig. 7.23 is quite remarkable. The state of lake at this time differs, however, from that shown in Fig. 7.40a, also corresponding to 1400h but on the previous day, in that the return westerly flows are not evident on the latter figure. The differences are probably the result of the wind history over Clear Lake. The easterly winds during day 142, 1999 (see Fig.7.7), had not only weakened the circulation generated by the strong northwesterly winds during the previous days, but they also failed to generate the horizontal temperature gradients driving the return flow along the north shore of the Oaks Arm.

7.6.4. Asymmetry in the longitudinal circulation of the Oaks Arm

The asymmetry in the longitudinal circulation of the Oaks Arm, generated by the wind variability and above all by the rotational effect of the Earth is particularly noticeable when one compares the evolution of the simulated velocity and temperature fields to the north and south of the longitudinal axis of the basin. In the first sub-section here (7.6.4.1) the time history of the velocity profiles at two stations located to the north and south of the longitudinal axis of the basin, half way between the Narrows and the mine site, is analyzed. The model coordinates are (185, 65) and (185, 56) for the north and south stations respectively. The stations will be identified as N (North) and S (South) from here





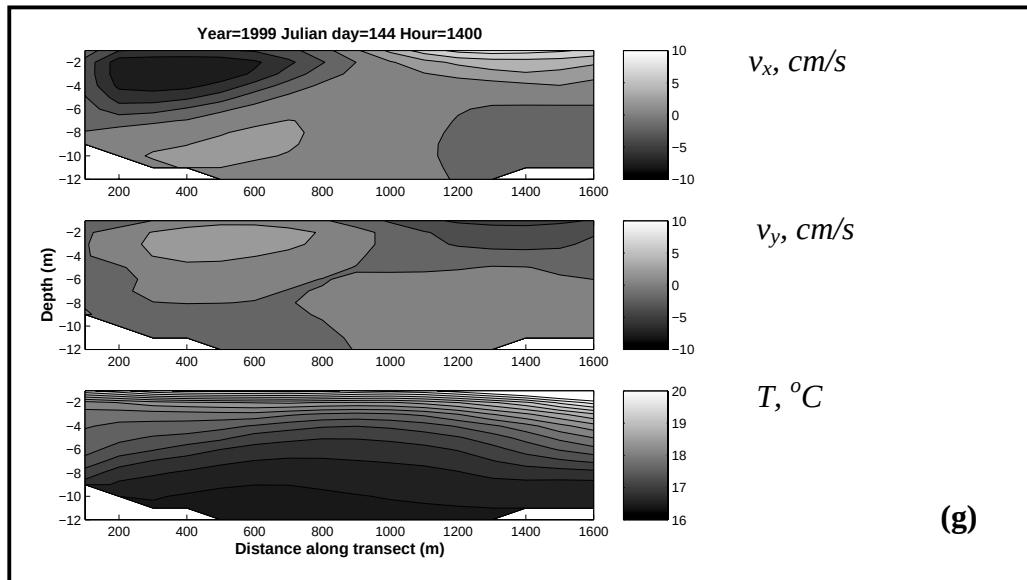


Figure 7.40. Sequence of simulated transects across the Oaks Arm of Clear Lake. Each frame (identified with a letter in the lower right corner) is composed of 3 subplots, illustrating the longitudinal (lower) and transverse (middle) components of the velocity field (cm/s), and the temperature distribution (bottom). The frames shown are 4 hours apart in a diurnal cycle. The velocity positive toward the east and north, and the contours are shown every 2 cm/s. The temperature is displayed in $^{\circ}\text{C}$, and the isotherms are shown every 0.25°C .

on. The differences are interpreted in terms of the relative strength of the wind-driven and baroclinically-driven longitudinal circulation on the north and south sides of the Oaks Arm. The residual velocity profile in the N and S stations, calculated by time averaging the velocity records at each depth during the period of simulation, is examined in section 7.6.4.2. In this section the residual profile is reproduced by using simple averaging arguments. In section 7.6.4.3 the numerical results are presented in two longitudinal sections to the north and south of the main axis of the Oaks Arm. The internal mechanisms that control the domains of influence of the density-driven and the wind-driven circulations are discussed in this last section.

7.6.4.1. Principal Component Analysis of longitudinal velocity profiles

To examine the evolution of the N and S velocity profiles Principal Component Analysis (PCA) was used. PCA is used here to separate different modes of motion and compare the strength of those modes of motion over the time span of the simulation. The theory of PCA and the justification of its use in the analysis of velocity profiles are given in Appendix C. Here the details of the application of the method to simulated east-west components of the velocity profiles are presented.

The time series of the longitudinal components of the velocity profiles at N and S are shown in Figs. 7.41(a) and 7.41(b). The series are presented in two figures to separate the periods when stratification is weak (Julian days 139 through 142) from those when the stratification is strong enough so as to affect the circulation patterns (Julian days 143 through 146).

All of the simulated velocities at each depth were averaged in time to obtain points for a single profile, which will be referred to as the residual profile. Figure 7.42 displays the residual profiles for N and S for the eight days of simulation. The residual profile at N is parabolic with westward velocities at all depths, a maximum near mid depth and minimal velocities at the surface and at the bottom. At S, the profile is logarithmic and eastward at all depths, with a maximum at the surface and a minimum or zero velocity at the bottom. In section 7.6.4.2 the differences in the mean profiles between N and S are examined and explained.

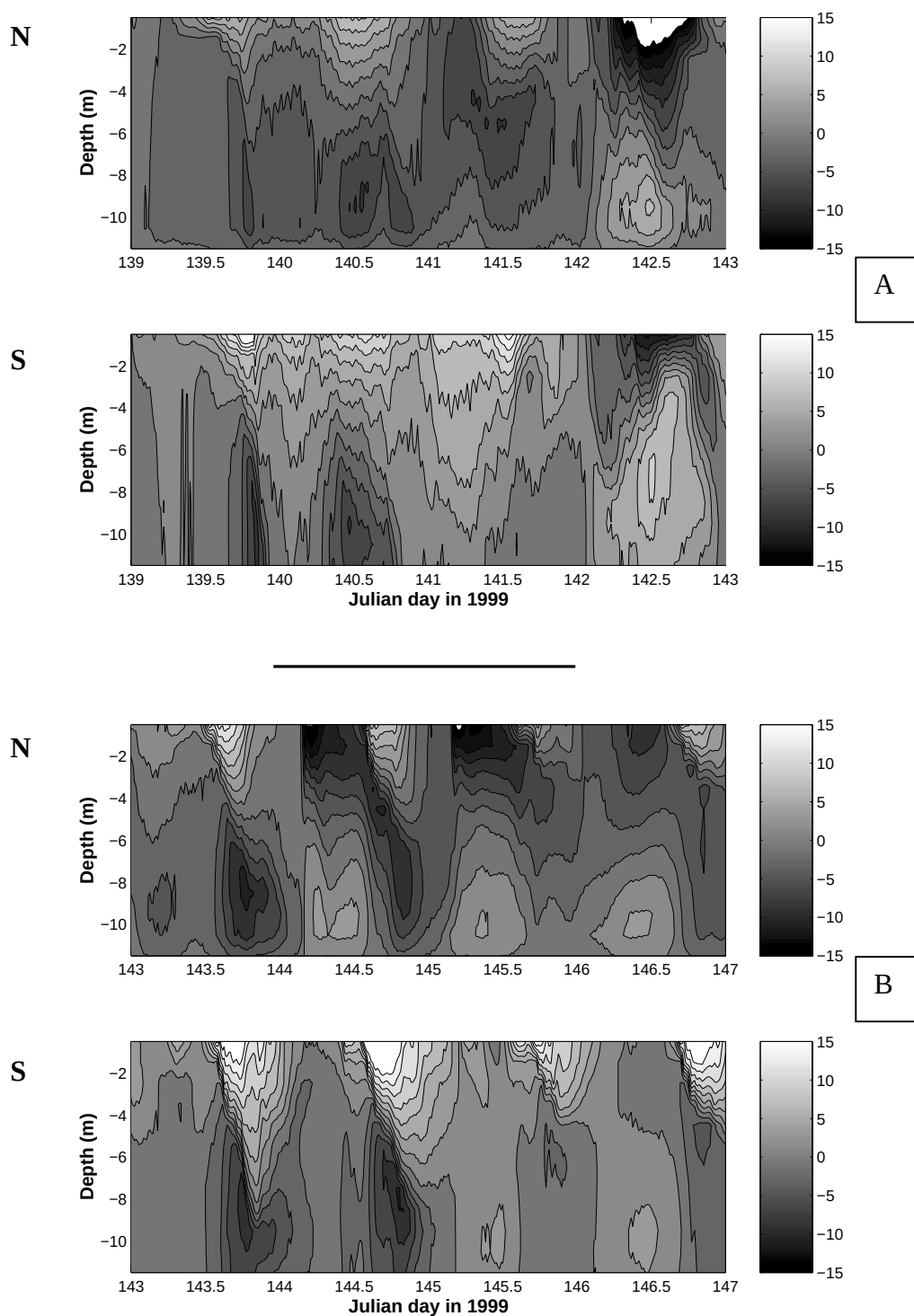


Figure 7.41. Time series of simulated longitudinal velocity profiles at two stations north (N station) and south (S station) of the longitudinal axis of the Oaks Arm. A: First four days of simulation with very weak stratification. B: Last four days of the simulation. The records at the N station are represented on top, those from station S in the bottom. Contours are shown every 2 cm/s.

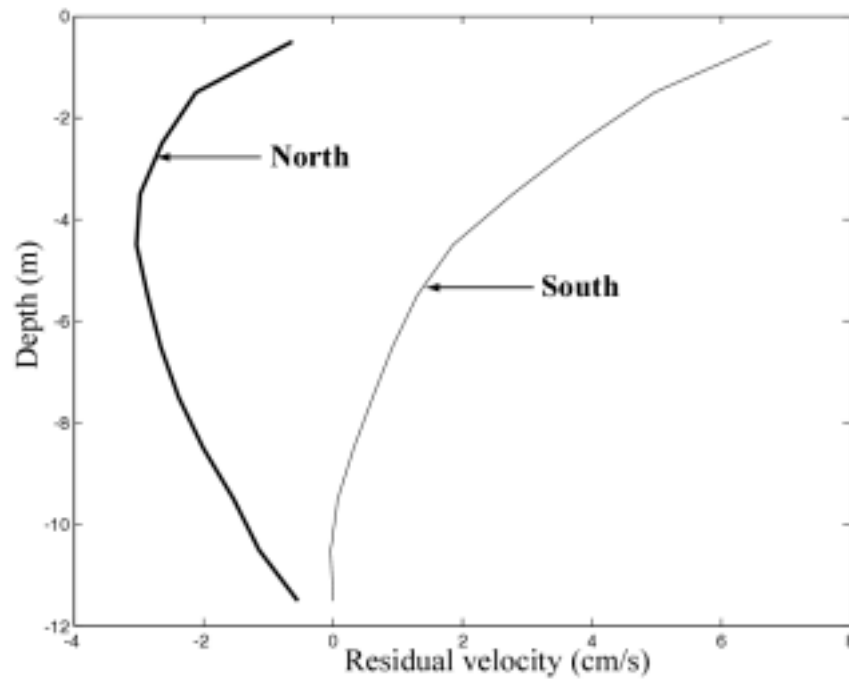


Figure 7.42. Residual velocity profiles from stations N and S. The residual profiles were calculated by averaging the longitudinal velocities at any given depth at each station during the eight days of simulation.

The residual profile was subtracted from each of the velocity profiles to center the velocity records at each layer on their means. The time series of the centered velocity profiles were organized in a matrix, where each column represents the model output at a given layer or depth cell in the profile. In the case of profiles N and S, the matrices have 192 rows ($p = 192$ records, one after each hour of simulation) and 12 columns (a total of $n = 12$ layers). Calling this matrix $\mathbf{Z}'$, the covariance matrix $\mathbf{C}$ is formed as

$$\mathbf{C} = \frac{1}{p} (\mathbf{Z}')^T \mathbf{Z}' \quad (7.14)$$

which is symmetric and positive definite. The eigenvectors $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ represent the vertical structure of the different modes of motion, and the eigenvalues $\{\lambda_1, \dots, \lambda_n\}$ represent the total variance of the data set explained by those modes (see appendix C). Let $\mathbf{E}$ be the $n \times n$ matrix whose columns are the eigenvectors, and $\mathbf{A}$ the $p \times n$ matrix obtained as

$$\mathbf{A} = \mathbf{Z} \mathbf{E} \quad (7.15)$$

Column j of matrix $\mathbf{A}$, $\mathbf{A}(j)$, represents the time series of the strength or amplitude of the mode j , and it is called the principal component (PC). The data matrix is then decomposed as

$$\mathbf{Z} = \mathbf{A} \mathbf{E}^T = \mathbf{A}(1) \mathbf{e}_1^T + \dots + \mathbf{A}(n) \mathbf{e}_n^T \quad (7.16)$$

Figure 7.43 shows the first and second modes for the N and S stations, and Fig. 7.44 compares the evolution of the amplitude of modes 1 and 2 at the north and south stations. These modes explain for both stations from 90 to 95% of the total variance of the simulated velocity profiles. The structure of the modes is quite similar at stations N and S, which suggest that similar mechanisms are at play. The first mode describes a recirculation pattern that can be driven either by wind or baroclinic forces. The wind-driven recirculation is represented by the first eigenmode affected by negative amplitudes when the lake is subject to westerly winds. If the wind is from the east, which is the case of Julian day 142 in 1999, the wind driven re-circulation will be described by using the same first mode, though affected by positive amplitudes (see Fig. 7.44). The first mode

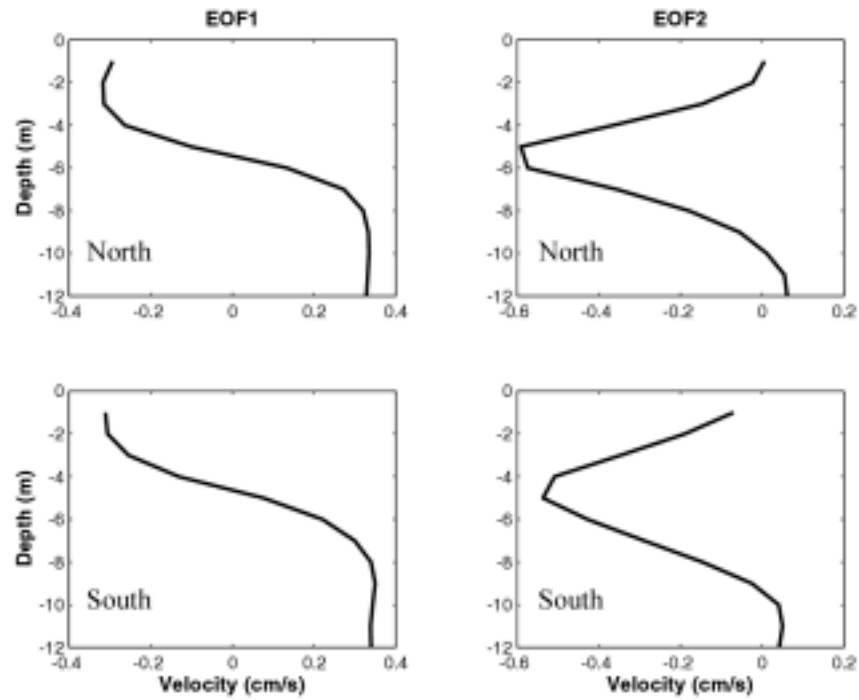


Figure 7.43. Empirical orthogonal functions of the time series for the simulated longitudinal velocity profile shown in Fig. 7.41(a) and 7.41(b).

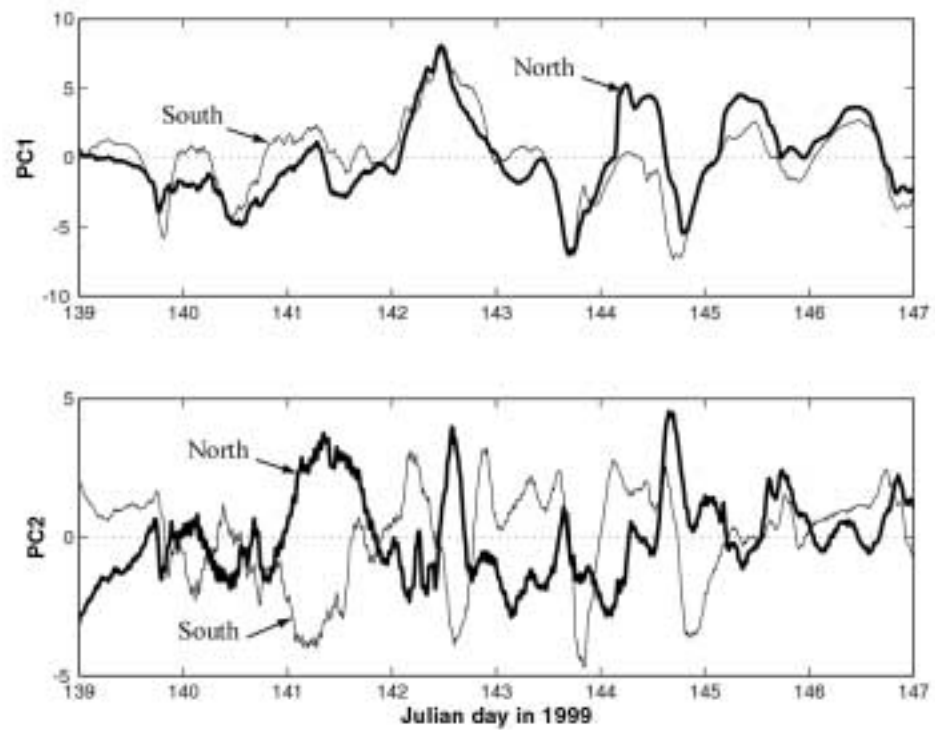


Figure 7.44. Time series of the amplitude (or principal component) of the first EOF (top) and second EOF (bottom) for stations N and S.

also describes the baroclinically-driven motions (see Fig. C.1.a in appendix C). However, in this case the amplitude of the first mode will be negative on days with westerly winds, and it will be positive if the wind blows from the east.

The second mode moves the shear region of the first mode upwards or downwards and controls the thickness of the surface layer. It will increase the thickness of the surface layer if its amplitude has the same sign as PC1. On Julian day 142, for example, the amplitude of the first mode is positive and has a similar magnitude at stations N and S, but the second mode has opposite signs (Fig. 7.44). At station N it is positive, indicating a stronger and deeper surface wind-driven layer, while at S it is negative, suggesting that the return flow is concentrated along the south shore. The second mode also acts at N to account for the effect of westerly winds on the velocity profiles existing during the calm phases of the diurnal wind cycle. This is characterized by the return westward flow in the entire water column under weakly stratified conditions (days 139 to 141), or by the baroclinic recirculation pattern with westward flow in the surface layers. For example, on Julian day 141, two days after the initiation of the simulations, the velocity profile during the early morning hours shows a logarithmic variation from zero at the bottom to the maximum westward flow at the top (Fig. 7.41.a). The westerly wind acts on the westward logarithmic velocity profile to cancel the momentum at the top, creating in this manner a parabolic velocity profile. In the parabolic profile negligible velocities exist at the top and at the bottom of the water column, and maximum westerly

flows are concentrated in the middle. This vertical velocity distribution is better represented by the structure of EOF2 than of EOF1, and consequently the amplitude of PC2 increases and that of PC1 decreases to become almost negligible.

The amplitude of the first mode has a unique peak during the evening hours of the first four days of the simulation, in all cases representing the wind driven circulation: windward flows at the top, and return flow at the bottom. The thickness of the surface layer with windward flows is larger at S than at N with westerly winds (days 139 to 141). The return flow is stronger at N (see Fig. 3.41.a). Easterly winds will produce larger and deeper windward flows at the N station. In all cases the depth-averaged circulation has a cyclonic nature or positive vorticity. As stratification builds up (the last four days of the simulation, shown in Fig.7.41.b), a second peak in PC1 develops during the night and early morning hours at both stations, representing the baroclinic response (Fig.7.43). However, the magnitude of the baroclinic peak at N is stronger than at S, and it can be of the same order as the evening peak representing the wind-driven circulation. At S the evening wind-driven peak dominates over the baroclinic peak, which has at most 10 to 20% of the strength of the wind-driven peak. Thus the baroclinic and the wind-driven circulation act with similar strength and duration at N, while at S the wind driven circulation dominates over the density-driven circulation. This relative contribution of forcing mechanisms at stations N and S will create a cyclonic depth-averaged circulation, as illustrated in the next section.

7.6.4.2. Residual velocity profiles

As described in the previous section, the residual profile at N during the 8 days of simulation was parabolic, with westward velocities at all depths, a maximum at mid-depth and minimal velocities at the surface and at the bottom. The residual profile at S, on the other hand, was logarithmic and eastward at all depths, with a maximum at the surface and a minimum or zero velocity at the bottom. The differences of the residual profiles at S and N reflect (1) the change in the relative contributions of the wind-driven and density-driven vertical circulation patterns, and (2) the differing strengths of the westward and eastward flows across the Oaks Arm which are due to rotational effects. It was shown in section 7.6.4.1 that N feels the baroclinic influence longer and with stronger intensity than at S. On the other hand, westward flows at N tend to be stronger than eastward flows, while the opposite holds true at S (Fig. 7.41.a). These observations will be used in what follows to generate, purely based on qualitative arguments, synthetic profiles that show the same features as those of the residual profiles in Fig. 7.42.

The wind-driven recirculation pattern in a closed basin is the result of balancing the wind-stress forcing, the barotropic forcing resulting from the wind setup of the water surface, and the bottom frictional effects, as shown schematically in Fig. 7.45.a. The density-driven circulation (also called baroclinic) is generated by a balance between the baroclinic pressure force, which increases from zero at the surface to a maximum at the

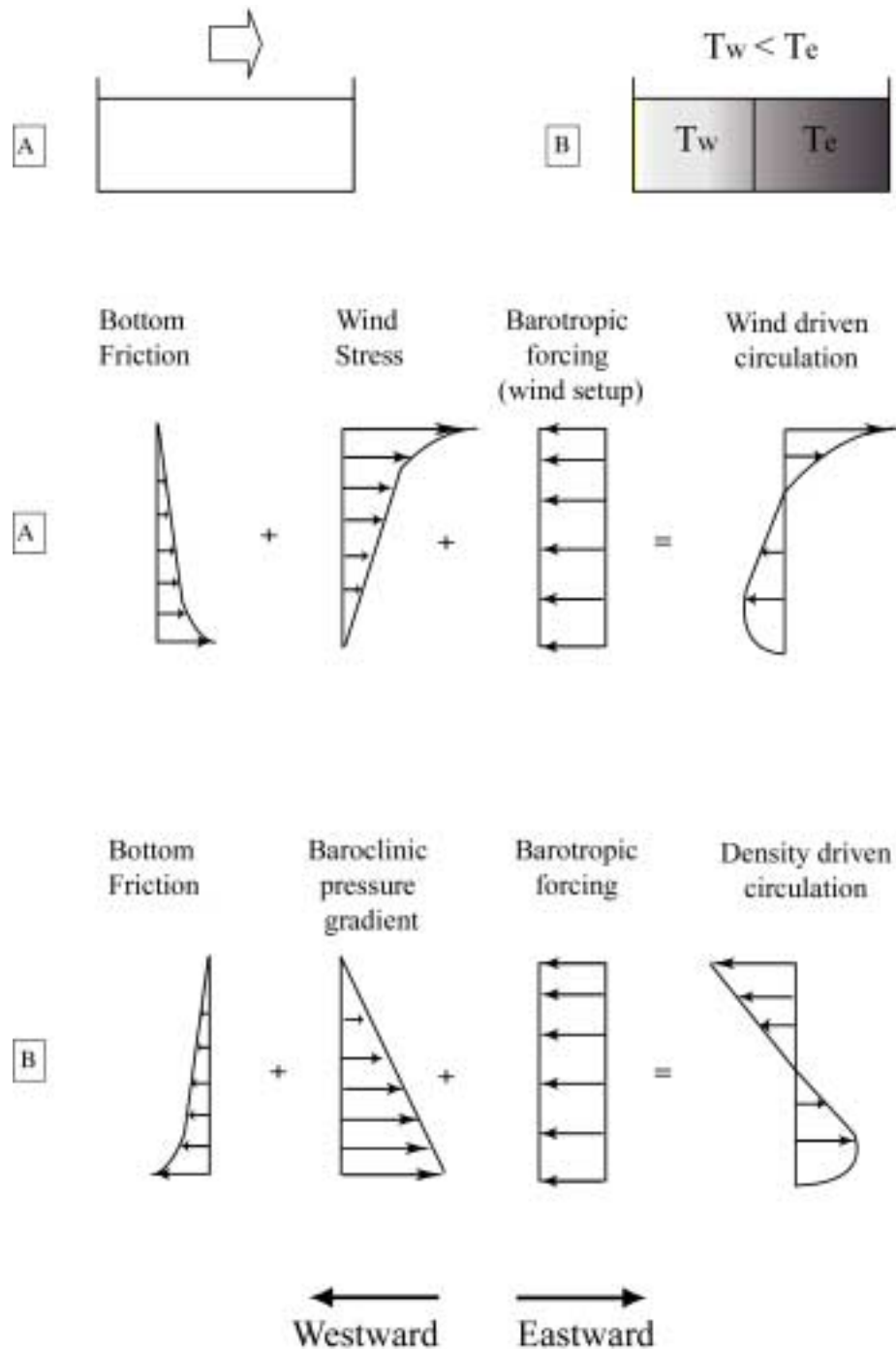


Figure 7.45. Forcing mechanisms in the generation of vertical circulation in a closed basin. (A) wind driven profiles and (B) density-driven or baroclinic profiles. The longitudinal axis is assumed in the east-west direction, and the basin is subject to westerly winds, as in the Oaks Arm. In the density-driven circulation warmer water is assumed to accumulate on the right of the figure (east).

bottom, the barotropic force arising from the tilt of the free surface, and the bottom frictional effects (Fig. 7.45.b). The tilt in the free surface is created by baroclinic pressure forces in the same manner as the wind stress creates the wind setup in a closed basin. In a narrow lake with no transverse variation, the contributions of the different forcing elements to the characteristic profile are such that the eastward and the westward transport are identical and the net depth-averaged longitudinal transport in the profile is zero. Zero net transport is obtained in the wind-driven profile by applying an equal weighting of the wind-stress and the barotropic forcing.

A procedure similar to that shown in Fig. 7.45 was followed here to generate synthetic profiles characterizing the wind-driven and the density-driven circulation at stations N and S. Net eastward or westward transport on either side of the Oaks Arm, which mimics the effects of rotation, can be obtained by changing the relative contribution of the forcing elements in Fig. 7.45. For example, the net eastward transport in the wind-driven profile at S can be reproduced by increasing the contribution of the wind-stress forcing relative to the barotropic forcing in Fig. 7.45.a. The characteristic wind-driven profile at N, with net westward transport, is obtained by increasing the relative contribution of the barotropic forcing. Also, to represent a stronger westward component in the baroclinic circulation at N, the contribution in the synthetic profiles of the barotropic component, acting towards the west, is increased relative to the baroclinic forcing acting towards the west. The relative contribution of the barotropic component is decreased at S to reproduce the stronger eastward component on the south side of the Oaks Arm. The

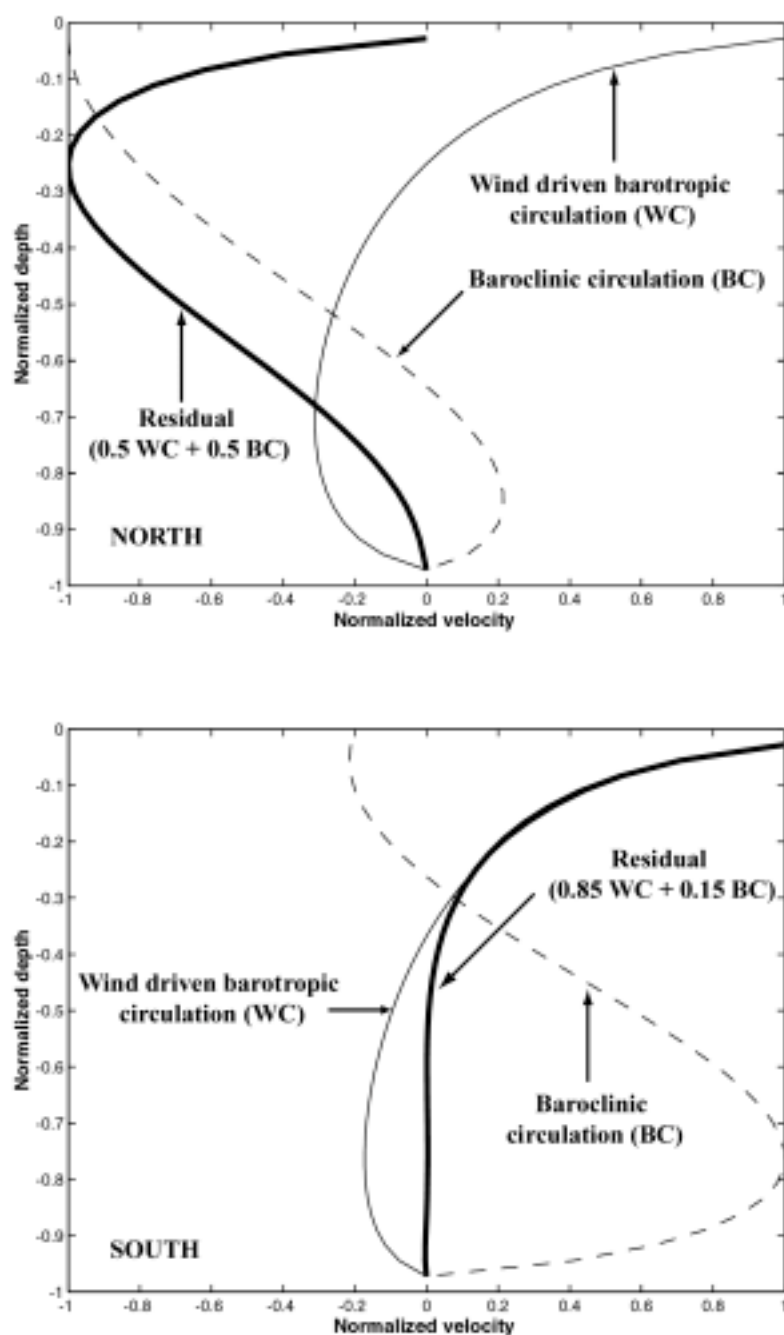


Figure 7.46. Synthetic residual profiles at N and S stations in the Oaks Arm, obtained by weighting the wind-driven and the density-driven vertical circulation patterns, according to the amount of time each circulation dominates in the course of a diurnal cycle.

relative contributions of the forcing elements to the synthetic profiles were chosen to reproduce the ratio of eastward to westward transport that is found in the simulated profiles at N and S. The synthetic wind- and density-driven profiles that are characteristic of each side of the Oaks Arm are displayed in Fig. 7.46.

The wind-driven vertical circulation at S was assumed to dominate the velocity profile 85% of the time. Otherwise (15%) the profile was assumed to be dominated by the density-driven circulation. That is, 0.85 and 0.15 are the relative contributions of the wind-driven and the density-driven circulation to the time averaged or “residual” profile at station S. Those weights lump together the differences in intensity and duration of both types of circulation at S. They are approximate values for the relative magnitude of the evening and morning peaks in the PC1 time series shown in Fig. 7.43.a. As discussed in the previous section, the evening peaks (local minima) in the PC1 time series represent the wind circulation, while the strongest baroclinic circulation coincides with the morning peaks (local maxima). The synthetic “residual” profile, obtained by weighting the contributions of the wind-driven and density-driven circulations, is shown in Fig. 7.46.a, and it displays the same features as the simulated residual profile shown in Fig 7.42, i.e. logarithmic with maximum easterly flows at the surface and a minimum or zero at the bottom.

At station N the synthetic residual profile was obtained by equally weighting the characteristic wind-driven and baroclinically-driven vertical circulations. The wind-driven circulation was assumed to dominate at N during 50% of the time. During the other 50%

of the time it was assumed that the circulation is better described by the density-driven profile. These weights are suggested by the similar magnitudes of the evening and morning peaks in the time series of PC1 (Fig. 7.43.a). The synthetic residual profile is shown in Fig. 7.46.b, parabolic with zero velocity at the top and bottom and with maximum westerly flows at mid-depth, the same features displayed by the simulated residual profile in Fig. 7.42.

7.6.4.3. Longitudinal sections in the Oaks Arm

In this section the results of the numerical simulations along two longitudinal sections parallel to the main axis of the Oaks Arm are presented. The sections are to the north and south of the axis and will be referred to as N' and S' sections respectively. In the plots of the longitudinal sections, the temperature field has been represented on the background with isotherms every 0.25 °C. A gray scale for the temperature has been included at the bottom of the figure. The sections are viewed from the south, with the Narrows to the left and the mine to the right. Overlaying the temperature field, the longitudinal and vertical velocities have been plotted in the form of arrows. The vertical velocity scale is exaggerated by a factor of 50 in relation to the scale of the horizontal velocity. Two arrows representing 0.1 cm/s in the vertical and 10 cm/s in the horizontal have been drawn near the lower left corner of the plot for reference. The zero on the horizontal distance axis is set at a cross section perpendicular to the main axis of the Oaks Arm at its western end, to the east of the channel connecting the Oaks and Lower Arms. The zero will be referred to as Windflower Point, after the name of the tip of the peninsula that separates the Oaks and Lower Arms. Figures 7.47, 7.48 and 7.49 present the N' and S'

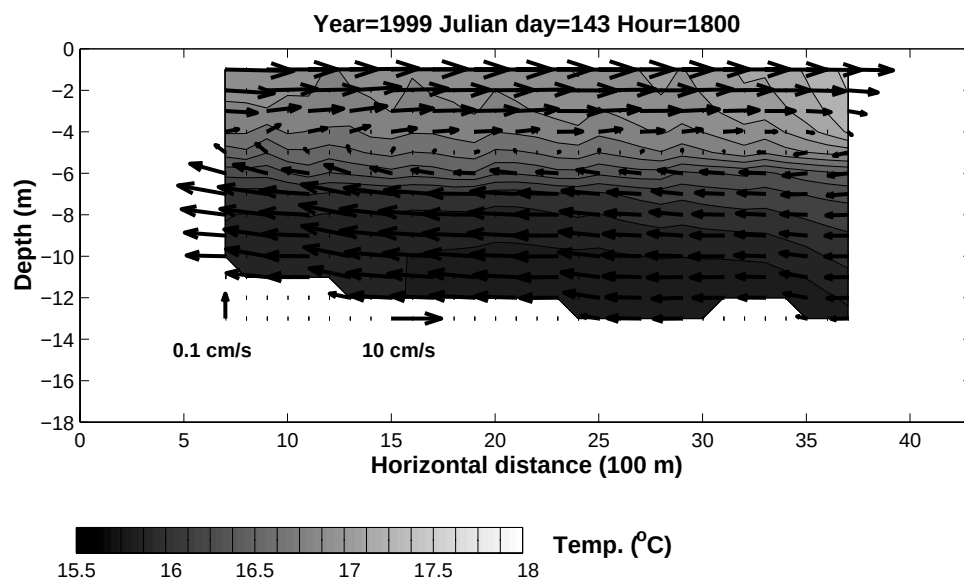


Figure 7.47.a. Longitudinal section N' in the Oaks arm at 1800h on Julian day 143. The temperature field has been represented both with a grayscale and with isotherms every 0.25 °C. The sections are viewed from the south. The longitudinal and vertical velocities have been plotted in the form of arrows.

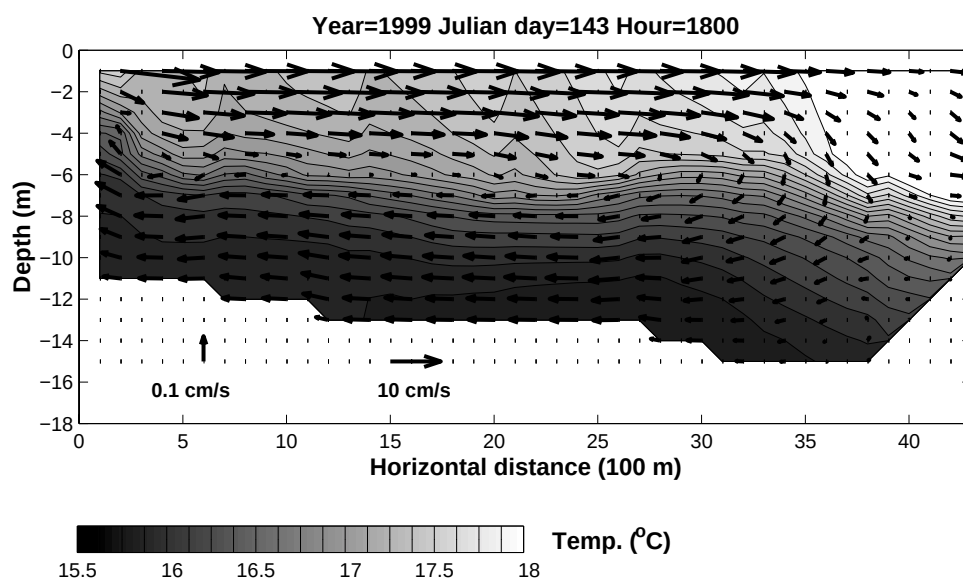


Figure 7.47.b. Longitudinal section S' in the Oaks arm at 1800h on Julian day 143.

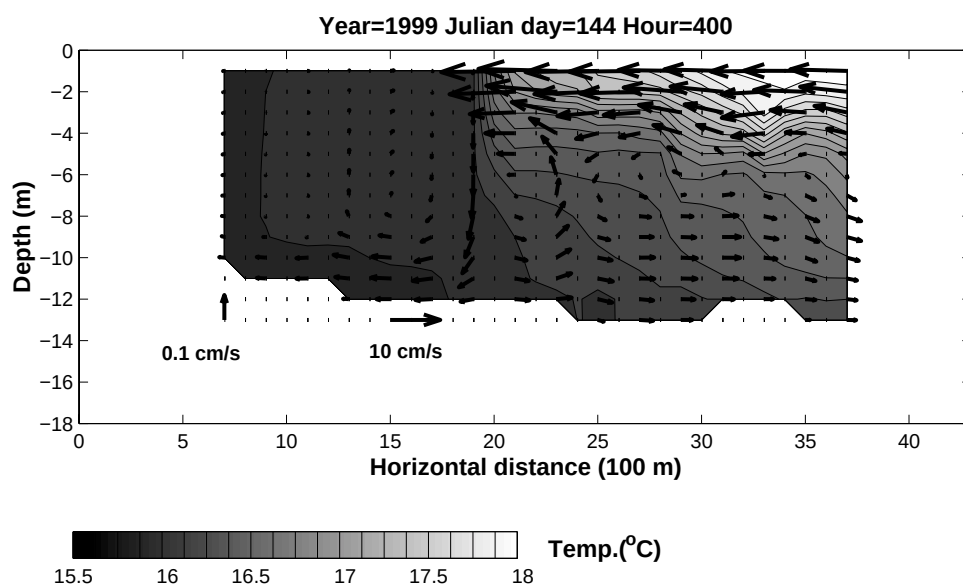


Figure 7.48.a. Longitudinal section N' parallel to the main axis of the Oaks arm, with simulated temperature and velocity field at 0400h on Julian day 144 .

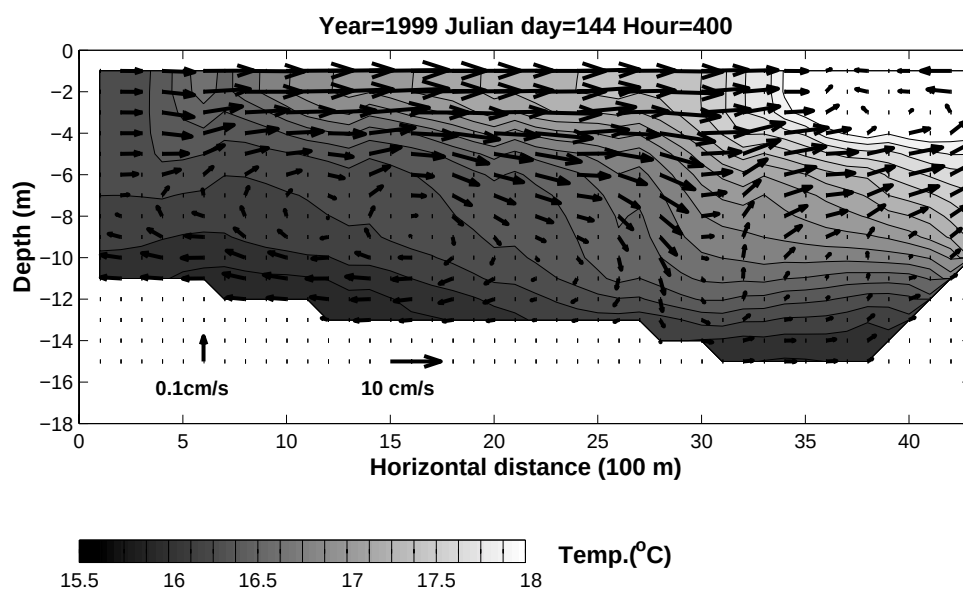


Figure 7.48.b. Longitudinal section S' parallel to the main axis of the Oaks arm, with simulated temperature and velocity field at 0400h on Julian day 144.

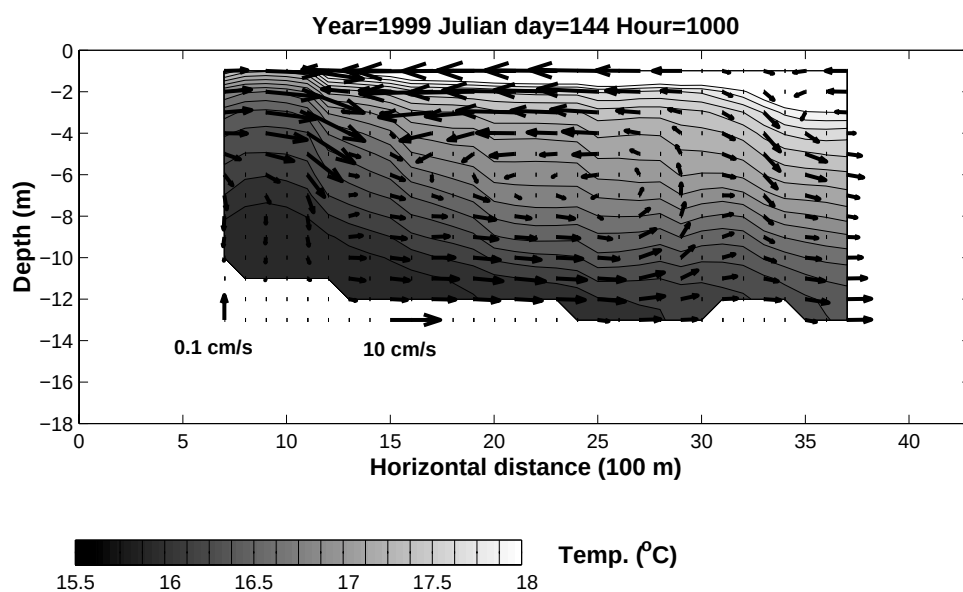


Figure 7.49.a. Longitudinal section N' parallel to the main axis of the Oaks arm, with simulated temperature and velocity field at 1000h on Julian day 144.

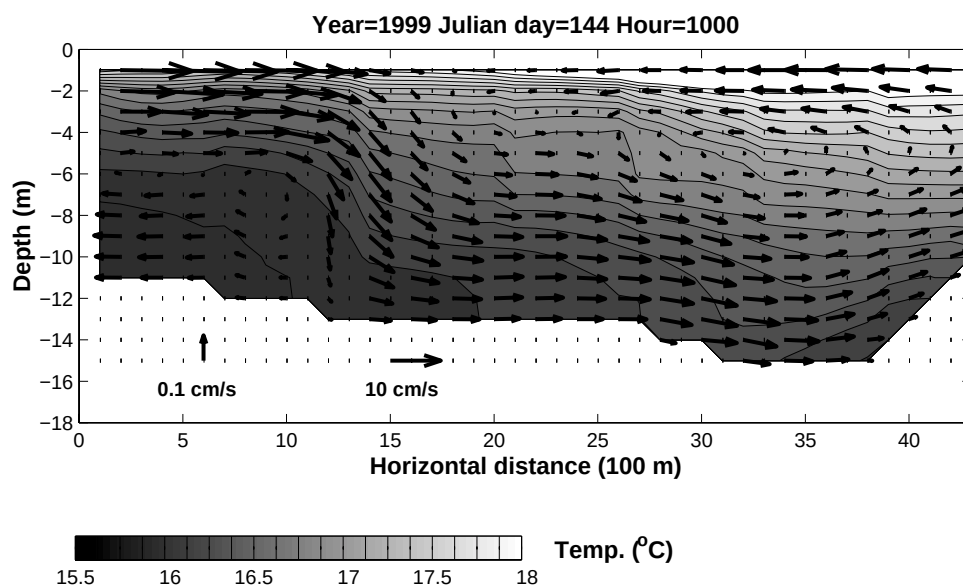


Figure 7.49.b. Longitudinal section S' parallel to the main axis of the Oaks arm, with simulated temperature and velocity field at 1000h on Julian day 144.

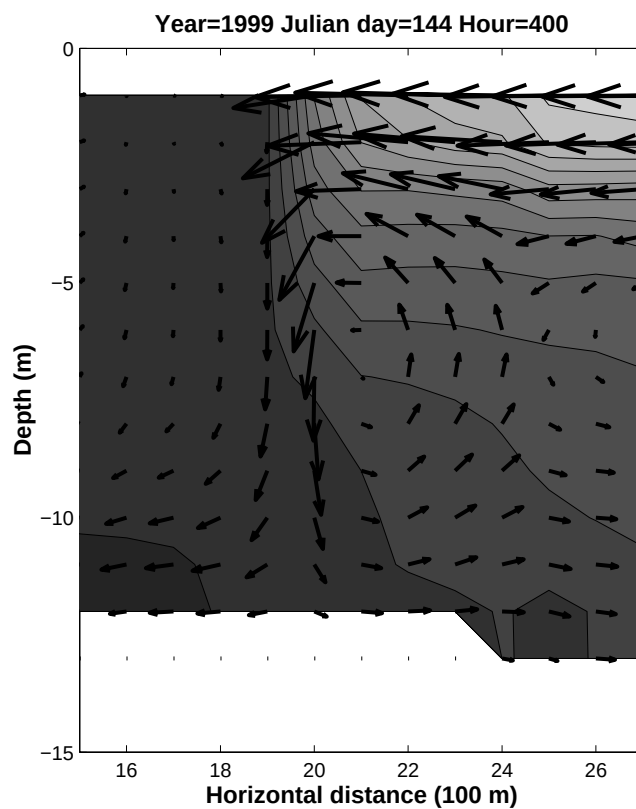


Figure 7.50. Detail view of the frontal region in Figure 7.48.a, showing the characteristic circulation in the vicinity of a front. Warmer water is lifted by a tongue of colder water that plunges and advances to the right following the bottom.

longitudinal sections at 1800h on Julian day 143, 400h on Julian day 144, and 1000h on Julian day 144, respectively.

The velocity field at 1800h on Julian day 143 (Figs. 7.47.a and 7.47.b) is dominated in both the N' and S' sections by the wind driven circulation: eastward flows at the top and return flow at the bottom, as shown in the top frame of Fig. 7.40.b. The temperatures are lower and the isotherms are more separated at the N' section (Fig. 7.47.a) in relation to S' section (Fig. 7.47.b) at about 2000 meters. The maximum temperature change in the N' section is at about 6 meters depth, and at 8 meters in the S' section, evidencing the lateral tilt of the thermocline caused by the lateral circulation (middle frame of Fig. 7.40.b.).

At 400h on Julian day 144, with calm conditions prevailing, the baroclinic pressures become the dominant forcing mechanism, and they start to weaken the wind-driven circulation and to generate a circulation that is driven by horizontal density gradients. The horizontal density gradients are created by the westerly winds that accumulate warm water to the east of the Oaks Arm and colder hypolimnetic water to the west. The density-driven circulation, hence, will be characterized by westward flow at the surface and eastward flow nearer the bottom. The area of influence of the density driven circulation is initially confined to the easternmost areas of the Oaks Arm, where, because of the presence of the boundary of the lake, the wind driven currents are weaker. At the location where the predominantly wind-driven and density-driven circulation patterns meet, the surface currents converge and the bottom currents diverge, creating a fringe with significant vertical motions that shall be referred as a circulation front. In Figs.

7.48.a and 7.48.b the longitudinal sections show the wind-driven circulation to the left and the density-driven circulation to the right, with an area of complex upward and downward motions in between that defines the circulation front. The front in the N' section is located between 1800 and 2200 meters to the east of Windflower Point, and it is characterized by almost vertical isotherms with the lower temperatures to the left. The circulation at this frontal region is very characteristic, with negative vertical velocities to the left – where the more dense water plunges – and positive vertical displacements to the right – where the warmer water is lifted. A detailed view of this frontal circulation for section N' is shown in Fig. 7.50. The location of the frontal region is less clear in section S' at 400h (Fig.48.b). In this section the bottom horizontal currents diverge at approximately 3000 m from Windflower Point. At the surface, however, the change in current direction occurs at about 3600 m. This indicates that the reversal of the current direction at a given location occurs last in those parts of the profile with the highest momentum, i.e. at the surface in the particular case of wind-driven profiles. The change in the sign of the vertical velocity is evident in the lower layers, though it is less pronounced than in section N', where the isotherms are steeper.

Figure 7.49 shows the temperature and velocity fields at 1000h on Julian day 144. The isotherms shown at this time are less tilted at section N' than they were at 400h, indicating that the temperature gradients have relaxed. Consequently the frontal circulation with strong vertical motions is not evident in Fig. 7.49.a. The circulation front, and hence the area of influence of the density-driven motions, has migrated to the west, and during the morning most of section N' is dominated by density-driven motions. The

region of surface converging flows at 1000h on day 144 is located about 1200 meters from Windflower Point in section N'. At that same time the surface front in section S' is 1000 meters further to the east (Fig. 7.49.b). This difference indicates that the migration rates of the density-driven circulation from its initial region of influence, at the eastern end of the Oaks Arm, towards the west are different on each side of the Oaks Arm. The phenomenon of a density-driven circulation cell advancing to the west in the Oaks Arm and substituting in its place a wind-driven circulation pattern has the character of a wave, in its most general sense, and this process will here be referred to as a wave of baroclinic circulation. The leading edge that separates the baroclinic circulation cell from the wind-driven flow has a persistent frontal nature. Understanding the differential propagation of the waves of baroclinic circulation on each side of the Oaks Arm requires the consideration of four factors: vertical stratification, bed friction, inertia and horizontal temperature gradients.

Friction at the bottom boundary causes a momentum loss in the flow near the bed which, due to vertical momentum transfer by turbulence, is propagated up the water column, away from the bottom. The result is a logarithmic velocity profile in a neutrally stratified channel flow. Wind, on the other hand, creates momentum in the water at the free surface. This additional momentum is, similarly, carried down the water column by turbulence. Stratification, however, can reduce the vertical turbulent momentum transfer, effectively decoupling the behavior at the top and bottom of the water column. The reduction in the turbulent transfer will be in proportion to the amount of stratification. The momentum imparted by the wind to a strongly stratified water column will accelerate

the surface layers, which will not feel the effects of bottom friction. Bottom friction will dominate the behavior of the bottom layers because of the reduced momentum transfer. Stratification, hence, greatly increases local shear. In a weakly stratified water column the momentum is more easily transported in the vertical, and the surface wind-driven layers will also experience the decelerating effects of friction. It has been shown in previous sections (see Fig. 7.40.c) that isotherms in the vertical direction compress at the south shore and stretch at the north end of the Oaks Arm in response to northwesterly winds. This behavior is also evident by comparing Fig. 7.48.a and Fig. 7.48.b. In section N' the surface windward momentum, after the wind has died, is rapidly cancelled as the strong westward momentum in the lower layers is transmitted through a weakly stratified column. In section S', on the other hand, the effects of friction and bottom return flow do not propagate into the surface layers due to the stronger stratification. Thus the surface layers will continue to flow eastward after the wind dies. The differential stratification induced by large-scale motions also produces higher windward velocities to the right of the wind direction (south for the dominant westerly winds in Clear Lake) and lower windward velocities to its left (north for the westerly winds of Clear Lake). During the wind events, the momentum imparted by the wind to the surface currents in the southern part of the Oaks Arm will not be effectively transmitted to the lower layers, nor will the lower layers flowing westward transmit their momentum to the surface layers. The surface layers will then slide more easily over the lower layers in the south. In the north of the Oaks Arm, however, the surface layers not only are accelerated by the wind but also feel the effects of friction and the return flow below them. The surface eastward velocities in the north are thus smaller than in the south.

The propagation of waves of baroclinic circulation involves the cancellation of wind-driven circulation patterns (Fig. 7.45.a) and the establishment of a density-driven circulation (Fig. 7.45.b). This transition requires a reversal of the flow direction at all depths. Reversal will occur last in those parts of the flow domain with the greatest momentum, because of their larger inertia. Since friction and turbulence rapidly cancel the momentum in the northerly section of the Oaks Arm, as discussed in the previous paragraph, the waves of baroclinic circulation will propagate faster through the north. In the southern section of the arm the momentum at the surface is higher because of the higher stratification (Fig. 7.40.c) and the reversal of flow will occur later than in the north. In the lower layers of the southern section the frictional effects will rapidly control the wind-driven flows, and hence the change in flow direction will take place earlier than in the surface layers. Phase differences in the vertical will hence arise in the southern section of the Oaks Arm due to differences in inertia and the presence of stratification. This effect is evident in Fig. 7.48.b between 3000 and 3500 meters from Windflower Point.

An additional factor that contributes to the asymmetric propagation rates of the waves of baroclinic circulation is the differences in the horizontal density gradients set up by the wind. It was shown in Fig. 7.34 that warmer water accumulates in the eastern and southern parts of the Oaks Arm, while colder water is piled up in the northwest sections of the basin. Greater longitudinal gradients are generated along the northern edge of the basin, and consequently the baroclinic pressure force that opposes the wind-driven

circulation patterns will be stronger in the north. The wind-driven currents will be cancelled in the north sooner than in the south, allowing for a faster propagation of the baroclinic pulse following the north shore.

7.6.5. Variation of mixing levels across the Oaks Arm

The analysis of section 7.6.4.3 showed that stratification and shear behave differently in the northern and southern regions of the Oaks Arm. Mixing processes, controlled by shear production and stratification, could consequently differ across the Oaks Arm. In this section the idea of differential mixing rates on opposing sides of the Oaks Arm is explored, using the numerical solution from the same two stations (N and S) studied in section 7.6.4. The analysis focuses on the results from the last four days of simulation, days 143 through 146 inclusive, when the lake was stratified and the wind was consistently from the northwest.

Figure 7.51 shows the evolution of the mean shear and the mean temperature gradient at stations N and S for Julian days 143 through 146. The velocity and temperature gradients were calculated for every consecutive pair of cells in the profile, and their absolute values were averaged to obtain a unique description of the intensity of the shear and stratification in the water column. The time series of mean shear at station N presents two maxima in each 24-hour period, especially evident in days 144 and 145 (top frame in Fig. 7.51). The first maximum, during the night and early hours, is related to the relaxation phase of the diurnal setup-relaxation cycle, described in section 7.6.3, and occurs abruptly in the records. Warm water accumulated at the eastern end of the Oaks Arm by westerly winds moves back to the west, driven by baroclinic pressure forces, and

overlays the colder waters near the north shore (see Figs. 7.48.a. and 7.40.e). The occurrence of this first maximum in the evolution of the shear at station N coincides with an abrupt increase in the stratification (Fig. 7.51 bottom), particularly evident on days 144 and 145. At station S, neither the mean shear nor the mean temperature gradient records show this peak.

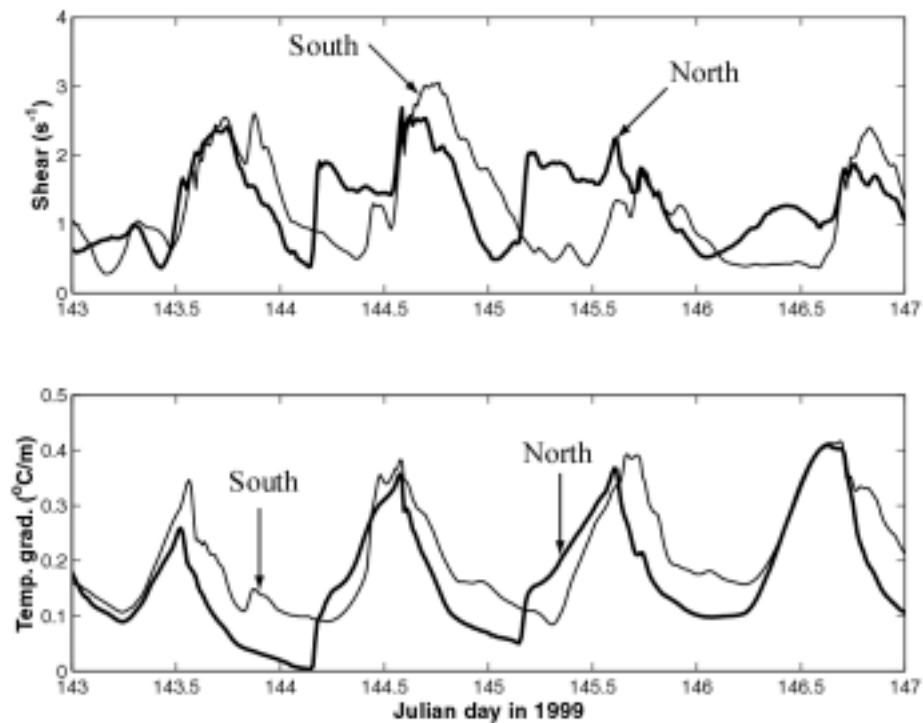


Figure 7.51. Time series of mean shear and mean temperature gradient in the water column at stations N and S, from simulated results.

The second maximum in the time series of shear at N is the result of the northwesterly winds acting on the surface of the lake. It coincides with the water column reaching the maximum stratification (bottom frame in Fig. 7.51). This peak marks the start of the wind event. During the initial stages of a wind event, the surface layers of a stratified water column will accelerate and shear will increase. The increased shear will reduce the

stratification soon after the onset of the wind event. On day 143, for example, the increase in shear at station N is accompanied by a decrease in temperature stratification. Station S displays a similar behavior. The shear and the mean temperature gradient, however, remain higher at this station than at N after the onset of the wind. As indicated in section 7.5.1 and illustrated in Fig. 7.40, the isotherms tend to stretch in the northwest of the Oaks Arm due to wind and rotational effects, while they tend to compress to the east and south. This behavior is well documented in the literature (Monismith and Fong 1996) and is driven by the large-scale vertical circulation in a stratified reservoir. As a consequence of increased compression of isotherms in the south end of the arm, the momentum transfer in the water column is reduced, and shear will therefore increase. The stretching of the isotherms in the north shore of the Oaks Arm reveals a decrease in stratification levels, which will facilitate the transfer of momentum. Hence the shear levels in the water column will not be as high at station N as they are at S.

Stratification and shear tend to have opposite effects on the level of mixing in the water column: shear enhances mixing, while stratification tends to suppress it. A non-dimensional parameter, the gradient Richardson number (Ri_g , defined in Eq. 5.51), expresses the intensity of those two opposing effects. Here Ri_g measures the ratio of the increase in potential energy by the turbulent buoyancy flux to the production of turbulent kinetic energy by shear. For Ri_g below a critical value (taken to be 0.25), vertical turbulent shear is reduced as Ri_g increases (see Eq. 5.56). If Ri_g is greater than 0.25 at some point in a velocity profile, then the bed stress-induced turbulent momentum is totally extinguished above that point. Values of Ri_g in the two water columns rarely

reached the critical value of 0.25, and the two columns mostly experienced values between 10^{-6} and 10^{-2} . Figure 7.52 is the histogram of Ri_g for stations N and S during the 4 days included in the analysis. The distribution of Ri_g at station N presents a maximum number of values between 10^{-5} and 10^{-4} . The distribution at station S is flatter, showing a maximum number of values in the range between 10^{-5} and 10^{-3} , suggesting that levels of mixing could show variations across the Oaks Arm, higher in the north.

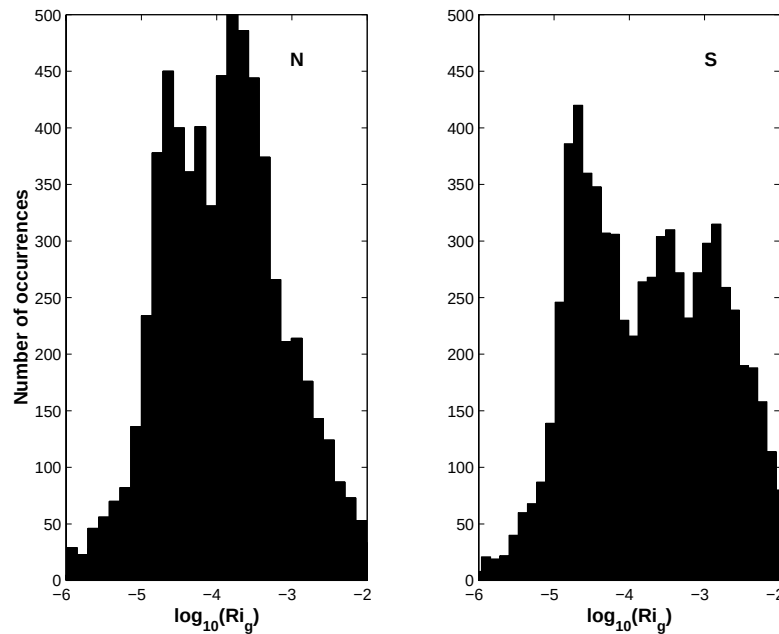


Figure 7.52. Histogram of the decimal logarithm of the gradient Richardson number (Ri_g) in the water column at stations N (left) and S (right). Simulated values of temperature and velocity from days 143 through 146 are used in the construction of Ri_g .

In Fig. 7.53 the evolution of the decimal logarithm of Ri_g in the water column at stations N (upper frame) and S (lower frame) is represented using shaded contours. Negative and zero numbers (unstable and neutrally stable profiles) are not represented. The Ri_g values in the lower layers at station S were higher than at station N during the last two days of the period under analysis (Julian days 145 and 146), and hence the stratification in the

southern part of the Oaks Arm affects mixing more than in the northern part. Coinciding with the occurrence of the wind events, the regions of lower Ri_g (higher mixing) grows from the surface, deepening more at station S than at N, as is the case, for example, at the end of Julian day 145. Below these surface layers of lower Ri_g , the mixing rates are reduced more significantly at the southern station than at the northern station, where Ri_g decreases but more slowly. This difference probably reflects the compression of isotherms in the south and their stretching in the north, already shown in Fig. 7.40.b or Fig. 7.40.c. The deepening of the surface layer is more pronounced after the wind event on Julian days 143 and 144. Even during those days of deeper mixing there exists at station S a layer of higher Ri_g values. In the layer below the wind-mixed surface region at station N, the values of Ri_g are lower than at station S. The contour plots reveal at both stations and especially during days 143 and 144, the development of regions of low Ri_g at the bottom of the profiles coinciding with the wind event. Those regions probably define the bottom boundary layer, and its thickness varies in proportion to the magnitude of the wind, which was higher during the evening wind events of days 143 and 144. During the night and early morning hours Ri_g at station S increases, reaching higher values than at station N. At station N the shear increases during the night hours as a tongue of warm water recedes towards the west after previously being pushed by the wind towards the east (see Fig. 7.48). As the warm water recedes, it cools, and the temperature difference between the surface and the bottom of the profiles is smeared, which explains the high values of Ri_g at station N during those hours.

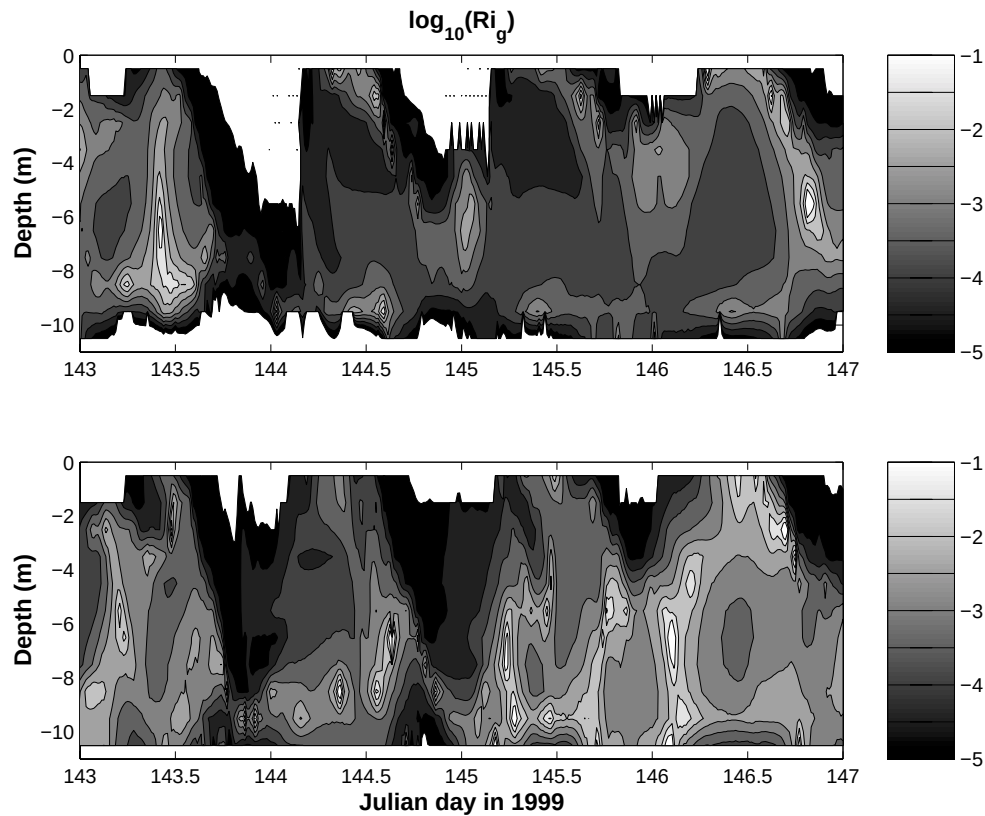


Figure 7.53. Evolution of simulated values of the decimal logarithm of the gradient Richardson number at stations N (top) and S (bottom). Contours are shown every 0.5 units in the logarithmic scale (color bars on the right).

The source of mixing in the code is directly and explicitly calculated as turbulent kinetic energy TKE by using the 2.5 Mellor-Yamada turbulence closure model, described in chapter 5. The evolution of the decimal logarithm of TKE in the water column, for days 143 through 146 inclusive, is shown in Fig. 7.54. Station N is shown on the top and S on the bottom, and the contours are drawn every 0.5 units on a logarithmic scale. The plots show the deepening of the wind-driven surface mixed layer during the evening hours at both stations. The deepening seems to be stronger and the levels of the TKE in the surface layer higher at station S during the wind events, and especially during days 143 and 144. The level of the TKE diminishes, and, the surface layer shrinks rapidly in the S

profile soon after the wind decays. At station N, however, mixing is still active throughout most of the water column, and the TKE slowly dissipates after the wind event. In summary, the results shown in Figs. 7.53 and 7.54 indicate that mixing in the water column has different spatial and temporal patterns in the north and the south. Those differences are mostly explained by the large-scale circulation features in the Oaks Arm, already studied in the previous sections.

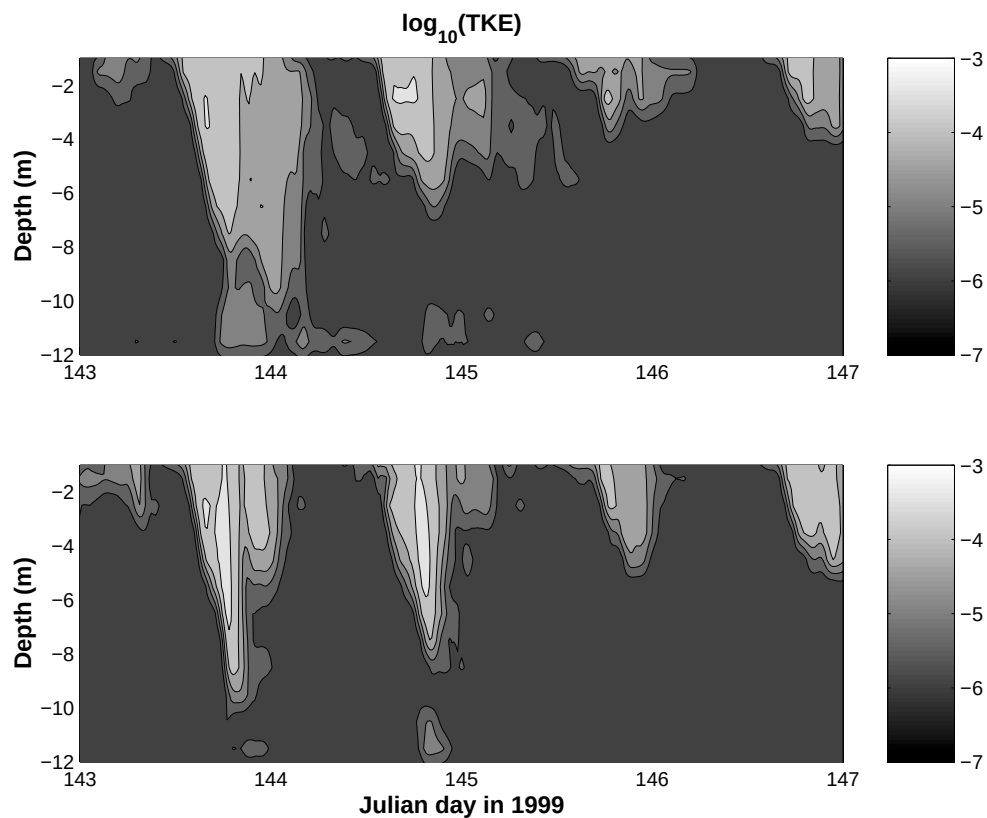


Figure 7.54. Evolution of simulated turbulent kinetic energy TKE at stations N (top) and S (bottom). The values shown are the decimal logarithm of the TKE calculated by the Mellor-Yamada 2.5 turbulence closure model built into SI3D-L. Contours are shown every 0.5 units on the logarithmic scale shown to the right of the plots.

7.7 Visualizing circulation patterns with particle tracking

7.7.1. Tracking particles in three-dimensional space

The particle tracking routines, developed in chapter 6, were employed to simulate the transport of particles that were initially located in the proximity of the Sulphur Bank Mercury Mine. The complete solution of the model was exported to a file every 30 minutes and was used to generate, by linearly interpolating in time, velocity fields every 30 seconds. The interpolated fields were then used to simulate the transport of a total of 100 particles released at 0100h on day 139, 1999, at the locations shown in Fig. 7.55. The locations of the particles in the vertical were randomly assigned using a Gaussian distribution centered at 1.5 m with a standard deviation of 1.5 m. In this series of experiments the particles were simply carried by the currents, and no random displacements were added to simulate the effects of turbulence. The locations of the particles were calculated every 30 seconds for 8 days and saved to a file every hour. These tracking exercises took about 1000 seconds to run on an SGI Octane with a 175 MHz IP30 R1000 processor and 642 Mbytes of memory.

Figure 7.56 shows the locations of the particles 8 days after their release. Some of them had migrated to the Upper and Lower Arms, almost 12-14 km from their original location. The mean distance covered by the particles was about 21 km. The trajectories of some of them support some of the features of the hydrodynamics of Clear Lake, described previously, and are used here to emphasize those aspects. To describe the particles' 3-D trajectories, plan-view (2-D) plots have been used, and the depth below the surface of the particle is indicated by the grayscale of the marker used to show its

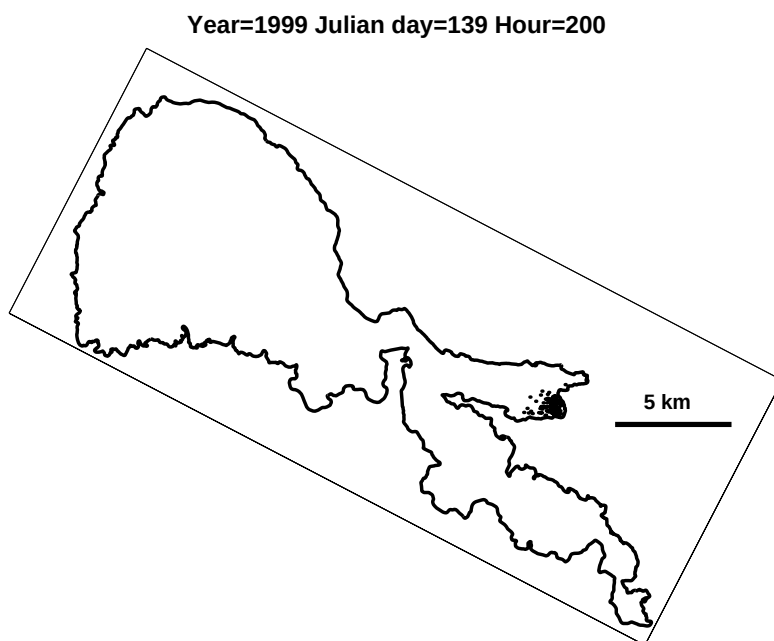


Figure 7.55. Location in the horizontal plane of particles at release time in the particle tracking exercise.

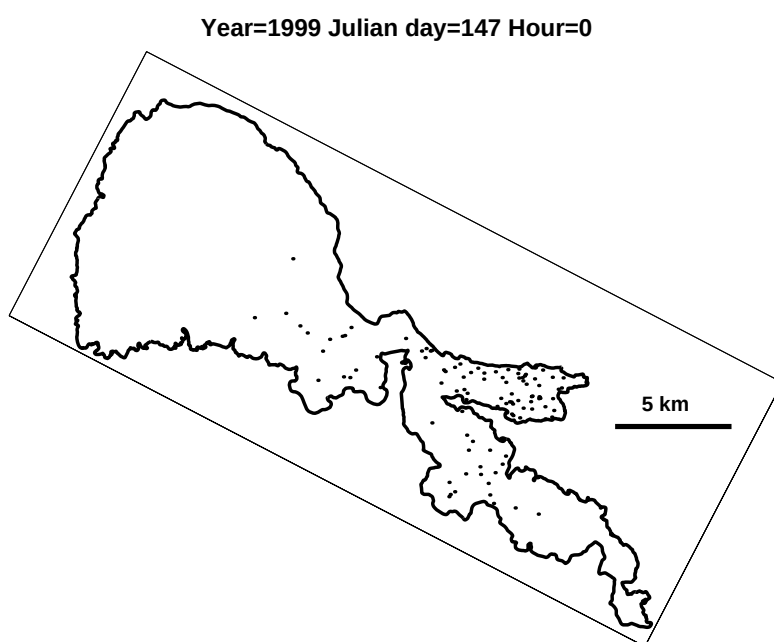
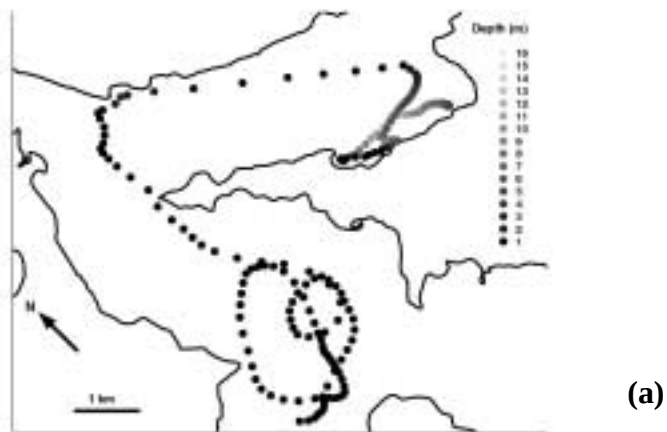


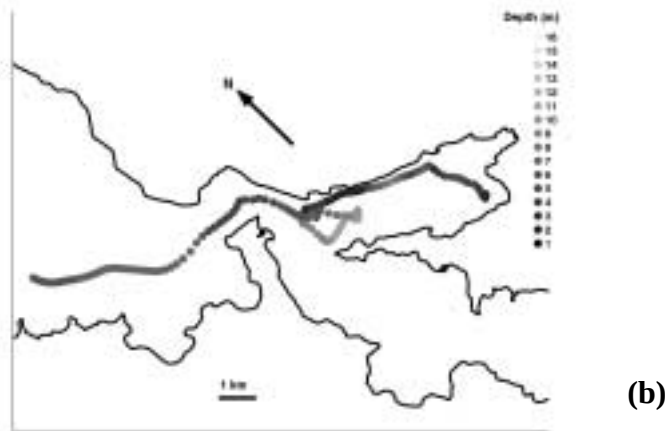
Figure 7.56. Particle distribution in plan view after 8 simulated days in the simulation of a stratified scenario in Clear Lake.

horizontal position. The positions are shown every hour, so an increased spacing between positions indicates higher velocities. Figure 7.57a shows the trajectory of a particle that, after plunging and rising twice at the downwind end of the Oaks Arm, is carried in day 142 by surface currents to the westernmost end of the Oaks Arm, and from there it enters the Lower Arm. The particle plunges following the wind-driven downwelling currents close to the mine, and it rises to the surface at the embayment located at the southwest edge of the Oaks Arm. In the Lower Arm it describes several clockwise loops, illustrating the wind-driven direct negative vorticity circulation in the Lower Arm. The trajectory shown in Fig. 7.57b is that followed by a particle that, once it reaches the bottom, follows the westward currents of the lower layers along the north shore of the Oaks Arm to end up 8 days later in the Upper Arm. A third particle (Fig. 7.57c) describes counterclockwise loops within the Oaks Arm, illustrating the positive vorticity of the sub-basin circulation. All particles that entered the Lower Arm from the Oaks Arm in the course of the simulation exercise did it through the surface layers, and all particles that entered the Upper Arm used the lower layers. This feature will be confirmed in section 7.8 by solving the Eulerian form of the transport equations describing the behavior of an ensemble of particles. It is further supported by observations during both experiments of the evolution and spatial structure of the exchange among the basins (section 7.5.5).

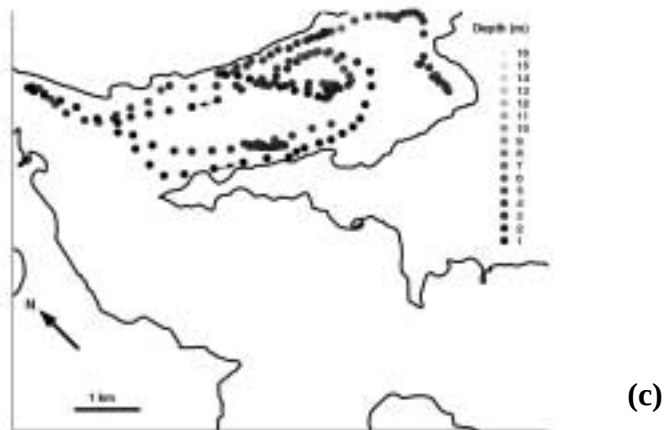
Figure 7.58(a) describes the trajectory of a particle affected by the baroclinically-driven return flow on day 144 that is confined to the north shore of the Oaks Arm (see Fig. 7.40). The particle covers approximately 2-3 km in 13 hours from site (1) in the figure at 0200h on day 144, to site (2) at 1500h under calm conditions. The time series of the



(a)



(b)



(c)

Figure 7.57. Trajectories followed by some particles in the particle tracking exercise. The arrows in subplot (c) indicate the direction of the particle motion (westward in the north, eastward in the south). Symbols are plotted one hour apart.

east-west velocity component of the particle is shown in Fig. 7.58(b) (top subplot). It exhibits first a change in the direction of motion from eastward to westward in the late evening of day 144, and then it experiences consistent negative, i.e. westerly, values during the evening and early morning hours of that same day. After eight days the particle ends up in the Upper Arm.

With a modified version of the hydrodynamic code, changes were deliberately introduced in the heat budget routines so as to reduce the heat flux at the surface. The stratification in the system was as a consequence rather weak, and the simulated conditions were close to isothermal. In fact, the maximum temperature difference attained in the 8 days of simulation with the modified code, measured at the deepest point of the Oaks Arm, was 0.7 degrees in 14-15 meters of water. The solution obtained with the “modified” code was then used to run the same particle tracking exercise as described above, to study the effects of stratification on the circulation in the Oaks Arm. Two aspects stand out. First, the mean transport rate of the 100 particles decreased considerably in this exercise in relation to that case with stratification. The mean distance covered by the particles was about 8-9, km in contrast with the 21 km calculated by using the realistic hydrodynamic simulations. None of the 100 particles released in the vicinity of the mine in the Oaks Arm entered either of the other two arms (Fig. 7.59). The second aspect is that, under very weak stratification, the circulation pattern in the Oaks Arm is that of a wind-driven re-circulation cell in a vertical plane that is oblique respect to the longitudinal axis of the Arm, as illustrated by the particle trajectory shown in Fig. 7.60. The downwelling region

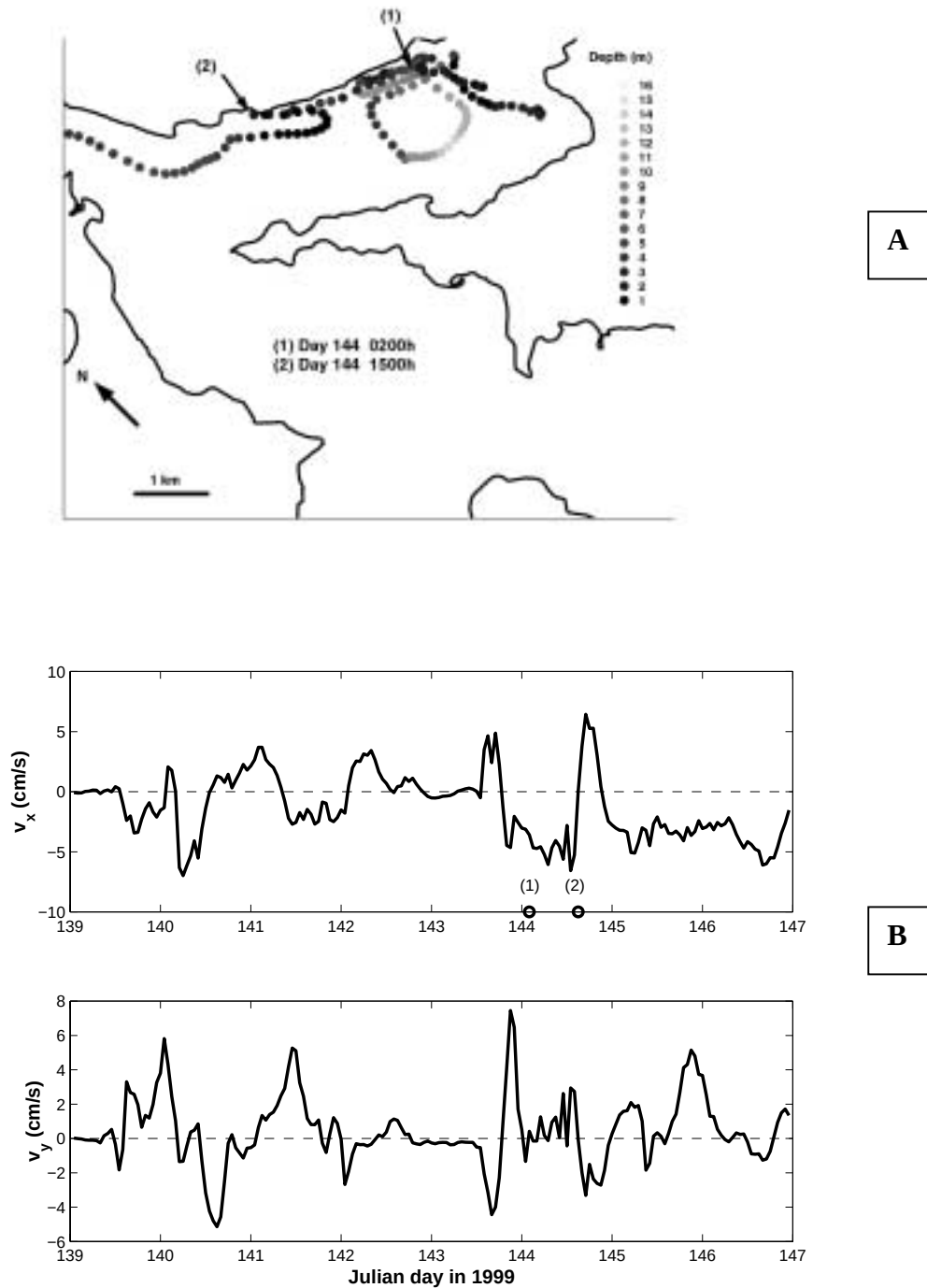


Figure 7.58. Trajectory of a particle illustrating the effect of baroclinic forcing during calm periods in the transport of particulate material. The top frame shows the trajectory of the particle, and the bottom frame shows a time series of the Lagrangian particle velocity. The locations of the particle at 200h on Julian day 144 and 13 hours later are marked in the top figure with arrows and identified as (1) and (2). Those two time marks are shown in the time series of longitudinal velocity as circles on the lower axis.

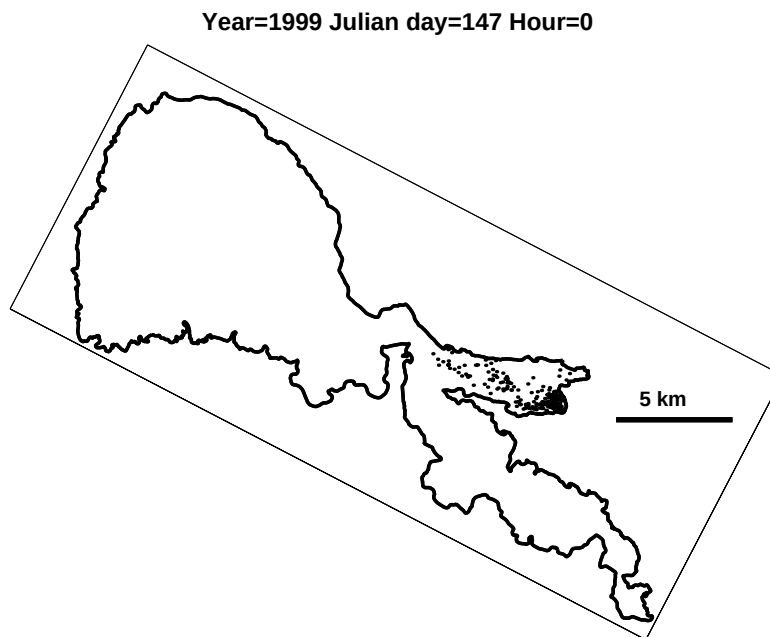


Figure 7.59. Distribution of particles after 8 days using hydrodynamic simulations with reduced heat budget and a very weak stratification in the system.



Figure 7.60. Particle tracking experiment on a weakly stratified Clear Lake. The particle shown in this figure reveals the re-circulation pattern in the Oaks Arm under a non-stratified scenario. The particle sinks at the eastern end of the Oaks Arm and surfaces at its northwest end, moving then to the southeast by being pushed by northwesterly winds.

is to the east and south in the Oaks Arm, and the upwelling region is at its northwest end, where the temperature station TII.1 was located during the second of the field experiments.

7.7.2. Simulating the behavior of the drogues in experiment II

The trajectories followed by the drogues in experiment II are individual realizations of a stochastic process. Not only the relatively small uncertainty of the release location of the drogues affects their trajectories, but the sub-grid-scale turbulent processes that are not resolved in SI3D-L can also induce random displacements in the drogue tracks. In trying to reproduce the Lagrangian experimental results with the particle-tracking algorithm, the stochastic nature of the trajectories ought to be considered. This has been done by (1) releasing a cloud of 100 particles with its center of mass located at the drogue's release location, and (2) by adding a random component to the advective or deterministic displacement of the particles calculated from the results of the hydrodynamic simulations. The random displacement is calculated, as explained in chapter 6, and by using a constant and uniform value of the horizontal eddy diffusivity of $1 \text{ m}^2/\text{s}$, which is typically employed in lake circulation models to describe horizontal turbulent processes (Cheng *et al.* 1976, Blumberg 1986). The particles were not affected by vertical velocities to simulate correctly the behavior of the drogues that remain at the depth at which they were released. The particle-tracking simulations started at 1100 h on Julian day 143 and continued until the end of day 146.

The simulated trajectories were examined one by one to identify those with features resembling the drogue tracks. The trajectories of some of those particles are shown in the series of plots in Fig. 7.61. The particle in Fig. 7.61.a (particle 1) was released at a depth of 8 meters and described the same back and forth motion as drogue 1. The time series of the Lagrangian velocity components of particle 1 (Fig. 7.62) reveals negative peaks (westward displacements) during the evening hours, as was the case of drogue 1 (Fig. 7.29.a), when the lake was subjected to northwesterly winds, and positive peaks (eastward displacements) after the wind decayed. The magnitude of the eastward velocity peak of the particle in the early hours of day 144, however, is smaller than what the drogue reveal, indicating that either SI3D-L underpredicted the velocity magnitude in this area of the lake or that more particle simulations should be analyzed.

The surface particles, each released 2.5 meters below the surface, all circled the Oaks Arm in a counterclockwise trajectory, as was the case for the drogues. Particle 2 (Fig. 7.61.b) reveals the vortex separation in the south embayment of the Oaks Arm; once it has reached the western end of the arm near the Narrows, it undergoes a southward displacement (compare with Fig. 7.28.b). Finally, particle 3 (Fig. 7.61.c) undergoes the back- and forth-motion that characterizes the displacement of the surface drogues (Fig. 7.28c and Fig. 7.28d) when they are located at the easternmost end of the Oaks Arm, and that is characteristic of the longitudinal response of the lake.

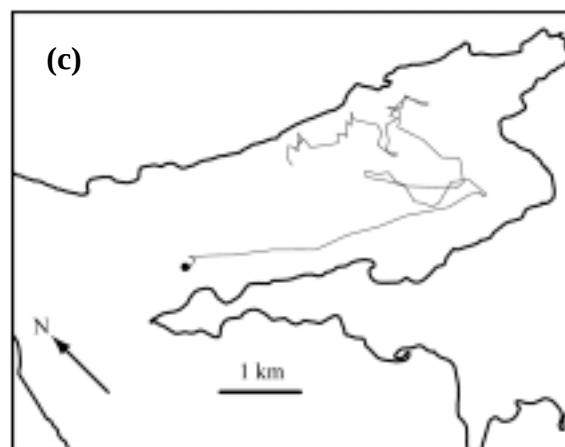
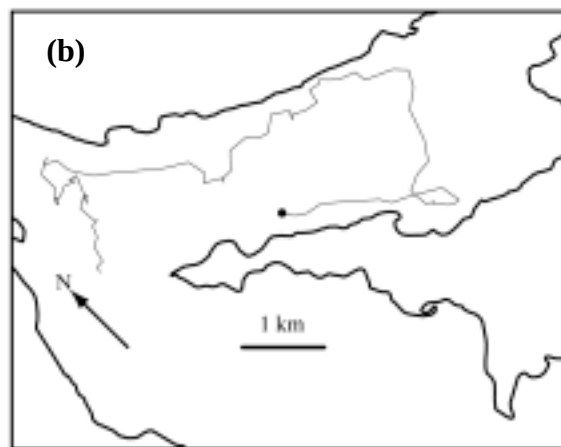
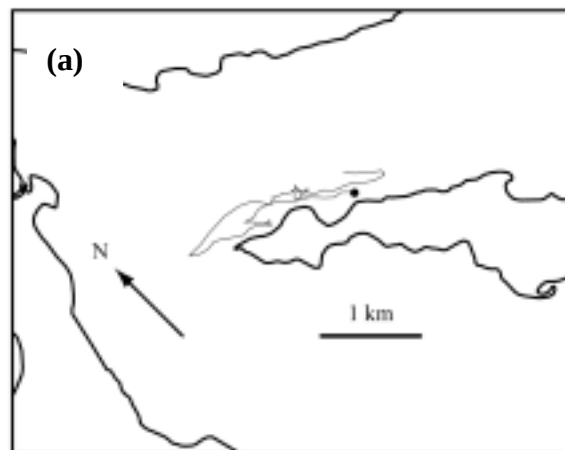


Figure 7.61. Simulated trajectories of buoyant particles released in the Oaks arm at 1100h on day 143, 1999. The dot indicates the release point, and the line its trajectory. Particle 1 (a) was released 8 meters below the free surface and shows the back and forth motion characteristic of the large-scale circulation of the Oaks arm. Particles 2 (b) and 3 (c) were released 2.5 meters below the free surface. They describe the large-scale cyclonic gyre that dominates the surface flows in the Oaks arm. Particle 2 (b) also reveals the separation vortex in the Wheelan Cove embayment. The behavior of these particles emulates that of the drogues.

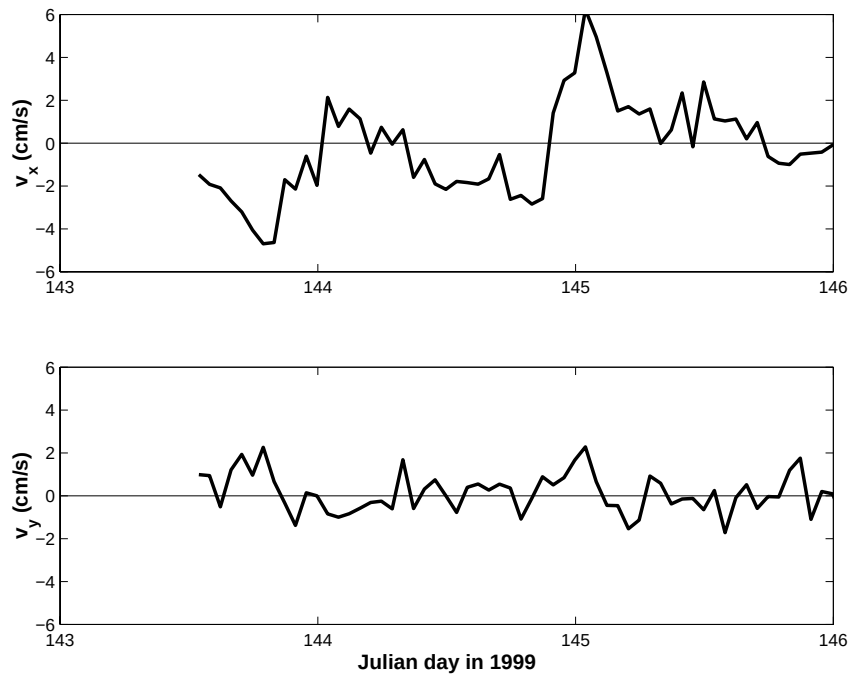


Figure 7.62. Lagrangian velocity of particle (a) in Fig. 7.61. It shows the periodic change in the sign of the longitudinal velocity, following the wind setup and relaxation cycle described in the text.

7.8. Numerical tracer simulations

The advection-diffusion equation for tracer plumes represents the behavior of an ensemble of particles being carried by currents and diffused by turbulent processes, i.e. its solution is equivalent to repeating a large number of times the tracking exercise presented in the last section. In this section the solution of the transport equation is presented for a passive tracer released in two locations in the Oaks Arm; this solution confirms some of the phenomena described in section 7.7 with individual particles, and it clearly portrays the circulation in the Oaks Arm. In these simulations the tracer was assumed to be immiscible (i.e. turbulent mixing processes are neglected) and was only advected by the mean lake currents.

In the first numerical tracer experiment (tracer experiment I) the initial concentration is a Gaussian plume with the center of mass at 1.5 m depth and at the horizontal location shown in Fig. 7.63a. Figures 7.63a through 7.63d show the decimal logarithm of the tracer concentration 1.5 meters below the surface at the release time (day 143, 1999 at 0100h) and at three later instants. The concentrations are normalized with the maximum concentration of the plume at its center. The intensity in the gray scale represents the normalized concentration, and it is only shown in Fig. 7.63a with the initial tracer distribution. The plots are rotated 27.5 degrees counterclockwise and display the same rotated geometry that was used in the 3-D code. Twenty four hours after the release, at 0100 on day 144, 1999, (Fig. 7.63.b) the tracer plume has been deformed, and some of it has been transported by wind-driven surface currents to the southeast end of the Oaks Arm. Ten hours later (Fig. 7.63c) the tracer is pushed back to the west, following the baroclinically generated return currents. The westerly return flow concentrated on the north shore projects a puff of tracer to the west, which illustrates quite remarkably the conceptual model of the setup and relaxation of temperature gradients influenced by the Earth's rotation, called here baroclinic pumping, as the mechanism driving the transport in the Oaks Arm during stratified conditions. Seventy-two hours after the release (Fig. 7.63d), the tracer has entered the Lower Arm and starts to move south following the north shore of that arm.

Figures 7.64a through 7.64d display the concentration field 7.5 meters below the surface in the tracer experiment I at the same times as shown in the series of plots in Fig. 7.63. The plume is released near the lake surface at the location shown in Fig. 7.63.a, and thus

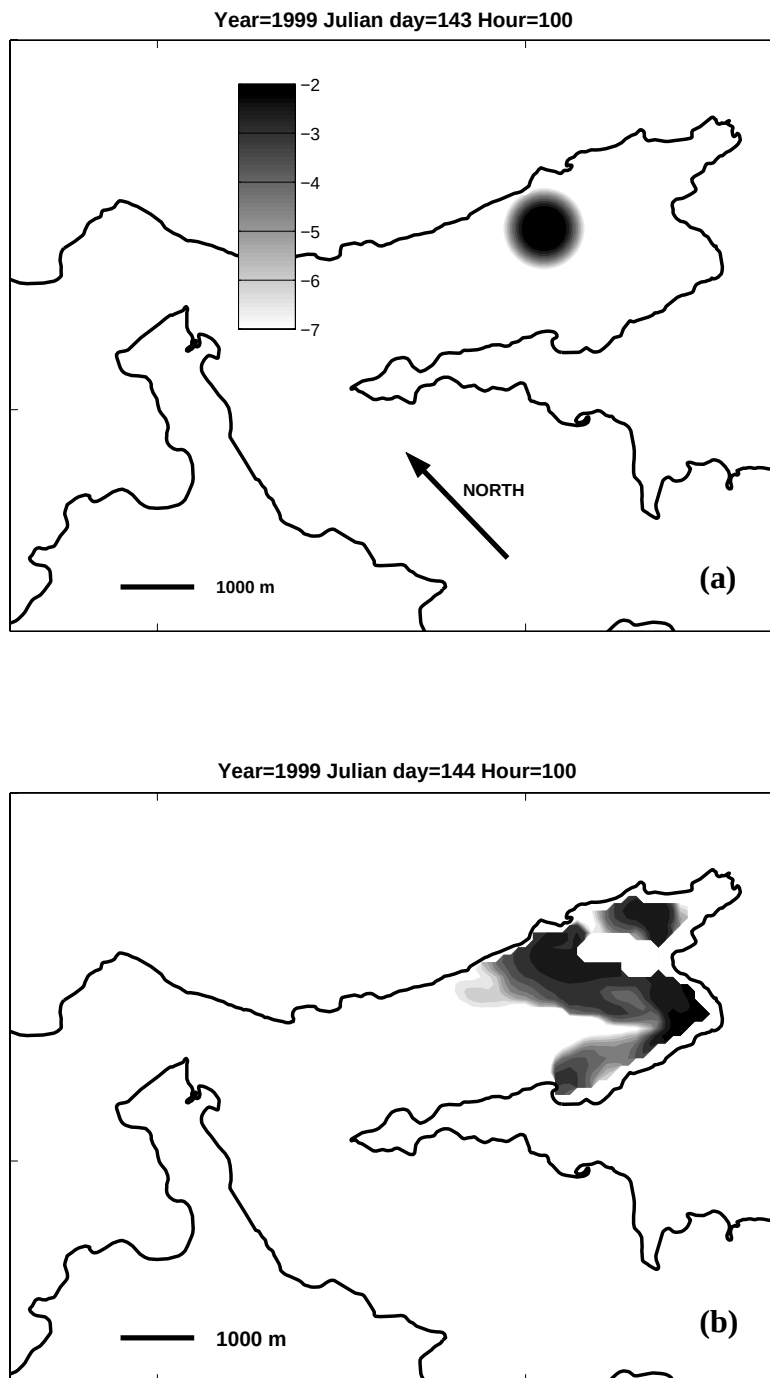


Figure 7.63. Tracer numerical experiment I. Tracer distribution 1.5 meters below the surface (a) at release time, (b) twenty-four hours later, (c) 10 hours after frame b; and (d) 72 hours after the release of the tracer, revealing the hydraulic connection of the Oaks Arm and the Lower Arm through surface currents. The gray scale shown in frame a represents values of the decimal logarithm of the tracer concentration. The geometry is rotated 27.5 degrees counterclockwise, as indicated in frame a.

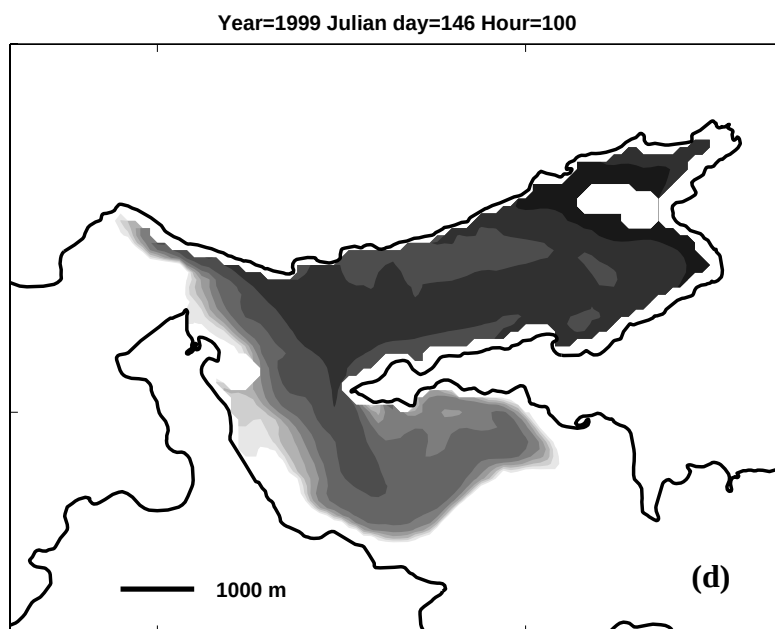
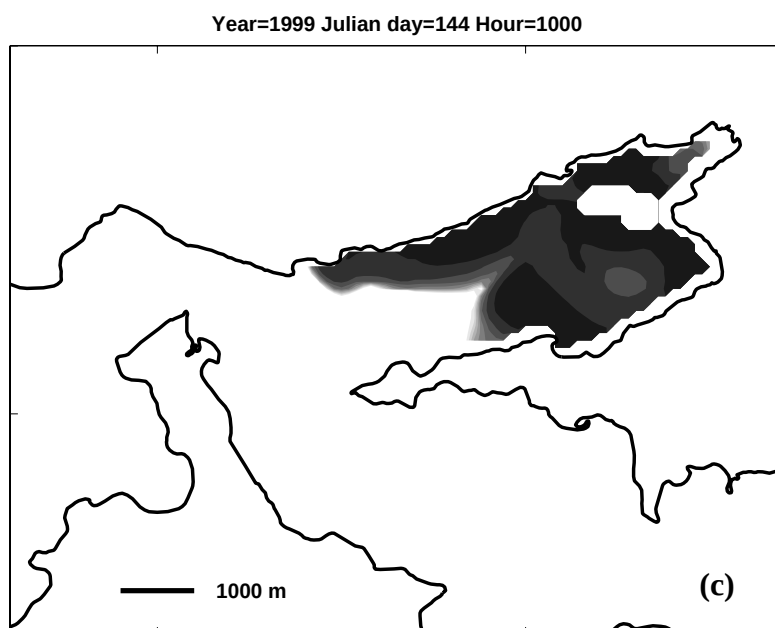


Figure 7.63 (continued).

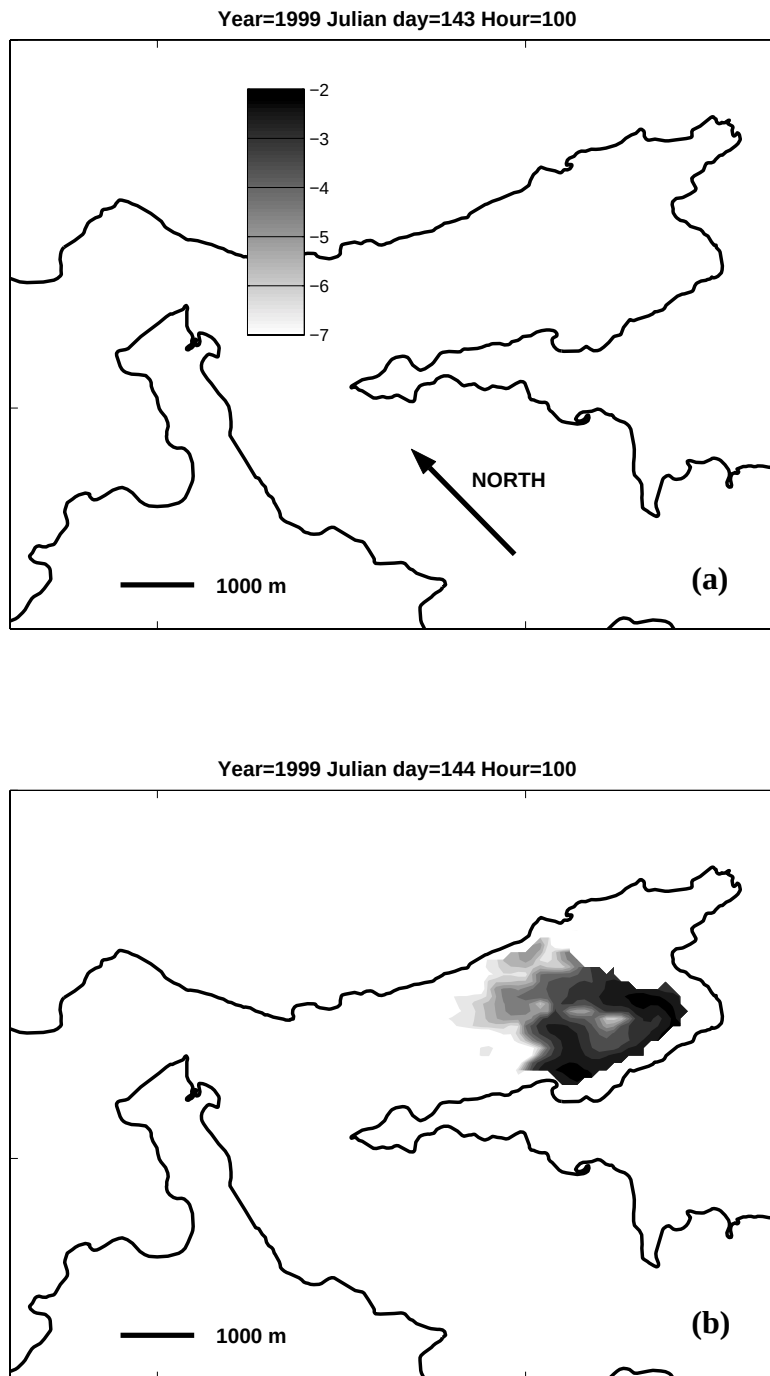


Figure 7.64. Tracer numerical experiment I. Tracer distribution 6.5 meters below the surface (a) at release time; (b) twenty-four hours later; (c) 10 hours after frame b; and (d) 72 hours after the release of the tracer revealing the hydraulic connection with the Upper Arm in the lower layers. Tracer concentration 6.5 meters below the surface. The gray scale shown in frame (a) represents values of the decimal logarithm of the tracer concentration. The geometry is rotated 27.5 degrees counterclockwise.

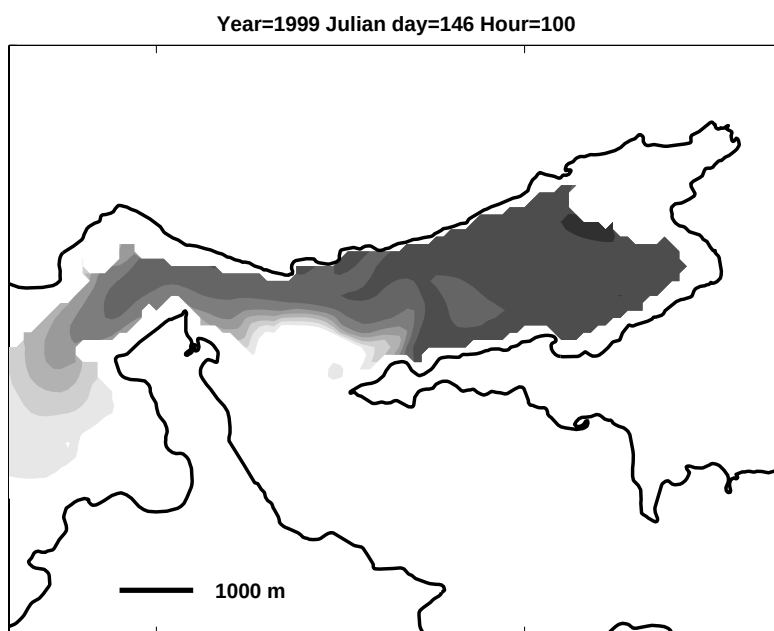
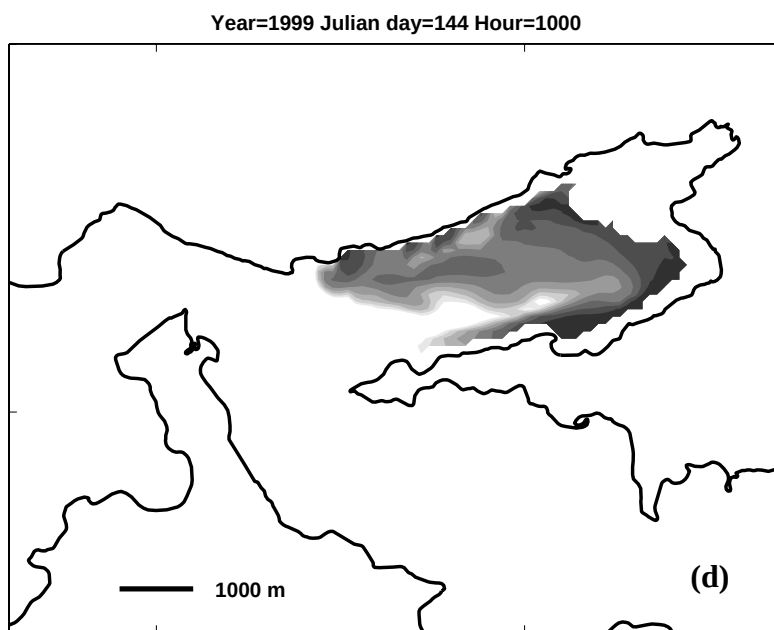


Figure 7.64 (continued).

the concentration of the tracer is initially zero at 7.5 meters (Fig. 7.44a). The tracer then plunges to this depth as a result of the action of the wind-driven downwelling currents at the easternmost end of the Oaks Arm (Fig. 7.64b). Ten hours later the tracer is pushed back to the west, mainly following the north shore (Fig. 7.64c), and after 3 days (Fig. 7.64d), the tracer at 7.5 meters has entered the Upper Arm. Figures 7.64d and 7.63d emphasize the idea, already presented in the last section, that the contaminant originating in the Oaks Arm reaches the Lower Arm mainly by transport with the surface currents, and enters the Upper Arm by transport with the deeper currents.

In a second experiment (tracer numerical experiment II), a tracer was released at 0100h on day 143, 1999, 1.5 meters below the surface at the northwest corner of the Oaks Arm (Fig. 7.65a), close to the Narrows, to illustrate the surface exchange patterns between the Upper Arm and the other two arms. The concentrations are again normalized on the peak concentration of the initial plume and are represented by the intensity of the gray scale. Figures 7.65b and 7.65c illustrate the tracer being transported east, following the south shore of the Oaks Arm and spreading into the Lower Arm.

7.9. Summary and conclusions

The hydrodynamics of Clear Lake, a relatively shallow polymictic lake, has been analyzed in this chapter in the light of observations collected during the course of two field experiments, and 3-D simulations conducted with the extended version of SI3D called SI3D-L. In particular, the research has focused on the Oaks Arm, its internal dynamics and the exchange with the other two basins. This strategy follows the interest

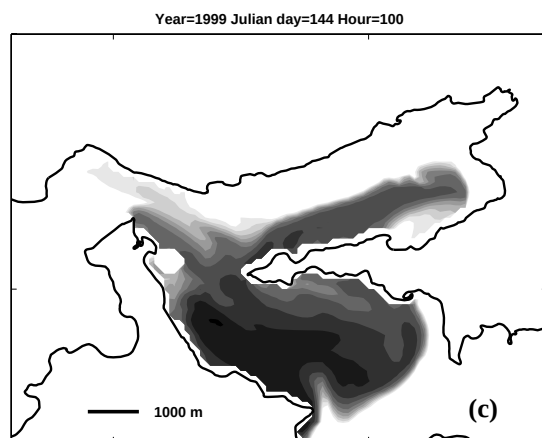
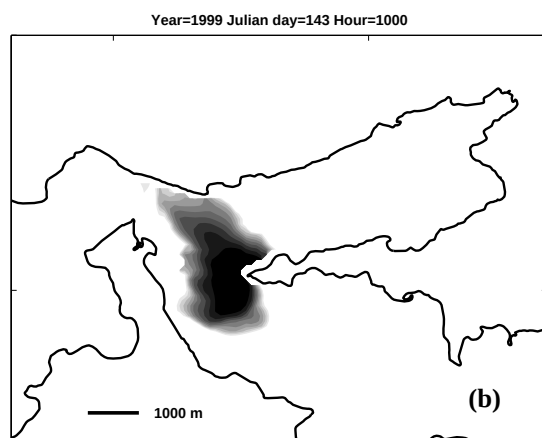
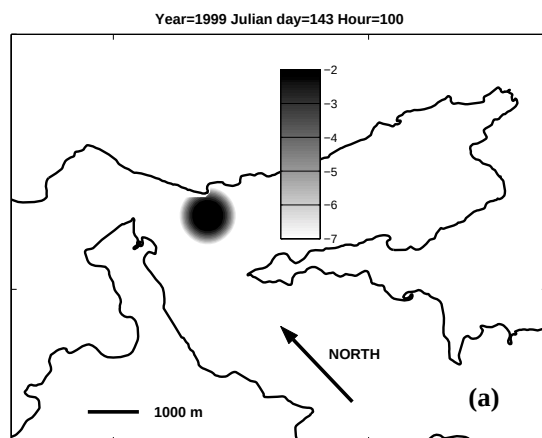


Figure 7.65. Tracer numerical experiment II. Tracer concentration 1.5 meters below the surface (a) at release time, (b) ten and (c) twenty- four hours later. The plots illustrate that water from the Upper Arm enters both the Oaks Arm following the south shore and the Lower Arm. The gray scale shown in frame (a) represents values of the decimal logarithm of the tracer concentration. The geometry of the lake is rotated 27.5 degrees counterclockwise, as indicated as well in frame (a)

other researchers of Clear Lake who wish to identify the paths of migration of mercury from its main source, the Sulphur Bank Mercury Mine in the Oaks Arm, to the rest of the lake. The dynamics in the Oaks Arm is characterized by a marked diurnal periodicity dictated by the wind regime. The typically northwesterly wind encounters a lake, that due to the surface heat flux during the morning and early afternoon hours, is stratified. The wind acts during the afternoon and evening hours on this weakly stratified system to generate horizontal temperature gradients both along and across the longitudinal axis of the Oaks Arm. During the night and early morning hours, when the wind forcing is negligible, the baroclinic pressure gradients that result from the horizontal differences in temperature become the dominant forcing mechanism in the system, driving currents of up to 10-15 cm/s westward at the surface and eastward near the bottom. This cycle of setup and relaxation of horizontal temperature gradients repeats in the stratified lake with a diurnal periodicity. The setup and relaxation processes are modulated by the influence of the earth's rotation and create a residual circulation. This circulation mechanism is here referred to as wind baroclinic pumping, or simply baroclinic pumping. The cyclonic residual circulation that results is superimposed on the direct wind-driven circulation, also of cyclonic nature, enhancing the rates of transport and facilitating the migration of mercury-laden particles to the other two arms of the lake.

Temperature observations gathered at three thermistor chains were used for validation of the three-dimensional model. The numerical results, after an initial spin-up period of two days, proved to be relatively close to the observations. The agreement between observations and simulation justify the analysis of the numerical solutions presented in

sections 7.6. The model was not subject to any form of calibration, though. In fact the model does not have any calibration parameters other than the value of the bottom drag coefficient. This suggests that the built-in algorithms are “good enough” representations of the natural processes being modeled, and could hence be applied to any other lake system if adequate information on the boundary and initial conditions are supplied. The execution time of the model on the platform used in this work, with a ratio of computational to real time of about 1:10, is good for relatively short runs of probably up to a month or even several months. Longer-term simulations of years would require the utilization of either faster or parallel platforms. However, the performance of the code in terms of computational time and quality of the results is very satisfactory.

The numerical simulations show that the factors driving the cyclonic baroclinic pumping circulation of the Oaks Arm are (1) stratification, (2) periodic and predominantly uniform westerly winds, and (3) Coriolis effects (Earth’s rotation). The spatial variability of the wind proved to be especially important in reproducing the anticyclonic circulation observed in the Lower Arm. The Earth’s rotation generates a remarkable asymmetry in the evolution of the temperature field across the longitudinal axis of the Oaks Arm, which is critical to an explanation of the existing residual cyclonic circulation and many of the observations gathered in the field. The asymmetric temperature stratification results in differences in the rate at which the vertical circulation switches from being driven by wind to being driven by baroclinic forces, along the north and south shores of the Oaks Arm. The change of vertical circulation starts at the easternmost end of the arm and moves to the west more rapidly along the north shore. Consequently there are stations on

the southwest side of the Oaks Arm where the baroclinic circulation rarely replaces the wind-driven circulation, while across the basin on the north shore the baroclinic circulation is felt more frequently and more intensely. This difference in the amount of time the baroclinically-driven vertical circulation dominates the wind-driven motions is reflected in the residual profiles. In the north the residual currents are westward, while in the south the residual currents are eastward. The influence of the asymmetric evolution of the temperature and velocity fields on the levels of mixing is studied in section 7.6.5. There is shown that differences exist in the mixing levels between the north and south stations. Those differences are related to the shear and stratification intensities, which are mostly controlled by large-scale processes. However, there could exist differences in the mechanisms producing mixing in the northern and southern portions of the Oaks Arm, that have not been investigated here but which deserve further consideration in future research.

The description here of the hydrodynamics of the Oaks Arm is sound and based on multiple sources of information, such as Eulerian observations of temperature from thermistor strings, Eulerian velocity observations with acoustic devices from fixed and mobile platforms, numerical 3-D simulations, and observations collected with Lagrangian current followers (drogues). The autonomous drogues are a simple and economical means of studying large-scale motions in large water bodies. In this study the trajectories of the drogues provided a magnificent and astonishing description of the large-scale residual cyclonic circulation of the Oaks Arm and the setup-relaxation cycles. They also provided evidence that shoreline irregularities enhance the rates of dispersion of scalars by

trapping and later releasing contaminants that would otherwise simply respond to the large-scale motions of the lake. The simple calculations presented in section 7.5.6 indicate that embayments of approximate 100 meters, which are not uncommon in the Oaks Arm, could even double the rates of longitudinal dispersion. The resolution of the computational grid used in the simulations of Clear Lake ($\Delta x = \Delta y = 100\text{m}$) is not fine enough to describe accurately the details of the circulation in those small embayments. If the calculations of section 7.5.6 are correct, the effects of the shoreline irregularities should somehow be parameterized. The trapping mechanism deserves further study in future research. The entrapment effect was reproduced with numerical particle tracking simulations in the largest of the embayments in the southeast corner of the Oaks Arm. All other features of the drogue tracks were successfully reproduced by using the 2-D version of the particle tracking routines developed in this work and presented in chapter 6, providing additional validation to the model results.

The exchange among the basins was investigated by using 3-D particle tracking routines and passive tracer releases. Those techniques also proved adequate in visualizing the baroclinic pumping mechanism driving the internal circulation in the Oaks Arm. The rate and patterns of mass exchange between the basins under stratified and non-stratified conditions were investigated in section 7.7. Clouds of particles and Gaussian plumes of passive tracers were released at several locations across the Oaks Arm and tracked for several days. The numerical results indicate that during non-stratified conditions the circulation in the Oaks Arm is such that very little exchange of particles exists with the other two basins. Under stratified conditions, however, the rate of transport is enhanced,

and the transport among the three arms of the lake is fairly active. Any contaminant released in the Oaks Arm would reach the other two basins in 3-7 days. The contaminant would penetrate into the Lower Arm following the surface currents or into the Upper Arm mainly through the bottom currents.

The observations, and particularly the numerical simulations, revealed that Clear Lake is a very active system that changes dramatically in response to the diurnal wind events. Because of the changing nature of the hydrodynamics of Clear Lake, observations gathered at a given moment in time may not be representative of the state of the lake 12 hours later. The lake hydrodynamics is also spatially variable, and the conditions at a given sampling station will not be representative of existing conditions only 1000 meters away. Basing any interpretation of field experiments on partial observations could lead to incorrect answers. The use of velocity transects to characterize the lake circulation, for example, may be very misleading, as there exists a notable bias to obtain observations during daylight hours and calm periods. It is during those hours when the lake adjusts from a type of circulation driven mainly by wind to another mainly driven by baroclinic pressure gradients, as shown in the 3-D simulations. Only one transect during the two experiments was performed under strong winds (shown in Fig. 7.17), and it was this one that probably provided the most important clue in the search for the mechanisms of motion in Clear Lake. These remarks challenge the strategy of many sampling campaigns where data, such as velocity, temperature, pH or other water quality parameters, are collected at weekly intervals in at most one or two places within lakes of similar characteristics. The sampling strategy should be adapted to the time and spatial scales of

change of the system. Given the impossibility to sample both at the right spatial and temporal scales of change for active systems such as Clear Lake, 3-D models should be employed to fill the gaps in the observational data. SI3D-L has been proven here to be an invaluable tool in the identification of the key transport processes in the Oaks Arm.

CHAPTER 8

MODELING BASIN-SCALE INTERNAL WAVES IN LAKE TAHOE

8.1. Motivation

Straddling the border between California and Nevada, Lake Tahoe (Fig. 8.1) is one of the largest alpine lakes in the United States, renowned for its beauty and remarkable transparency. The watershed geology, large volume, and limited drainage area are largely responsible for the lake's sparse algal population and clear waters (Goldman 1988). Few nutrients are naturally available to fertilize its waters. However, human development in the Tahoe Basin over the last five decades has been quite damaging. The lake has responded to increased nutrient and particulate loading from the streams, atmosphere and groundwater with steadily increasing algal growth (*cultural eutrophication*) and a progressive reduction of clarity. The lake has lost approximately 10 m of transparency⁵⁰ during the past 27 years, a rate that, if maintained, will result in the loss of Tahoe's famous clarity in about 30 years (Goldman 1988).

Basin-scale internal waves provide a driving force for vertical and horizontal fluxes in a stratified lake below the wind-mixed layer, and these waves should be modeled accurately for the correct simulation of the dynamics of nutrients and particulates, which

⁵⁰ The transparency is measured with the help of a secchi disk, a white or white-black disk that is lowered into the water to measure the depth at which it cannot be distinguished any more. That depth defines the secchi depth and is used as an indication of water clarity. The Tahoe Research Group at the University of California in Davis (see <http://trg.ucdavis.edu>) has been monitoring the secchi depth in Lake Tahoe since 1967, and their data reveals the trend of declining transparency in Tahoe waters.

controls the water clarity. The energy imparted by wind to a lake leads to basin-scale waves that steepen due to non-linear processes and shoaling, so that the energy at the largest basin scales cascades through a spectrum of wave frequencies (from 10^{-6} Hz to 10^{-2} Hz) to the buoyancy frequency (Imberger 1998). As reviewed by Imberger (Imberger 1998), most of the energy of internal waves in the range from 10^{-5} to 10^{-2} Hz arriving at sloping boundaries is dissipated, leading to the re-suspension and mixing of particles and nutrients in the water column. This phenomenon is referred to as boundary mixing (e.g. MacIntyre *et al.* 1999) and is probably the most important mixing mechanism below the wind-mixed layer (Fischer *et al.* 1979).

The second application of the 3-D hydrodynamic model consists of the simulation of internal waves in Lake Tahoe, and this simulation is the first step towards the modeling of mixing mechanisms (boundary mixing) below the surface mixed-layer. The simulations in this work are particularly aimed at explaining the nature of the oscillations observed in temperature records collected in Lake Tahoe during winter 1999-2000. The performance of SI3D-L is checked against the results of both (1) a theoretical analysis of the initial conditions, using simplified models of the geometry of Lake Tahoe, and (2) the Fourier analysis of the observations from two thermistor chains. A brief description of the temperature data set employed for the validation of the numerical results is given in section 8.3. The theoretical analysis is presented in section 8.4, and it provides considerable insight into the types of free waves that the lake stratification should be able to support and the 3-D hydrodynamic model should be capable of representing. The characteristic features of the different types of internal waves are also discussed in that

section. The measured and simulated results will be presented in sections 8.5 and 8.6 respectively.

8.2. Physical characteristics of Lake Tahoe

Lake Tahoe (Fig. 8.1) is located along the crest of the Sierra Nevada mountain range between California and Nevada at an altitude of 1898 m (Goldman 1988). It has a maximum depth of 505 m, at the highest water surface elevation, and an average depth of 313 m. Its surface area is 501 km², and that of the watershed is 812 km². The relatively small and forested watershed, which is drained by 63 small streams, is mostly formed from decomposed granite and volcanic formations. The Lower Truckee River is the lake's only outlet. This combination has resulted in a lake of extremely low fertility and high transparency (Goldman 1988). Lake Tahoe contains 156 km³ of water, and its hydraulic residence time is approximately 650 years. The lake is monomictic and does not freeze. Typically, thermal stratification commences in the period from February to April and reaches a maximum in August, at which time the thermocline is approximately 20 m below the surface. Slow weakening of the stratification may lead to approximately isothermal conditions by February.

8.3. Brief description of the field data set

Two thermistor chains were deployed in Lake Tahoe during winter 1999-2000 to provide continuous record of the winter temperature and density structure. One of the chains was deployed at the INDEX station, near the west central shore (N 39° 6.744', W 120° 8.932'), on a steep slope where the lake is approximately 125 m. The second thermistor chain was

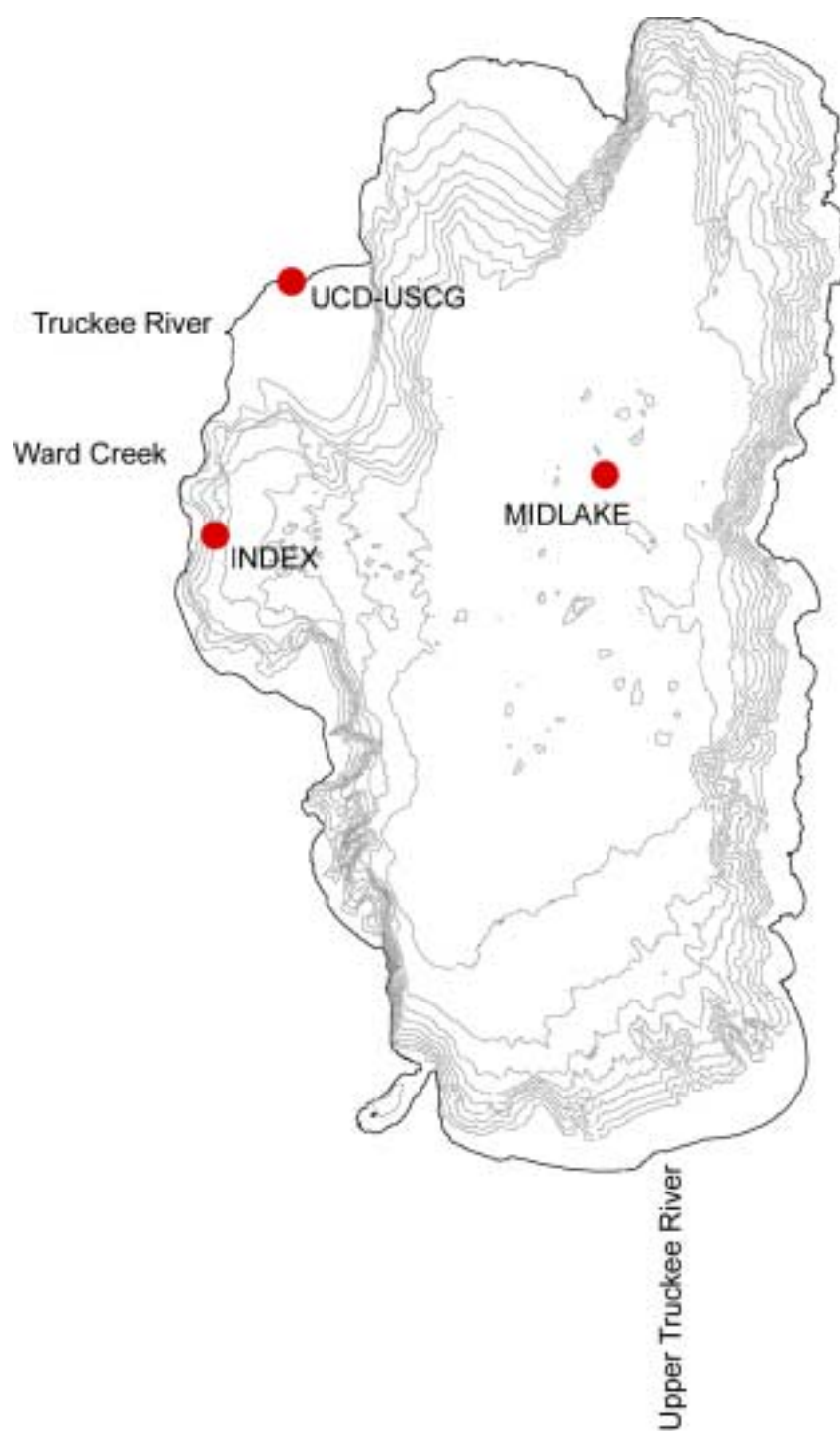


Figure 8.1. Lake Tahoe bathymetry and deployment locations. Bathymetry is derived from recent U.S.G.S. data (Gardner et al. 1998) and contoured on 50 m intervals. Thermistor chain locations are identified in the map as MIDLAKE and INDEX stations. The UCD-USCG label indicates the location of the meteorological station on the Coast Guard Dock.

deployed near the center of the lake, at the MIDLAKE station (N $39^{\circ} 7.535'$, W $120^{\circ} 0.731'$), where the depth is approximately 463 m. Figure 8.1 indicates the location of these two stations within Lake Tahoe. The two time series of temperature analyzed in this work are segments of 20 days each, from the larger data set collected at INDEX and MIDLAKE stations during the winter months of 1999-2000. The time series start on December 19, 1999 and finish on January 9, 2000, shortly before the occurrence of an upwelling event that considerably modified the temperature structure of Lake Tahoe. In the records, water temperature was sampled at 2 minute intervals, using Oregon Environmental Instruments Inc. (OEI) 9311 and Richard Brancker Research Ltd. (RBR) TR-1000F data loggers. The loggers were attached to the chains at 10 to 40 m intervals, with the closer spacings near the surface. The meteorological data were collected at the dock of the Tahoe U.S. Coast Guard station, located at (N $39^{\circ}10.838'$, W $120^{\circ}07.157'$). Figure 8.2 displays the time series of the data collected at the meteorological station, that includes the period subject to analysis. All the experimental data analyzed in this work were collected by Palmarsson (Palmarsson 2000), as part of his work on the effect of boundary mixing on Lake Tahoe water clarity. For further details on the thermistor chains and the weather station, the reader is referred to Thompson 2000.

8.4. Analysis of the initial temperature profile

The temperature records at MIDLAKE station were averaged in time during the first three days of the 20-day period to provide a description of the initial vertical temperature distribution that would be relatively free of internal wave influences. The temperature profile that results from that averaging exercise is shown in Fig. 8.3. It is characterized by

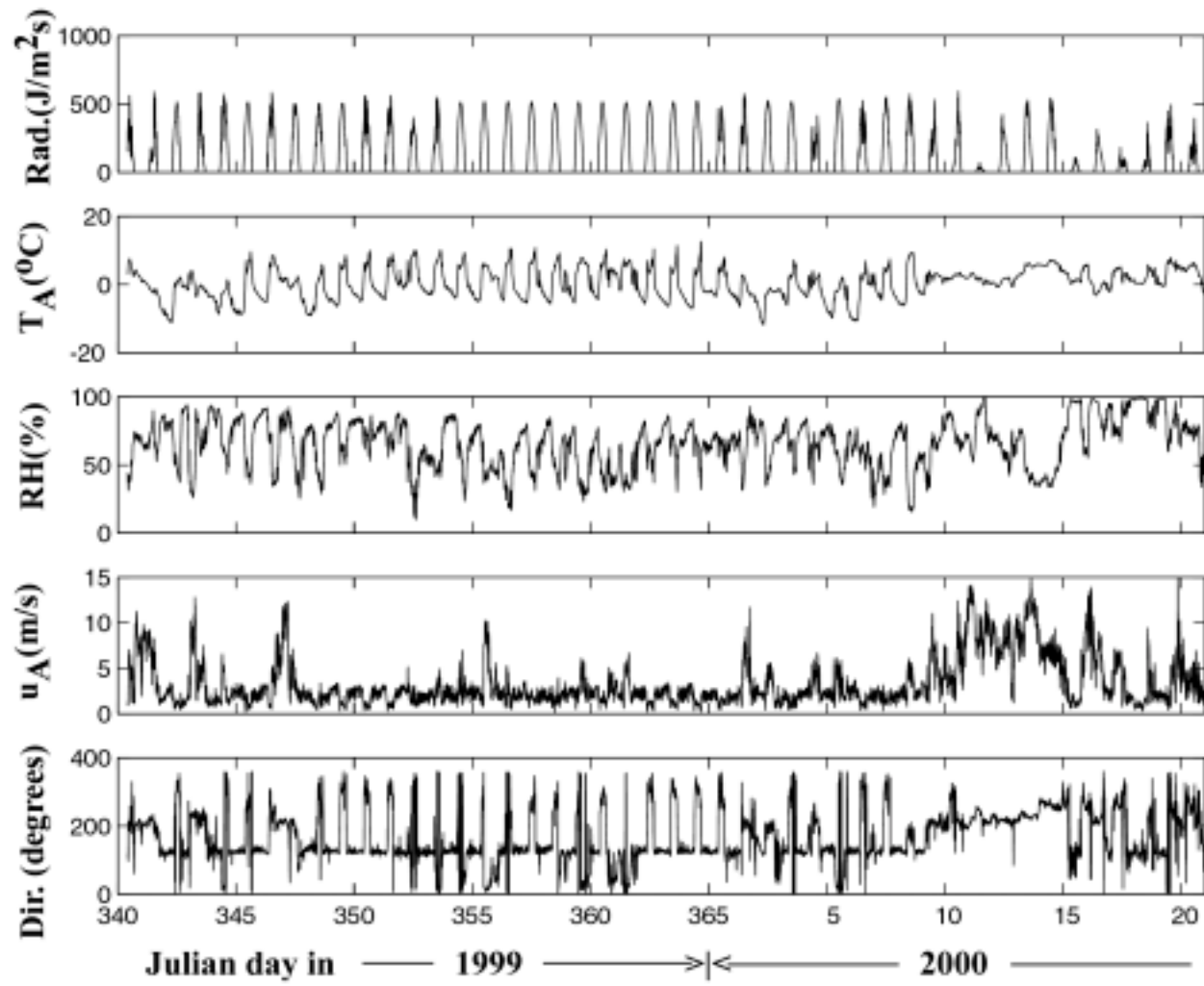


Figure 8.2. Time series of selected meteorological variables that encompasses the period analyzed in this work. This period started on Julian day 353, 1999, and ended on Julian day 10 in year 2000.

a relatively smooth gradation in temperatures from the epilimnion, with an almost uniform temperature of 7.5 °C in the first 50 meters of the water column, to the hypolimnion starting at about 150 m below the water surface and having a uniform temperature of less than 5 °C . The analysis of this initial temperature profile can provide considerable insight into the internal wave climate existing in Lake Tahoe during the 20 days of winter 1999-2000 that will be examined in this work. It is first necessary, for the purposes of this analysis, to assume that Lake Tahoe can be approximated as a rectangular basin, with a flat bottom of dimensions 34000m x 19600m x 500 m (box model), or as a circular flat-bottomed basin with a diameter of 19600m (circular model). Due to its latitude ($\theta \sim 39^\circ$), the inertial period of oscillation is 19.07 hours (inertial frequency $\sim 9.189 \times 10^{-5} \text{ s}^{-1}$).

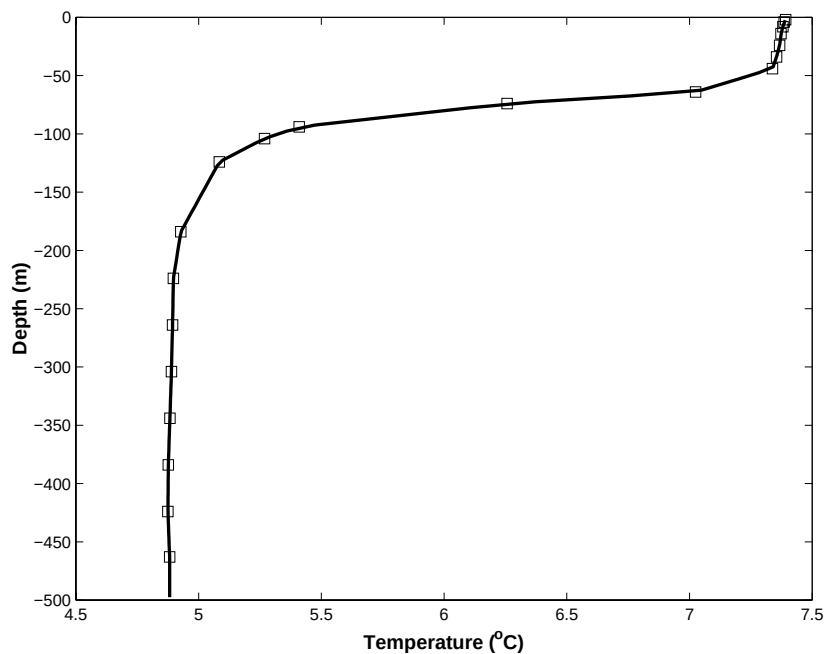


Figure 8.3. Three-day average temperature profile from MIDLAKE station used to characterize the initial vertical temperature distribution in the 20-day period starting on December 19, 1999. The location of the temperature loggers in the thermistor chain is identified with square symbols along the profile (□).

8.4.1. Speed of propagation of internal waves

The stratification existing in Lake Tahoe on December 19, 1999 (Fig. 8.3) can be closely represented as a two-layer system. As such, the speed of propagation of perturbations at the interface (internal waves) is given by

$$c = \sqrt{g' \frac{h_1(H - h_1)}{H}} \quad (8.1)$$

in which h_1 is the thickness of the upper layer, H is the total depth of the water column, and $g' = g\Delta\rho/\rho_0$ is the reduced gravity. For Lake Tahoe it is appropriate to take $h_1 = 100$ m and $H = 500$ m. In addition, $\Delta\rho/\rho_0 (= \alpha\Delta T)$ is 9.45×10^{-5} (using $\Delta T = 2.1^\circ\text{C}$ and the coefficient of thermal expansion, $\alpha = 45 \times 10^{-6} \text{ }^\circ\text{C}^{-1}$), and $g' = 9.26 \times 10^{-4}$. The speed of propagation that comes out of Eq. 8.1 is about 0.27 m/s.

A similar result is obtained by using the continuous stratification from Fig. 8.3 and a modal decomposition approach (Kundu 1990); this result is shown in Table 8.1. The governing equation for internal waves in a continuously stratified system is (Kundu 1990)

$$\frac{\partial^2}{\partial t^2} \nabla^2 w + N^2 \nabla_H^2 w = 0 \quad (8.2)$$

where

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}, \quad \nabla_H^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}, \quad \text{and} \quad N^2 = \frac{g}{\rho} + \frac{\partial \rho}{\partial z}. \quad (8.3)$$

Here w is the vertical velocity component and (x, y, z) are Cartesian coordinates. The vertical coordinate is measured upwards with its origin at the free surface, and the boundary conditions are $w(0) = w(-H) = 0$. In the modal decomposition one assumes that the vertical velocity is a periodic function and can be described by

$$w(x, y, z, t) = W(z) \exp[i(kx + ly - \omega t)] \quad (8.4)$$

where $W(z)$ represents the structure of the vertical velocity in the vertical direction z , k and l are wave numbers in the horizontal plane, and ω is the frequency of oscillations. It can be assumed without loss of generality that the oscillations represented in Eq. 8.4 are aligned with the x -axis, i.e. $l = 0$. The simplified Eq. 8.4 once substituted in Eq. 8.2, then gives

$$\frac{d^2 W}{dz^2} + \left(\frac{N^2}{\omega^2} - 1 \right) k^2 W = 0 \quad (8.5)$$

This equation only has non-trivial solutions for discrete values of ω and k . In the particular case of standing waves k , is an integer part of twice the length of the lake ($2L$) and hence

$$k_m = m \frac{\pi}{L} \quad (8.6)$$

Equation 8.5, with k_m given by Eq. 8.6, constitutes an eigenvalue problem that is discretized and solved using linear algebra tools (see appendix D). The results for the first modes are given in Table 8.1.

<i>NORMAL MODE</i>	<i>Speed of propagation (m/s)</i>
1	0.2480
2	0.0599
3	0.0387

Table 8.1. Speed of propagation for vertical modes 1, 2 and 3, as obtained through modal decomposition using the initial temperature profile provided to SI3D-L.

8.4.2. Internal Rossby Radius of Deformation

The Rossby radius of deformation defines a length scale, above which the rotation of the earth introduces significant effects into the hydrodynamics of an aquatic system. For barotropic processes it is well above the dimensions of Lake Tahoe, signifying that the rotation of the earth will not introduce significant effects on the barotropic behavior of the lake. For stratified systems the internal Rossby radius of deformation is defined as

$$\Lambda = c / f \quad (8.7)$$

where f is the Coriolis parameter (inertial frequency) and c is the speed of propagation of internal waves. For Lake Tahoe and first vertical mode internal waves the internal Rossby radius is about 3 km, well below the width of the basin (~ 20 km). This suggests that, even with the weak stratification existing in Lake Tahoe during the period of time analyzed, the earth's rotation will exert a considerable influence on the internal waves. These internal waves may be of Kelvin or Poincaré form.

8.4.3. Internal Kelvin waves

Perturbations introduced in a stratified lake may travel cyclonically (counterclockwise) around the basin, with the coast on the right. These are internal Kelvin waves (KW). The

amplitude of the perturbations shows an exponential offshore decay with a scale of $O(\Lambda)$, the internal Rossby radius of deformation. The velocity fields show maxima at the boundaries (where the motion is rectilinear), with current vectors in the lake interior rotating counterclockwise.

The parameter governing the possible existence of internal Kelvin waves in circular basins is given by (Csanady 1967)

$$\beta = \frac{f^2 R^2}{c_i^2} \quad (8.8)$$

where c_i is the speed of propagation of internal waves of vertical mode i . This parameter is the square of the ratio of the radius of the basin to Λ . For each azimuthal wave number n ($= 0, 1, 2, \dots$), there is one Kelvin mode provided that the following conditions applies

$$\beta > n(n+1) \quad (8.9)$$

In a circular model of Lake Tahoe with radius $R_0 \sim 10$ km and a depth of 500 m, for the temperature profile in Fig. 8.3 and the first vertical mode, β is ~ 12 , and Eq. 8.9 is satisfied for the first vertical mode and azimuthal wave numbers $n = 1, 2, 3$. The frequency σ of the Kelvin wave for a circular model would be given by:

$$\frac{R_0}{R^*} \cdot I_{n-1}\left(\frac{R_0}{R^*}\right) + n\left(\frac{f}{\sigma} - 1\right) \cdot I_n\left(\frac{R_0}{R^*}\right) = 0 \quad (8.10)$$

where I_n is the modified Bessel function of order n , and R^* is given by the expression

$$R^* = \frac{c_i}{\sqrt{f^2 - \sigma^2}} \quad (8.11)$$

At a fixed station near the shore the Kelvin wave will be perceived as a wave with length $\lambda = 2\pi R_0/n$ and period $2\pi/\sigma$. The apparent velocity is $c_K = R_0\sigma/n$ and is always close to the velocity of propagation of an internal wave in the absence of rotation. Hence, an estimate of the period of an internal Kelvin wave with an azimuthal wave number of 1 is given by

$$T_K = \frac{P}{c_i} \quad (8.12)$$

In this equation P is the perimeter of the basin, and c_i is the speed of propagation of the baroclinic wave of vertical mode i ($i = 1, 2, \dots$). Using a wave speed of 0.27 m/s (Table 8.1) and the approximate perimeter of Lake Tahoe (115 km), the first vertical mode, first horizontal mode internal Kelvin wave would have a period on the order of 4 or 5 days with the existing conditions in winter 1999-2000. Kelvin waves of higher horizontal wave numbers have periods that are an integer fraction of the period shown in Eq. 8.2. For example, a first vertical mode, second horizontal mode Kelvin wave will have a period of approximately 2 to 2 ½ days in Lake Tahoe in winter conditions.

8.4.4. Poincaré Waves

A different mode of free internal oscillations is the Poincaré wave or rotating internal seiche (Csanady 1967). In the Poincaré mode the perturbations propagate clockwise. The velocity vectors rotate clockwise with maxima at the center of the basin. As is the case

for the Kelvin modes, the Poincaré modes are progressive waves. In a circular model of a lake, the frequency of Poincaré waves is governed by

$$\frac{1}{n} \cdot J_n^* \left(\frac{R_0}{R^{**}} \right) = 1 - \frac{f}{\sigma} \quad (8.13)$$

where

$$J_n^* \left(\frac{R_0}{R^{**}} \right) = \frac{R_0}{R^{**}} J_{n-1} \left(\frac{R_0}{R^{**}} \right) / J_n \left(\frac{R_0}{R^{**}} \right), \text{ and } R^{**} = \sqrt{\frac{c_i^2}{\sigma^2 - f^2}} \quad (8.14)$$

An infinite number of solutions can be found for any n (azimuthal wave number). For fixed n ($\sim$ number of nodal diameters) the individual solutions may be distinguished by the number m of nodal circles present in the corresponding pattern of surface and interface elevations. The spatial pattern of the Poincaré waves will hence consist of cells of radial dimensions $r_j - r_{j-1}$ (r_j = radius of the j th nodal circle) and of angular width $(\pi + 2k\pi)/2n$, with $k = 0, 1, 2, \dots$ (Csanady 1967). For the circular model of Lake Tahoe, the solution of the frequency equation indicates that only the lowest modes have a frequency close to the inertial frequency ($\sigma/f \sim 1.15$ for $n = 1, m = 0$).

In a channel of infinite length with width B and depth H , the Poincaré waves represent transverse oscillations, and their frequencies satisfy a much simpler dispersion relation

$$\sigma_n^2 = f^2 + \left(n\pi \frac{c}{b} \right)^2 \quad (8.15)$$

where n indicates the horizontal mode, and c the speed of propagation. Applied to Lake Tahoe, Eq. 8.15 gives the periods shown in Tables 8.2 and 8.3. The frequency of the internal Poincaré waves approaches the inertial frequency f for the lowest modes of oscillation. This same period ($2\pi/f$) characterizes the well-known simple mode of motion in a rotating fluid far from any boundaries. In contrast to this simple model, in a Poincaré wave there are also significant surface or thermocline motions (Csanady 1975).

Table 8.2. Characteristic frequency of oscillations along the longitudinal axis (N-S) for the box model of Lake Tahoe ($b = 34$ km), from Eq. 8.15

<i>Mode</i>	σ/f	σ (cph)	T (hours)
1	1.1014	0.0580	17.245
2	1.3609	0.0717	13.956
3	1.7080	0.0899	11.120
4	2.0996	0.1105	9.046
5	2.5151	0.1324	7.552
6	2.9443	0.1550	6.451
7	3.3820	0.1781	5.616

Table 8.3. Characteristic frequency of oscillations along the transverse (E-W) axis for the box model of Lake Tahoe ($b = 19$ km), from Eq. 8.15.

<i>Mode</i>	σ/f	σ (cph)	T (hours)
1	1.0362	0.0546	18.330
2	1.1379	0.0599	16.692
3	1.2897	0.0679	14.727
4	1.4763	0.0777	12.866
5	1.6861	0.0888	11.265
6	1.9114	0.1006	9.937
7	2.1475	0.1131	8.844

The problem of estimating the frequencies of the Poincaré waves may be addressed by using the modal decomposition of the governing equations again, but now including the Coriolis terms. The problem then becomes an eigenvalue problem given by

$$\frac{d^2 W}{dz^2} + m^2(z)W = 0 \quad (8.16)$$

where

$$m^2(z) = \frac{(k^2 + l^2)(N^2(z) - \omega^2)}{\omega^2 - f^2} \quad (8.17)$$

Equations 8.16 and 8.17 can be similarly discretized and solved by using standard eigenvalue solvers. In the particular case of f , the Coriolis parameter, being zero, Eq. 8.16 becomes the same as Eq. 8.5. The same notation is used in Eqs. 8.16 and 8.5. In appendix D we give the MATLAB script which implements the numerical solution of the eigenvalue problem in Eq. 8.16, and hence in Eq. 8.5.

Table 8.4.a. Period (hours) of first vertical mode (V1) Poincaré waves for Lake Tahoe in winter 1999-2000, resulting from modal decomposition. m = wave number in the longitudinal direction (N-S). l = wave number in the transverse direction (E-W).

$m \setminus l$	0	1	2	3
0	18.9937	17.4510	14.4077	11.6495
1	18.4365	17.0161	14.1604	11.5179
2	17.0200	15.8840	13.4886	11.1484
3	15.2469	14.4147	12.5549	10.6049

Table 8.4.b. Period of second vertical mode (V2) Poincaré waves in Lake Tahoe in winter 1999-2000, resulting from modal decomposition.

$m \setminus l$	0	1	2	3
0	18.9937	18.8713	18.5179	17.9712
1	18.9527	18.8311	18.4800	17.9366
2	18.8315	18.7122	18.3677	17.8339
3	18.6345	18.5190	18.1849	17.6667

Tables 8.4.a and 8.4.b show the period of the different horizontal and vertical Poincaré modes for Lake Tahoe, as given by the numerical solution of Eq. 8.16. This analysis

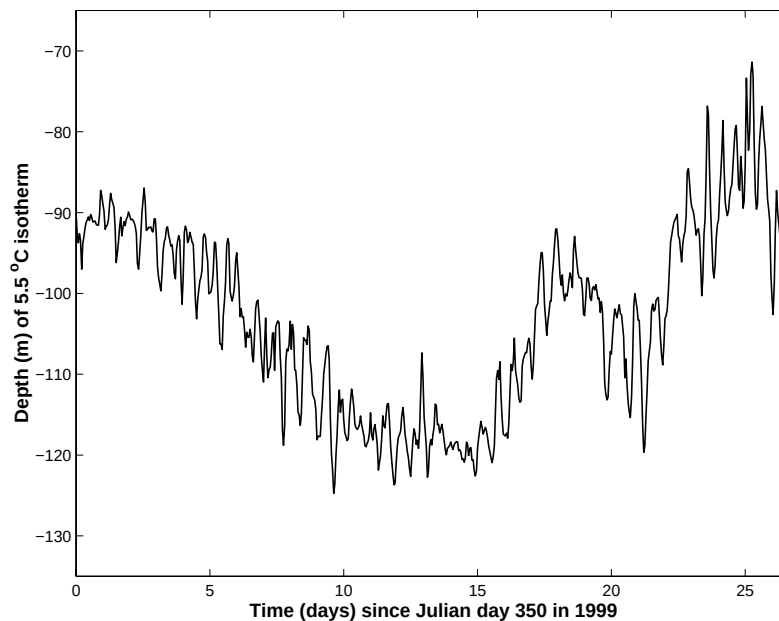
shows that the frequency of many internal Poincaré waves remains close to the inertial value (about 19 hours), and hence one would expect internal wave activity to be concentrated in a frequency range surrounding the inertial frequency.

8.5. Analyzing experimental results

The time series of the 5.5 °C isotherm displacement was obtained through linear interpolation of the temperature records in the thermistor chains, and its power spectrum was calculated to identify the characteristic frequencies of oscillation of Lake Tahoe during the 20-day period starting on December 19, 1999. The 5.5 °C isotherm is approximately located at base of the metalimnion, as shown Fig. 8.3, and it will experience significant oscillations due to internal waves. The 2-minute data were then smoothed by using a moving-average filter with 180 points, so as to remove high-frequency oscillations and give 6-hour averages. Figure 8.4 displays the evolution of the smoothed 5.5 °C isotherm location, measured from the free surface at MIDLAKE station. It reveals the presence of sub-inertial or near-inertial oscillations in the isotherm location, together with very low frequency changes.

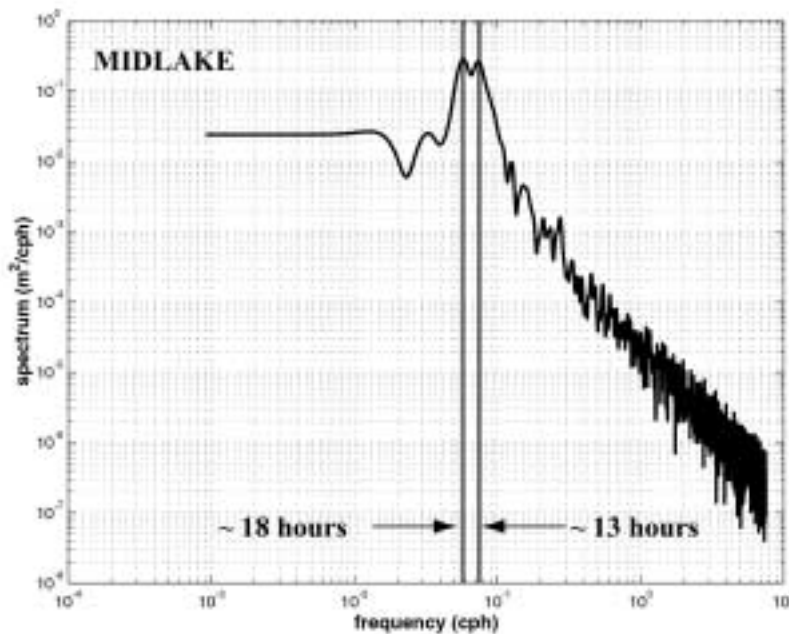
First order differences⁵¹ were applied to the time series of isotherm depth to remove, or at least minimize, the non-stationary components. It was on the differenced and smoothed series that a Fourier analysis was applied. Figure 8.5 displays the results of the Fourier analysis in the differenced and smoothed 5.5 °C isotherm from the MIDLAKE station. The absence of peaks in the low frequency range (periods higher than two days) is quite

⁵¹ First order differences are successive operations of the form $\Delta x_i = x_i - x_{i-1}$ which are applied to the original time series $\{x_i, i = 1, n\}$ to eliminate or minimize the effects of non-stationarity.



*Figure 8.4.
Time series of
the 5.5 °C
isotherm
location
derived from
thermistor
chain data at
the MIDLAKE
station.*

remarkable and illustrates the effectiveness of the first order differences in removing the non-stationary components from a time series. The spectrum shows a clear peak of energy at about 13 to 18 hours (frequency = 0.0556 to 0.0769 cph), which, according to the analysis of the initial temperature profile, would represent the characteristic signature of Poincaré waves, and most likely the first vertical mode, first horizontal mode Poincaré wave. This interpretation contrasts with that given by Thompson (Thompson 2000) to oscillations of similar periodicity in the temperature records collected in Lake Tahoe during the winter periods of four years. In that study those oscillations were attributed to lateral internal seiches. However, given the magnitude of the Rossby radius of deformation during typical winter conditions in the lake, the Earth's rotation has a considerable influence on the behavior of the internal waves.



*Figure 8.5.
Power spectrum
of the differenced
time series of the
5.5 °C isotherm
displacement at
INDEX station.*

The same procedure was followed to obtain and analyze the 5.5 °C isotherm at the INDEX station. The results of the Fourier analysis are shown in Fig. 8.6. The spectrum reveals in this case the presence of oscillations with sub-inertial and super-inertial frequencies. The peak with super-inertial frequency can be attributed to Poincaré waves with a characteristic period of approximately 18 hours. In the sub-inertial range at least two peaks are evident: one with a period of 2 to 2.5 days, and a second one with a diurnal or quasi-diurnal periodicity. Another peak with a sub-inertial frequency shows up in the spectrum when the Fourier analysis is applied to the non-differenced time series of isotherm displacement. The time series was smoothed with a moving-average filter of 360 points to eliminate the components with higher than semi-diurnal frequency. The temperature records were then resampled to give a time series with data points every hour, to replicate the output frequency from the 3-D model (see section 8.6.1). Figure 8.7 displays the result of the Fourier analysis applied to the hourly time series, revealing a peak with a period of 4 to 5 days. This spectral peak would represent the signature of the

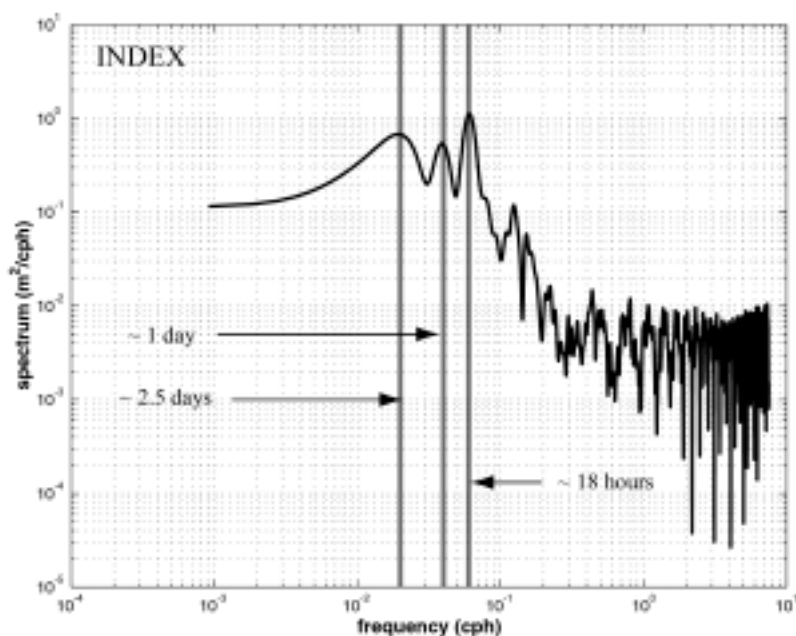


Figure 8.6. Power spectrum of the first-order differenced time series of the 5.5 °C isotherm displacement at the INDEX station.

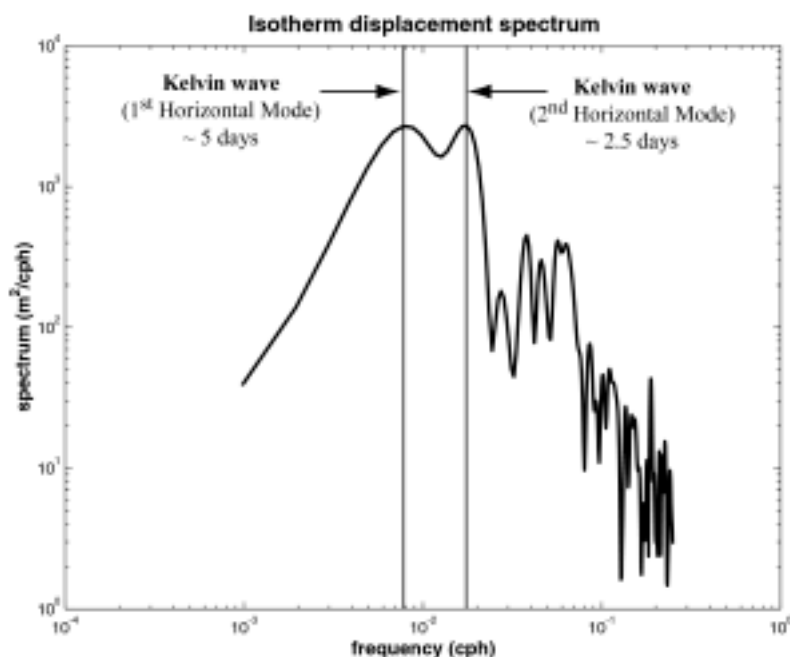


Figure 8.7. Power spectrum of the non-differenced time series of the 5.5 °C isotherm displacement at the INDEX station. The time series was smoothed to eliminate frequencies higher than semi-diurnal and resampled to yield an hourly time step.

first vertical mode, first horizontal mode Kelvin wave described in section 8.4.3. It is not clear, however, from simple inspection of the data the origin of the peak at 1.7×10^{-2} cph (2-2.5 day period). Its period agrees with that expected for a first vertical mode, second azimuthal mode Kelvin wave in the circular model of Lake Tahoe. However, this interpretation should be taken very cautiously. Studies of the internal wave field of Lake Biwa (Saggio and Imberger 1998) and Lake Kinneret (Hodges *et al.* 2000) have only identified Kelvin waves of the first horizontal mode and up to the second vertical mode. However, the size of Lake Tahoe and the relatively weak stratification in winter periods are such that second horizontal Kelvin wave modes, and even third horizontal modes, can be supported. The interpretation of the $2\frac{1}{2}$ day period peak as representing a Kelvin wave of horizontal mode 2 will be confirmed in the next section by using the results of numerical simulations. The third super-inertial peak, at about 4×10^{-1} cph (24 hour period), probably reflects the diurnal periodicity of the wind events that energize the internal wave climate in Lake Tahoe. The quasi-diurnal periodicity of the wind events is evident from the records at the USCG station shown in Fig. 8.2.

8.6. Analyzing the numerical results

8.6.1 Model setup

Lake Tahoe was discretized using a grid with 500 m resolution in the horizontal directions and 5 m in the vertical direction. Bathymetric information at 500 m resolution was extracted from the detailed bathymetric survey of Lake Tahoe from the U.S. Geological Survey (Gardner *et al.* 1998). The extraction was carried out by using the spatial analyst tool of ArcView (Esri, Inc.). Figure 8.8 shows the discretized version of

Lake Tahoe bathymetry as supplied to the 3-D model. The total number of computational cells was 325136, of which almost 80% are wet (active in the computations). The time step was set to $\Delta t = 300$ seconds (5 minutes) and the results were saved every hour. In each time step, two trapezoidal iterations were performed after the initial leapfrog step. No instabilities were observed in the course of simulation, indicating that the level of spatio-temporal discretization was appropriate. For the analysis of the internal waves, a period of 20 days starting on December 19, 1999, was simulated with the 3-D model.

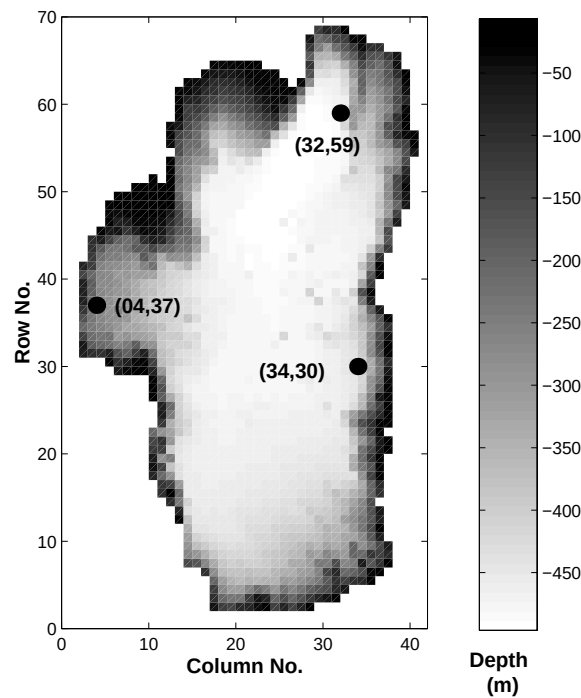


Figure 8.8. Bathymetry of Lake Tahoe model. The computational cells are 500 m in the horizontal directions and 5 m in the vertical. The grid has a total of 325136 cells, and almost 80% of them are wet (or active). In the plot three stations are identified where the time series of temperature and/or velocity were saved and analyzed.

To represent the attenuation of short-wave solar radiation in the water column, the values of R , L_1 and L_2 in Eq. 5.10 were set to 0.58, 0.35 m and 23 m respectively. These values correspond to Type I water or clear water, according to the classification proposed by

Jerlov (referenced in Rosati and Miyakoda 1988). The code was driven with meteorological data collected at the dock of the Tahoe U.S. Coast Guard station, located at (N 39° 10.838', W 120° 07.157'). Figure 8.2 displays the time series of the data collected at the meteorological station, which includes the simulated period. The incoming long-wave radiation was calculated indirectly by assuming a cloud cover of 0.15, since the long-wave radiation sensor malfunctioned during the period of the simulations. The wind drag coefficient was set to a constant value of 2.51×10^{-3} . This value is higher than those recommended in the literature ($= 1.3 \times 10^{-3}$, Fischer *et al.* 1979) and tries to account for fetch effects in the wind speed and wind stress (Taylor and Salmon 1993, Perrie and Toulany 1990). The meteorological station is located on the windward side of the lake and could be partially sheltered by the surrounding topography. Furthermore, the wind sensor is located approximately 6 m above the lake surface, while the values of the drag coefficients suggested in the literature are for wind speed measurements taken at 10 m.

The bottom drag coefficient selected is 0.002, as suggested for a mud bottom (Smith 1997). Given the depth of the basin, this choice will probably not affect the results. The horizontal eddy viscosity was set to a constant value of $1 \text{ m}^2/\text{s}$. The code was initialized with a horizontally homogeneous temperature field. The vertical distribution of temperatures was obtained by averaging the records from each thermistor at the MIDLAKE station during three days from December 16 to December 19, 1999. The average thermistor records were then linearly interpolated to the centers of the computational cells, regularly spaced in the vertical every 5 meters.

8.6.2. Spectral analysis of isotherm displacements

The location of the 5.5 °C isotherm in the water column was obtained at every time step from temperature profiles calculated by the model at the cells in row 30 and column 34 of the computational grid. Station (34,30) is located on the east side of the deep central basin of Lake Tahoe; it was selected as an observation station so that large excursions in the 5.5 °C could easily be detected and possible effects from an irregular shoreline would not exist. As discussed in section 8.5, the 5.5 °C isotherm will most probably experience significant oscillations due to internal waves, as it is initially located approximately at the base of a region of maximum temperature gradients. The result of the Fourier analysis, applied to the simulated time series of the 5.5 °C isotherm, is displayed in Fig. 8.9. The

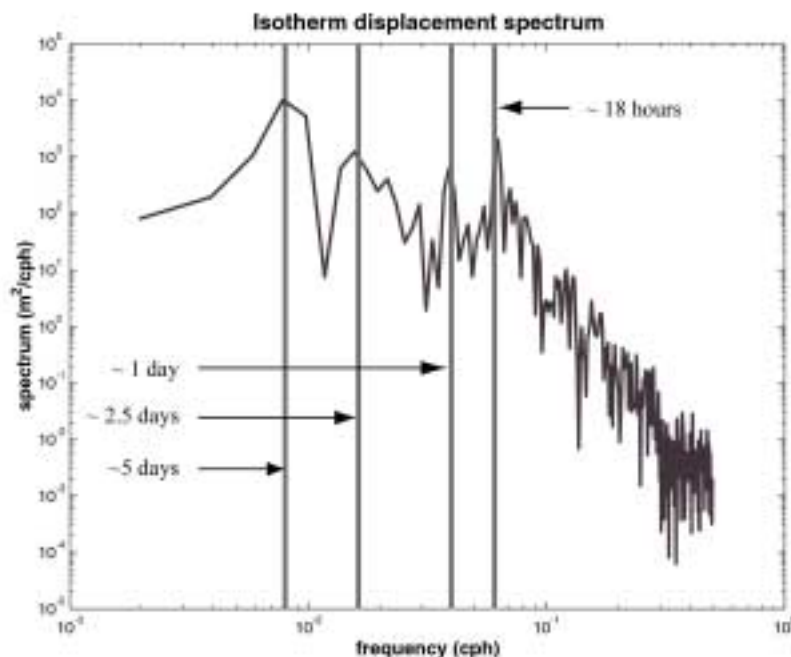


Figure 8.9.
Power
spectrum of the
5.5 °C isotherm
displacement,
from node
(x,y)= (34,30)
in the
numerical
model, near
MIDLAKE
station.

power spectrum density of the simulated displacements reveals the same 4 peaks encountered in the analysis of the thermistor chain records: three peaks in the sub-inertial range of frequency, with periods of approximately 4-5, 2-2.5 and one days, respectively, and one peak in the super-inertial range with a period of approximately 18 hours. The peaks were attributed in section 8.5 to: (1) Kelvin waves of the first vertical and first horizontal modes with a 5-day period; (2) Kelvin waves of the first vertical mode and second horizontal mode, having a period of 2 to 2.5 days; (3) the diurnal periodicity of wind events; and (4) Poincaré waves with a period of approximately 18 hours.

8.6.3. Rotary spectrum of velocity data

As discussed in section 8.4, the velocity vectors describe very characteristic rotations in a Poincaré or a Kelvin wave field. The velocity vectors in a Poincaré wave field rotate clockwise with the characteristic frequency of the wave, while in a Kelvin wave field the direction of rotation is anti-clockwise. The rotary spectrum of vector time series (Fofonoff 1969, Mooers 1973, Gonella 1972) provides estimates of characteristic frequencies of oscillation and the direction of the vector's rotation. In this section a rotary spectrum analysis is applied to the simulated time series of velocity vectors to give additional support to the results of the previous sections, suggesting that Kelvin waves and Poincaré waves could have been responsible for the oscillations in the temperature records.

Figure 8.10 displays the rotary spectrum of the velocity records from node (32,59,30) located at 150 meters below the free surface. The nodes or computational cells are

identified by using their column number (x-direction), row number (y-direction) and layer number (z-direction, increasing downward). The characteristic frequencies that are found in the spectrum of isotherm displacements are shown with vertical lines and agree, in general, with the peaks of the rotary spectrum. The oscillations with a period of about 18 hours have a very strong clockwise rotating component, following the behavior expected for a Poincaré wave. The counterclockwise rotating components of the velocity time series, on the other hand, dominate in the sub-inertial range of frequencies that are associated with the Kelvin modes.

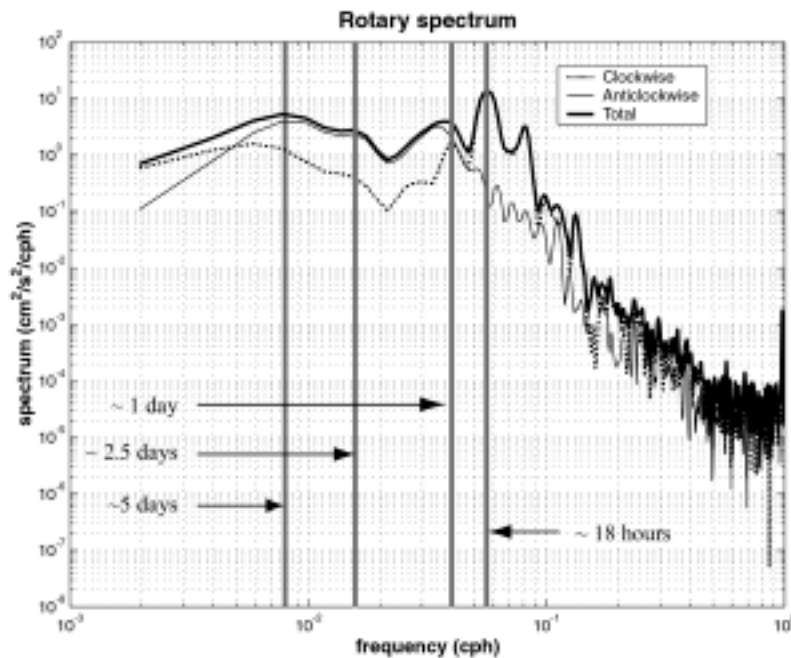


Figure 8.10. Rotary spectrum analysis of velocity data at node (32,59,30) located at approximately 150 meters below the surface.

8.6.4. Fingerprints of internal waves in temperature time series

The fingerprints of the first horizontal, first vertical mode Kelvin waves, and the Poincaré waves, are particularly evident in the simulated time series of temperature. The spatial structure of the waves is revealed by comparing the time series calculated from two

separate points located close to shore and located at the same depth. In Fig. 8.11 the time series of simulated temperature from computational cells (04, 37, 22) and (32, 59, 22) are displayed. Since the two cells used here are both centered at 107.5 meters below the free surface, the third coordinate will henceforth be ignored in the identification of these two stations. The station (04, 37) is located in the proximity of the INDEX station, in a large embayment on the west edge of Lake Tahoe, and station (32, 59) is located in the northeast corner of the lake. The thin trace in the plots represents the raw temperature data, and the thicker trace represents a filtered time series. The latter was obtained after smoothing the raw data with a moving-average filter that eliminates the oscillatory components with periods lower than 3 days. The filtered series capture the influence of the first vertical and horizontal mode Kelvin wave on the temperature field. The difference between the raw and the filtered data, which shall be referred to here as the residual component, has a period of approximately 18 hours, revealing the influence of the Poincaré waves. The residual components of the time series at stations (04, 37) and (32, 59) are nearly 180 degrees out of phase, as can be seen at the time marked by the thin vertical line on the right side of the figure. The opposition in phase is evidence of the transverse oscillations in the temperature field characteristic that are characteristic of the Poincaré waves. The lag in the filtered time series is caused by the time that is required by the Kelvin wave to cover the distance along the shore between stations (04, 37) and (32, 59), propagating at the speed of the first vertical mode internal waves in Lake Tahoe under the existing conditions in winter 1999-2000.

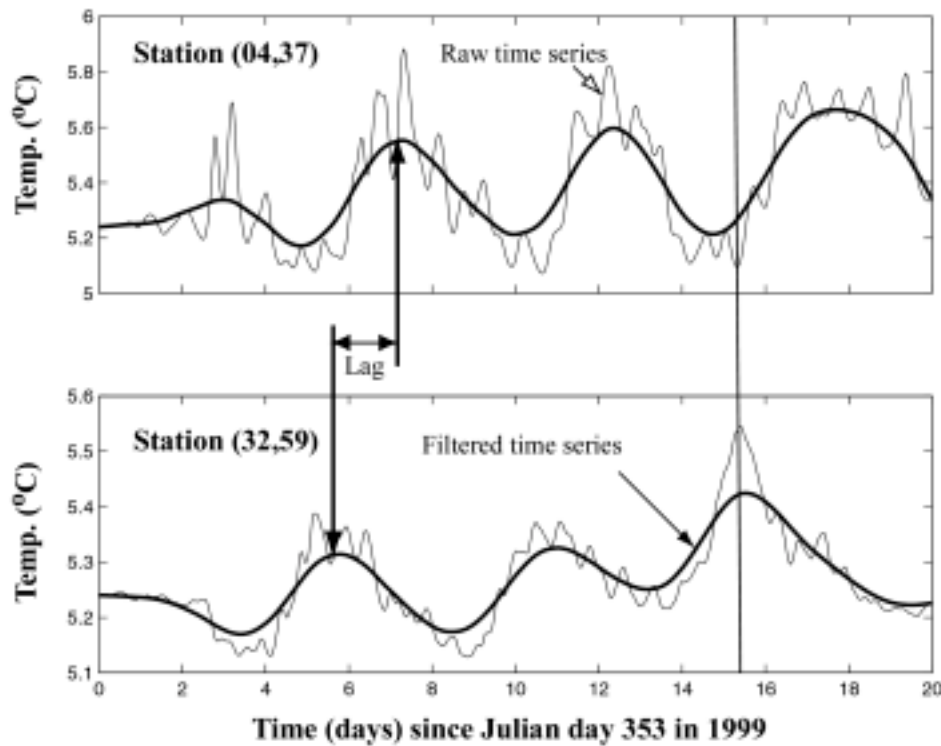


Figure 8.11. Filtered and raw time series of simulated temperature 107 m below the free surface, from two stations in the model of Lake Tahoe.

8.6.5. Spatial structure of internal waves

To visualize the spatial structure and evolution of the Kelvin and Poincaré waves in Lake Tahoe, SI3D-L was set to calculate and record the distance from the free surface of the 5.5 °C isotherm at every horizontal position and at every time step. Butterworth digital bandpass filters of order 10 were later used to isolate the oscillations with the characteristic frequencies revealed in the spectra. The filter passband that was used to isolate the 18-hour period oscillations was $0.1035 < \omega \text{ (cph)} < 0.1246$. The passband of the filter used to isolate the 48-hour oscillations was $0.0306 < \omega \text{ (cph)} < 0.0543$. For the oscillations with a 5-day period, the location of the isotherm was not filtered, as the raw data already revealed the spatial structure of the wave. The results are displayed in

sequences of 3-D plots showing isotherm displacement surfaces (or isotherm depth for the 5-day period) at instants in time during the cycle of the waves. In those plots the magnitude of the displacement (in meters) is represented on the vertical axis, and the lake is viewed from its southwest corner, which is the origin for the horizontal cell numbering scheme adopted in the code. The axis to the right of the origin shows the column number (or east-west direction), and that on the left shows the row number (or north-south direction). The time is displayed on each frame at the top.

Figure 8.12 displays a sequence of plots, separated by 2 hours, illustrating the oscillations of the 5.5 °C isotherm caused by the 18-hour-period internal waves. The isotherm surface oscillates in the east-west direction, having a node along the North-South axis of Lake Tahoe. These transverse oscillations of the isotherms are very characteristic of Poincaré waves, as described in Csanady 1975.

Figure 8.13 shows a sequence of plots illustrating the effects of a first horizontal mode, first vertical mode Kelvin wave at the location of the 5.5 °C isotherm. The plots are separated by 24 hours. The vertical axis, in this case, shows the depth in meters of the 5.5 °C isotherm relative to the free surface. The plots illustrate a Kelvin wave traveling cyclonically around Lake Tahoe, with the shore on its right. The Kelvin waves simulated with SI3D-L produce isotherm excursions of approximately 15 m. The isotherm displacements are particularly strong when the diurnal and usually westerly winds act in phase with the Kelvin wave. The amplitude decreases during the 4-day period and is again excited when the wave induces upwelling at the upwind end of Lake Tahoe

(western shore). These waves were repeatedly observed in the Lake Tahoe during the four year study of Thompson (Thompson 2000), but with periodicities that increased from the 2.5 to 3 days intervals in the early Fall to values closer to those shown in this work later in winter.

The isotherm displacements caused by the 2½-day-period internal wave are displayed in the sequence of plots in Fig. 8.14. The spatial structure of the wave characterizes the second horizontal mode first vertical mode Kelvin wave with two maxima along the perimeter of the lake separated by two minima, leaving a saddle point in the middle. This “saddle” structure rotates around Lake Tahoe with a periodicity of about 54 hours, approximately half the period of the first horizontal model Kelvin wave. These plots, based on the 3-D model output, were critical in clarifying the origin of the 54-hour peak in the spectrum of simulated and measured isotherm displacements, since the existence of the horizontal mode 2 Kelvin wave in Lake Tahoe has not been previously documented.

8.7. Summary and conclusions

In this chapter the 3-D model has been used to help in the identification of the nature of the dominant basin-scale internal waves of Lake Tahoe under the conditions existing in winter 1999-2000. During that period of time the lake was stratified with a well-defined metalimnion between 50 and 150 meters below the free water surface. Three modes of oscillation were isolated and identified in the internal wave field of Lake Tahoe. Two of them have sub-inertial frequencies and were identified as vertical mode one Kelvin waves traveling counterclockwise around the perimeter of the lake with periodicities of either

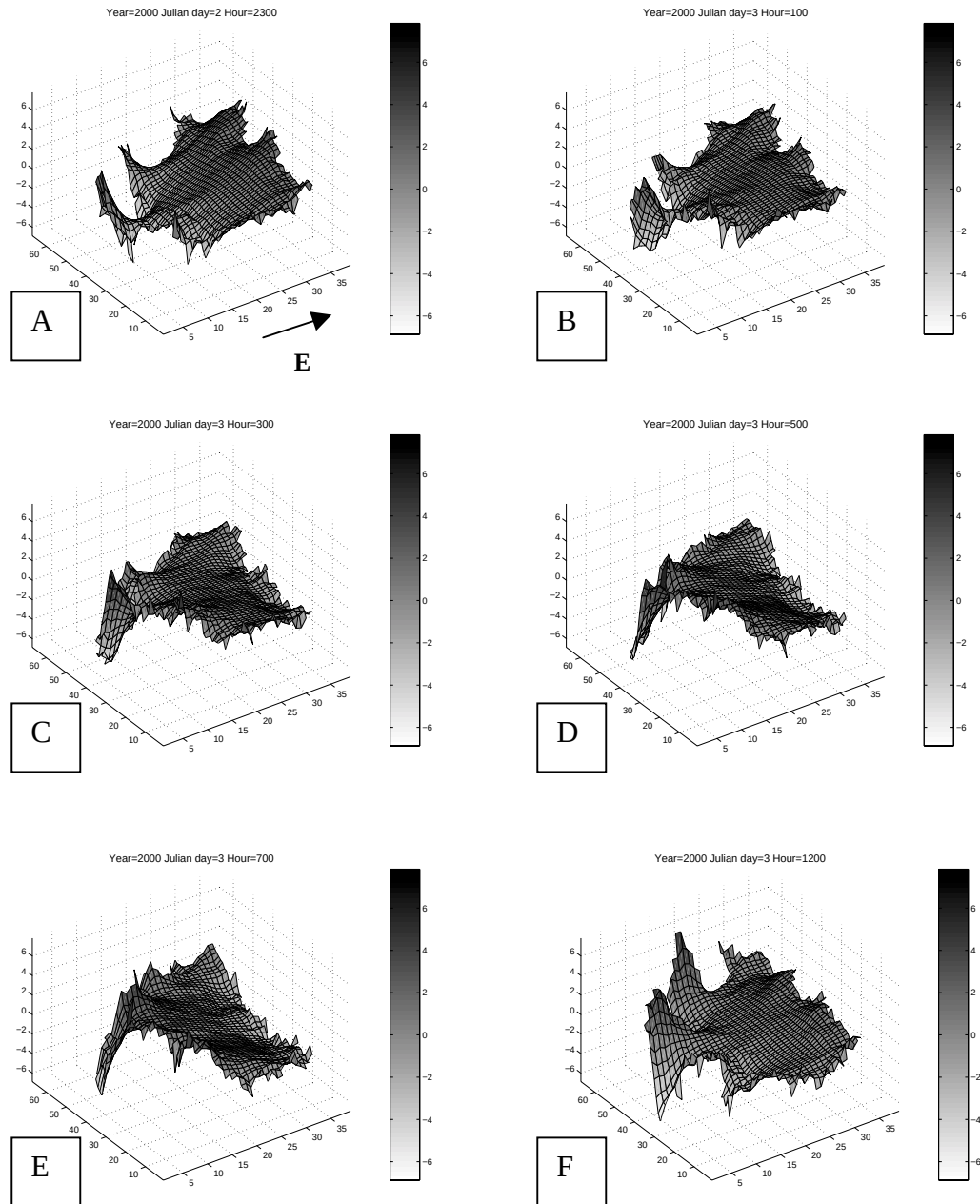


Figure 8.12. Evolution of the 5.5°C isotherm displacement in Lake Tahoe, filtered to isolate 18-hour-period oscillations. The plots are separated by 2 hours. The vertical axis represents the displacement (m) of the isotherm from its mean position, and the horizontal axes display the horizontal location in the computational grid. The East direction on the plots is shown in frame A. The gray intensity reinforces the idea of the magnitude of displacement. The figure displays the transverse oscillations that characterize Poincaré waves.

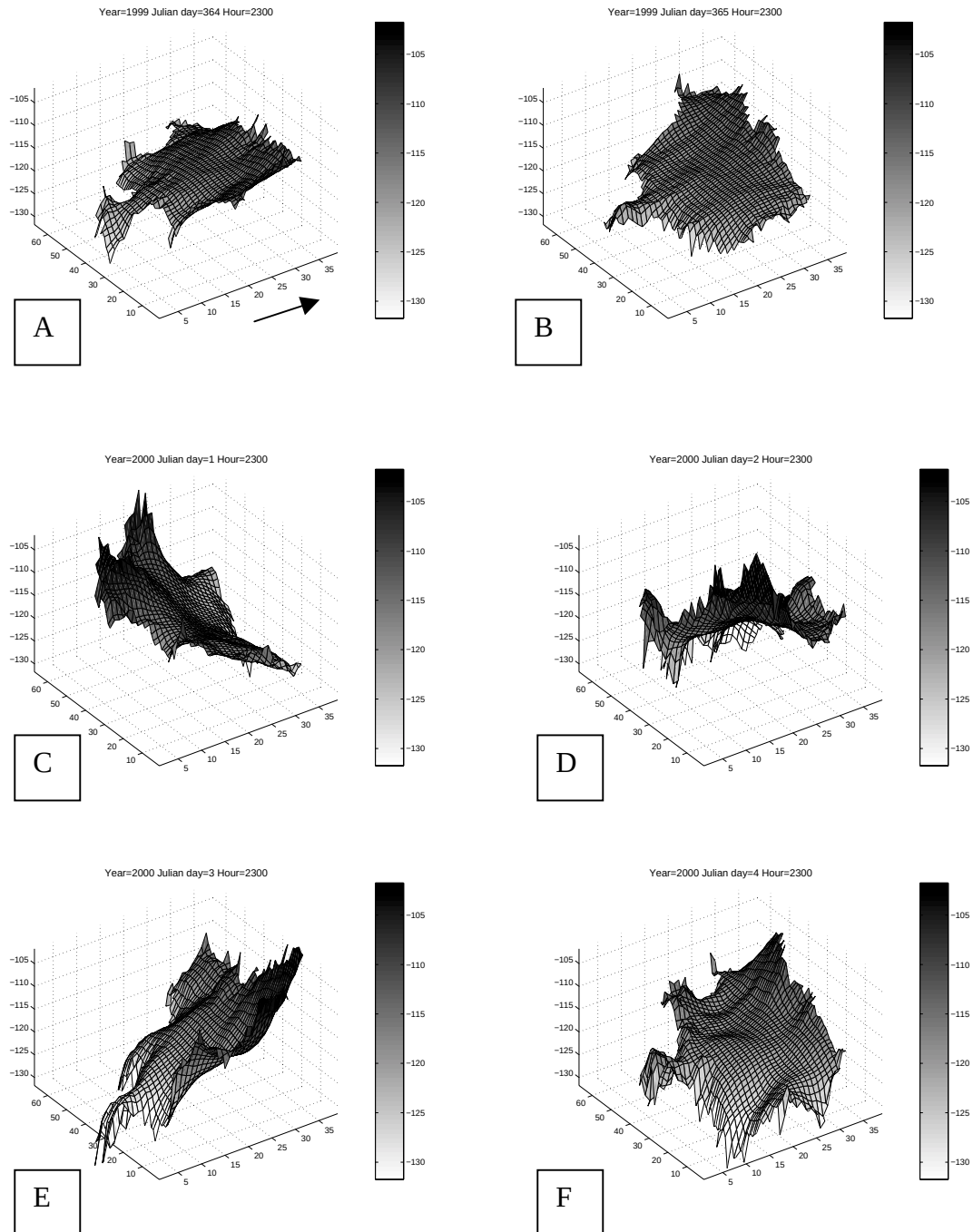


Figure 8.13. Evolution of 5.5 °C isotherm location in the water column, illustrating a Kelvin wave traveling counterclockwise with the shore on its right. The plots are separated by 1 day. The data are not filtered, and the vertical axis represents the depth of the isotherm from the free surface (m). The horizontal axes represent the row and column number where the isotherm depth is calculated. The East direction is shown in frame A.

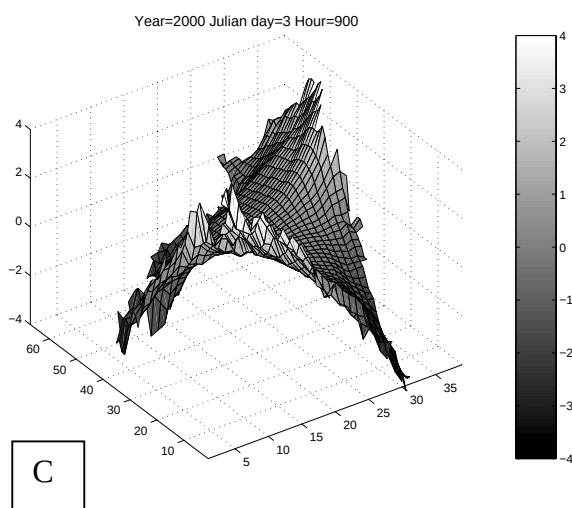
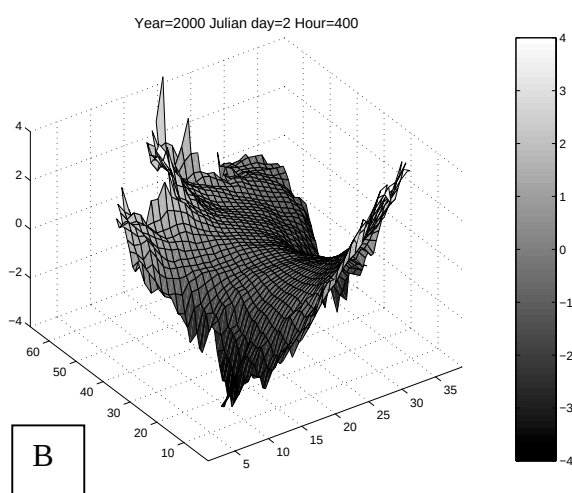
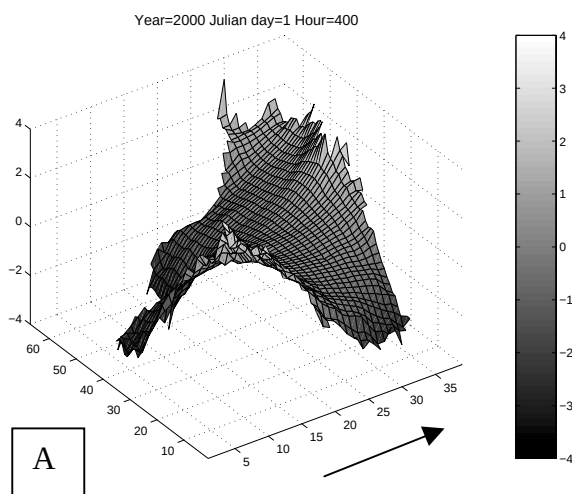


Figure 8.14. Evolution of 5.5 °C isotherm displacement, filtered to isolate oscillations with a period of 2 to 2.5 days. The plots are separated by approximately half a cycle (24-29 hours). The vertical axis represents the displacement (m) of the isotherm from its mean position and the horizontal axes display the row and column numbers in the computational grid where the isotherm location was calculated. The East direction on the plots is shown in frame A. The spatial structure of the displacement is characteristic of a second horizontal mode Kelvin wave, with two maxima along the perimeter separated by two minima, forming a saddle structure.

5 days, approximately, or 54 hours (Fig. 8.5). The 5-day period Kelvin wave has horizontal mode 1, and the 54-hour period Kelvin wave has horizontal mode 2. The isotherm displacements caused by the Kelvin waves are graphically displayed in the sequence of plots in Fig. 8.11 and Fig. 8.12. The third internal wave mode is a Poincaré wave having a frequency of about 18 hours, causing transverse oscillations of the isotherms, as illustrated in the series of plots in Fig. 8.10.

The description of the internal wave field in Lake Tahoe, presented in this work, has very strong support, as it is based on various kinds, and separate sources, of information, all giving very consistent results. The theoretical analysis of the initial conditions, presented in chapter 8.4, has been traditionally used in the interpretation of the spectra of temperature field observations. Here it was used to present the characteristic features of the different modes of internal oscillations that the analyst should be looking for when examining field or numerical results. The spectra of the temperature and velocity numerical simulations were first examined in sections 8.6.2 and 8.6.3 to establish whether the same spectral peaks as those found in the analysis of the field observations were present. The rotary spectrum additionally provides information on the direction of rotation of the velocity vector at each frequency, which can be used to discriminate between Kelvin and Poincaré waves. The filters and the 3-D visual tools, used in this chapter to display the numerical isotherm displacements, probably constitute the most powerful methodology to interpret the spectral peaks in the Fourier analysis of the thermistor chain records. In the case of Lake Tahoe, they proved to be particularly helpful

in identifying the nature of the 54-hour period oscillations, which was only clear after visualization of the evolution of the filtered isotherm displacement.

Although the application of the model to identify the internal waves in Lake Tahoe was successful there are several aspects that should be considered in future applications. The first aspect is the computational cost associated with the model runs. The ratio of real to computational time in the 20-day simulation presented here was approximately 20:1, which is probably satisfactory for relatively short simulations on the order of several months at most, but not for longer simulation periods. Those long-term simulations are probably necessary in Lake Tahoe to represent phenomena on seasonal time scales that might be involved in the evolution of the lake density structure. One such phenomenon is the lateral intrusion of cold water masses caused by differential cooling. Thompson (Thompson 2000) gives evidence of the occurrence of such phenomenon in Lake Tahoe, using thermistor chain data collected during four winter periods, and argues that this process could be responsible for deep winter mixing. Given its seasonal time scale, 1-2 month simulations will not be able to reproduce the effects of those lateral intrusions. It will be necessary to increase the computational efficiency of the model to conduct multi-year simulations in a reasonable time. The most expensive feature in the simulation of Lake Tahoe was not the solution of the free surface problem, as it was in the case of Clear Lake, but the solution of the tridiagonal systems of equations associated with the vertical turbulent transfer of momentum or scalars. Those calculations are performed independently in each water column and hence are perfect candidates for parallelization. Alternatively, the method used to represent the vertical transfer of momentum and scalars

could be changed, and mixed-layer type models could be used instead of the expensive turbulence closure models. No comparison exists in the literature of the relative expense of these two types of turbulence parameterizations. Nor is it clear how the mixed-layer models could handle cases of strong upwelling events with surfacing of the thermocline. It could be argued that the rapid advancement of computer speed will of its own nature allow multi-year simulations to be conducted in reasonable time. However, as one moves beyond the need to resolve basin-scale motions and instead focuses on more detailed processes (of which lateral intrusions are but one), the need for a higher spatial resolution will by itself account for much of the gain from any improvement in computer hardware.

The occurrence of seasonal time-scale processes in Lake Tahoe that influence the temperature structure challenges the approach used in this chapter to characterize the stratification of the lake on December 19, 1999, using only the information from one thermistor chain. The vertical density structure of the lake is probably not uniform across the lake and might be continuously varying. As seen in the time series of the 5.5 °C isotherm location at MIDLAKE station, there are very low frequency components, not attributable to internal waves, which were eliminated from the analysis by using first-order differences. Fourier analysis is probably not the best tool to analyze long time series since the characteristic frequencies might be evolving. Other tools, such as wavelets or a local Fourier analysis, might be more appropriate to analyze seasonal time series of temperatures. Fourier analysis is probably “good enough” for the analysis of the relatively short period of 20 days used in this chapter, but the width of the peaks might reveal small drifts of the characteristic frequencies in response to the slowly changing

stratification. Providing the model a uniformly vertical temperature distribution is probably erroneous and might compromise the numerical results, if a detailed comparison of temperature or velocity records were to be done. To such a comparison in the time domain, and not in the frequency domain as was done here, long-term simulations should be conducting, starting in the late winter, when the lake is reasonably well-mixed and no lateral structures exist in the density field. Additionally, spatially varying input data should be supplied to the model. The lake is roughly 20 km wide, and synoptic variations of meteorological variables are expected. As shown in Strub and Powell 1986, the spatial variability of the wind is a key feature if the surface currents and temperatures in Lake Tahoe are to be reproduced with numerical tools. Very detailed surface temperature information from recently launched NASA satellites, together with several thermistor chains, could provide the necessary observational data for validation purposes.

CHAPTER 9

SUMMARY AND CONCLUSIONS

The goal of this work was to construct a comprehensive, efficient and accurate three-dimensional computational tool that could be used to improve one's understanding of the physical processes of water circulation, transport and mixing in lakes. The availability of a relatively large number of three-dimensional flow solvers suggested at the outset that there was a greater need for formulating a methodology by which such codes could be evaluated, rather than the development of a new code. To this end, both a finite difference and a finite element flow solvers were used to develop this methodology, and then they were compared for their suitability for applications in lakes. While the governing equations that pertain to lakes are identical to those for other surface water systems, lakes (and reservoirs) are singular in that the main forcing term (wind) is highly variable and may be totally absent for prolonged periods of time. Furthermore, the enclosed nature of lakes results in the responses to forcing being wave-like in nature. This is in general compounded by the presence of thermal stratification that gives rise to both an external (barotropic) and an internal (baroclinic) response. Complex bathymetry and complex topography also contributes to the variability of the wind field. Thus lakes represent a somewhat unique situation when compared with rivers and estuaries, and it can not be taken for granted that solvers that suffice in those systems will be suitable in lakes.

Two lakes of very contrasting internal dynamics were used to test and validate the completed model. The first, Clear Lake, is a multi-basin, shallow, polymictic and highly eutrophic lake in northern California. The second, Lake Tahoe, is a deep monomictic and oligotrophic lake straddling the border between California and Nevada. The internal dynamics of Clear Lake are characterized by a cycle of setup and relaxation of horizontal density gradients controlled by strong, periodic wind events. In contrast, the dominant features in the internal dynamics of Lake Tahoe are the basin-scale internal waves.

9.1. Evaluating alternative modeling approaches

A series of benchmark problems have been proposed to evaluate the ability of existing three-dimensional models to reproduce efficiently and accurately a wide range of characteristic hydrodynamic phenomena in natural lakes. Those problems consist of the numerical simulation of free surface oscillations (barotropic seiches), wind-driven circulation and oscillations of the isopycnals around an equilibrium position (baroclinic or internal seiches) in either homogenous or stratified basis of simple and complex geometry. All are linear problems with a known analytical solution. Numerical experimentation offers a straightforward method to evaluate the performance of 3-D algorithms used in the solution of problems in the hydrodynamics of lakes. The experiments not only constitute an alternative, and sometimes unique, means of evaluating classical measures of accuracy, but they further allow the analysis of conservation properties of the schemes (mass and energy conservation), the effects of complex geometries and forcing, and the computational expense involved in the solution of problems. They further allow the inexperienced modeler to become familiar with the

numerical algorithms in simple problems before embarking in the usually more difficult task of modeling real systems.

Even though they do not cover the complete range of physical processes occurring in lakes, the test problems proved to be very effective tools in exposing deficiencies in the model formulations and hence in comparing alternative algorithms. Two models were compared in this work on the basis of their performance in the solution of those test problems. One uses a classical finite element approach and is particularly suited to represent domains with a complex geometry. The second solves the 3-D governing equations using a highly efficient, semi-implicit finite difference discretization on an orthogonal structured grid. The test problems revealed significant sources of numerical dissipation in the finite element model, not only related to a non-centered time discretization scheme, which could seriously compromise its performance in the simulation of stratified flows. In addition, and in order to stabilize the method, relatively large values of the horizontal eddy viscosity were needed, introducing additional sources of dissipation that are not strictly related to the physics of the problem. The results also exposed deficiencies in the matrix solver of the finite element model that significantly slows down the computations in 3-D problems. Its low computational efficiency can be partly overcome by reducing either the horizontal or the vertical grid resolution and hence decreasing the accuracy of the method.

The 3-D semi-implicit leapfrog-trapezoidal finite difference model, SI3D (Smith 1997), was selected on the basis of the results of the test problems to provide the basis for the

3-D model for lake environments. The semi-implicit method used in SI3D is based on treating the gravity wave and vertical diffusion terms implicitly while treating all other terms, including advection, explicitly. In this manner the method is guaranteed to be stable without the time-step limitation imposed by the gravity wave CFL condition. Horizontal diffusion terms are not needed to stabilize the method. The chief advantage of SI3D was its computational efficiency and the absence of numerical diffusion. Any dissipation represented in the model will therefore come from sub-models representing the physical effects of turbulence or friction. The need for a high spatial resolution in the grid to represent correctly complex domains not only does not pose a serious problem to the performance of the code, given its high efficiency, but it also increases the accuracy in the representation of the wave phases.

9.2. Development of SI3D-L

SI3D was extensively modified and extended to incorporate features needed for the accurate simulation of lake hydrodynamic phenomena. The extended version of the model was termed SI3D-L. The extended model uses spatially and temporally variable wind fields to drive the large-scale circulation, and it calculates the momentum and scalar turbulent transfer coefficients with a high-order turbulence closure model (Mellor-Yamada Level 2.5). The two equations governing the turbulent kinetic energy and the turbulent macroscale in the Mellor-Yamada closure model are discretized using a 3-level fully implicit algorithm (Fletcher 1991; Ferziger and Peric 1996). The algorithm has a truncation error of $O(\Delta t^2, \Delta x^2)$, is A-stable and damps the spurious oscillations associated with stiff problems. It proved to be a stable and robust method in all tests of it and later in applications of the code to real systems. The mixing model further incorporates a

parameterization of the effects of internal waves, as proposed by Kantha and Clayson (Kantha and Clayson 1994). The wind drag coefficient, used in the estimation of the wind stress, is calculated as a function of the wind speed, to account for the effects of the free surface roughness. A formulation initially proposed by Amorochio and DeVries (Amorochio and DeVries 1980), which assumes adiabatic conditions, is followed in the model. The extension of the wind drag formulation to incorporate the effects of atmospheric stability will most probably require some form of iteration, increasing the computational cost of the model, and hence has not been attempted here.

A new two-time-level transport model for an active scalar (temperature) was implemented, abandoning the original leapfrog-trapezoidal method, which was prone to developing spurious oscillations that could interfere in the calculations of turbulent transfer coefficients. Those oscillations appear, in particular, in the vicinity of discontinuities and poorly resolved gradients. Such conditions are ubiquitous in lakes because of the occurrence of both frontal structures and upwelling. The new transport model uses 1-D flux limiters and directional splitting methods to discretize the 3-D advection operator and incorporates new features such as the calculation of heat fluxes at the free surface and its redistribution within the water column. Special care was taken in the design of the transport model to achieve consistency with continuity (Gross 1997) in order to conserve mass.

A stand-alone Lagrangian particle-tracking model was also developed and used as a post-processing tool to visualize circulation patterns, or to simulate the transport of particulate

matter in lake environments. The model follows the algorithm originally proposed by Pollock (Pollock 1988) to calculate the deterministic displacements of the particles caused by mean currents. The stochastic displacement that was introduced to simulate the effects of turbulence is based on the eddy diffusivity values calculated in SI3D-L. Very small time steps are necessary, however, to avoid erratic and long displacements in the random step that would be able to send particles outside the computational domain. This, most probably, occurs near unstable density profiles, where the Mellor-Yamada model increases considerably the eddy coefficients to mimic the powerful mixing mechanisms caused by convection. A reflecting algorithm was developed to confront these cases though, in order to be true to the statistics and better represent the turbulent processes, the percentage of particles experiencing the reflection from solid walls should be minimized.

The extended model is free from calibration parameters other than the bottom drag coefficient. As originally formulated, SI3D uses a unique value of the drag coefficient, typically dictated by the nature of the bottom boundary. The usually lengthy and laborious process of calibration against field observations is avoided in this manner. In fact, the model was not calibrated either in the Clear Lake application or in the Lake Tahoe study. Nevertheless, the results proved to be close to the observed values. The value of the drag coefficient was selected on the basis of recommended values for mud bottoms. Field observations are needed to validate the model results, however, and to establish the validity of assumptions made in the construction of the initial conditions and the boundary conditions for the model.

9.3. Clear Lake

Clear Lake is a multi-basin, shallow, polymictic, highly eutrophic, mercury-contaminated lake in northern California. The first application of the model addresses the identification of the driving mechanism and large-scale circulation patterns in Clear Lake. The model results were used, together with field observations, to provide a solid description of the internal dynamics and circulation of the Oaks Arm and the exchange of mass with the other two basins of Clear Lake, the Lower and the Upper Arms. The field observations were performed by the author, in collaboration with others. The analysis of these field data (that is presented here) was the work of the author.

Because of its rather complex geometry, the modeling of Clear Lake posed an especially severe test to SI3D-L. A high-resolution grid is needed in the horizontal to represent the lake boundaries reasonably, resulting in a large computational burden. Close to half a million nodes were used to discretize the 3-D domain of Clear Lake. The resolution of the computational grid ($\Delta x = \Delta y = 100$ m, and $\Delta z = 1$ m) is more than sufficient to represent the geometry of the Upper Arm, the largest of the three basins with an approximately circular form and a length of 18 km. For the other two arms, though, and in particular the Oaks Arm, whose maximum width is approximately 2000 m, it is “just” sufficient to resolve lateral structures in the large-scale circulation. Most of the computational costs in the simulations were associated at each time step with the free surface solution, which required a small time step (3 minutes) to converge within a reasonable number of iterations. Nevertheless, the ratio of real to computational time, using two trapezoidal iterations after the initial leapfrog computation in each time step, was approximately

10:1. Future efforts should be directed towards increasing the efficiency of the model, thereby facilitating long-term simulations within a reasonable time. Possible lines of research in this respect are suggested in section 9.5.

The internal dynamics of the Oaks Arm under stratified conditions was found to be very characteristic with cycles of setup and relaxation of horizontal temperature gradients driven by strong, diurnal winds. The typically northwesterly wind encounters a lake that, due to the surface heat flux during the morning and early afternoon hours, is stratified. The wind acts during the afternoon and evening hours on this weakly-stratified system to generate horizontal temperature gradients both along and across the longitudinal axis of the Oaks Arm. During the night and early morning hours, when the wind forcing is negligible, the baroclinic pressure gradients that result from the horizontal differences in temperature become the dominant forcing mechanism in the system, driving currents of up to 10-15 cm/s westward at the surface following the north shore and eastward at the bottom. This circulation mechanism creates a residual cyclonic circulation and is here referred to as wind baroclinic pumping, or simply baroclinic pumping. Internal waves do not exist, as the stratification conditions rapidly change and even disappear in a fraction of the time the waves would take to complete one cycle. Baroclinic pumping is in all likelihood not exclusive to Clear Lake, but it probably exists in other large and relatively shallow lakes forced by strong and almost periodic wind events where basin-scale internal waves do not play a major role in the dynamics of the system.

The numerical simulations show that the factors driving the cyclonic baroclinic pumping circulation of the Oaks Arm are (1) stratification, (2) periodic and predominantly uniform northwesterly winds, and (3) Coriolis effects (Earth's rotation). The Earth's rotation generates a notable asymmetry in the evolution of the temperature field across the longitudinal axis of the Oaks Arm, which is critical to explain the existing residual cyclonic circulation and many of the observations gathered in the field. The asymmetric temperature stratification results in differences in the rate at which the vertical circulation switches from being driven by wind to being driven by baroclinic forces along the north and south shores of the Oaks Arm. The change of vertical circulation starts at the easternmost end of the arm and moves to the west more rapidly following the north shore than along the south shore. Consequently there are stations in the southwest of the Oaks Arm where the baroclinic circulation rarely replaces the wind-driven circulation while, across the basin, in the north shore the onset of baroclinic circulation is felt frequently as a rapidly advancing front. This difference in the amount of time the baroclinically driven vertical circulation dominates the wind driven motions is reflected in the residual – or time averaged – profiles. Along the north shore the residual currents are westward, while in the south the residual currents are eastward. The asymmetric evolution of the temperature and velocity fields also influences the levels of mixing, as shown in this work. The variation in mixing levels between the northern and southern portions of the Oaks Arm are related to the differing intensities of the shear and the stratification, which are mostly controlled by large-scale processes.

The spatial variability of the wind proved to be especially important in reproducing the anti-cyclonic circulation observed in the Lower Arm in previous works. The wind field is shaped by the surrounding topography and shows a complex and very characteristic spatial structure. The analysis of the spatial distribution of vorticity associated with the wind field revealed the presence of two cells, one of positive vorticity over the Oaks Arm and another of opposite sign over the Lower Arm. The creation of vorticity in the wind field is most probably associated with the deflecting and sheltering effects of Mount Konocti, a dormant volcano rising 1000 m above the lake level, on an otherwise uniform boundary layer flow. These results suggest that (1) the spatial variability of the wind field must be accurately characterized to carry out realistic simulations in lakes, and (2) the number of wind stations needed for that purpose should vary according to the size of the lake and the complexity of the surrounding topography.

The exchange among the basins was investigated by using 3-D particle-tracking routines and passive tracer releases. The Lagrangian particle-tracking model was validated against data obtained with GPS-tracked autonomous drogues. The results of the Lagrangian model in 3-D indicate that the rate and patterns of mass exchange between the basins is highly dependent on the stratification conditions. Despite the large and shallow nature of Clear Lake, the water column undergoes periods of time of variable length (usually weeks) when it is not well mixed. It is during those periods that stratification exerts a considerable influence on the lake hydrodynamics, modifying the circulation patterns and intensifying the transport rate within the system. Under non-stratified conditions particles released in the Oaks Arm remained in that basin during the eight days of simulation,

simply describing circles in the vertical. Under stratified conditions, however, the simulations suggest that any contaminant released in the Oaks Arm would reach the other two basins in 3-7 days. The contaminant would penetrate into the Lower Arm following the surface currents or into the Upper Arm mainly through the bottom currents. These results suggest that the transport of mercury out of the Oaks Arm, associated with a flocculent material, most probably occurs intermittently, with periods of low transport and hence accumulation, alternating with periods – when the lake is stratified – during which the mercury-loaded floc would be exported to the other arms of Clear lake.

The description in this thesis of the hydrodynamics of the Oaks Arm is sound and consistent, based on multiple sources of information. Eulerian observations of temperature from thermistor strings, Eulerian velocity observations with acoustic devices from fixed and mobile platforms, numerical 3-D simulations, and observations collected with Lagrangian current followers (drogues), were used together to build a conceptual model of the large-scale circulation within the Oaks Arm and increase one's understanding of the exchange patterns of water masses between the three arms of Clear Lake. None of the sources of information provide alone a complete description of the hydrodynamics. Used together, they become very powerful tools for the analysis of natural systems.

9.4. Lake Tahoe

The second application of the 3-D model consisted of the simulation of internal waves in Lake Tahoe, which provide the most important driving mechanism for vertical and horizontal fluxes below the wind-mixed layer. The simulations were particularly aimed at

explaining the nature of the oscillations observed in temperature records collected by others in Lake Tahoe during winter 1999-2000. A theoretical analysis of the initial conditions, using simplified models of the geometry of Lake Tahoe, was additionally used to support the interpretation of the characteristic frequencies of the observed spectrum. During that period of time the lake was stratified with a well-defined metalimnion between 50 and 150 meters below the free water surface. Three modes of oscillations were isolated and identified in the internal wave field of Lake Tahoe. Two of the modes have sub-inertial frequencies and were identified as vertical mode one Kelvin waves traveling counterclockwise around the perimeter of the lake with periodicities of about 5 days and 60 hours. The 5-day-period-Kelvin wave has horizontal mode 1, and the 60-hour-period Kelvin wave has horizontal mode 2. The third internal wave mode is a Poincare wave having a frequency of about 18 hours; it causes transverse oscillations of the isotherms. These latter two modes had never been previously identified in Lake Tahoe. The mode-2 Kelvin wave has not previously been reported in any lake.

The numerical solution of SI3D-L in Lake Tahoe was first filtered to isolate the oscillations associated with each of the characteristic frequencies, and then visualized using sequences of 3-D plots of isotherm surfaces. This procedure is an extremely powerful tool to interpret the spectral peaks in the Fourier analysis of the thermistor chain records. It proved to be particularly useful in identifying the nature of the 60-hour-period oscillations in the spectrum of measured and simulated temperature records. The theoretical analysis of the initial conditions, using simplified models of the lake geometry, has been the traditional tool that is used in the interpretation of the spectra of

temperature field observations. And it was used here for such purposes, with results that are consistent with those obtained with SI3D-L. However, the visual tools provide a clearer and more direct means of presenting and explaining the results to researchers from completely different fields. In this manner, the communication of ideas within a multidisciplinary group, of biologists, chemists, engineers and others, can be considerably facilitated.

The simulations of Lake Tahoe pose a very severe problem to the 3-D model, different in nature from the problem posed by Clear Lake. In Clear Lake the boundaries of the basin are very complex and required a very large number of computational cells for its representation. In contrast, the surface geometry of Lake Tahoe is fairly regular, almost ellipsoidal, which can be represented with a relatively low number of cells. The free surface solution, hence, will be expensive in Clear Lake but not in Lake Tahoe. The problem posed by the simulations of Lake Tahoe is caused by its depth. The lake is 500 m deep at its deepest point, and requires of a large number of cells in the vertical to correctly represent the stratification. Clear Lake is only 15-16 m deep, and it can be adequately resolved with 16 layers, each 1 m thick. The most straining components in the Lake Tahoe computations are the treatment of the vertical transfer of momentum and scalars (temperatures, turbulent kinetic energy and turbulent macroscale). The code performance in the simulations of internal waves in Lake Tahoe was satisfactory for simulations of up to months. However, longer simulations will be needed to correctly represent the evolution of the stratification, which has seasonal time scales, according to previous studies of Lake Tahoe (Thompson 2000).

9.5. Recommendations for future research

Although the 3-D model gave satisfactory results in the simulation of simple test cases, and also in the more complicated applications to Clear Lake and Lake Tahoe, there are some features that should be improved in SI3D-L to increase further its efficiency and the range of applicability. There are also aspects in the hydrodynamics of Clear Lake and Lake Tahoe that ought to be addressed in some future research.

All simulations in this dissertation were limited in time, and hence they were not able to capture the whole range of hydrodynamic processes occurring in the two lakes where the code was applied. In Clear Lake only 8 days were simulated in the month of May, during a period of time of increasing stratification when adequate information on the boundary conditions was available. The simulation period was quite sufficient to characterize the diurnal cycle of the lake under stratified conditions. However, it was not adequate to capture the changes that Clear Lake probably undergoes over a complete stratification cycle, which is relatively short but still on the order of several weeks. Clear Lake could also undergo significant changes over a yearly cycle, which were not possible to characterize. In Lake Tahoe the simulation time was longer than in Clear Lake, close to a month. This length of time was adequate to describe basin-scale internal waves, though it was not long enough to characterize the slow seasonal changes that could have a very significant impact on the density structure. Longer simulation runs should thus be conducted in both Clear Lake and Lake Tahoe to characterize seasonal changes in the systems.

In order to undertake such a task, the numerical model should be modified. New routines to represent boundary wetting and drying processes should be added to the model to account for changes in the water elevation. The water level in Clear Lake, for example, undergoes significant changes in the course of the summer period when the lake water is used for irrigation purposes. The magnitudes of those changes are probably enough to dry the first and most superficial layer in the numerical grid, something that the present formulation of the code is not asked to model.

The efficiency of the code should be increased further. The most expensive feature in the simulations of Lake Tahoe was not the solution of the free surface problem, as it was in the case of Clear Lake, but the solution of tridiagonal systems of equations associated with the vertical turbulent transfer of momentum or scalars. Those calculations are performed independently in each water column and hence are perfect candidates for parallelization. Alternatively, the method of representing the vertical transfer of momentum and scalars could be changed, and mixed-layer type models could be used instead of the expensive turbulence closure models. At present, the 3-D model only allows layers of uniform thickness throughout the water column. By eliminating this restriction it would be possible to increase the speed of the code, since it would allow one to use a smaller number of layers to discretize the deeper portions of the Tahoe basin where the stratification is not significant at all. The increase in efficiency in Clear Lake should come, in contrast, from the modification of the free surface solution. The iterative solver presently used in the code (Preconditioned Conjugate Gradient) seems to experience convergence problems associated with the dimensionality of the problem and

the geometric shape of the wet domain. A new solver should, hence, be implemented in the model to increase the speed of convergence. The incorporation of multigrid techniques (Briggs 1987) may provide a very efficient way to increase the convergence speed of the iterative solvers, and its implementation should hence be explored. The use of a Cartesian grid with a fixed cell size is a very restricting limitation, since high-resolution grids are needed to adequately represent complex geometries. The use of unstructured grids (e.g. Casulli and Walters 2000) or Cartesian grids with several levels of refinement (e.g. Almgren *et al.* 1997) could provide significant computational advantages.

Longer simulations will also entail the collection of extensive and more accurate wind and meteorological information to drive the model, and more water temperature and velocity for model validation. The atmospheric information should ideally be obtained at several locations throughout the lake to characterize adequately the spatial scales of the variation of the wind and meteorological fields. Some of this information will preferably be collected over the water surface to represent truly the boundary conditions over the free surface. In Clear Lake that information could be used in constructing a wind model that accounts for the deflecting and sheltering effects of the topographic features of the landscape. The wind model, once calibrated, could be used to drive the 3-D hydrodynamic model in routine and real-time simulations, using a lower number of wind stations that are at present operational in the area.

Together SI3D-L and the Lagrangian particle-tracking model should become the foundations for the future development of a model that is able to represent and predict the fate and transport of mercury associated with an alumino-silicate flocculent material in Clear Lake. Such a model would be an invaluable tool in assessing remediation schemes proposed in the lake to reduce the levels of mercury contamination. Although the particles, in the present formulation of the Lagrangian model, are neutrally buoyant, additional features can be added to simulate settling, resuspension or any other of the physical, and possibly chemical, characteristics of the floc. Further studies to understand the physicochemical characteristics of the focculent material and the conditions controlling the rate of its incorporation into the Clear Lake system should be undertaken, prior to any modification to the numerical algorithm. The transport and fate of dissolved contaminant species in Clear Lake, or in any other lake system in fact, can be addressed by using the advection-diffusion routines which are already incorporated into the code to simulate tracer plumes. Source-sink terms and interactions among different species could be easily incorporated. The same algorithms could be applied to Lake Tahoe to represent particle and contaminant dynamics. In this case, however, boundary mixing processes must be parameterized to simulate correctly the introduction of sources of contaminants or particulates into the body of the lake.

REFERENCES

- Almgren, A. S., J. B. Bell, P. Colella, and T. Marthaler (1997). "A Cartesian grid projection method for the incompressible Euler equations in complex geometries." *SIAM Journal on Scientific Computing*, 18(5), 1289-1309.
- Amorocho, J., and J. J. DeVries (1980). "A new evaluation of the wind stress coefficient over water surfaces." *Journal of Geophysical Research*, 85, 433-442.
- Arfken, G. B., and H. J. Weber (1995). *Mathematical Methods for Physicists*, Academic Press.
- Baines, P. G. (1995). *Topographic Effects in Stratified Fluids*, Cambridge University Press, Cambridge, United Kingdom.
- Bale, A. E. (1995). "Modeling mercury transport and transformation in the aquatic environment." Ph.D. Dissertation, University of California, Davis.
- Banner, M. L., and W. L. Peirson (1998). "Tangential stress beneath wind-driven air-water interfaces." *Journal of Fluid Mechanics*, 364, 115-145.
- Barnes, S. L. (1973). "A technique for maximizing details in numerical weather map analysis." *Journal of Applied Meteorology*, 3, 396-409.
- Blumberg, A. F. (1986). "Turbulent mixing processes in lakes, reservoirs and impoundments." Physics-based modelling of lakes, reservoirs and impoundments, W. G. Gray, ed., ASCE, New York, 79-104.
- Blumberg, A. F., and G. L. Mellor (1987). "A description of a three-dimensional coastal ocean circulation model." Three-Dimensional Coastal Ocean Models, S. Heaps, ed., American Geophysical Union, Washington, DC., 1-16.
- Briggs, W. L. (1987). *A Multigrid Tutorial*, Society for Industrial and Applied Mathematics.
- Brutsaert, W. (1975). "On a derivable formula for long-wave radiation from clear skies." *Water Resources Research*, 11, 742-744.
- Bryan, K., and M. D. Cox (1967). "A numerical investigation of the oceanic general circulation." *Tellus*, 19, 54-80.
- Casulli, V., and E. Cattani (1994). "Stability, accuracy and efficiency of a semi-Implicit method for three-dimensional shallow water flow." *Computers and Mathematics with Applications*, 27(4), 99-112.
- Casulli, V., and R. T. Cheng (1992). "Semi-implicit finite difference methods for three-dimensional shallow water flow." *International Journal for Numerical Methods in Fluids*, 15, 629-648.
- Casulli, V., and R. A. Walters (2000). "An unstructured grid, three-dimensional model based on the shallow water equations." *International Journal for Numerical Methods in Fluids*, 32, 331-348.
- Cathers, B., and B. A. O'Connor (1985). "The group velocity of some numerical schemes." *International Journal of Numerical Methods in Fluids*, 5, 201-224.
- Chamberlin, C. E., R. Chaney, G. Finney, M. Hood, P. Lehman, M. McKee, and R. Willis (1990). "Abatement and control study: Sulfur Bank Mine and Clear Lake. Prepared for the California Regional Water Quality Control Board." Environmental Resources Engineering Department, Humboldt State University.

- Charnock, H. (1955). "Wind stress on a water surface." *Quarterly Journal Royal Meteorological Society*, 81, 639.
- Cheng, R. T., and V. Casulli (1982). "On Lagrangian residual currents with applications in south San Francisco Bay, California." *Water Resources Research*, 18.
- Cheng, R. T., T. M. Powell, and T. M. Dillon. (1976). "Numerical models of wind-driven circulation in lakes." *Applied Mathematical Modelling*, 1, 141-159.
- Chrystal, G. (1904). "On the hydrodynamical theory of seiches." *Transactions Royal Society Edinburgh*, 41(3), 599-649.
- Cook, C. B. (2000). "Internal dynamics of a terminal basin lake: a numerical model for management of the Salton Sea." Ph.D. Dissertation, University of California, Davis.
- Cox, R. M., J. Sontowski, R. N. Fry Jr., C. M. Dougherty, and T. J. Smith (1998). "Wind and diffusion modeling for complex terrain." *Journal of Applied Meteorology*, 37, 996-1009.
- Cressman, G. P. (1959). "An operational objective analysis system." *Monthly Weather Review*, 87, 367-374.
- Csanady, G. T. (1967). "Large-scale motion in the Great Lakes." *Journal of Geophysical Research*, 72(16), 4151-4162.
- Csanady, G. T. (1975). "Hydrodynamics of Large Lakes." *Annual Review of Fluid Mechanics*, 7, 357-386.
- Csanady, G. T. (1980). *Turbulent Diffusion in the Environment*, D.Reidel, Norwell, Mass.
- Csanady, G. T. (1982). *Circulation in the Coastal Ocean*, D. Reidel Publishing Company, Dordrecht, Holland.
- Cushman-Roisin, B. (1994). *Introduction to Geophysical Fluid Dynamics*, Prentice-Hall, Inc., Englewood Cliffs, New Jersey.
- Davies, A. M., J. E. Jones, and J. Xing (1997). "Review of recent developments in tidal hydrodynamic modeling. I-Spectral models." *ASCE Journal of Hydraulic Engineering*, 123(4), 278-292.
- de Goede, E. D. (1991). "A time-splitting method for the three-dimensional shallow water equations." *International Journal for Numerical Methods in Fluids*, 13(4), 519-534.
- Dimou, K. N., and E. E. Adams (1993). "A random-walk, particle tracking model for well-mixed estuaries and coastal waters." *Estuarine, Coastal and Shelf Science*, 37, 99-110.
- Dunsbergen, D. W., and G. S. Stelling (1993). "A 3-D particle model for transport problems in transformed coordinates." 93-7, Delft University of Technology. Department of Civil Engineering. Hydraulic and Geotechnical Engineering Division. Hydromechanics Group., Delft, Netherlands.
- Durbin, P. A. (1984). "Comments on papers by Wilson et al. (1981) and Legg and Raupach (1982)." *Boundary-Layer Meteorology*, 29, 409-411.
- Durrant, D. R. (1999). *Numerical Methods for Wave Equations in Geophysical Fluid Dynamics*, Springer.
- Efron, B., and G. Gong (1983). "A leisurely look at the bootstrap, the jackknife and cross-validation." *The American Statistician*, 37(1), 36-48.

- Eliason, D. E., and A. J. Bourgeois (1997). "Validation of numerical shallow water models for stratified seiches." *International Journal for Numerical Methods in Fluids*, 24, 771-786.
- Erasmus, D. A. (1986). "A model for objective simulation of boundary-layer winds in an area of complex terrain." *Journal of Climate and Applied Meteorology*, 25, 1832-1841.
- Ferziger, J. H., and M. Peric (1996). *Computational Methods for Fluid Dynamics*, Springer-Verlag.
- Fischer, H. B., E. J. List, R. C. Y. Koh, J. Imberger, and N. H. Brooks (1979). *Mixing in Inland and Coastal Waters*, Academic Press.
- Fletcher, C. A. J. (1991). *Computational Techniques for Fluid Dynamics*, Springer.
- Fofonoff, N. P. (1969). "Spectral characteristics of internal waves in the ocean." *Deep Sea Research*, 16 Suppl., 59-71.
- Freeman, N. J., A. M. Hale, and M. B. Danard (1972). "A modified sigma equations approach to the numerical modeling of the Great Lakes hydrodynamics." *Journal of Geophysical Research*, 77, 1050-1060.
- Galperin, B., L. H. Kantha, S. Hassid, and A. Rosati (1988). "A quasi-equilibrium turbulent energy model for geophysical flows." *Journal of the Atmospheric Sciences*, 45(1), 55-62.
- Gardiner, C. W. (1990). *Handbook of Stochastic Methods for Physics, Chemistry and the Natural Sciences*, Springer-Verlag, New York.
- Gardner, J. V., L. A. Mayer, and J. Hughes-Clarke (1998). "The bathymetry of Lake Tahoe, California-Nevada." *Open -File Report 98-509*, U.S.G.S.
- Geernaert, G. (1990). "The theory and modeling of wind stress with applications to air-sea interaction and remote sensing." *Reviews in Aquatic Sciences*, 2(1), 125-149.
- Geernaert, G. L., S. E. Larsen, and F. Hansen (1987). "Measurements of the wind stress, heat flux and turbulence intensity during storm conditions over the North Sea." *Journal of Geophysical Research*, 92, 13127-13139.
- Giroux, H., and J. Lawton (1980). "The Geysers KGRA micro-meteorological network. Environmental Research and Design." Prepared for Lake County Air Pollution Control District.
- Goldman, C. R. (1988). "Primary productivity, nutrients and transparency during the early onset of eutrophication in ultra-oligotrophic Lake Tahoe, California-Nevada." *Limnology and Oceanography*, 33(6), 1321-1333.
- Golub, G. H., and C. F. V. Loan (1996). *Matrix Computations*, The John Hopkins University Press, Baltimore, U.S.
- Gonella, J. (1972). "A rotary-component method for analysing meteorological and oceanographic vector time series." *Deep Sea Research*, 19, 833-846.
- Gordon, A., W. Grace, P. Schwerdtfeger, and R. Byron-Scott (1998). *Dynamic Meteorology: a Basic Course*, Arnold, London.
- Gray, W. G., and D. R. Lynch (1979). "Time-stepping schemes for finite element tidal computations." *Advances in Water Resources*, 1(2), 83-95.
- Gresho, P. M., and R. L. Sani (1998). *Incompressible Flow and the Finite Element Method: Advection-Diffusion and Isothermal Laminar Flow*, Wiley, Chichester, New York.

- Gross, E. S. (1997). "Numerical modeling of hydrodynamics and scalar transport in an estuary." Ph.D. Dissertation, Stanford University.
- Gross, E. S., J. R. Koseff, and S. G. Monismith (1999). "Three-dimensional salinity simulations of south San Francisco Bay." *ASCE Journal of Hydraulic Engineering*, 125(11), 1199-1209.
- Haidvogel, D. B., J. L. Wilkin, and R. Young (1991). "A semi-spectral primitive equation ocean circulation model using vertical sigma and orthogonal curvilinear horizontal coordinates." *Journal of Computational Physics*, 94, 151-185.
- Hardy, R. L. (1971). "Multiquadric equations of topography and other irregular surfaces." *Journal of Geophysical Research*, 76, 1905-1915.
- Hathhorn, W. E. (1997). "Simplified approach to particle tracking methods for contaminant transport." *ASCE Journal of Hydraulic Engineering*, 123(12), 1157-1160.
- Heaps, N. S. (1972). "On the numerical solution of the three-dimensional hydrodynamical equations for tides and storm surges." *Memoires de la Societe Royale des Sciences de Liege. Series 6*, 2, 143-180.
- Heaps, N. S. (1984). "Vertical structure of current in homogeneous and stratified waters." *Hydrodynamics of lakes*, K. Hutter, ed., Springer Verlag, Wien-New York, 153-207.
- Heath, R. E., A. M. Teeter, G. E. Freeman, and W. L. Boyt (1999). "Ashtabula River, Ohio, Sedimentation Study - Report 4 - Numerical Model." CHL-99-9-4, U.S. Army Engineer District, Buffalo.
- Heemink, A. W. (1990). "Stochastic modelling of dispersion in shallow water." *Stochastic Hydrology and Hydraulics*, 4, 161-174.
- Henderson-Sellers, B. (1982). "A simple formula for vertical eddy diffusion coefficients under conditions of nonneutral stability." *Journal of Geophysical Research*, 87(C8), 5860-5864.
- Henderson-Sellers, B. (1984). *Engineering Limnology*, Pitman Advanced Publishing Program, Boston.
- Henderson-Sellers, B. (1986). "Calculating the surface energy balance for lake and reservoir modeling: a review." *Reviews of Geophysics*, 24(3), 625-649.
- Hirsch, C. (1990). *Numerical Computation of Internal and External Flows*, John Wiley & Sons, Chichester, England.
- Hodges, B. R., J. Imberger, A. Saggio, and K. B. Winters (2000). "Modeling basin-scale motions in a stratified lake." *Limnology and Oceanography*, 45(7), 1603-1620.
- Holman, J. P. (1990). *Heat Transfer*, McGraw-Hill.
- Horne, A. J. (1975). "The ecology of Clear Lake Phytoplankton." , Clear Lake Algal Research Unit, Lakeport, California, 116.
- Hsieh, C., G. Katul, J. Schieldge, J. T. Sigmon, and K. K. Knoerr (1997). "The Lagrangian stochastic model for fetch and latent heat flux estimation above uniform and nonuniform terrain." *Water Resources Research*, 33(3), 427-438.
- Huang, J. C. K., and P. W. Sloss (1981). "Simulation and verification of Lake Ontario's mean state." *Journal of Physical Oceanography*, 11, 1548-1566.
- Hunter, J. R. (1987). "The application of Lagrangian particle-tracking techniques to modelling of dispersion in the sea." *Numerical modelling: applications to marine systems*, J. Noye, ed., Elsevier Science Publishers B.V., North-Holland.

- Hunter, J. R., P. D. Craig, and H. E. Phillips (1993). "On the use of random walk models with spatially variable diffusivity." *Journal of Computational Physics*, 106, 366-376.
- Imberger, J. (1985). "The diurnal mixed layer." *Limnology and Oceanography*, 30(4), 737-770.
- Imberger, J. (1998). "Flux paths in a stratified lake: a review." *Physical Processes in Lakes and Oceans*, J. Imberger, ed., AGU, 1-17.
- Imberger, J., and J. C. Patterson (1990). "Physical Limnology." *Advances in Applied Mechanics*, 27, 303-475.
- Jerlov, N. G. (1968). *Optical oceanography*, Elsevier.
- Jin, K.-R., J. H. Hamrick, and T. Tisdale (2000). "Application of three-dimensional hydrodynamic model for Lake Okeechobee." *ASCE Journal of Hydraulic Engineering*, 126(10), 758-771.
- Kantha, L. H., and C. A. Clayson (1994). "An improved mixed layer model for geophysical applications." *Journal of Geophysical Research*, 99(C12), 25235-25266.
- Kato, H., and O. M. Phillips (1969). "On the penetration of a turbulent layer into stratified fluid." *Journal of Fluid Mechanics*, 37(4), 643-655.
- Kawahara, M., M. Kobayashi, and K. Nakata (1982). "A three-dimensional multiple level finite element method considering variable water density." *Finite Element in Fluids*, R. H. Gallagher, D. H. Norrie, J. T. Oden, and O. C. Zienkiewicz, eds., Wiley, Chichester, 129-156.
- Kincaid, D. R., T. C. Oppe, and W. D. Joubert (1989). "An introduction to the NSPCG software package." *International Journal of Numerical Methods in Engineering*, 27(3), 589-608.
- King, I. P. (1985). "Strategies of finite element modeling of three dimensional hydrodynamic systems." *Advances in Water Resources*, 8, 69-76.
- King, I. P. (1993). "A finite element model for three-dimensional density stratified flow. Sidney Deepwater Outfalls Environmental Monitoring Program Post-commissioning Phase." 93/01/01, Australian Water and Coastal Studies (AWACS) PTY LTD, Sidney.
- King, I. P. (1996). "RMAGEN. A program for generation of finite element networks."
- King, I. P. (1998). "A finite element model for stratified flow. RMA-10."
- King, I. P. (1999). "Personal communication."
- Kinzelbach, W. (1988). "The random walk method in pollutant transport simulation." *Groundwater flow and quality modelling*, E. Custodio, ed., D. Reidel Publ., Rotterdam, The Netherlands, 227-245.
- Kitaigorodskii, S. A., and M. M. Zaslavskii (1974). "A dynamical analysis of the drag conditions at the sea surface." *Boundary Layer Meteorology*, 6, 53-61.
- Kitanidis, P. K. (1994). "Particle-tracking equations for the solution of the advection-dispersion equation with variable coefficients." *Water Resources Research*, 30(11), 3225-3227.
- Kraus, E. B., and J. S. Turner (1967). "A one-dimensional model of the seasonal thermocline: II. The general theory and its consequences." *Tellus*, 19, 98-106.
- Kundu, P. J. (1980). "A numerical investigation of mixed-layer dynamics." *Journal of Physical Oceanography*, 10, 220-236.

- Kundu, P. K. (1990). *Fluid Mechanics*, Prentice Hall.
- Kundu, P. K., J. S. Allen, and R. L. Smith (1975). "Modal decomposition of the velocity field near Oregon Coast." *Journal of Physical Oceanography*, 5(October), 683-704.
- LaBolle, E. M., G. E. Fogg, and A. F. B. Thompson (1996). "Random-walk simulation of transport in heterogeneous porous media: local mass-conservation problem and implementation methods." *Water Resources Research*, 32(3), 583-593.
- Lallatin, R. D. (1975). "Clear Lake water quality data." Department of Water Resources - Northern District, 321.
- Lamb, H. (1945). *Hydrodynamics*, Dover Publications, New York.
- Large, W. G., J. C. McWilliams, and S. C. Doney (1994). "Oceanic vertical mixing: a review and a model with a nonlocal boundary layer parameterization." *Reviews of Geophysics*, 32(4), 363-403.
- Lee, R. L., and E. Naslund (1998). "Lagrangian stochastic particle tracking model simulations of turbulent dispersion around buildings." *Atmospheric Environment*, 32(4), 665-672.
- Leendertse, J. J. (1989). "A new approach to three-dimensional free-surface flow modeling." *R-3712-NETH/RC*, Rand Corporation, Santa Monica.
- Leendertse, J. J., R. C. Alexander, and S. K. Liu (1973). "A three-dimensional model for estuaries and coastal seas-Volume I, principles of computation." *R-1417-OWRR*, Rand Corporation, Santa Monica.
- Leendertse, J. J., and S. K. Liu (1975). "A three-dimensional model for estuaries and coastal seas-Volume II, aspects of computation." *R-1764-OWRT*, Rand Corporation, Santa Monica.
- Legg, B. J., and M. R. Raupach (1982). "Markov-chain simulation of particle dispersion in inhomogeneous flows: the mean drift velocity induced by a gradient in eulerian velocity variance." *Boundary-Layer Meteorology*, 24, 3-13.
- Lemmin, U., and N. D'Adamo (1996). "Summertime winds and direct cyclonic circulation: observations from Lake Geneva." *Annales Geophysicae*, 14, 1207-1220.
- Leonard, B. P. (1991). "The ULTIMATE conservative difference scheme applied to unsteady one-dimensional advection." *Computer Methods in Applied Mechanics and Engineering*, 88, 17-74.
- LeVeque, R. J. (1996). "High-resolution conservative algorithm for advection in incompressible flow." *SIAM Journal of Numerical Analysis*, 33, 627-665.
- Lin, B., and R. A. Falconer (1997). "Tidal flow and transport modeling using ULTIMATE QUICKEST scheme." *ASCE Journal of Hydraulic Engineering*, 123(4), 303-314.
- Liu, P. C., and D. J. Schwab (1987). "A comparison of methods for estimating u_* from given u_z and air-sea temperature differences." *Journal of Geophysical Research*, 92(C6), 6488-6494.
- Lucquin, B., and O. Pironneau (1998). *Introduction to Scientific Computing*, John Wiley & Sons Ltd., Chichester, UK.
- Ludwig, F. L., J. M. Livingston, and R. M. Endlich (1991). "Use of mass conservation and critical dividing streamline concepts for efficient objective analysis of winds in complex terrain." *Journal of Applied Meteorology*, 30, 1490-1499.

- Lynch, D. R. (1986). "Basic hydrodynamic equations for lakes." Physics-based modelling of lakes, reservoirs and impoundments, W. G. Gray, ed., ASCE, New York, 17-53.
- Lynch, D. R., and F. E. Werner (1991). "Three-dimensional hydrodynamics on finite elements. Part II: non-Linear time-stepping model." *International Journal of Numerical Methods in Fluids*, 12, 507-533.
- Lynch, M. G. (1996). "Seasonal variations in lake mixing, Clear Lake, California." Master's Thesis, University of California, Davis.
- Lynch, M. G., and S. G. Schladow (1996). "Clear Lake current study: mixing and transport dynamics. Special report. Prepared for the United States Environmental Protection Agency." *EDL96-01MGL*, Department of Civil and Environmental Engineering, University of California, Davis.
- MacIntyre, S., K. M. Flynn, R. Jellison, and J. R. Romero (1999). "Boundary mixing and nutrient fluxes in Mono Lake, California." *Limnology and Oceanography*, 44(3), 512-529.
- Martin, P. J. (1985). "Simulation of the mixed layer at OWS November and Papa with several models." *Journal of Geophysical Research*, 90(C1), 903-916.
- Maxworthy, T., and S. G. Monismith (1988). "Differential mixing in a stratified fluid." *Journal of Fluid Mechanics*, 189, 571-598.
- McCormick, M. J., and G. A. Meadows (1988). "An intercomparison of four mixed layer models in a shallow inland sea." *Journal of Geophysical Research*, 93(C6), 6774-6788.
- Mellor, G. L., and T. Yamada (1982). "Development of a turbulence closure model for geophysical fluid problems." *Reviews of Geophysics and Space Physics*, 20(4), 851-875.
- Mellor, G. T., and T. Yamada (1974). "A hierarchy of turbulence closure models for planetary boundary layers." *Journal of Atmospheric Sciences*, 31, 1791-1806.
- Monismith, S. G. (1999). "Personal Communication."
- Monismith, S. G., and D. A. Fong (1996). "A simple model of mixing in stratified flows." *Journal of Geophysical Research*, 101(C12), 28583-28559.
- Montero, G., R. Montenegro, and J. M. Escobar (1998). "A 3-D diagnostic model for wind field adjustment." *Journal of Wind Engineering*, 74-76, 249-261.
- Mooers, C. N. K. (1973). "A technique for the cross spectrum analysis of pairs of complex-valued time series, with emphasis on properties of polarized components and rotational invariants." *Deep Sea Research*, 20, 1129-1141.
- Munk, W. H., and E. R. Anderson (1948). "Notes on a theory of the thermocline." *Journal of Marine Research*, 1(276-295).
- Munnich, M. (1996). "The Influence of bottom topography on internal seiches in stratified media." *Dynamics of Atmospheres and Oceans*, 23, 257-266.
- Naslund, E., J. C. Rodean, and J. S. Nasstrom (1994). "A comparison between two stochastic diffusion models in a complex three-dimensional flow." *Boundary-Layer Meteorology*, 67, 369-384.
- Nelson, D. C. (2000). "On the nature of the bottom of Clear Lake." .
- Neuman, G., and W. J. Pierson (1966). *Principles of Physical Oceanography*, Prentice-Hall, Englewood Cliffs, N.J.

- Niiler, P. P. (1975). "Deepening of the wind-mixed layer." *Journal Marine Research*, 33, 405-422.
- North, G. R., T. L. Bell, and R. F. Cahalan (1982). "Sampling errors in the estimation of empirical orthogonal functions." *Monthly Weather Review*, July, 699-706.
- Nuss, W. A., and D. W. Titley (1994). "Use of multiquadric interpolation for meteorological objective analysis." *Monthly Weather Review*, 122, 1611-1631.
- Obled, C., and J. D. Creutin (1986). "Some developments in the use of empirical orthogonal functions for mapping meteorological fields." *Journal of Climate and Applied Meteorology*, 25(9), 1189-1204.
- Okubo, A. (1973). "Effect of shoreline irregularities on streamwise dispersion in estuaries and other embayments." *Netherlands Journal of Sea Research*, 6(1-2), 213-224.
- Oton, S., S. G. Schladow, J. Clark, B. Hudson, and A. Dvorak (1998). "Hydrological transport between the Sulphur Bank Mercury Mine and Clear Lake using gas tracer." *Second Annual Clear Lake Science and Management Symposium*, Lakeport, California, 119-124.
- Palmarsson, S. O. (2000). "Personal Communication."
- Parker, G. J., and J. Imberger (1986). "Differential mixed-layer deepening in lakes and reservoirs." *Limnology in Australia*, P. de Decker and W. D. Williams, eds., W. Junk, Melbourne, Victoria, 63-92.
- Perrie, W., and B. Toulany (1990). "Fetch relations for wind-generated waves as a function of wind-stress scaling." *Journal of Physical Oceanography*, 20, 1666-1681.
- Pielke, R. A. (1985). "The use of mesoscale numerical models to assess wind distribution and boundary-layer structure in complex terrain." *Boundary-Layer Meteorology*, 31, 217-231.
- Pinder, G. F., and W. G. Gray (1977). *Finite Element Simulation in Surface and Subsurface Hydrology*, Academic Press.
- Pollard, R. T., P. B. Rhines, and R. O. R. Y. Thompson (1973). "The deepening of the wind-mixed layer." *Geophysical Fluid Dynamics*, 3, 381-404.
- Pollock, D. W. (1988). "Semianalytical computation of path lines for finite difference models." *Groundwater*, 26(6), 743-750.
- Press, W. H., S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery (1992). *Numerical Recipes in Fortran. The Art of Scientific Computing.*, Cambridge University Press.
- Price, J. F. (1979). "On the scaling of stress-driven entrainment experiments." *Journal of Fluid Mechanics*, 90(3), 509-529.
- Reddy, J. N. (1993). *An Introduction to the Finite Element Method*, McGraw-Hill.
- Riddle, A. M. (1998). "The specification of mixing in random walk models for dispersion in the sea." *Continental Shelf Research*, 18, 441-456.
- Rosati, A., and K. Miyakoda (1988). "A general circulation model for upper ocean simulation." *Journal of Physical Oceanography*, 18, 1601-1626.
- Saggio, A., and J. Imberger (1998). "Internal wave weather in a stratified lake." *Limnology and Oceanography*, 43(8), 1780-1795.
- Sawford, B. L. (1985). "Lagrangian statistical simulation of concentration mean and fluctuation fields." *Journal of Climate and Applied Meteorology*, 24, 1152-1166.

- Schaefer, J. T., and C. A. Doswell (1979). "On the interpolation of a vector field." *Monthly Weather Review*, 107, 458-476.
- Scheibe, T. D., and C. R. Cole (1994). "Non-Gaussian particle tracking: application to scaling of transport processes in heterogeneous porous media." *Water Resources Research*, 30(7), 2027-2039.
- Sheng, Y. P. (1986). "Finite-difference models for hydrodynamics of lakes and shallow seas." *Physics-based modelling of lakes, reservoirs and impoundments*, W. G. Gray, ed., ASCE, New York, 146-128.
- Sheng, Y. P., W. Lick, R. T. Gedney, and F. B. Molls (1978). "Numerical computation of three-dimensional circulation in Lake Erie - A comparison of a free-surface model and a rigid-lid model." *Journal of Physical Oceanography*, 8, 713-727.
- Sheppard, P. A. (1956). "Air flow over mountains." *Quarterly Journal Royal Meteorological Society*, 82, 528-529.
- Sherman, F. S., J. Imberger, and G. M. Corcos (1978). "Turbulence and mixing in stably stratified waters." *Annual Review of Fluid Mechanics*, 10, 267-288.
- Simons, T. J. (1974). "Verification of numerical models of Lake Ontario. Part I: circulation in spring and early summer." *Journal of Physical Oceanography*, 4, 507-523.
- Smith, P. E. (1997). "A three-dimensional, finite-difference model for estuarine circulation." Ph.D. Dissertation, University of California, Davis.
- Smith, P. E., and B. E. Larock (1993). "A finite-difference model for 3-D flow in bays and estuaries." *Hydraulic Engineering '93 ASCE Hydraulics Division Conference*, July 25-30, 1993, San Francisco, CA, 2166-2122.
- Smith, P. E., and B. E. Larock (1997). "Semi-implicit, numerical schemes for 3-D flow modeling." *Environmental and Coastal Hydraulics: Protecting the Aquatic Habitat. Theme B. Water for a Changing Global Community. The 27th Congress of the International Association for Hydraulic Research*, 773-778.
- Smolarkiewicz, P. K. (1984). "A fully multidimensional positive-definite advection transport algorithm with small implicit diffusion." *Journal of Computational Physics*, 54, 325-362.
- Song, Y., and D. Haidvogel (1994). "A semi-implicit ocean circulation model using a generalized topography-following coordinate system." *Journal of Computational Physics*, 115, 228-244.
- Sontek (1996). "Sontek Technical Notes - Acoustic Doppler Profiler (ADP). Principles of Operation."
- Spigel, R. H., and J. Imberger (1980). "The classification of mixed layer dynamics in lakes of small to medium size." *Journal of Physical Oceanography*, 10, 1104-1121.
- Spigel, R. M. (1978). "Wind mixing in lakes." Ph.D. Dissertation, University of California, Berkeley.
- Strub, P. D., and T. M. Powell (1986). "Wind-driven surface transport in stratified closed basins: direct versus residual circulations." *Journal of Geophysical Research*, 91(C7), 8497-8508.
- Stull, R. B. (1994). *An Introduction to Boundary Layer Meteorology*, Kluwer Academic Publishers, Dordrecht, The Netherlands.

- Suchanek, T. H., L. H. Mullen, B. H. Lamphere, P. J. Richerson, C. E. Woodmansee, D. G. Slotton, E. J. Harner, and L. A. Woodward (1998). "Redistribution of mercury from contaminated lake sediments of Clear Lake, California." *Water, Air and Soil Pollution*, 104(1/2), 77-102.
- Suchanek, T. H., P. J. Richerson, L. J. Mullen, L. L. Brister, J. C. Becker, A. Maxon, and D. G. Slotton (1997). "The role of the Sulphur Bank Mercury Mine site (and associated hydrogeological processes) in the dynamics of mercury transport and bioaccumulation within the Clear Lake aquatic ecosystem." Prepared for the USEPA, Region IX Superfund Program.
- Survey, U. S. C. G. (1948-1949). "Advance Copy. Hydrographic Survey. Clear Lake, California." , Washington, D.C.
- Taner, M. T., F. Koehler, and R. E. Sheriff (1979). "Complex seismic trace analysis." *Geophysics*, 44(6), 1041-1063.
- Taylor, P. A., and R. J. Lee (1984). "Simple guidelines for estimating wind speed variations due to small-scale topographic features." *Climatol. Bulletin*, 18, 3-32.
- Taylor, P. A., and J. R. Salmon (1993). "A model for the correction of surface wind data for sheltering by upwind obstacles." *Journal of Applied Meteorology*, 32, 1683-1694.
- Tennekes, H., and J. L. Lumley (1972). *A First Course in Turbulence*, MIT Press, Cambridge, Massachusetts.
- Tennessee Valley Authority T.V.A (1972). "Heat and mass transfer between a water surface and the atmosphere." 0-6803.
- Thompson, K. L. (2000). "Winter mixing dynamics and deep mixing in Lake Tahoe." Master's Thesis, University of California, Davis.
- Thomson, D. J. (1987). "Criteria for the selection of stochastic models of particle trajectories in turbulent flows." *Journal of Fluid Mechanics*, 180, 529-556.
- Tompson, A. F. B., and L. W. Gelhar (1990). "Numerical simulation of solute transport in three-dimensional, randomly heterogeneous porous media." *Water Resources Research*, 26(10), 2541-2562.
- Uffink, G. J. M. (1988). "Modeling of solute transport with the random walk method." Groundwater flow and quality modelling, E. C. e. al., ed., D. Reidel Publ., Rotterdam, The Netherlands, 247-265.
- Uittenbogaard, R. E., J. A. T. M. v. Kester, and G. S. Stelling (1992). "Implementation of three turbulence models in TRISULA for rectangular horizontal grids." , Delft Hydraulics, Delft, Netherlands.
- Van den Boogaard, H. F. P., M. J. J. Hoogkamer, and A. W. Heemink (1993). "Parameter identification in particle models." *Stochastic Hydrology and Hydraulics*, 7(2), 109-130.
- Vreugdenhil, C. B. (1994). *Numerical Methods for Shallow-Water Flow*, Kluwer Academic Publishers, Dordrecht.
- Walters, R. A. (1983). "Numerically induced oscillations in finite element approximations to the shallow water equations." *International Journal of Numerical Methods in Fluids*, 3, 591-604.
- Walters, R. A., and G. F. Carey (1983). "Analysis of spurious oscillation modes for the shallow water and Navier-Stokes equations." *Computers and Fluids*, 11(1), 51-68.

- Wilson, B. W. (1972). "Seiches." *Advances in Hydrosiences*, Academic Press, New York, 1-94.
- Wilson, J. D., and B. L. Sawford (1996). "Review of Lagrangian stochastic models for trajectories in the turbulent atmosphere." *Boundary-Layer Meteorology*, 78, 191-210.
- Winters, K. B., P. N. Lombard, J. J. Riley, and E. A. D'Asaro (1995). "Available potential energy and mixing in density-stratified fluids." *Journal of Fluid Mechanics*, 289, 115-128.
- Wu, J. (1985). "Parameterization of wind-stress coefficients over water surfaces." *Journal of Geophysical Research*, 90(C5), 9069-9072.
- Zalesak, S. T. (1979). "Fully multidimensional flux-corrected transport algorithms for fluids." *Journal of Computational Physics*, 31, 335-362.

APPENDIX A

LIST OF SYMBOLS AND ABBREVIATIONS FOR CHAPTER 3

a, χ	parameters in definition of truncated parabolic geometry
c	gravity wave speed = $\sqrt{gH}$
c_p	parameter in the solution to free oscillations in parabolic basin
g	acceleration of gravity
h	displacement of water surface from its equilibrium position
h_0	maximum amplitude of water elevation displacement
s	number of nodal diameters in barotropic oscillations in circular basin
t	time
(u, v, w)	velocity vector
u^*	friction velocity produced by wind forcing
x_c	distance from center of rectangular basin
(x, y, z)	cartesian coordinates; z-coordinate directed upwards from bottom
CN	gravity wave Courant number = $c\Delta t/\Delta x$
COSF/SINF	Cosine and sine functions as defined in Chrystal 1904
CTA	Complex trace analysis
H	depth of rectangular basin with flat bottom
H_{max}	maximum depth in rectangular basin with truncated parabolic bottom
K_z	vertical momentum transfer coefficient
L	Wavelength
$L/\Delta x$	Non-dimensional wave length
L_0	length of rectangular basin
ND	Mode of barotropic oscillation in a circular basin with one nodal diameter
NC	Mode with one nodal circle
Q_e	Estimated celerity ratio from numerical experiments
Q_t	Theoretical celerity ratio (Table 3.3)
R_0	Radius of circular basin
U	Horizontal transport = $\int u dz$
ε_A	Dissipation factor
η	x_c/a in parabolic basin
σ	frequency of oscillations
$(\Delta x, \Delta y, \Delta z)$	spacing between consecutive computational points
Δt	time step

APPENDIX B

DETERMINING THE FREQUENCY OF BAROTROPIC SEICHES IN LAKES OF COMPLEX GEOMETRY. APPLICATION TO CLEAR LAKE, CALIFORNIA.

B.1. Introduction

Many systems in nature can invoke a built-in *restoring force* for re-establishing its equilibrium position once it has been displaced from it. If the system also possesses sufficient *inertia* that causes it to overshoot its equilibrium position, the system will perform *free oscillations*, provided the disturbing forces responsible for the initial displacement are not sustained. The free oscillations are characteristics of the system only and are independent of the exciting force, except of course for the initial magnitude. A seiche, or more specifically a barotropic seiche, is such an oscillation of the *free surface* performed in bodies of fluid enclosed wholly or partially by boundaries (Wilson 1972). The restoring force is provided by the action of gravity, tending always to return the fluid surface towards its horizontal equilibrium position, and the disturbances are caused by wind and/or pressure variations. The characteristics of the seiche are mainly determined by the geometry of the containing basin and type of fluid within it.

For the simplest geometry, a rectangular basin of length L and uniform depth H , the hydrodynamic theory of seiches was developed as early as 1824 by Merian (referenced in

Wilson 1972). The period of the dominant mode of oscillation in those simple basins obeys the expression

$$T = \frac{2L}{\sqrt{gH}} \quad (\text{B.1})$$

referred to as Merian's formula, in which g is the acceleration of gravity. Although Eq. B.1 can be used to estimate the period of the dominant seiche in bodies of water of irregular depth⁵², its applicability to lakes with multiple basins or a complex shape is very limited. It is the objective in this appendix to present a finite element model that solves the mathematical problem governing the frequency and the spatial structure of seiches in enclosed basins of arbitrary shape and bathymetry. After a brief presentation of the hydrodynamic theory of seiches (section B.2), the finite element model is introduced and tested against the analytical solutions of the problem in a simple rectangular basin (section B.3). The numerical algorithm is then applied to study the barotropic seiches in Clear Lake, a multiple-basin lake in northern California. The results from the numerical model for Clear Lake are analyzed together with observations gathered in the field during several data collection campaigns conducted in 1997 and 1999 (see chapter 7).

B.2. Hydrodynamic theory of seiches

B.2.1. Standing waves

Standing waves form as a result of the superposition of two progressive waves of the same amplitude A , wavelength λ and period T that propagate in opposite directions. In the

⁵² One only needs to use the mean geometric properties of the basin in Eq. B.1.

particular case of waves traveling in a 1-D domain, the surface displacement ζ that results from the superposition of two such progressive gravity waves is

$$\zeta = A\cos(kx - \omega t) + A\cos(kx + \omega t) = 2A\cos(kx)\cos(\omega t) \quad (\text{B.2})$$

where ω ($= 2\pi/T$) is the frequency and k ($= 2\pi/\lambda$) is the wave number. It follows from Eq. B.2 that the surface displacement is zero for $kx = \pm \pi/2, \pm 3\pi/2, \dots$ and will be maximum at $kx = 0, \pm \pi, \pm 2\pi, \dots$. Points (or lines in 2-D) of zero surface displacement are called *nodes*, while those other points where the surface undergoes oscillations of maximum amplitude are referred to as *antinodes*. The free surface does not propagate but simply oscillates up and down with frequency ω , keeping the nodal points fixed.

In limited water bodies, standing waves in the free surface form by the reflection of long progressive surface gravity waves from the boundaries. A seiche is a special case of standing wave that develops as a result of long waves reflecting at two vertical end boundaries, and in which only certain discrete values of wavelengths and frequencies are allowed (Kundu 1990, p. 208). In the particular case of a rectangular basin of length L and uniform depth H the allowable wavelengths of the seiches are

$$\lambda = \frac{2L}{n+1} \quad n = 0, 1, 2, \dots \quad (\text{B.3})$$

The allowed periods can be found through Eq. B.4 linking the phase speed c of long surface gravity waves ($= \sqrt{gH}$), their wavelength λ and period T (Kundu 1990, p.203):

$$T = \frac{\lambda}{c} = \frac{\lambda}{\sqrt{gH}} = \frac{2L}{(n+1)\sqrt{gH}} \quad n = 0, 1, 2, \dots \quad (\text{B.4})$$

For the largest wavelength seiche ($\lambda = 2L$), Eq. B.4 becomes Merian's formula. This seiche is the fundamental mode of oscillation in lakes and has its unique node located at $x = L/2$. The oscillations would be binodal if $\lambda = L$, trinodal if $\lambda = 2L/3$, or, in general, n -nodal if $\lambda = 2L/n$. Many of these higher modes of oscillation coexist with the fundamental seiche in lakes.

B.2.2. Governing equations

The equations governing the motion of a water particle in a natural lake are the continuity (or mass conservation) equation for incompressible fluids and the Navier-Stokes equations for momentum conservation (Lynch 1986). Those equations can be dramatically simplified by invoking assumptions such as the hydrostatic or Boussinesq approximations. For the study of surface gravity waves those equations can be integrated over the depth to yield the 2-D shallow water equations. If one further assumes that frictional and Coriolis effects are negligible, and one restricts the search to oscillations of small amplitude, the governing continuity and momentum equations reduce to

$$\begin{aligned} \frac{\partial U}{\partial t} + gH \frac{\partial \zeta}{\partial x} &= 0 \\ \frac{\partial V}{\partial t} + gH \frac{\partial \zeta}{\partial y} &= 0 \\ \frac{\partial \zeta}{\partial t} + \frac{\partial U}{\partial x} + \frac{\partial V}{\partial y} &= 0 \end{aligned} \quad (\text{B.5})$$

where ζ is the deviation of the water surface elevation from its equilibrium position, H is the depth of the water column when the surface is in equilibrium, g is the acceleration of gravity, U is the depth integrated velocity in the x direction and V is the depth integrated velocity in the y direction. The governing equations may be combined to yield a single equation in ζ :

$$\frac{\partial^2 \zeta}{\partial t^2} = g \left[\frac{\partial}{\partial x} \left(H \frac{\partial \zeta}{\partial x} \right) + \frac{\partial}{\partial y} \left(H \frac{\partial \zeta}{\partial y} \right) \right] \quad (\text{B.6})$$

If the solution is assumed to be of the form $\zeta = \zeta_0 \exp[i \omega t]$, with $\Gamma^2 = -1$, Eq.B.6 becomes,

$$\frac{\partial}{\partial x} \left(H \frac{\partial \zeta}{\partial x} \right) + \frac{\partial}{\partial y} \left(H \frac{\partial \zeta}{\partial y} \right) + \left(\frac{\omega^2}{g} \right) \zeta = 0 \quad (\text{B.7})$$

where ω is the natural frequency of oscillation of the basin ($= 2\pi/T$ where T is the period of the oscillation). The governing equation is subject to appropriate boundary condition, which in the case of enclosed basins (lakes) is given by

$$\frac{\partial \zeta}{\partial n} = 0 \quad (\text{B.8})$$

where n denotes the normal direction; this boundary condition states that there can be no flow across the vertical boundary surrounding the basin. Equations B.7 and B.8 describe an eigenvalue problem, which can be solved by first discretizing the governing differential equations into a set of algebraic equations and then solving for the eigenvalues and eigenvectors of the resulting matrix using standard procedures from

linear algebra. The problem is fully determined by the shape of the basin and the depth and type of fluid within it.

Once the spatial structure of the water surface oscillations is determined, Eqs. B.5 give the structure of the oscillation in the velocity field. In general, the nodes in the water surface oscillations correspond to the antinodes of the velocity. Similarly, the velocity and water surface elevation at a given point in space are 90 degrees out of phase.

B.3. Numerical finite element solution for seiches in complex basins

The differential eigenvalue problem posed by Eqs. B.7 and B.8 was discretized by using the finite element method (FEM), since this method allows for (1) a more accurate representation of complex domains with fewer computational nodes than with finite difference methods, and (2) a natural and straightforward implementation of the zero Neuman boundary conditions in Eq. B.8. The model uses a standard Galerkin weighted residual formulation for the finite element method (Fletcher 1991, Reddy 1993). In the FEM the computational domain Ω is represented as a collection of simple domains called elements. Over each element, an approximate solution to the problem is constructed in the form

$$\hat{\zeta}(x, y, t) = \sum_{j=1}^n \zeta_j(t) \cdot N_j(x, y), \quad (\text{B.9})$$

where $N_j(x, y)$ are basis functions (also known as shape functions in the engineering literature) and $\zeta_j(t)$ are the approximate values of variable ζ at n points called nodes. The basis functions are linear, and the elements can be either triangles or quadrilaterals, with

the computational nodes located at the vertices. The elements are defined by using an isoparametric approximation, i.e. the basis or interpolation functions are used to map an irregular element in the physical domain into a “master” element of fixed and regular shape.

When the approximate solution (Eq. B.9) is substituted for ζ in the governing equation, Eq. B.7, it will give a residual of the form

$$R(x, y) = \frac{\partial}{\partial x} \left(g \frac{\partial \hat{\zeta}}{\partial x} \right) + \frac{\partial}{\partial y} \left(g \frac{\partial \hat{\zeta}}{\partial y} \right) + \left(\frac{\omega^2}{g} \right) \hat{\zeta} = 0 \quad (\text{B.10})$$

The residual in the Galerkin method is made orthogonal to all of the interpolation functions $N_i(x, y)$, i.e.

$$\int_{\Omega} \left[\frac{\partial}{\partial x} \left(H \frac{\partial \hat{\zeta}}{\partial x} \right) + \frac{\partial}{\partial y} \left(H \frac{\partial \hat{\zeta}}{\partial y} \right) + \left(\frac{\omega^2}{g} \right) \hat{\zeta} \right] \cdot N_i(x, y) \cdot dx \cdot dy = 0 \quad i = 1, \dots, N_{nodes} \quad (\text{B.11})$$

where N_{nodes} represents the number of nodes in the computational domain. Using Green's first identity, the above expression can be simplified to the following form

$$\begin{aligned} \int_{\Omega} R(x, y) N_i(x, y) dx dy &= - \int_{\Omega} g \cdot \left[\frac{\partial \hat{\zeta}}{\partial x} \frac{\partial N_i}{\partial x} + \frac{\partial \hat{\zeta}}{\partial y} \frac{\partial N_i}{\partial y} + \frac{\omega^2}{g} \hat{\zeta} N_i \right] dx dy + \\ &\int_{\partial\Omega} g \cdot \left[\frac{\partial \hat{\zeta}}{\partial x} n_x + \frac{\partial \hat{\zeta}}{\partial y} n_y \right] N_i d\sigma = 0 \quad i = 1, \dots, N_{nodes} \end{aligned} \quad (\text{B.12})$$

where $\delta\Omega$ is the boundary of the domain where normal flux boundary conditions are specified, σ is a generalized variable representing distance along the boundary in a counterclockwise sense, and n_x and n_y are the components of an outwardly directed unit vector that is normal to $\delta\Omega$. For enclosed basins the boundary condition that must be satisfied along the whole boundary of the solution domain is zero normal flux, and thus the second integral term is identically zero.

Combining Eq. B.9 and Eq. B.12, the following algebraic problem results:

$$\mathbf{A}\mathbf{v} = \mathbf{B}\mathbf{v}\mathbf{D} \quad (\text{B.13})$$

Here $\mathbf{A}$ is referred to as the stiffness matrix and $\mathbf{B}$ is the mass matrix. Their elements obey the expressions

$$\begin{aligned} A_{ij} &= -\int_{\Omega} g \left[\frac{\partial N_i}{\partial x} \frac{\partial N_j}{\partial x} + \frac{\partial N_i}{\partial y} \frac{\partial N_j}{\partial y} \right] dx dy \\ B_{ij} &= -\int_{\delta\Omega} g [N_i N_j] dx dy \end{aligned} \quad (\text{B.14})$$

Such an algebraic problem is only satisfied for N_{nodes} values of ω^2/g (the eigenvalues) and N_{nodes} vectors $\mathbf{v}$ (eigenvectors). In Eq. B.13 $\mathbf{D}$ is a diagonal matrix with the eigenvalues along its main diagonal. They determine the characteristic periods for the different modes of barotropic oscillations in the enclosed basin. The components of the eigenvectors ($\mathbf{v}$) represent the unknown amplitudes of the water surface elevation displacements at the nodes of the finite element grid (ζ_j). The integration in the evaluation of the elements of matrices $\mathbf{A}$ and $\mathbf{B}$ is performed numerically using a 2x2 Gauss-Legendre quadrature rule

(Reddy 1993) in the quadrilateral elements and 3-point quadrature for the triangular isoparametric elements. The numerical scheme was implemented in MATLAB® (The Mathworks, Inc.).

The index j in the eigenvector identifies the location of the node in the physical domain through the numbering of the nodes in the numerical grid. The finite element mesh in the applications that follow was generated using RMAGEN (King 1996), a finite element mesh generation program. Other triangular mesh grid generators are freely available on the web (see <http://www-users.informatik.rwth-aachen.de/~roberts/software.html>) and could be used instead. In the case of RMAGEN, the numbering of the nodes in RMAGEN is not designed to optimize the bandwidth of the global stiffness or mass matrix (it is actually designed to optimize the performance of a frontal matrix solver), and the resulting matrices can be extraordinarily ill-conditioned to the search for eigenvalues and eigenvectors. The Cuthill-McKee algorithm (Lucquin and Pironneau 1998) was used to reduce the band width of the matrix, which proved especially useful when dealing with grids that had suffered several modifications on complex geometries. Once reordered, the eigenvalues and eigenvectors are searched using the function EIG in MATLAB® (The Mathworks, Inc.).

The performance of the code was initially verified against the analytical solution for longitudinal seiches in a rectangular basin that was 8192 m long, 512 m wide and 12 m deep. The periods of the different modes of barotropic oscillations follow Eq. B.4. The basin was discretized with triangular equilateral elements with sides 512 m long,

resulting in a total of 32 nodes (16 along the main axis of the basin). Figure B.1 represents the frequency in cycles per hour (cph) of the different modes calculated with the finite element model and with Eq. B.4. The agreement is relatively good for the fundamental and lower oscillation modes and deteriorates for the higher harmonics. In what follows, the code is called FEMSEICHE, which stands for Finite Element Model Seiche program.

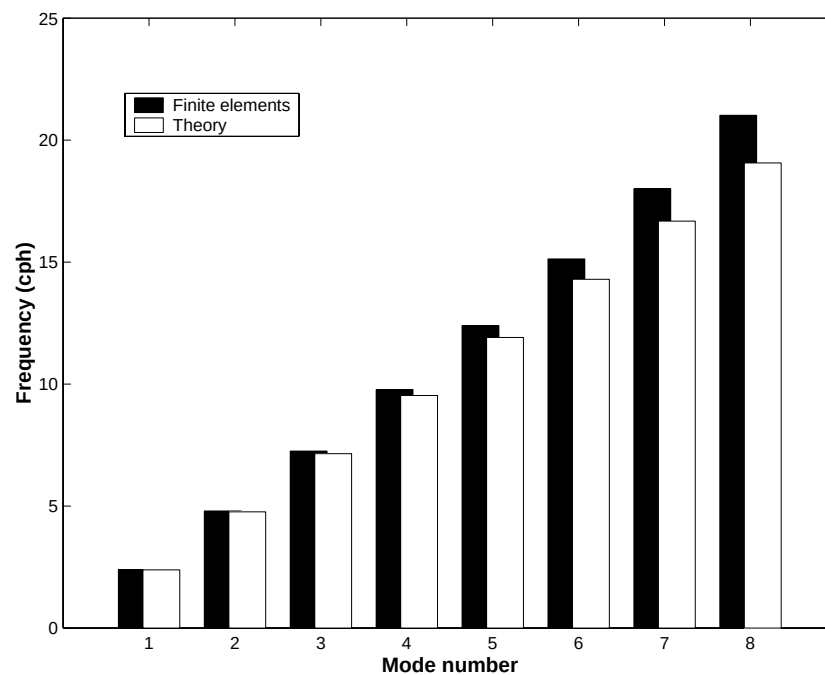


Figure B.1. Frequency (cycles per hour or cph) for the different modes of oscillation of the free surface in a rectangular basin of uniform depth. Dark columns represent the frequency for different modes of oscillation, as calculated with the finite element model, and the white columns represent the frequency calculated with Merian's formula (Eq. B.1).

B.4. The seiches of Clear Lake

The numerical algorithm is applied here to analyze barotropic seiches in Clear Lake.

Given the complex geometry of the lake, this application is a very severe test for the

model. The results of the numerical model are validated by using observations gathered in Clear Lake during several data collection campaigns carried out in 1997 and 1999.

B.4.3. Fingerprinting barotropic seiches in velocity records

The problem of identifying the characteristic frequency of the modes of oscillations, and their spatial structure, was initially motivated by the need to interpret velocity observations gathered in 1995 at the Narrows (e.g. Lynch and Schladow 1996). A strong peak with a periodicity of about 3 hours was revealed in the Fourier analysis of the velocity records. The 3-hour periodicity was originally interpreted (see Lynch and Schladow 1996, p. 75) as the signature of counter rotating vortices being shed by topographic features and advected through the Narrows⁵³. None of the fundamental seiches for each of the arms, estimated with Merian's formula and their mean geometric characteristics, was near the 3-h periodicity observed in the data. The same peak appeared in the power spectrum density of the velocity records from the ADP and the Argonauts located at the Narrows (stations SI.4 and SI.5) during August 1997. A careful analysis of the 1997 data set showed that the 3-h signal in the velocity records, isolated using a band-pass Butterworth filter, was in phase for all depths of the water column (Fig. B.2). The oscillations were excited by the wind events, decreasing their magnitudes thereon.

⁵³ This hypothesis was tested and rejected using velocity information collected at the Narrows during the experiment in August 1999. In that experiment an ADP was located at the southernmost end of the strait, and two point velocity sensors (Argonauts) were deployed about 150 m north of the ADP. The filtered 3-h signals from the north and the south, which should have been out of phase if the hypothesis of vortex shedding had been correct, did appear to be in phase.

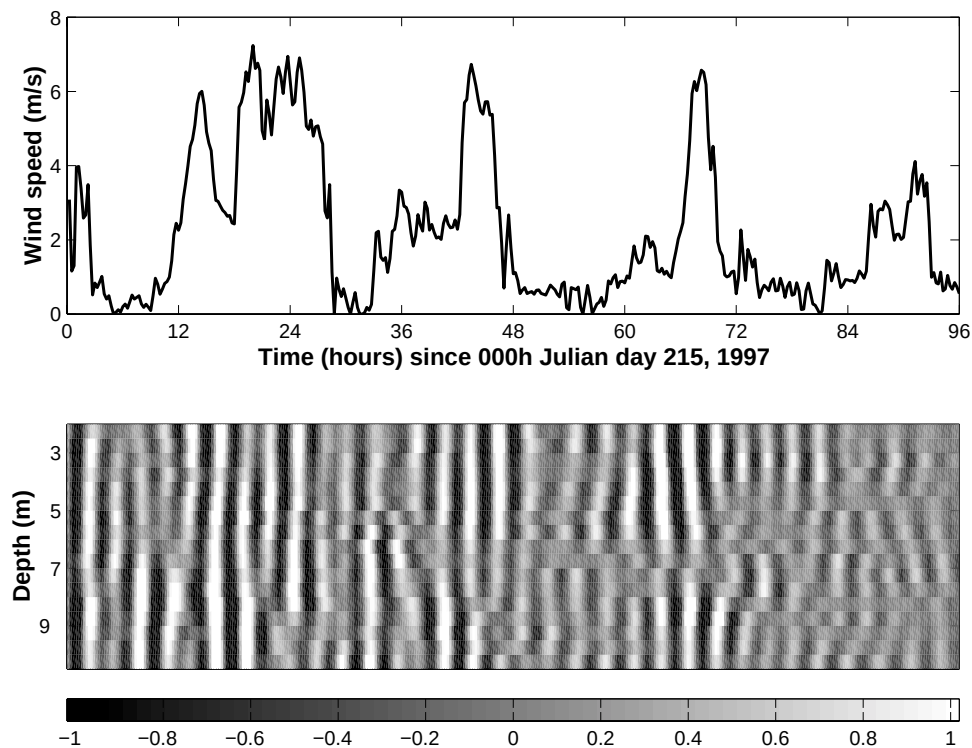


Figure B.2. Time series of wind speed (top) and bandpass filtered velocity profiles at the Narrows in Clear Lake (bottom) during 4 days in August 1997. The magnitude of the filtered velocity component is normalized and is represented by the gray scale.

One approach to the problem of identifying the nature of the oscillations would be to run a complete 3-D or 2-D vertically-averaged hydrodynamic model. As an example RMA10 (King 1998) was run in 2-D depth-averaged mode using a constant and uniform wind forcing. As shown by Csanady 1982, a constant and uniform wind excites a series of standing oscillations of the same characteristics as the seiches (free oscillations). In Fig. B.3 the total energy of the rotary power spectrum density RPSD (e.g. Gonella 1972) of the simulated velocity signal at the Narrows is compared with that of the velocity measured between 2 and 3 meters below the surface, showing in both cases a peak at a frequency of about 1/3 cycle per hour corresponding to a period of approximately 3

hours. The peak at 24 hours in the RPSD of the velocity measurements reflects the diurnal periodicity of the wind in Clear Lake. Other peaks (with periods of about 1 h and 0.5 h) were evident in the RPSD of the simulated signals.

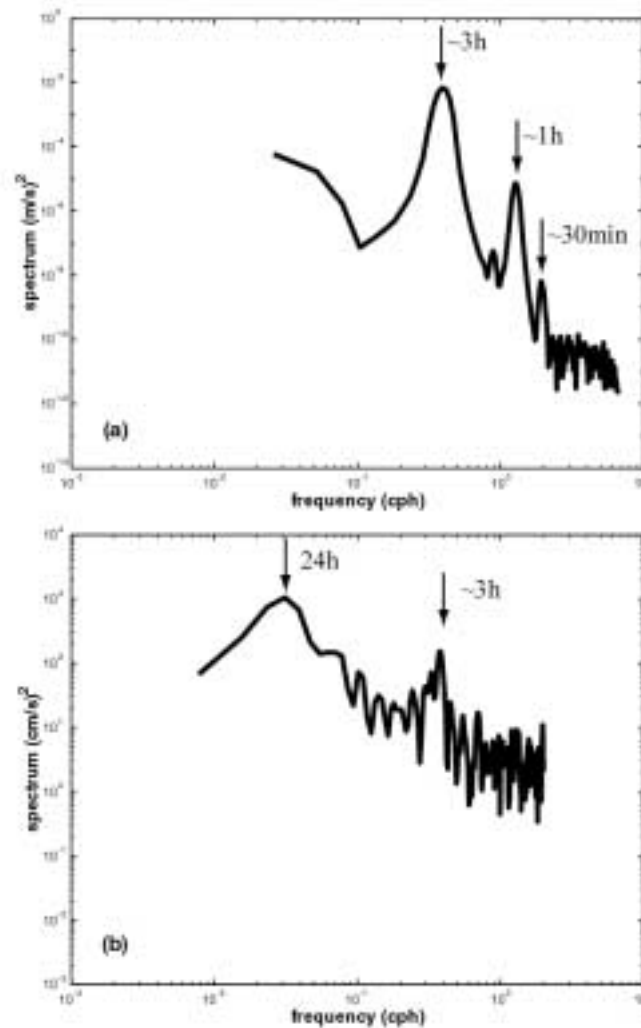


Figure B.3. Total energy in the rotary power spectrum density of velocity at the Narrows. (1) Simulated 2-D depth-averaged velocity. (b) Observed velocity between 2 and 3 meters below the surface.

The simulated water surface elevation and velocity signals were 90 degrees out of phase, suggesting that seiche motions are responsible for those peaks. The spatial structure of the seiches can be seen by analyzing the velocity or water surface elevation time series at

each of the nodes of the computational mesh to define the amplitude of the 3-h period oscillations. Mapping the amplitudes of the oscillations in the free surface elevations reveals in this manner the location of the nodes (lines with zero amplitude oscillations) and antinodes (lines with maximum amplitude oscillations). However, this approach is time consuming.

B.4.4. Results and validation of the numerical model

The computational grid for FEMSEICHE was created with RMAGEN (King 1996) by using the bathymetric information for Clear Lake given in (Bale 1995). A total number of 418 computational nodes were employed to define the geometry of the basin. The frequencies of the first 6 modes are defined in Table B.2, and their spatial structure is plotted in the series of images beginning with Fig. B.4 and continuing to Fig. B.9. The inverse gray intensity represents the amplitude of the free surface oscillations. Primary interest is in the relative magnitude of the oscillations. The dark areas denote areas where the water surface is subject to low, or even zero, amplitude oscillations and depict the nodes of the seiches. The clear areas define areas of pronounced free surface oscillations or antinodes.

<i>MODE</i>	<i>Period (hours)</i>
I	2.49
II	1.09
III	0.84
IV	0.78
V	0.61
VI	0.52

Table B.2. Period (hours) of the first modes of barotropic oscillations in Clear Lake, given by the finite element code described in this appendix.

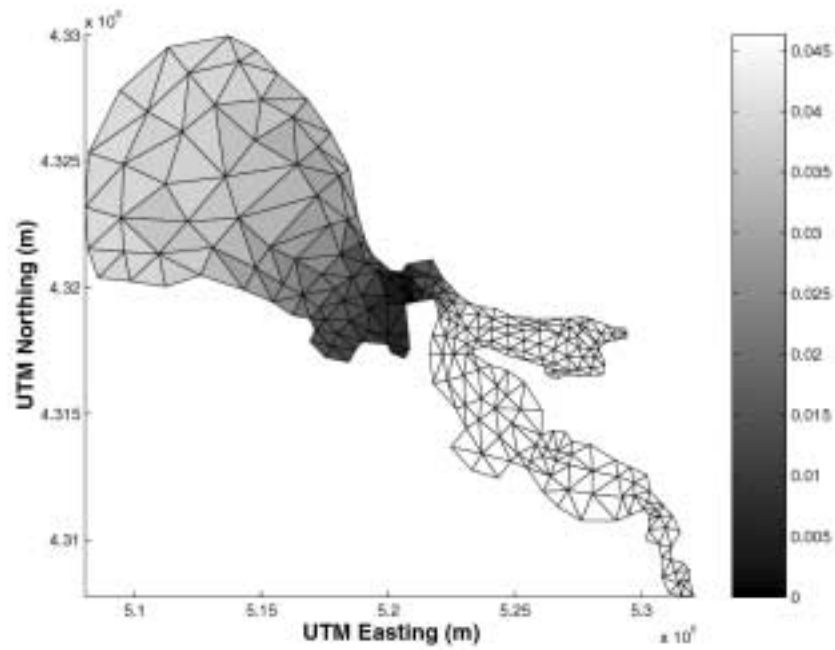


Figure B.4. Spatial distribution of the amplitude of the free surface oscillations for the fundamental seiche (period ~ 2.50 hours), as calculated with the finite element model.

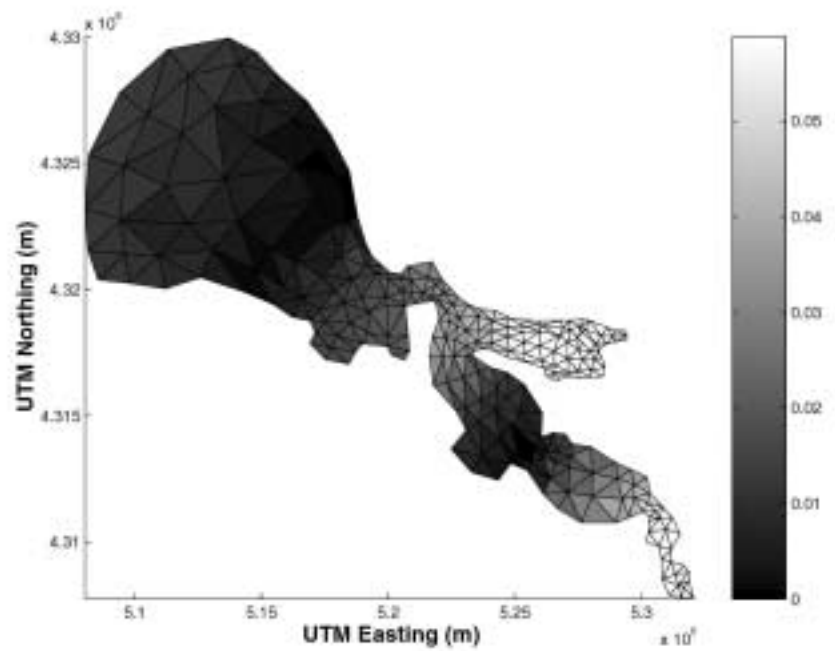


Figure B.5. Spatial distribution of the amplitude of the free surface oscillations for the second barotropic mode (period ~ 1.10 hours), as calculated with the finite element model.

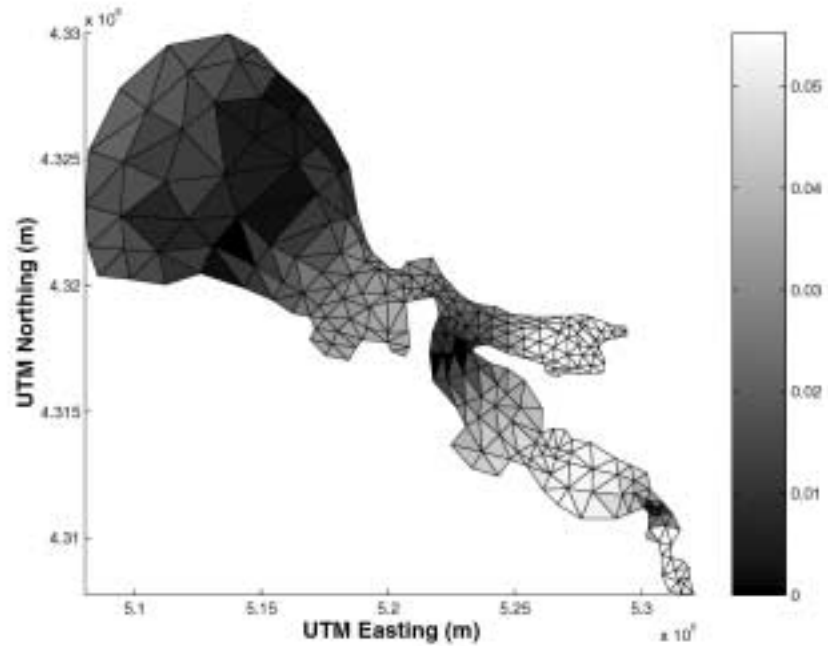


Figure B.6. Spatial distribution of the amplitude of the free surface oscillations for the third barotropic mode (period ~ 0.84 hours), as calculated with the finite element model.

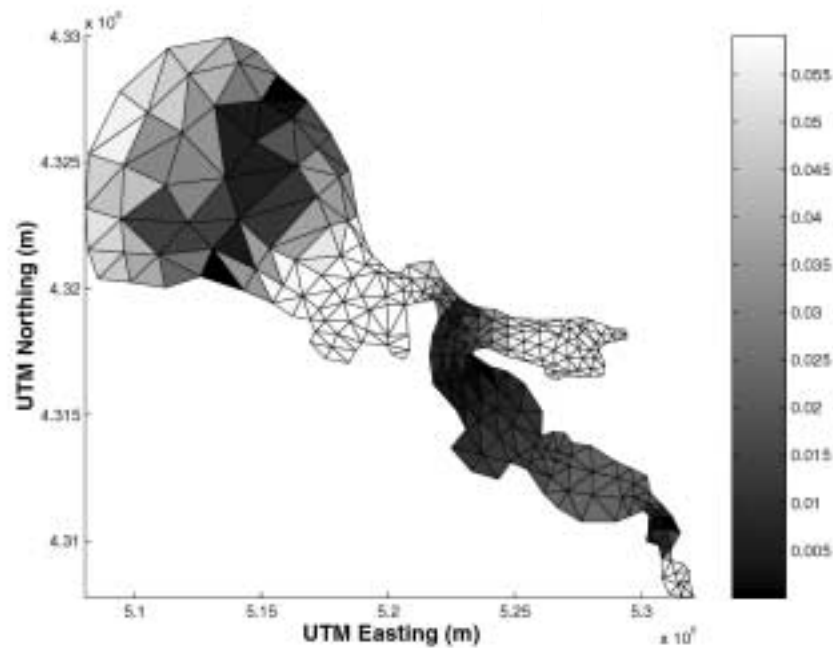


Figure B.7. Spatial distribution of the amplitude of the free surface oscillations for the fourth barotropic mode (period ~ 0.8 hours), as calculated with the finite element model.

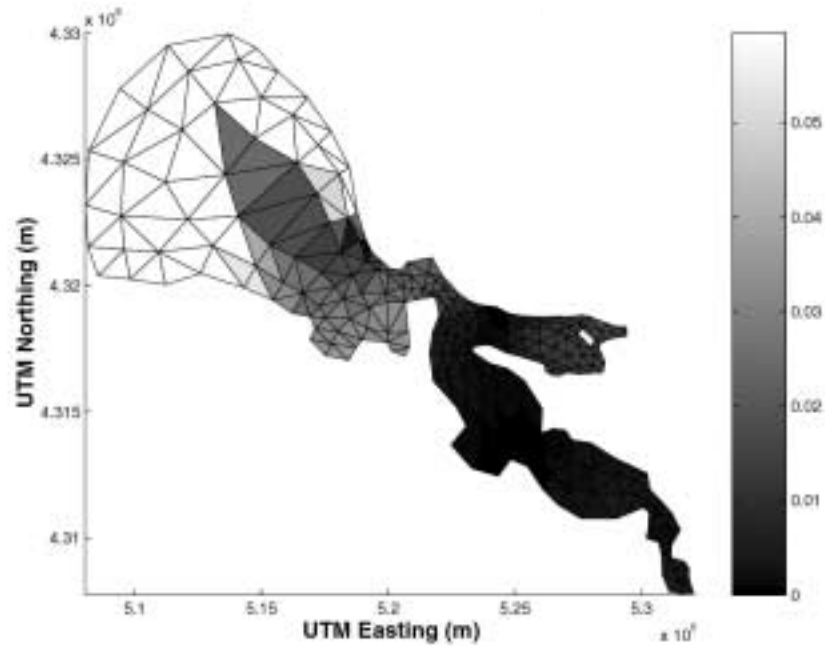


Figure B.8. Spatial distribution of the amplitude of the free surface oscillations for the fifth barotropic mode (period ~ 0.61 hours), as calculated with the finite element model.

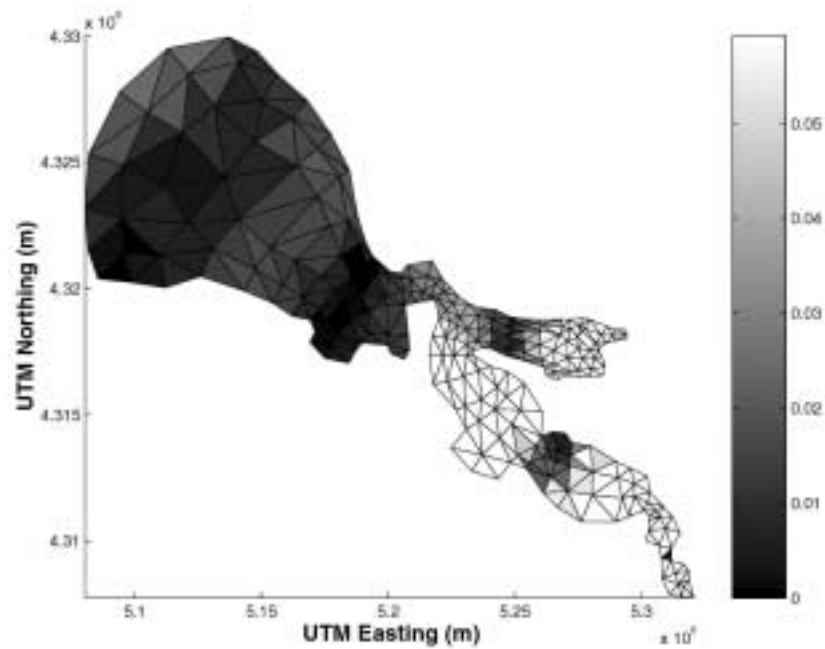


Figure B.9. Spatial distribution of the amplitude of the free surface oscillations for the sixth barotropic mode (period ~ 0.52 hours), as calculated with the finite element model.

Data collected in May 1999 supports the results obtained with the FEMSEICHE model. Some of the Alpha-Omega temperature probes described in section 7.3.2 (Fig. 7.6) were also equipped with pressure sensors. Figures B.10 through B.12 show the power spectrum density of the water surface elevation records collected at the 3 stations (TII.1, TII.2 and TII.3) of experiment II. The pressure records from stations TII.1, TII.2 and TII.3 will be referred to here as P1, P2 and P3 respectively. An additional pressure sensor was located in experiment II at the Narrows (not shown in Fig. 7.6) and will be identified with the symbol P4. The power spectrum of these pressure records at P4 is shown in Fig. B.13. The vertical bars at the top of the figures represent the frequencies of the seiches calculated with the model, which agree with the location of the peaks in the spectrum. In all of the pressure records there exist two peaks in the spectrum with periods of 24 and about 2.5 or 3 hours. The 24-hour peak is dictated by the diurnal wind events, mostly from the northwest, which pushes water downwind and causes the surface to rise at the eastern end of the Oaks Arm. This peak is especially strong at station P3, at the downwind end of the lake. The 3-hour peak corresponds to the first barotropic mode or fundamental seiche. Figure B.4 shows the node for water surface oscillations located in the vicinity of the Narrows, and accordingly the power of the 3-hour peak at station P4 (Fig. B.13) is the lowest among the four stations presented here. Other peaks can be similarly identified in the spectrum, which correspond to higher modes of oscillations. The only periodicity that the code identifies, but which does not appear in the pressure records, is at 0.61 hours. This mode could be either an artifact of the solver or, as shown in Fig. B.8, the amplitude of the oscillations where the pressure sensors are located is minimum and would not appear in the power spectrum. The surface mode of oscillation

with 0.52-hour period (Fig. B.9) provides an excellent example of the spatial variation of the magnitude of the oscillations captured by an array of pressure sensors. A node of this mode is located across the Oaks Arm at its center, near stations P1 and P2, and accordingly the peak at 2 cph does not appear in the spectra of the water surface elevations at stations P1 and P2, though it is quite conspicuous in the spectra from stations P3 and P4.

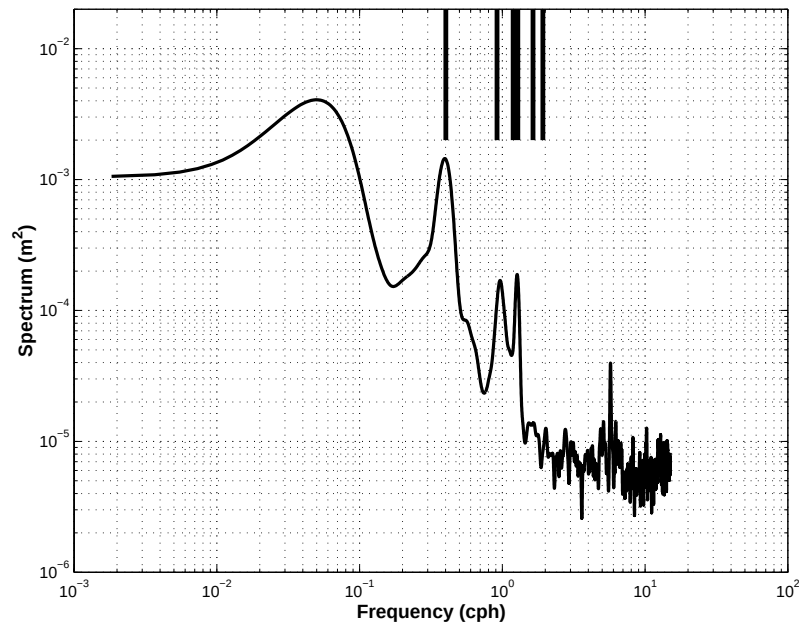


Figure B.10. Power spectrum of the free surface elevation signal at station P1.

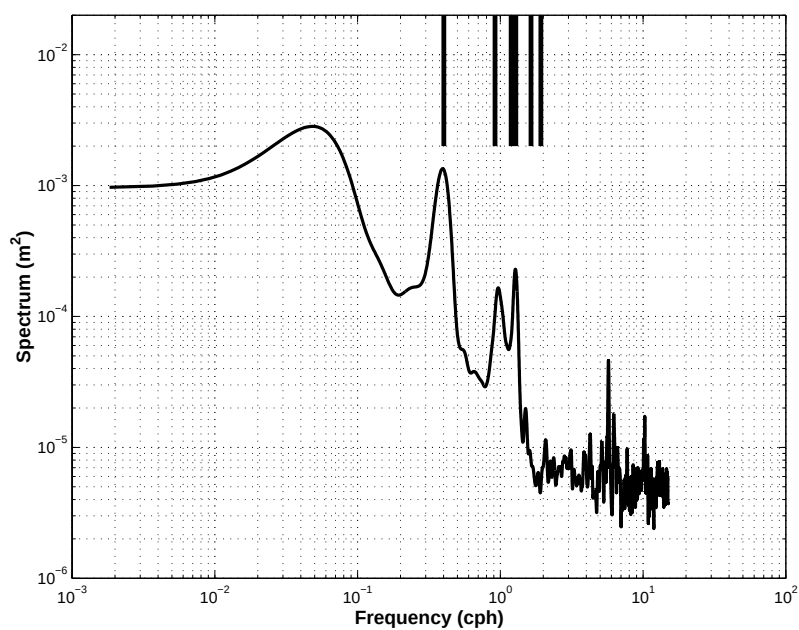


Figure B.11. Power spectrum of the free surface elevation signal at station P2.

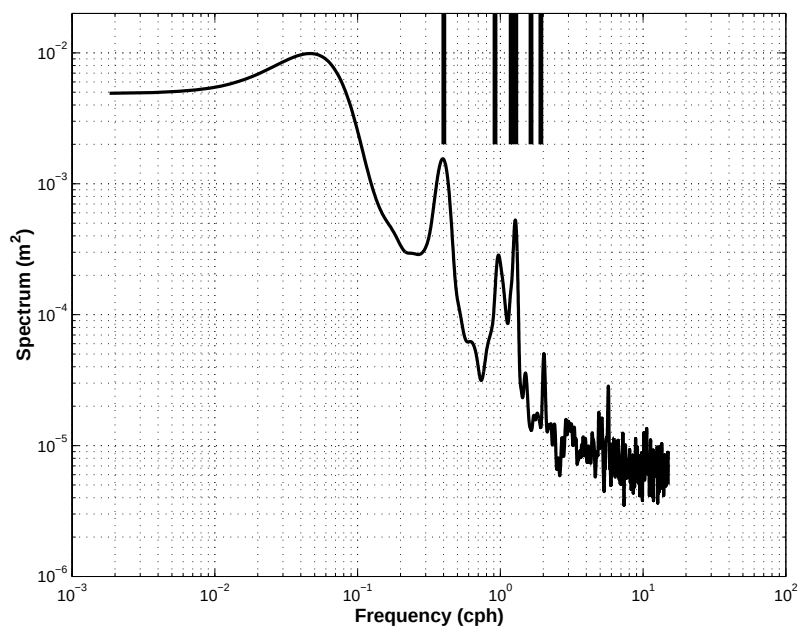


Figure B.12. Power spectrum of the free surface elevation signal at station P3.

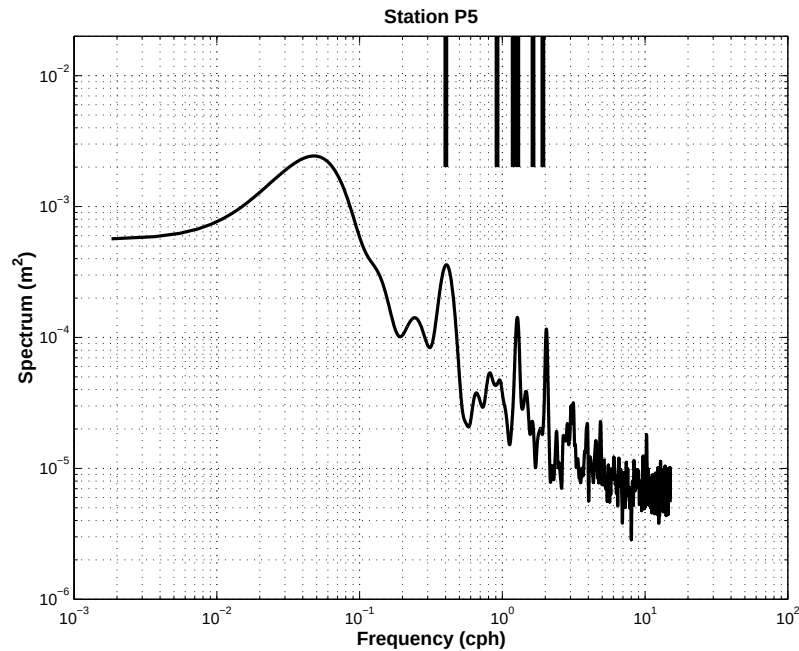


Figure B.13. Power spectrum of the free surface elevation signal at station P4.

B. 5. Summary and conclusions

This appendix presents a numerical finite element method to approximate the frequency (or period) and spatial structure of the standing oscillations (seiches) that the free surface of lakes experiences in response to disturbing mechanisms such as wind or variations in atmospheric pressure. The algorithm FEMSEICHE is particularly useful for the analysis of seiches in lakes of complex geometry where the classical Merian formula (Eq. B.1) with mean geometric properties of the basin does not apply. As an example of its utility, the algorithm is used jointly with observations gathered with velocity and pressure sensors at Clear Lake, California. The code proved in this application to be a valuable tool (1) to identify the nature of some of the spectral peaks that were encountered, and (2) to single out regions in the lake where the sensors should feel the effects of the different modes of free surface oscillations. This analysis may be done with considerably less

effort and time than would be required to run a complete 2-D or 3-D model. As such, it is a useful tool in the planning stage of experiments to determine the best placement of instruments, as well as later in the analysis of data.

APPENDIX C

PRINCIPAL COMPONENT ANALYSIS

C.1. Introduction

Principal component analysis, also known as empirical orthogonal function (EOF) analysis, is widely used in oceanography and in atmospheric sciences to help in the interpretation of the sources of variability within data sets. The technique is based on the eigenvalue decomposition of the correlation matrix of the data set. The theory of the Principal Components Analysis is presented in section C.2 as the discrete version of the well-known Karhunen-Loeve expansion in functional analysis. The mathematical properties of the technique are discussed, and its use in applications such as those presented in this dissertation (chapter 7), i.e. the analysis of velocity profiles, is justified in section C.3. The details of the application to simulated velocity profiles are given in chapter 7.

C.2. The theory of Principal Component Analysis

C.2.1. Analysis of random fields with continuous support

Consider a random field, $T(w, x)$, where x defines the support of the process (time or distance in one dimension, or geographical coordinates in two dimensions) and w indexes the realization in the sample space of the events. Further, it will be assumed that the mean of $T(w, x)$ has been subtracted so that $\langle T(x) \rangle \equiv 0$. The angular brackets $\langle \rangle$ denote expectation value over an ensemble of realizations, i.e.

$$\langle T(x) \rangle = \int T(w, x) f_x(w) dw \quad (C.1)$$

where $f_x(w)$ is the probability density function of the random process at location x . The covariance γ of the field between any two points x and x' is in general a continuous function and is defined as

$$\gamma(x, x') = \langle T(x)T(x') \rangle \quad (C.2)$$

In the **Karhunen-Loeve expansion** each realization of the field $T(w, x)$ is expanded into an infinite series of basis functions $\Phi_j(x)$

$$T(w, x) = \sum_{j=1}^{\infty} T_j(w) \cdot \Phi_j(x) \quad (C.3)$$

where

$$T_j(w) = \frac{1}{A} \int_A T(w, x) \cdot \Phi_j(x) dx \quad (C.4)$$

The basis functions in the Karhunen-Loeve expansion are the eigenfunctions, $\Phi_j(x)$, of the Fredholm integral equation

$$\frac{1}{A} \int_A \gamma(x, x') \cdot \Phi_j(x') \cdot dx' = \lambda_j \cdot \Phi_j(x) \quad (C.5)$$

Here A is the area of the domain, λ_j is the eigenvalue associated with the eigenfunction $\Phi_j(x)$, and j is an integer index. Since the covariance γ (Eq. C.2) is bounded, continuous

and symmetric in its arguments, the following conditions are satisfied (Obled and Creutin 1986), which guarantee the convergence of the series C.3:

1. The eigenvalues λ_j are positive and discrete, although some eigenfunctions may have the same eigenvalue. In general there will be infinitely many.
2. The functions $\Phi_j(x)$ may be normalized, so that they form an orthonormal set

$$\frac{1}{A} \int_A \Phi_i(x) \cdot \Phi_j(x) \cdot dx = \delta_{ij} \quad (\text{C.6})$$

3. The functions $\Phi_j(x)$ are a complete set:

$$\sum_{j=1}^{\infty} \Phi_j(x) \cdot \Phi_j(x') = \delta(x - x') \quad (\text{C.7})$$

In Eq. C.7 $\delta(x-x')$ is the Dirac delta function, and δ_{ij} in Eq. C.6 is the Kronecker delta, both standard operators in functional and tensor analysis (Arfken and Weber 1995).

The probability distribution of the coefficients $T_j(w)$ of Eq. C.3 satisfies

$$\langle T_j \rangle = 0; \quad \langle T_i \cdot T_j \rangle = \lambda_j \delta_{ij} \quad (\text{C.8})$$

i.e., the different components of Eq. C.3 are uncorrelated with each other, and the variance of mode j is the eigenvalue λ_j . From Eqs. C.3 and C.8 it is straightforward to show that the variance of the field at a point is

$$\langle T(x)^2 \rangle = \sum_{j=1}^{\infty} \lambda_j \Phi_j^2(x) \quad (\text{C.9})$$

The average variance integrated over the domain is

$$\frac{1}{A} \int_A \langle T(x)^2 \rangle dx = \sum_{j=1}^{\infty} \lambda_j \quad (\text{C.10})$$

which indicates that the eigenvalue λ_j can be thought as the portion of the total variance ‘explained’ by the eigenfunction $\Phi_j(x)$.

It is convenient to label the eigenvalues in descending order, i.e., $\lambda_1 \geq \lambda_2 \geq \dots$. If only the first n terms in the series (C.3) are retained, this ordering convention guarantees that the largest contributors to the total variance (C.10) are kept. The property of the Karhunen-Loeve decomposition that makes it special is that $\{\Phi_j(x), j = 1, 2, 3 \dots\}$ is the optimum set of basis functions for a given truncation n . That is, for a given n , no other basis set can explain more of the average variance, i.e., generate a larger sub-sum approximating (C.10).

C.2.2. Analysis of random fields with discrete support

When dealing with measurements of a random field $T(w, x)$ at a discrete set of points $\{x_i, i = 1, 2, \dots, m\}$, each realization is expanded in terms of the eigenvectors of the covariance matrix $\{\Phi_j, j = 1, 2, \dots, m\}$, i.e.

$$T(w, x_i) = \sum_{j=1}^m T_j(w) \cdot \Phi_j(x_i) \quad i = 1, 2, \dots, m \quad (\text{C.11})$$

In the particular case of the analysis of a time series of velocity profiles, the discrete set of points is the set of cells in the profile, and each measurement at a given time can be thought of as a realization of a random process. The integral equation (Eq. C.5) is approximated in the discrete case by a finite sum

$$\frac{1}{A} \sum_{k=1}^m \gamma(x_i, x_k) \Delta A_k \Phi_j(x_k) = \lambda_j \Phi_j(x_i) \quad (\text{C.12})$$

where x_i , $i = 1, 2, \dots, m$ are the sequence of m individual grid points in which the domain is split, γ is the covariance matrix, and ΔA_k is an area element containing a point x_k . If one assumes $\Delta A_k = \Delta A$, $k = 1, \dots, m$, Eq. C.12 becomes

$$\sum_{k=1}^m \gamma(x_i, x_k) \Phi_j(x_k) = \frac{A}{\Delta A} \lambda_j \Phi_j(x_i) = l_j \Phi_j(x_i) \quad (\text{C.13})$$

This is the equation that solves for the eigenvectors $\{\Phi_j, j = 1, \dots, m\}$ of the covariance matrix γ of an m -variate random process $\underline{x} = (x_1, \dots, x_m)'$. In that basis the covariance matrix of the data is diagonal, indicating that the random components $T_j(w)$ are uncorrelated with each other and the variance of the component j is the eigenvalue λ_j .

This technique is what the literature refers to as Principal Component Analysis (PCA). The eigenvectors of the covariance matrix (the basis of the expansion) are here referred to as Empirical Orthogonal Functions (EOF), and the expansion coefficients are called Principal Components (PC). All the properties described in this Appendix for the Karhunen-Loeve expansion apply as well to its discrete counterpart. In particular, the following properties are worthy of mention:

1. Different components are uncorrelated with each other.
2. The eigenvalue λ_j can be thought as the portion of the total variance in the data set ‘explained’ by the eigenfunction $\Phi_j(\mathbf{x})$.
3. The Principal Component decomposition is optimal, in the sense that, for any given $n < m$, no other basis set can explain more of the average variance of the discrete random process.

C.3. Why use Principal Component Analysis?

The main reason for using empirical orthogonal decomposition in the analysis of velocity profiles is the fact for linear systems observed at a fixed intervals after evolving from random initial conditions that the normal modes are the EOF (Kundu *et al.* 1975, North *et al.* 1982). The concept of normal modes is briefly introduced in this section, and an example of normal modes of motion for a simplified case is given.

The linearized equations governing the motion of large water bodies with no friction effects are

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0 \quad (\text{C.14})$$

$$\frac{\partial u}{\partial t} - fv = -\frac{1}{\rho_0} \frac{\partial p}{\partial y} \quad (\text{C.15})$$

$$\frac{\partial v}{\partial t} + fu = -\frac{1}{\rho_0} \frac{\partial p}{\partial x} \quad (\text{C.16})$$

$$0 = -\frac{\partial p}{\partial z} - \rho g \quad (\text{C.17})$$

$$\frac{\partial p}{\partial t} - \frac{\rho_0 N^2}{g} w = 0 \quad (\text{C.18})$$

where x, y, z, t represent spatial and temporal coordinates, u, v, w stand for east-west, north-south and vertical velocities, f is the Coriolis parameter, p and ρ represent perturbations of pressure and density from the state of rest (p_0 and ρ_0), and N is the Brunt-Vaisala frequency (a measure of the degree of stratification of the fluid). Using the method of separation of variables, any variable q in Eq. C.14 through Eq. C.18 can be written as

$$q = \sum q_n(x, y, t) \cdot \Phi_n(z) \quad (\text{C.14})$$

i.e., as a sum of various vertical “modes”, which are called normal modes because they are orthogonal to each other. The vertical structure of the mode is described by Φ_n , and q_n describes the horizontal and temporal propagation of the mode.

The normal modes of motion can be found analytically only under very simplified conditions, for example in the absence of topographic variations and assuming a constant variation of density with depth. If the rigid lid approximation (vertical velocity is zero at the surface) is further adopted, the normal modes are (Kundu 1990)

$$\Phi_n(z) = \cos \frac{n\pi z}{H} \quad (\text{C.15})$$

where H represents the depth of the water. The first mode ($n = 0$) is called the barotropic mode, and it represents the behavior of an unstratified fluid responding to spatial gradients in the water surface elevation, when the water moves with the same velocity throughout the depth. The remaining modes are called baroclinic modes and are driven by

the stratification of the fluid. In Fig. C.1 the first baroclinic modes from Eq. C.15 are represented.

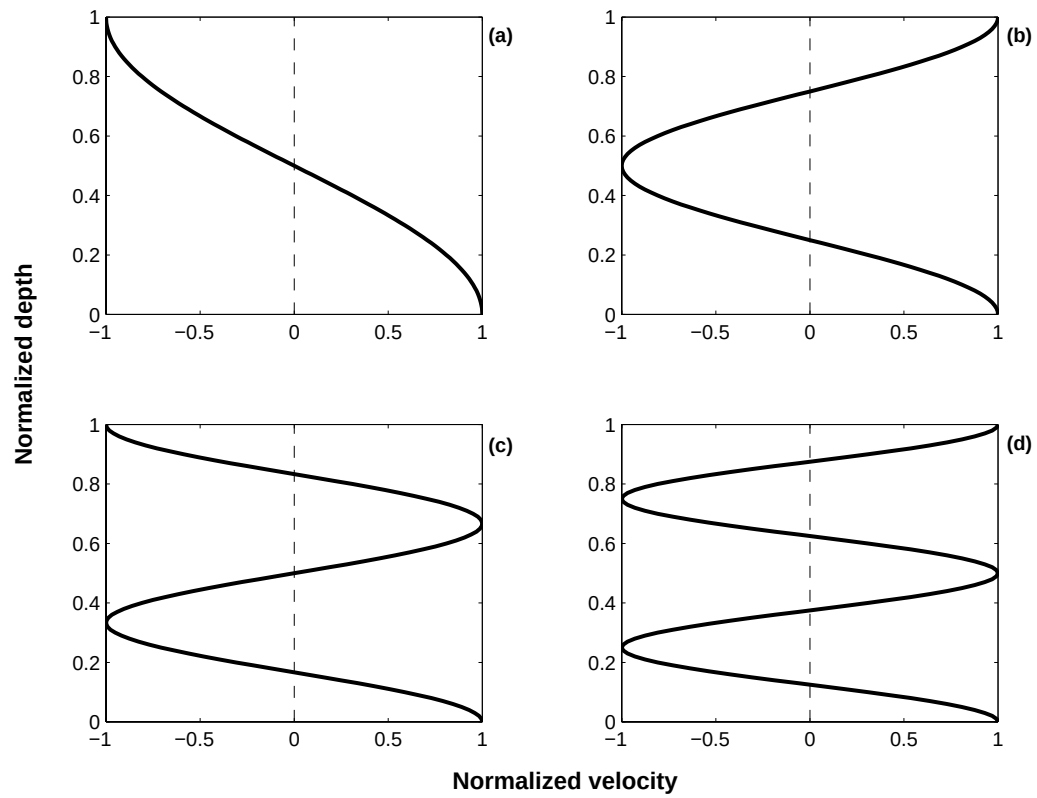


Figure C.1. Vertical structure of baroclinic modes of motion in a linear and frictionless water body: (a) mode 1, (b) mode 2, (c) mode 3, and (d) mode 4.

APPENDIX D

MATLAB SCRIPT FOR DETERMINING THE FREQUENCY OF BASIN-SCALE INTERNAL WAVES

D.1. Introduction

In most lakes the internal dynamics is characterized by the occurrence of oscillatory motions with some well-defined frequency, referred to as internal waves, that cause isopycnals to tilt and oscillate around an equilibrium position (Imberger 1998). At the basin-scale they are identified as Kelvin or Poincaré waves or simple gravitational seiching, and their frequency is mainly determined by the mean stratification properties of the fluid and the geometry of the basin. In this appendix a MATLAB script is presented that discretizes and solves the eigenvalue problem given by Eqs. 8.16 and 8.17. It is assumed here that the stratified water body is contained in a rectangular flat-bottom basin, which constitutes the simplest possible geometry. An analysis of the effects of the basin geometry on the internal wave frequency can be found in Munnich 1996. For further details on the derivation of the governing equations, the reader is referred to Kundu 1990 or Cushman-Roisin 1994.

D.2. The MATLAB script

```
function[IWF, TIW]= modaldecmp(th,z,L,W,f,m,l)

% Given a density profile the script gives the periods for the
% ViH1 mode internal waves in a rotating framework (Coriolis
% parameter = f). L and W are the length and width of the basin
% z coordinate is assumed upwards with zero at the free surface,
% and th is the temperature records for the depths defined in z
%
% Francisco Rueda, January 1999
```

```

% Estimate density from temperature
% -----
rho = 516. * ( 1.93993+temp.*(5.588599e-5-1.108539e-5*temp));
%
% Compute values of Brunt-Vaisala frequency
% -----
g=9.8;rho0=mean(rho);%
g=9.8;rho0=min(rho);%
N2=-g/rho0*diff(rho)/dz;
%
% Form matrix corresponding to 2nd difference operator
% -----
n=length(N2);
D=sparse(1:n,1:n,-2*ones(1,n),n,n);
E=sparse(2:n,1:n-1,ones(1,n-1),n,n);
S=E+D+E';
F=full(S);
%
% Wave numbers for m-th l-th horizontal modes
% -----
km=pi*m/L;
kl=pi*l/W;
wn=sqrt(km^2+kl^2);
%
% Form matrices representing the internal wave eigenvalue problem
% -----
B0=S-dz^2*wn^2*D/(-2);B=-1*full(B0);
A=f^2*S-diag(N2)*dz^2*wn^2;A=-1*full(A);
%
% Solve the generalized eigenvalue problem by QZ method
% -----
[AA, BB, Q, Z, V] = qz(A,B);
ew=(sqrt(diag(AA)./diag(BB)));
IWF=real(ew);
IWF=sort(IWF);TIW=((2*pi)./IWF);

```